

## RM68200 Data Sheet

Single Chip Driver with 16.7M color  
for 800RGBx1280 a-Si TFT LCD driver

**Revision : 2.0**

**Date : Aug. 8, 2014**

## Revision History

| Version No. | Date       | Description                                                                                                                  | Modified By | Checked By |
|-------------|------------|------------------------------------------------------------------------------------------------------------------------------|-------------|------------|
| 0.0         | 2014/01/27 | First Release                                                                                                                | Kurt Huang  | -          |
| 1.0         | 2014/02/21 | 2 <sup>nd</sup> version                                                                                                      | Kurt Huang  | -          |
| 1.1         | 2014/04/08 | Revise for typo                                                                                                              | Dylan Lee   | SP Hung    |
| 1.2         | 2014/04/17 | 1. Remove GAMSET in 0x0D00[2:0]<br>2. Remove register setting 0x3A00<br>3. Add porch setting<br>4. Add dummy pin description | Dylan Lee   | SP Hung    |
| 1.3         | 2014/05/16 | remove 5.3 DSI commend mode                                                                                                  | Kurt Huang  | SP Hung    |
| 2.0         | 2014/08/08 | 3 <sup>rd</sup> version:<br>Revise typo<br>Add bump and pad information<br>Add MCS command description                       | Stern Hsieh | SP Hung    |
|             |            |                                                                                                                              |             |            |
|             |            |                                                                                                                              |             |            |
|             |            |                                                                                                                              |             |            |
|             |            |                                                                                                                              |             |            |
|             |            |                                                                                                                              |             |            |
|             |            |                                                                                                                              |             |            |
|             |            |                                                                                                                              |             |            |
|             |            |                                                                                                                              |             |            |
|             |            |                                                                                                                              |             |            |
|             |            |                                                                                                                              |             |            |
|             |            |                                                                                                                              |             |            |
|             |            |                                                                                                                              |             |            |
|             |            |                                                                                                                              |             |            |

## Table of Contents

|                                                                             |    |
|-----------------------------------------------------------------------------|----|
| Table of Contents.....                                                      | 3  |
| 1. General Description.....                                                 | 2  |
| 2. Features .....                                                           | 7  |
| 3. Block Diagram .....                                                      | 9  |
| 4. Pin Description.....                                                     | 10 |
| 4.1 Power Supply Pins.....                                                  | 10 |
| 4.2 SPI Interface Pins.....                                                 | 10 |
| 4.3 MIPI Interface Pins.....                                                | 11 |
| 4.4 Interface Logic Pins .....                                              | 12 |
| 4.5 CABC Control Pins.....                                                  | 13 |
| 4.6 DC/DC Converter Pins .....                                              | 13 |
| 4.7 Driver Output Pins (Pins for Panel) .....                               | 14 |
| 4.8 Test Pins .....                                                         | 14 |
| 5. Chip and Bump Information .....                                          | 15 |
| 5.1 Alignment Mark and Bump .....                                           | 16 |
| 5.2 Pad Coordinates .....                                                   | 17 |
| 6. Function Description .....                                               | 35 |
| 6.1 Serial-Parallel Interface.....                                          | 35 |
| 6.1.1 Write Cycle and Sequence .....                                        | 35 |
| 6.1.2 Read Cycle and Sequence .....                                         | 36 |
| 6.1.3 Break and Pause Sequence .....                                        | 37 |
| 6.2 Display Serial Interface (DSI) .....                                    | 38 |
| 6.2.1 Protocol.....                                                         | 39 |
| 6.2.2 Processor-to-Peripheral Transactions.....                             | 41 |
| 6.2.3 Peripheral-to-Processor LP Transmission .....                         | 45 |
| 6.2.4 Appropriate Response to Commands and ACK Requests .....               | 46 |
| 6.2.5 Error Reporting Format .....                                          | 47 |
| 6.2.6 Peripheral-to-Processor Transaction – Detail Format Description ..... | 48 |
| 6.2.7 Error Correction Code (ECC).....                                      | 49 |
| 6.3 Tearing Effect Output.....                                              | 50 |
| 6.3.1 Tearing Effect Mode.....                                              | 50 |
| 6.3.2 Timing of Tearing Effect .....                                        | 51 |
| 6.4 Panel Type.....                                                         | 52 |
| 6.4.1 Normal Type.....                                                      | 52 |
| 6.4.2 Zig-Zag Type.....                                                     | 53 |
| 6.5 Independent Gamma Correction Function.....                              | 55 |
| 6.6 Dynamic Backlight Control.....                                          | 56 |
| 6.6.1 Content Adaptive Brightness Control (CABC).....                       | 56 |
| 6.6.2 PWM Control Architecture .....                                        | 57 |
| 6.6.3 Dimming Function for Manual Brightness Control .....                  | 59 |
| 6.6.4 PWM Signal Setting for CABC .....                                     | 62 |
| 6.7 MTP Programming .....                                                   | 64 |
| 6.7.1 Available Pages for MTP Write .....                                   | 64 |
| 6.7.1 MTP Trim Flow.....                                                    | 65 |
| 6.8 Power On/Off Sequence.....                                              | 66 |
| 6.8.1 2-Power mode with Power IC or PFM mode.....                           | 66 |
| 6.8.2 3-Power IC mode .....                                                 | 67 |
| 6.8.3 Uncontrolled Power Off .....                                          | 68 |
| 7. DCS Command .....                                                        | 69 |
| 7.1 DCS Command List.....                                                   | 69 |
| 7.2 DCS Command Description .....                                           | 71 |
| NOP (0000h).....                                                            | 71 |
| SWRESET(0100h): Software Reset .....                                        | 72 |
| RDDID(0400h): Read Display ID.....                                          | 73 |
| RDNUMED(0500h): Read Number of Errors on DSI.....                           | 74 |

|     |                                                                |     |
|-----|----------------------------------------------------------------|-----|
|     | RDDPM (0A00h): Read Display Power Mode .....                   | 75  |
|     | RDDMADCTR (0B00h): Read Display MADCTR .....                   | 76  |
|     | RDDCOLMOD (0C00h): Read Display Pixel Format .....             | 77  |
|     | RDDIM (0D00h): Read Display Image Mode .....                   | 78  |
|     | RDDSM (0E00h): Read Display Signal Mode.....                   | 79  |
|     | RDDSDR (0F00h): Read Display Self-Diagnostic Result.....       | 80  |
|     | SLPIN (1000h): Sleep In .....                                  | 81  |
|     | SLPOUT (1100h): Sleep Out.....                                 | 82  |
|     | PTLON (1200h): Partial Display Mode On.....                    | 83  |
|     | NORON (1300h): Normal Display Mode On .....                    | 84  |
|     | INVOFF (2000H): Display Inversion Off .....                    | 85  |
|     | INVON (2100H): Display Inversion On.....                       | 86  |
|     | ALLPOFF (2200H): All Pixel Off.....                            | 87  |
|     | ALLPON (2300H): All Pixel On.....                              | 88  |
|     | DISPOFF (2800h): Display Off.....                              | 89  |
|     | DISPON (2900h): Display On.....                                | 90  |
|     | PTLAR (3000h): Partial Area.....                               | 91  |
|     | TEOFF (3400h): Tearing Effect Line OFF .....                   | 93  |
|     | TEON (3500h): Tearing Effect Line ON.....                      | 94  |
|     | MADCTR (3600h): Scan Direction Control.....                    | 95  |
|     | IDMOFF (3800h): Idle Mode Off.....                             | 97  |
|     | IDMON (3900h): Enter_idle_mode.....                            | 98  |
|     | STESL(4400h): Set Tear Scan line .....                         | 100 |
|     | GSL (4500h): Get Scan Line .....                               | 101 |
|     | DSTBON (4F00h): Deep Standby Mode On .....                     | 102 |
|     | WRDISBV (5100h): Write Display Brightness .....                | 103 |
|     | RDDISBV (5200h): Read Display Brightness .....                 | 104 |
|     | WRCTRLD (5300h): Write CTRL Display.....                       | 105 |
|     | RDCTRLD (5400h): Read CTRL Display Value .....                 | 107 |
|     | WRCABC (5500h): Write Content Adaptive Brightness Control..... | 109 |
|     | RDCABC (5600h): Read Content Adaptive Brightness Control.....  | 111 |
|     | WRCE (5800h): Set color enhance.....                           | 112 |
|     | RDCE (5900h): Read color enhance .....                         | 113 |
|     | WRCABCMB (5E00h): Write CABC Minimum Brightness.....           | 114 |
|     | RDCABCMB (5F00h): Read CABC Minimum Brightness .....           | 115 |
|     | RDDDBS(A100h) : Read_DDB_Start.....                            | 116 |
|     | RDDDBC(A800h): Read DDB Continuous .....                       | 118 |
|     | RDFCS(AA00h): Read First Checksum .....                        | 120 |
|     | RDCCS(AF00h): Read Continue Checksum.....                      | 121 |
|     | RDID1(DA00h) : Read ID1.....                                   | 122 |
|     | RDID2(DB00h) : Read ID2 .....                                  | 123 |
|     | RDID3(DC00h) : Read ID3.....                                   | 124 |
|     | CMD_Page (FE00h): CMD Mode Switch.....                         | 125 |
|     | CMD_Status(FF00h): Read CMD Page Status.....                   | 127 |
| 8.  | MCS Command .....                                              | 129 |
| 8.1 | MCS Command List .....                                         | 129 |
| 8.2 | MCS Command Description .....                                  | 131 |
|     | DPRSLCTR (0E40h) .....                                         | 131 |
|     | DPRSLCTR (0F40h): DPRSLCTR.....                                | 132 |
|     | STBCTR(1040h~1140h): TE1 Output Setting .....                  | 133 |
|     | STBCTR(1240h): TE1 Output Setting and Zig-Zag Connection ..... | 134 |
|     | TMPSCCTR(1340h): Temperature Sensing Settings.....             | 135 |
|     | TMPSCCTR (1440h) Temperature Sensing Settings.....             | 136 |
|     | SDEQCTR(1740h~1940h): Source EQ Settings.....                  | 137 |
|     | INVCTR(1B40h): Inversion Type.....                             | 138 |
|     | EXT PWR IC Control (2440h~2540h) External PWR IC Control ..... | 139 |
|     | SETAVDD(2740h): PFM Control for AVDD Output .....              | 140 |
|     | SETAVEE(2940h): PFM Control for AVEE Output .....              | 140 |

|       |                                                                   |     |
|-------|-------------------------------------------------------------------|-----|
|       | BT2CTR(2B40h): DDVDL Charge Pump Control.....                     | 141 |
|       | BT3CTR (2F40h): VGH Charge Pump Control.....                      | 142 |
|       | BT3CTR(3140h): VGH Charge Pump Clock.....                         | 143 |
|       | TMPSVGHCTR(3240h~3340h):VGH Temperature Compensation .....        | 144 |
|       | BT4CTR(3440h): VGL Charge Pump Control.....                       | 145 |
|       | BT4CTR(3640h): VGL Charge Pump Clock.....                         | 146 |
|       | TMPSVGLCTR(3740h~3840h): VGL Temperature Compensation .....       | 147 |
|       | BT2CTR(3940h): VCL Charge Pump Control.....                       | 148 |
|       | VCMCTR(3D40h): VCOM Control .....                                 | 149 |
|       | VCMCTR (4540h~4940h): VCOM Output Level Control .....             | 150 |
|       | SETVGMN (5240h): VGMN Control .....                               | 151 |
|       | SETVGSN (5340h): VGSN Control .....                               | 151 |
|       | SETVGMP (5440h): VGMP Control .....                               | 152 |
|       | SETVGSP (5540h): VGSP Control.....                                | 152 |
|       | SETVG (5640h): MSB of VGMP / VGSN / VGMN / VGSP .....             | 153 |
|       | SW_CTRL (5F40h): Interface Control for PFM and MIPI .....         | 153 |
|       | WRDDB (E140h~E440h): Read/Write Supplier ID Code Information..... | 154 |
|       | WRDID (E540h~E740h): Read/Write Device ID Code.....               | 154 |
|       | RDMTPFLG (EE40h): Page Status for MTP Programming .....           | 155 |
|       | MTPCTR (F040h): MTP Program Page Enable Control .....             | 156 |
|       | WMTP (F240h): MTP Programming Start and Power Selection .....     | 156 |
|       | MTPW (F340h~F540h): Key Check for MTP Programming.....            | 157 |
|       | MGRDMTP (F640h): MTP Read Mode Selection.....                     | 157 |
|       | Gamma VP1~VP16(6080h~6F80h): Positive Gamma Setting .....         | 158 |
|       | Gamma VN1~VN16(7080h~7F80h): Negative Gamma Setting.....          | 159 |
| 9.    | Electrical Characteristics .....                                  | 160 |
| 9.1   | Absolute Maximum Ratings.....                                     | 160 |
| 9.2   | ESD Protection Level.....                                         | 160 |
| 9.3   | Latch-Up Protection Level .....                                   | 160 |
| 9.4   | DC Characteristics.....                                           | 161 |
| 9.4.1 | Basic Characteristics .....                                       | 161 |
| 9.5   | MIPI Characteristics .....                                        | 162 |
| 9.5.1 | DC Specifications High-Speed Receiver Specification.....          | 162 |
| 9.5.2 | Forward high speed transmissions.....                             | 163 |
| 9.5.3 | Low power transceiver specifications .....                        | 164 |
| 9.5.4 | DSI Timing Characteristics .....                                  | 165 |
| 9.5.5 | DSI Video Mode Interface Timing .....                             | 168 |
| 9.6   | SPI Timing Characteristics .....                                  | 169 |
| 9.7   | Reset Timing Characteristics .....                                | 170 |

## **1. General Description**

RM68200 is a single-chip solution for a-Si TFT LCD that incorporates gate drivers and is capable of driving different panel resolutions as shown in the following section. This device does not comprise of internal CGRAM that is often used as frame buffer. A timing controller, level-shifters for different power domains, and power supply generation circuit for driving liquid crystal are all built in this device.

RM68200 supports both MIPI Interface and serial peripheral interfaces (SPI) .It is also able to make gamma correction settings for RGB dots separately so as to allow begin adjustments according to panel characteristics, resulting in higher display qualities. Both sleep mode and deep standby mode are also supported for lower power consumption.

This LSI is suitable for small or medium-sized portable mobile solutions requiring long-term driving capabilities, including bi-directional pagers, digital audio players, cellular phones and handheld PDA.

CONFIDENTIAL

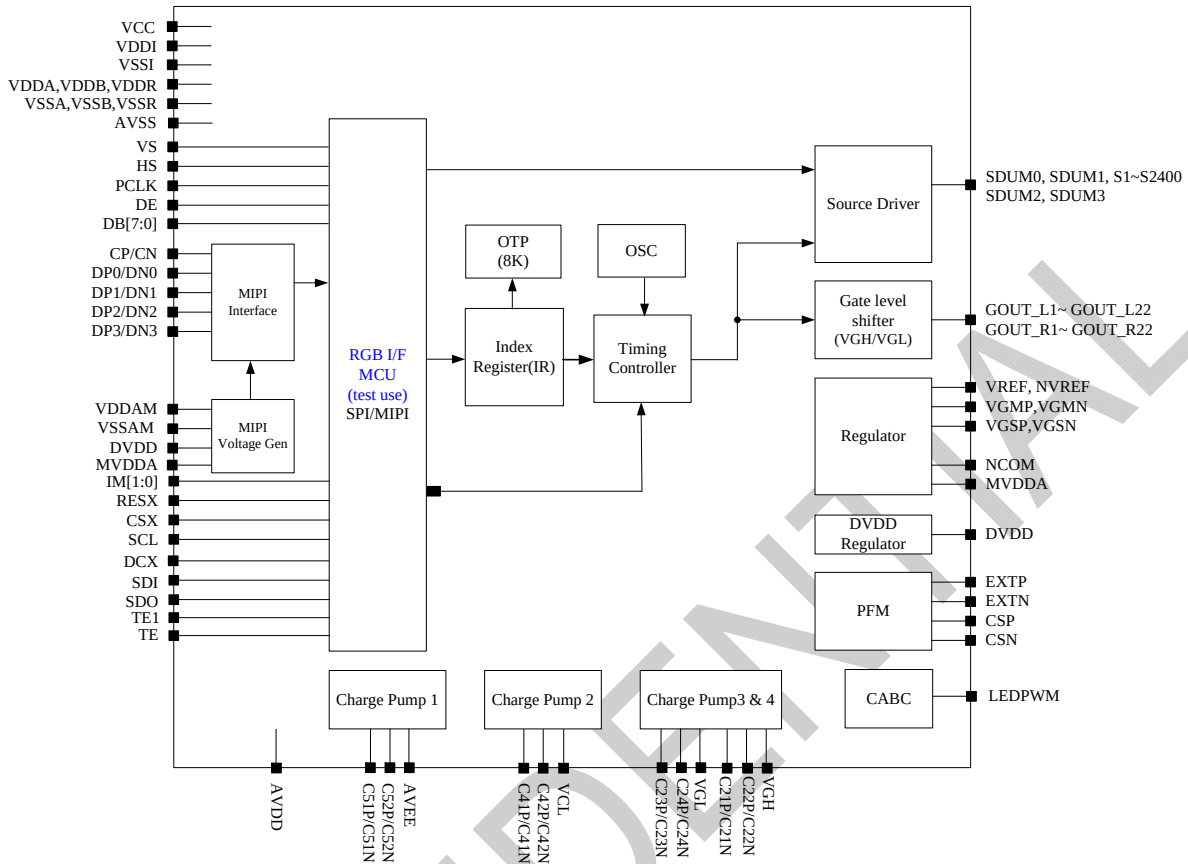
## 2. Features

- Single chip WXGAA-Si TFT LCD controller/driver
- Display resolution option
  - 800RGB x (480+4xNL)
  - 768RGB x (480+4xNL)
  - 720RGB x (480+4xNL)
  - 640RGB x (480+4xNL)
  - 600RGB x (480+4xNL)
  - 540RGB x (480+4xNL)
- Display mode (Color mode)
  - Full color mode: 16.7M-colors (8/8/8)
  - Reduce color mode: 262K colors (6/6/6)
  - Reduce color mode: 65K colors (5/6/5)
  - Idle mode: 8-colors
- Interface
  - System interface
    - ◆ serial peripheral interface(SPI)
  - Display interface
  - MIPI Display Serial Interface (4 data lane pairs, single lane: 850Mbps)
    - ◆ Support 2, 3 & 4 lane
    - ◆ Maximum total bit rate is 1.7Gbps of 2 data lanes 24-bit data format/  
1.27Gbps of 2 data lane 18-bit data format/ 1.13Gbps of 2 data lane 16-bit data format
    - ◆ Maximum total bit rate is 2Gbps of 3 data lanes 24-bit data format/  
1.5Gbps of 3 data lane 18-bit data format/ 1.33Gbps of 3 data lane 16-bit data format
    - ◆ Maximum total bit rate is 2Gbps of 4 data lanes 24-bit data format/  
1.5Gbps of 4 data lane 18-bit data format/ 1.33Gbps of 4 data lane 16-bit data format
- Data de-compression function
  - MTK 1/2 decompression algorithm
- Abundant color display and drawing functions
  - Programmable  $\gamma$ -correction function for 16.7 million color display
  - Built-in digital separate RGB Gamma
- Low power consumption architecture
  - Deep standby mode
  - Sleep mode
  - 8-color display function
- Color enhancement
- CABC
- Sunlight readable function
- Temperature compensation for VGH/VGL
- Support 1-, 2-, 3-, 4-dot, column and zigzag inversion with DC VCOM
- Support "STB output(TE1) for Touch sync control"

- On chip
  - Internal oscillator for display clock
  - 2400ch source output
  - Supports control signals for gate driver on panel
- Built-in OTP function to adjust panel setting
- Built-in OTP trim power for OTP write
- Logic / interface power supply voltage VDDI = 1.65V ~ 3.3V
- Power supply to the internal logic power regulator circuit VCC = 1.65V ~ 4.8V
- Power Supply for MIPI regulator circuit VDDAM = 1.65V ~ 4.8V
- Analog power supply voltage VDD (=VCI) = 2.5V ~ 4.8V
- Output voltage levels
  - Positive gate driver voltage range for VGH: 13V ~ 20V
  - Negative gate driver voltage range for VGL: -7V ~ -15V
  - Step-up 1,2 output voltage range for AVDD: 4.5V ~ 6.5V, AVEE: -4.5V ~ -6.5V
  - Positive Gamma high/low voltage range for VGMP: 3.0V ~ 6.3V , VGSP: 0V, 0.3V ~ 3.7V
  - Negative Gamma high/low voltage range for VGMPN: -3.0 ~ -6.3V , VGSPN: 0V, -0.3V ~ -3.7V
  - Common electrode voltage range for VCOM: 0.0 ~ -3.5V
- Package: COG
- Bump size versus product naming:
  - Bump size 14um\*86um without bump offset: RM68200
  - Bump size 16um\*64um without bump offset: RM68200GA0
  - Bump size 16um\*64um with bump offset: RM68200GA1



### 3. Block Diagram



- **Interface**

RM68200 supports MIPI DSI interface. MIPI DSI can access both internal command and display data.

- **Grayscale Voltage Generating Circuit**

Grayscale voltage generating circuit generates a drive voltage, which corresponds to grayscale level set in the  $\gamma$  correction register. RM68200 displays 16.7M colors at the maximum.

- **Power Supply Circuit**

The power supply circuit generates supply voltages to drive TFT-LCD panel.

- **Timing Generating**

The timing generator generates timing signals for internal circuits such as the display timing.

- **Oscillator**

RM68200 incorporates an RC oscillator circuit and can be changeable by command settings.

- **Panel Driver Circuit**

The TFT-LCD display driver circuit consists of 2400 source drivers (S1~S2400). Display data are latched when 2400 data are input to the source driver. The shift direction of source output data can be changed by setting commands. Both left-hand side and right-hand side of RM68200 have a set of gate outputs pin, GOUT1 ~ GOUT22. These GOA outputs are 2-level HV logic with high level VGH and low level VGL. Indeed, ground level is the third level for GOA output but only for charge sharing.

## 4. Pin Description

### 4.1 Power Supply Pins

| Signal  | I/O | Function                                                                                                            |
|---------|-----|---------------------------------------------------------------------------------------------------------------------|
| Vddb    | P   | Power supply for DC/DC converter<br>Please refer the description of pin BSTM[2:0] for the voltage connection.       |
| Vdda    | P   | Power supply for analog system<br>Please refer the description of pin BSTM[2:0] for the voltage connection.         |
| VDDR    | P   | Power supply for regulator system<br>Please refer the description of pin BSTM[2:0] for the voltage connection.      |
| VDDAM   | P   | Power supply for MIPI analog regulator system                                                                       |
| VDDI    | P   | Power supply for interface system except MIPI interface                                                             |
| VCC     | P   | Power for DVDD regulator                                                                                            |
| VSSB    | P   | System ground for DC/DC converter                                                                                   |
| VSSA    | P   | System ground for analog system                                                                                     |
| VSSR    | P   | System ground for regulator system                                                                                  |
| VSSAM   | P   | System ground for internal MIPI/MDDI analog system                                                                  |
| VSSI    | P   | System ground for interface system except MIPI interface                                                            |
| DVSS    | P   | System ground for internal digital system                                                                           |
| AVSS    | P   | System ground for source OP system.                                                                                 |
| MTP_PWR | P   | MTP programming power supply pin (7.5V typical)<br>Please left this pin floating if MTP programming is not executed |

### 4.2 SPI Interface Pins

| Signal | I/O | Function                                                                  |
|--------|-----|---------------------------------------------------------------------------|
| CSX    | I   | If not use SPI interface, please connect to VSSI or VDDI or open this pin |
| DCX    | I   | If not use SPI interface, please connect to VSSI or VDDI or open this pin |
| SDI    | I/O | If not use SPI interface, please connect to VSSI or VDDI or open this pin |
| SDO    | O   | If not use SPI interface, please open this pin                            |
| SCL    | I   | If not use SPI interface, please connect to VSSI or VDDI or open this pin |

## 4.3 MIPI Interface Pins

| Signal                   | I/O | Function                                                                                                                                                                                                         |
|--------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| HSSI_CLK_P<br>HSSI_CLK_N | I   | These are differential clock input pins of MIPI interface. Ensure the trace length is shortest so that the COG resistance is less than 10 ohm. If not used, please connect these pins to VSSAM.                  |
| HSSI_D0_P<br>HSSI_D0_N   | I/O | These are physical Lane-0's differential data input pins of MIPI interface. Ensure the trace length is shortest so that the COG resistance is less than 10 ohm. If not used, please connect these pins to VSSAM. |
| HSSI_D1_P<br>HSSI_D1_N   | I   | These are physical Lane-1's differential data input pins of MIPI interface. Ensure the trace length is shortest so that the COG resistance is less than 10 ohm. If not used, please connect these pins to VSSAM. |
| HSSI_D2_P<br>HSSI_D2_N   | I   | These are physical Lane-2's differential data input pins of MIPI interface. Ensure the trace length is shortest so that the COG resistance is less than 10 ohm. If not used, please connect these pins to VSSAM. |
| HSSI_D3_P<br>HSSI_D3_N   | I   | These are physical Lane-3's differential data input pins of MIPI interface. Ensure the trace length is shortest so that the COG resistance is less than 10 ohm. If not used, please connect these pins to VSSAM. |

DSWAP[1:0]  
PSWAP

I

Digital input pins to select HSSI\_D0/D1/D2/D3 data lane's sequence and polarity for either high-speed or low-power interface.

| DSWAP<br>[1:0] | PSWAP | HSSI_D<br>0_N | HSSI_D<br>0_P | HSSI_D<br>1_N | HSSI_D<br>1_P | HSSI_C<br>LK_N | HSSI_C<br>LK_P | HSSI_D<br>2_N | HSSI_D<br>2_P | HSSI_D<br>3_N | HSSI_D<br>3_P |
|----------------|-------|---------------|---------------|---------------|---------------|----------------|----------------|---------------|---------------|---------------|---------------|
| 00             | 0     | D3-           | D3+           | D2-           | D2+           | CLK-           | CLK+           | D1-           | D1+           | D0-           | D0+           |
|                | 1     | D3+           | D3-           | D2+           | D2-           | CLK+           | CLK-           | D1+           | D1-           | D0+           | D0-           |
| 01             | 0     | D0-           | D0+           | D1-           | D1+           | CLK-           | CLK+           | D2-           | D2+           | D3-           | D3+           |
|                | 1     | D0+           | D0-           | D1+           | D1-           | CLK+           | CLK-           | D2+           | D2-           | D3+           | D3-           |
| 10             | 0     | D2-           | D2+           | D1-           | D1+           | CLK-           | CLK+           | D0-           | D0+           | D3-           | D3+           |
|                | 1     | D2+           | D2-           | D1+           | D1-           | CLK+           | CLK-           | D0+           | D0-           | D3+           | D3-           |
| 11             | 0     | D3-           | D3+           | D0-           | D0+           | CLK-           | CLK+           | D1-           | D1+           | D2-           | D2+           |
|                | 1     | D3+           | D3-           | D0+           | D0-           | CLK+           | CLK-           | D1+           | D1-           | D2+           | D2-           |

## 4.4 Interface Logic Pins

| Signal    | I/O                                         | Function                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |               |                                                                                           |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
|-----------|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------------------------------------------------------------------------------------|----|---------------------------------------------|------|--------------------------------------------|----|--------------------------------------------|----|---------------------------------------------|---|---|---|---|-----------------------------------|---|---|---|---|----------|---|---|---|---|----------------------------|---|---|---|-----|---------------------------------------------------------|-----|----------------------------------------------------------|-----|----------|-----|-----------------------------------------------------------------------|-----|-------------------------------------------------------------------------------------------|-----|-------------------------------------------------------------------------------|-----|-----------------------------------------------------------------------|
| RESX      | I                                           | This signal will reset the device and must be applied to properly initialize the chip. Signal is active low.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |               |                                                                                           |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
| IM[1:0]   | I                                           | <div>Interface type selection(Default=2'b00):<table><tr><th>IM[1:0]</th><th>Selected Interface</th></tr><tr><td>00</td><td>MIPI DSI ( HSSI_D0P/N~,HSSI_D3P/N ) + SPI16</td></tr><tr><td>01</td><td>MIPI DSI ( HSSI_D0P/N~,HSSI_D3P/N ) + SPI3</td></tr><tr><td>10</td><td>MIPI DSI ( HSSI_D0P/N~,HSSI_D3P/N ) + SPI4</td></tr><tr><td>11</td><td>MIPI DSI ( HSSI_D0P/N~,HSSI_D3P/N ) + SPI16</td></tr></table></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | IM[1:0]       | Selected Interface                                                                        | 00 | MIPI DSI ( HSSI_D0P/N~,HSSI_D3P/N ) + SPI16 | 01   | MIPI DSI ( HSSI_D0P/N~,HSSI_D3P/N ) + SPI3 | 10 | MIPI DSI ( HSSI_D0P/N~,HSSI_D3P/N ) + SPI4 | 11 | MIPI DSI ( HSSI_D0P/N~,HSSI_D3P/N ) + SPI16 |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
| IM[1:0]   | Selected Interface                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |                                                                                           |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
| 00        | MIPI DSI ( HSSI_D0P/N~,HSSI_D3P/N ) + SPI16 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |                                                                                           |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
| 01        | MIPI DSI ( HSSI_D0P/N~,HSSI_D3P/N ) + SPI3  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |                                                                                           |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
| 10        | MIPI DSI ( HSSI_D0P/N~,HSSI_D3P/N ) + SPI4  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |                                                                                           |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
| 11        | MIPI DSI ( HSSI_D0P/N~,HSSI_D3P/N ) + SPI16 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |                                                                                           |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
| BSTM[2:0] | I                                           | <div>Digital I/O pins for boost mode selection:<table><tr><th colspan="3">BSTM[2:0]</th><th>PFM_TYPE[2:0]</th><th>Mode</th></tr><tr><td>0</td><td>0</td><td>0</td><td>X</td><td>Reserved</td></tr><tr><td>0</td><td>X</td><td>1</td><td>X</td><td>3-power mode (VDDI / AVDD / AVEE)</td></tr><tr><td>0</td><td>1</td><td>0</td><td>X</td><td>Reserved</td></tr><tr><td>1</td><td>X</td><td>0</td><td>X</td><td>Power IC mode (VDDI / VCI)</td></tr><tr><td rowspan="7">1</td><td rowspan="7">X</td><td rowspan="7">1</td><td>000</td><td>2-power mode (VDDI / VCI)<br/>2x power MOS + 1x inductor</td></tr><tr><td>001</td><td>2-power mode (VDDI / VCI)<br/>2x power MOS + 2x inductors</td></tr><tr><td>010</td><td>Reserved</td></tr><tr><td>011</td><td>2-power mode (VDDI / VCI)<br/>AVDD: By PFM<br/>AVEE: By internal C-pump</td></tr><tr><td>100</td><td>2-power mode (VDDI / AVDD)<br/>VCI=AVDD: External power supply<br/>AVEE: By internal C-pump</td></tr><tr><td>101</td><td>2-power mode (VDDI / AVDD)<br/>VCI=AVDD: External power supply<br/>AVEE: By PFM</td></tr><tr><td>110</td><td>2-power mode (VDDI / VCI)<br/>AVDD: By PFM<br/>AVEE: By external C-pump</td></tr></table><div>Note:<br/>(1) PFM_TYPE[2:0] is set by command table and please refer to register 0x5F40[6:4].<br/>(2) Please refer to 0x2440[7:6] for different power IC control schemes.</div></div> | BSTM[2:0]     |                                                                                           |    | PFM_TYPE[2:0]                               | Mode | 0                                          | 0  | 0                                          | X  | Reserved                                    | 0 | X | 1 | X | 3-power mode (VDDI / AVDD / AVEE) | 0 | 1 | 0 | X | Reserved | 1 | X | 0 | X | Power IC mode (VDDI / VCI) | 1 | X | 1 | 000 | 2-power mode (VDDI / VCI)<br>2x power MOS + 1x inductor | 001 | 2-power mode (VDDI / VCI)<br>2x power MOS + 2x inductors | 010 | Reserved | 011 | 2-power mode (VDDI / VCI)<br>AVDD: By PFM<br>AVEE: By internal C-pump | 100 | 2-power mode (VDDI / AVDD)<br>VCI=AVDD: External power supply<br>AVEE: By internal C-pump | 101 | 2-power mode (VDDI / AVDD)<br>VCI=AVDD: External power supply<br>AVEE: By PFM | 110 | 2-power mode (VDDI / VCI)<br>AVDD: By PFM<br>AVEE: By external C-pump |
| BSTM[2:0] |                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | PFM_TYPE[2:0] | Mode                                                                                      |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
| 0         | 0                                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | X             | Reserved                                                                                  |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
| 0         | X                                           | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | X             | 3-power mode (VDDI / AVDD / AVEE)                                                         |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
| 0         | 1                                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | X             | Reserved                                                                                  |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
| 1         | X                                           | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | X             | Power IC mode (VDDI / VCI)                                                                |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
| 1         | X                                           | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 000           | 2-power mode (VDDI / VCI)<br>2x power MOS + 1x inductor                                   |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
|           |                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 001           | 2-power mode (VDDI / VCI)<br>2x power MOS + 2x inductors                                  |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
|           |                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 010           | Reserved                                                                                  |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
|           |                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 011           | 2-power mode (VDDI / VCI)<br>AVDD: By PFM<br>AVEE: By internal C-pump                     |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
|           |                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 100           | 2-power mode (VDDI / AVDD)<br>VCI=AVDD: External power supply<br>AVEE: By internal C-pump |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
|           |                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 101           | 2-power mode (VDDI / AVDD)<br>VCI=AVDD: External power supply<br>AVEE: By PFM             |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
|           |                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 110           | 2-power mode (VDDI / VCI)<br>AVDD: By PFM<br>AVEE: By external C-pump                     |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
| TE        | O                                           | Tearing effect output pin to synchronize MCU to frame writing, activated by S/W command. When this pin is not activated, this pin is output low. If not used, please open this pin.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |               |                                                                                           |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |
| TE1       | O                                           | The pulse output per scan line signal for noise sensing of TP. This pin is output low when it is not activated. If not used, please open this pin.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |               |                                                                                           |    |                                             |      |                                            |    |                                            |    |                                             |   |   |   |   |                                   |   |   |   |   |          |   |   |   |   |                            |   |   |   |     |                                                         |     |                                                          |     |          |     |                                                                       |     |                                                                                           |     |                                                                               |     |                                                                       |

## 4.5 CABC Control Pins

| Signal | I/O | Function                                                                                                                                                                               |
|--------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| LEDPWM | O   | It is a PWM signal for brightness control of the LED backlight. The width of LEDPWM signal is set from 256 values between 0% (Low) and 100% (High). If not used, please open this pin. |

## 4.6 DC/DC Converter Pins

| Signal                    | I/O | Function                                                                                                                                                   |
|---------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AVDD                      | O   | Output voltage from PFM, power IC, or external power supply. Please connect a capacitor for voltage stabilization.                                         |
| AVEE                      | O   | Output voltage from PFM, power IC, internal charge pump step-up circuit 2, or external power supply. Please connect a capacitor for voltage stabilization. |
| C51P, C51N<br>C52P, C52N  | IO  | Nodes of flying capacitors used for internal step-up circuit 2. Please leave these pins open when step-up circuit 2 is not used.                           |
| VCL                       | O   | Output voltage from internal step-up circuit 3, whose power input is VDDB. Please connect a capacitor for voltage stabilization.                           |
| C41P, C41N<br>C42P, C42N  | IO  | Nodes of flying capacitors used for internal step-up circuit 3. Please leave these pins open when step-up circuit 2 is not used.                           |
| VGH                       | O   | Output voltage from internal step-up circuit 4. Please connect a capacitor for voltage stabilization.                                                      |
| C21P, C21N<br>C22P, C22N, | IO  | Nodes of flying capacitors used for internal step-up circuit 4. Please leave these pins open when step-up circuit 2 is not used.                           |
| VGL                       | O   | Output voltage from internal step-up circuit 5. Please connect a capacitor for voltage stabilization.                                                      |
| C23P, C23N<br>C24P, C24N  | IO  | Nodes of flying capacitors used for internal step-up circuit 5. Please leave these pins open when step-up circuit 5 is not used.                           |
| VGMP                      | O   | Regulator output for positive gamma high voltage generator.                                                                                                |
| VGMPN                     | O   | Regulator output for negative gamma high voltage generator.                                                                                                |
| VREF                      | O   | Regulator output for internal reference voltage. Please connect capacitor for voltage stabilization.                                                       |
| EXTP, EXTN                | O   | Control signal for external PFM circuit or power IC.                                                                                                       |
| DVDD                      | O   | Regulator output for internal logic circuits. Please connect a capacitor for voltage stabilization.                                                        |
| MVDDA                     | O   | Regulator output for internal MIPI's HS operation. Please connect a capacitor for voltage stabilization.                                                   |
| VRGH                      | O   | This pin is used as power test pin and connected to VGH internally. Please leave this pin open.                                                            |
| VGLO1,<br>VGLO2           | O   | These pins are used as power test pins and connected to VGL internally. Please leave them open.                                                            |

**4.7 Driver Output Pins (Pins for Panel)**

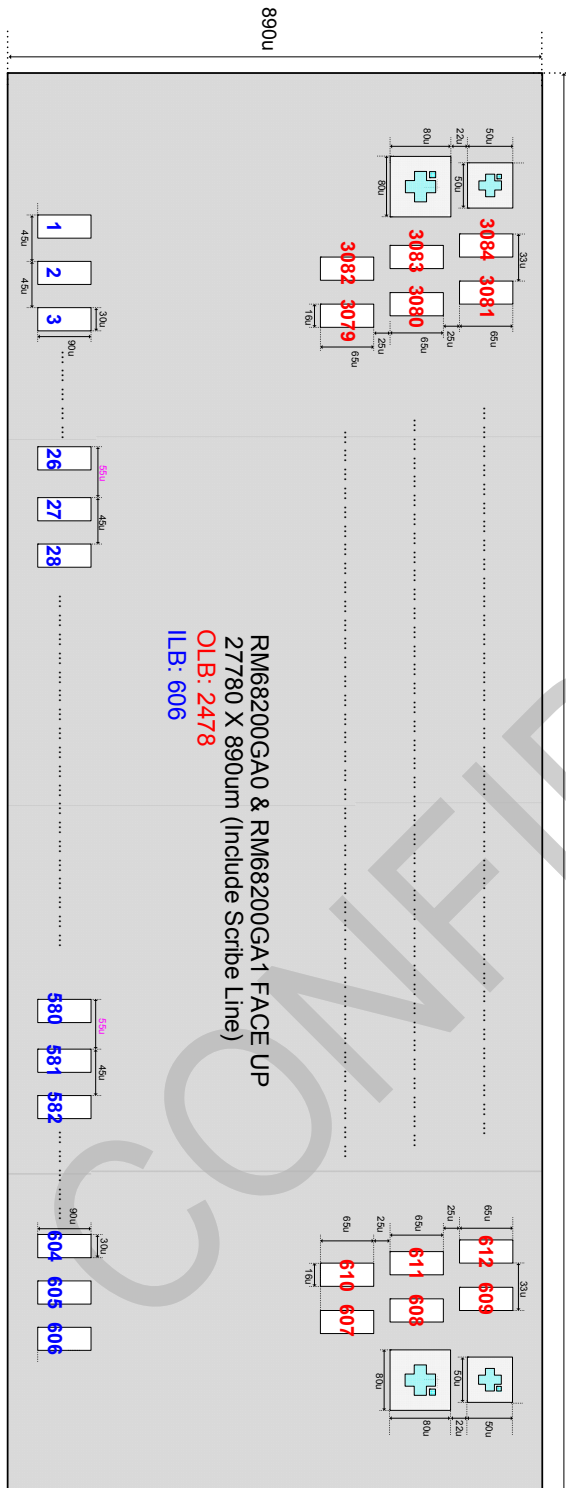
| Signal                                    | I/O | Function                                                                                                    |
|-------------------------------------------|-----|-------------------------------------------------------------------------------------------------------------|
| S1 ~ S2400                                | O   | Pixel electrode driving output.                                                                             |
| SDUM0 ~ SDUM3                             | O   | Dummy source output pins. Please leave them open if no used.                                                |
| GOUT_L1 ~ GOUT_L22,<br>GOUT_R1 ~ GOUT_R22 | O   | Gate control signals for panel                                                                              |
| VCOM                                      | O   | Regulator output for common voltage of panel. Please connect a holding capacitor for voltage stabilization. |

**4.8 Test Pins**

| Signal | I/O | Function                                                                |
|--------|-----|-------------------------------------------------------------------------|
| PL     | I   | Test pin, not release to user, please leave it open or connect to VSSI. |
| VS     | I   | Test pin, not release to user, please leave it open or connect to VSSI. |
| HS     | I   | Test pin, not release to user, please leave it open or connect to VSSI. |
| DE     | I   | Test pin, not release to user, please leave it open or connect to VSSI. |
| D[7:0] | I/O | Test pin, not release to user, please leave it open or connect to VSSI. |
| CSP    | O   | Test pin, not release to user, please leave it open or connect to VSSI. |
| CSN    | O   | Test pin, not release to user, please leave it open or connect to VSSI. |
| DUMMY  | O   | Test pin, not release to user, please leave it open or connect to VSSI. |
| DUMMYN | O   | Test pin, not release to user, please leave it open or connect to VSSI. |
| VCOMR  | O   | Test pin, not release to user, please leave it open or connect to VSSI. |

## 5. Chip and Bump Information

The chip top view for RM68200GA0 and RM68200GA1 is briefly depicted as follows. As to the RM68200, since its size of output bump is different from RM68200GA0 /RM68200GA1, its top-view is a little different from the drawing below.



Chip size: 27780um x 890um

Chip thickness: 240um

AL bump height: 9um or 12um

AL bump input size: 30um x 90um

AL bump output size:

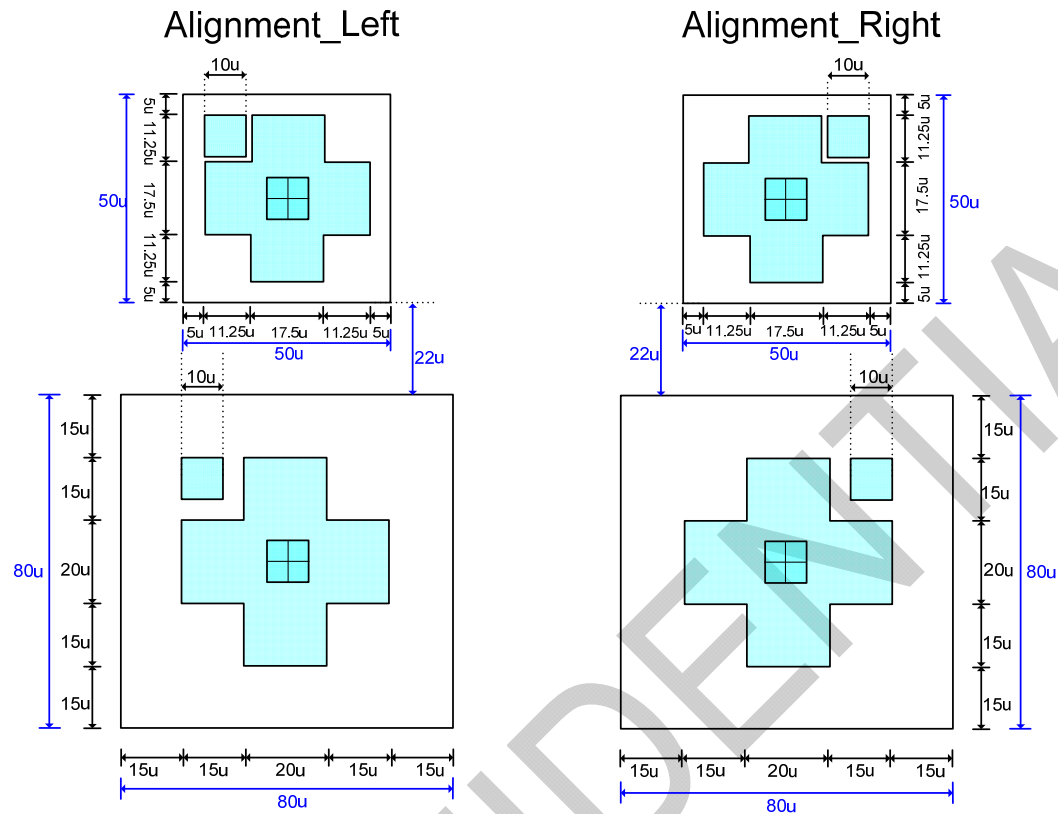
RM68200: 14um x 86um

RM68200GA0: 16um x 65um

RM68200GA1: 16um x 65um

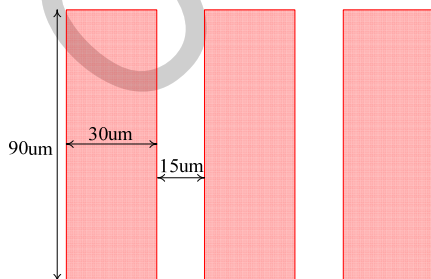
## 5.1 Alignment Mark and Bump

### ● Alignment Mark:

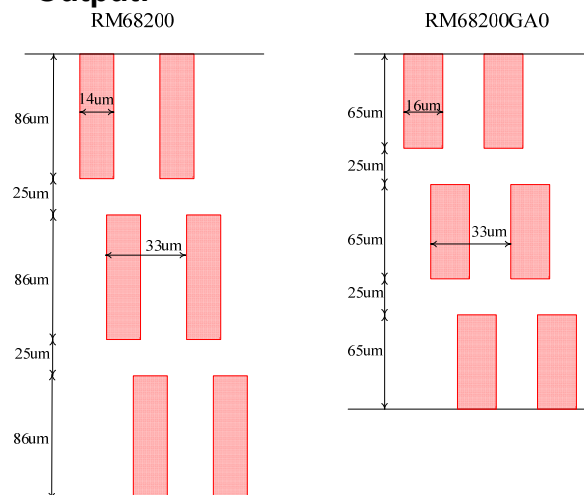


### ● Bump Arrangement:

Input:



Output:





The pad coordinate list below is for RM68200GA0 only. For RM68200, the bump size is 14um\*86um rather than 16um\*65u, so the pad coordinate of output pads are much different from the RM68200GA0. For RM68200GA1, the bump coordinate is modified to compensate the thermal expansion due to high temperature processing for IC bonding. For both cases, though input pad coordinates are complete the same, the output pad coordinates are different and please contact **Raydium** for detailed output pad information.

| NO. | Pad Coordinate |          |        | 61  | HSSI_D1_P  | -10912.5 | -361 | 123 | MVDDA | -8122.5 | -361 |
|-----|----------------|----------|--------|-----|------------|----------|------|-----|-------|---------|------|
|     | PAD NAME       | X-axis   | Y-axis |     |            |          |      |     |       |         |      |
| 1   | DUMMY1         | -13622.5 | -361   | 63  | HSSI_D1_P  | -10822.5 | -361 | 125 | MVDDA | -8032.5 | -361 |
| 2   | GOUT_L1        | -13577.5 | -361   | 64  | HSSI_D1_P  | -10777.5 | -361 | 126 | MVDDA | -7987.5 | -361 |
| 3   | GOUT_L2        | -13532.5 | -361   | 65  | VSSAM      | -10732.5 | -361 | 127 | MVDDA | -7942.5 | -361 |
| 4   | GOUT_L3        | -13487.5 | -361   | 66  | HSSI_CLK_N | -10687.5 | -361 | 128 | VDDAM | -7897.5 | -361 |
| 5   | GOUT_L4        | -13442.5 | -361   | 67  | HSSI_CLK_N | -10642.5 | -361 | 129 | VDDAM | -7852.5 | -361 |
| 6   | GOUT_L5        | -13397.5 | -361   | 68  | HSSI_CLK_N | -10597.5 | -361 | 130 | VDDAM | -7807.5 | -361 |
| 7   | GOUT_L6        | -13352.5 | -361   | 69  | HSSI_CLK_N | -10552.5 | -361 | 131 | VDDAM | -7762.5 | -361 |
| 8   | GOUT_L7        | -13307.5 | -361   | 70  | HSSI_CLK_N | -10507.5 | -361 | 132 | VDDAM | -7717.5 | -361 |
| 9   | GOUT_L8        | -13262.5 | -361   | 71  | HSSI_CLK_N | -10462.5 | -361 | 133 | VDDAM | -7672.5 | -361 |
| 10  | GOUT_L9        | -13217.5 | -361   | 72  | HSSI_CLK_P | -10417.5 | -361 | 134 | VDDAM | -7627.5 | -361 |
| 11  | GOUT_L10       | -13172.5 | -361   | 73  | HSSI_CLK_P | -10372.5 | -361 | 135 | VDDAM | -7582.5 | -361 |
| 12  | GOUT_L11       | -13127.5 | -361   | 74  | HSSI_CLK_P | -10327.5 | -361 | 136 | VDDAM | -7537.5 | -361 |
| 13  | GOUT_L12       | -13082.5 | -361   | 75  | HSSI_CLK_P | -10282.5 | -361 | 137 | VDDAM | -7492.5 | -361 |
| 14  | GOUT_L13       | -13037.5 | -361   | 76  | HSSI_CLK_P | -10237.5 | -361 | 138 | VDDAM | -7447.5 | -361 |
| 15  | GOUT_L14       | -12992.5 | -361   | 77  | HSSI_CLK_P | -10192.5 | -361 | 139 | VDDAM | -7402.5 | -361 |
| 16  | GOUT_L15       | -12947.5 | -361   | 78  | VSSAM      | -10147.5 | -361 | 140 | VCC   | -7357.5 | -361 |
| 17  | GOUT_L16       | -12902.5 | -361   | 79  | HSSI_D2_N  | -10102.5 | -361 | 141 | VCC   | -7312.5 | -361 |
| 18  | GOUT_L17       | -12857.5 | -361   | 80  | HSSI_D2_N  | -10057.5 | -361 | 142 | VCC   | -7267.5 | -361 |
| 19  | GOUT_L18       | -12812.5 | -361   | 81  | HSSI_D2_N  | -10012.5 | -361 | 143 | VCC   | -7222.5 | -361 |
| 20  | GOUT_L19       | -12767.5 | -361   | 82  | HSSI_D2_N  | -9967.5  | -361 | 144 | VCC   | -7177.5 | -361 |
| 21  | GOUT_L20       | -12722.5 | -361   | 83  | HSSI_D2_N  | -9922.5  | -361 | 145 | VCC   | -7132.5 | -361 |
| 22  | GOUT_L21       | -12677.5 | -361   | 84  | HSSI_D2_N  | -9877.5  | -361 | 146 | VCC   | -7087.5 | -361 |
| 23  | GOUT_L22       | -12632.5 | -361   | 85  | HSSI_D2_P  | -9832.5  | -361 | 147 | VCC   | -7042.5 | -361 |
| 24  | VCOM           | -12587.5 | -361   | 86  | HSSI_D2_P  | -9787.5  | -361 | 148 | VCC   | -6997.5 | -361 |
| 25  | VCOM           | -12542.5 | -361   | 87  | HSSI_D2_P  | -9742.5  | -361 | 149 | VCC   | -6952.5 | -361 |
| 26  | VCOM           | -12497.5 | -361   | 88  | HSSI_D2_P  | -9697.5  | -361 | 150 | VCC   | -6907.5 | -361 |
| 27  | AVSS           | -12442.5 | -361   | 89  | HSSI_D2_P  | -9652.5  | -361 | 151 | VCC   | -6862.5 | -361 |
| 28  | AVSS           | -12397.5 | -361   | 90  | HSSI_D2_P  | -9607.5  | -361 | 152 | VCC   | -6817.5 | -361 |
| 29  | AVSS           | -12352.5 | -361   | 91  | VSSAM      | -9562.5  | -361 | 153 | VCC   | -6772.5 | -361 |
| 30  | AVSS           | -12307.5 | -361   | 92  | HSSI_D3_N  | -9517.5  | -361 | 154 | VCC   | -6727.5 | -361 |
| 31  | AVSS           | -12262.5 | -361   | 93  | HSSI_D3_N  | -9472.5  | -361 | 155 | DVDD  | -6682.5 | -361 |
| 32  | AVSS           | -12217.5 | -361   | 94  | HSSI_D3_N  | -9427.5  | -361 | 156 | DVDD  | -6637.5 | -361 |
| 33  | AVSS           | -12172.5 | -361   | 95  | HSSI_D3_N  | -9382.5  | -361 | 157 | DVDD  | -6592.5 | -361 |
| 34  | AVSS           | -12127.5 | -361   | 96  | HSSI_D3_N  | -9337.5  | -361 | 158 | DVDD  | -6547.5 | -361 |
| 35  | AVSS           | -12082.5 | -361   | 97  | HSSI_D3_N  | -9292.5  | -361 | 159 | DVDD  | -6502.5 | -361 |
| 36  | AVSS           | -12037.5 | -361   | 98  | HSSI_D3_P  | -9247.5  | -361 | 160 | DVDD  | -6457.5 | -361 |
| 37  | AVSS           | -11992.5 | -361   | 99  | HSSI_D3_P  | -9202.5  | -361 | 161 | DVDD  | -6412.5 | -361 |
| 38  | AVSS           | -11947.5 | -361   | 100 | HSSI_D3_P  | -9157.5  | -361 | 162 | DVDD  | -6367.5 | -361 |
| 39  | VSSAM          | -11902.5 | -361   | 101 | HSSI_D3_P  | -9112.5  | -361 | 163 | DVDD  | -6322.5 | -361 |
| 40  | HSSI_D0_N      | -11857.5 | -361   | 102 | HSSI_D3_P  | -9067.5  | -361 | 164 | DVDD  | -6277.5 | -361 |
| 41  | HSSI_D0_N      | -11812.5 | -361   | 103 | HSSI_D3_P  | -9022.5  | -361 | 165 | DVDD  | -6232.5 | -361 |
| 42  | HSSI_D0_N      | -11767.5 | -361   | 104 | VSSAM      | -8977.5  | -361 | 166 | DVDD  | -6187.5 | -361 |
| 43  | HSSI_D0_N      | -11722.5 | -361   | 105 | VSSAM      | -8932.5  | -361 | 167 | DVDD  | -6142.5 | -361 |
| 44  | HSSI_D0_N      | -11677.5 | -361   | 106 | VSSAM      | -8887.5  | -361 | 168 | DVDD  | -6097.5 | -361 |
| 45  | HSSI_D0_N      | -11632.5 | -361   | 107 | VSSAM      | -8842.5  | -361 | 169 | DVDD  | -6052.5 | -361 |
| 46  | HSSI_D0_P      | -11587.5 | -361   | 108 | VSSAM      | -8797.5  | -361 | 170 | DVSS  | -6007.5 | -361 |
| 47  | HSSI_D0_P      | -11542.5 | -361   | 109 | VSSAM      | -8752.5  | -361 | 171 | DVSS  | -5962.5 | -361 |
| 48  | HSSI_D0_P      | -11497.5 | -361   | 110 | VSSAM      | -8707.5  | -361 | 172 | DVSS  | -5917.5 | -361 |
| 49  | HSSI_D0_P      | -11452.5 | -361   | 111 | VSSAM      | -8662.5  | -361 | 173 | DVSS  | -5872.5 | -361 |
| 50  | HSSI_D0_P      | -11407.5 | -361   | 112 | VSSAM      | -8617.5  | -361 | 174 | DVSS  | -5827.5 | -361 |
| 51  | HSSI_D0_P      | -11362.5 | -361   | 113 | VSSAM      | -8572.5  | -361 | 175 | DVSS  | -5782.5 | -361 |
| 52  | VSSAM          | -11317.5 | -361   | 114 | VSSAM      | -8527.5  | -361 | 176 | DVSS  | -5737.5 | -361 |
| 53  | HSSI_D1_N      | -11272.5 | -361   | 115 | VSSAM      | -8482.5  | -361 | 177 | DVSS  | -5692.5 | -361 |
| 54  | HSSI_D1_N      | -11227.5 | -361   | 116 | MVDDA      | -8437.5  | -361 | 178 | DVSS  | -5647.5 | -361 |
| 55  | HSSI_D1_N      | -11182.5 | -361   | 117 | MVDDA      | -8392.5  | -361 | 179 | DVSS  | -5602.5 | -361 |
| 56  | HSSI_D1_N      | -11137.5 | -361   | 118 | MVDDA      | -8347.5  | -361 | 180 | DVSS  | -5557.5 | -361 |
| 57  | HSSI_D1_N      | -11092.5 | -361   | 119 | MVDDA      | -8302.5  | -361 | 181 | DVSS  | -5512.5 | -361 |
| 58  | HSSI_D1_N      | -11047.5 | -361   | 120 | MVDDA      | -8257.5  | -361 | 182 | DVSS  | -5467.5 | -361 |
| 59  | HSSI_D1_P      | -11002.5 | -361   | 121 | MVDDA      | -8212.5  | -361 | 183 | DVSS  | -5422.5 | -361 |
| 60  | HSSI_D1_P      | -10957.5 | -361   | 122 | MVDDA      | -8167.5  | -361 | 184 | DVSS  | -5377.5 | -361 |

|     |        |         |      |     |        |         |      |     |        |        |      |
|-----|--------|---------|------|-----|--------|---------|------|-----|--------|--------|------|
| 185 | DUMMYN | -5332.5 | -361 | 247 | D0     | -2542.5 | -361 | 309 | VSS    | 247.5  | -361 |
| 186 | DUMMYN | -5287.5 | -361 | 248 | HS     | -2497.5 | -361 | 310 | DUMMY  | 292.5  | -361 |
| 187 | DUMMYN | -5242.5 | -361 | 249 | VS     | -2452.5 | -361 | 311 | DUMMY  | 337.5  | -361 |
| 188 | DUMMYN | -5197.5 | -361 | 250 | DE     | -2407.5 | -361 | 312 | VDDI   | 382.5  | -361 |
| 189 | DUMMYN | -5152.5 | -361 | 251 | PL     | -2362.5 | -361 | 313 | VDDI   | 427.5  | -361 |
| 190 | DUMMYN | -5107.5 | -361 | 252 | DUMMY  | -2317.5 | -361 | 314 | DUMMY  | 472.5  | -361 |
| 191 | DUMMYN | -5062.5 | -361 | 253 | DUMMY  | -2272.5 | -361 | 315 | DUMMY  | 517.5  | -361 |
| 192 | DUMMYN | -5017.5 | -361 | 254 | DUMMY  | -2227.5 | -361 | 316 | VSS    | 562.5  | -361 |
| 193 | AVDD   | -4972.5 | -361 | 255 | DUMMY  | -2182.5 | -361 | 317 | VSS    | 607.5  | -361 |
| 194 | AVDD   | -4927.5 | -361 | 256 | IM0    | -2137.5 | -361 | 318 | DSWAP1 | 652.5  | -361 |
| 195 | AVDD   | -4882.5 | -361 | 257 | IM0    | -2092.5 | -361 | 319 | DSWAP1 | 697.5  | -361 |
| 196 | AVDD   | -4837.5 | -361 | 258 | IM1    | -2047.5 | -361 | 320 | VDDI   | 742.5  | -361 |
| 197 | AVDD   | -4792.5 | -361 | 259 | IM1    | -2002.5 | -361 | 321 | VDDI   | 787.5  | -361 |
| 198 | AVDD   | -4747.5 | -361 | 260 | DUMMY  | -1957.5 | -361 | 322 | BSTM0  | 832.5  | -361 |
| 199 | AVDD   | -4702.5 | -361 | 261 | DUMMY  | -1912.5 | -361 | 323 | BSTM0  | 877.5  | -361 |
| 200 | VSS    | -4657.5 | -361 | 262 | DUMMY  | -1867.5 | -361 | 324 | BSTM1  | 922.5  | -361 |
| 201 | VSS    | -4612.5 | -361 | 263 | DUMMY  | -1822.5 | -361 | 325 | BSTM1  | 967.5  | -361 |
| 202 | VSS    | -4567.5 | -361 | 264 | DCX    | -1777.5 | -361 | 326 | BSTM2  | 1012.5 | -361 |
| 203 | VSS    | -4522.5 | -361 | 265 | DCX    | -1732.5 | -361 | 327 | BSTM2  | 1057.5 | -361 |
| 204 | VSS    | -4477.5 | -361 | 266 | CSX    | -1687.5 | -361 | 328 | VDDA   | 1102.5 | -361 |
| 205 | VSS    | -4432.5 | -361 | 267 | CSX    | -1642.5 | -361 | 329 | VDDA   | 1147.5 | -361 |
| 206 | VSS    | -4387.5 | -361 | 268 | SCL    | -1597.5 | -361 | 330 | VDDA   | 1192.5 | -361 |
| 207 | VSS    | -4342.5 | -361 | 269 | SCL    | -1552.5 | -361 | 331 | VDDA   | 1237.5 | -361 |
| 208 | C51P   | -4297.5 | -361 | 270 | SDI    | -1507.5 | -361 | 332 | VDDA   | 1282.5 | -361 |
| 209 | C51P   | -4252.5 | -361 | 271 | SDI    | -1462.5 | -361 | 333 | VDDA   | 1327.5 | -361 |
| 210 | C51P   | -4207.5 | -361 | 272 | SDO    | -1417.5 | -361 | 334 | VDDA   | 1372.5 | -361 |
| 211 | C51P   | -4162.5 | -361 | 273 | SDO    | -1372.5 | -361 | 335 | VDDA   | 1417.5 | -361 |
| 212 | C51P   | -4117.5 | -361 | 274 | LEDPWM | -1327.5 | -361 | 336 | VDDA   | 1462.5 | -361 |
| 213 | C51P   | -4072.5 | -361 | 275 | LEDPWM | -1282.5 | -361 | 337 | VDDA   | 1507.5 | -361 |
| 214 | C51N   | -4027.5 | -361 | 276 | LEDPWM | -1237.5 | -361 | 338 | DVDD   | 1552.5 | -361 |
| 215 | C51N   | -3982.5 | -361 | 277 | LEDPWM | -1192.5 | -361 | 339 | DVDD   | 1597.5 | -361 |
| 216 | C51N   | -3937.5 | -361 | 278 | TE     | -1147.5 | -361 | 340 | DVDD   | 1642.5 | -361 |
| 217 | C51N   | -3892.5 | -361 | 279 | TE     | -1102.5 | -361 | 341 | DVDD   | 1687.5 | -361 |
| 218 | C51N   | -3847.5 | -361 | 280 | TE     | -1057.5 | -361 | 342 | DVDD   | 1732.5 | -361 |
| 219 | C51N   | -3802.5 | -361 | 281 | TE     | -1012.5 | -361 | 343 | DVDD   | 1777.5 | -361 |
| 220 | VSS    | -3757.5 | -361 | 282 | TE     | -967.5  | -361 | 344 | DVDD   | 1822.5 | -361 |
| 221 | VSS    | -3712.5 | -361 | 283 | TE     | -922.5  | -361 | 345 | DVDD   | 1867.5 | -361 |
| 222 | C52P   | -3667.5 | -361 | 284 | TE1    | -877.5  | -361 | 346 | DVDD   | 1912.5 | -361 |
| 223 | C52P   | -3622.5 | -361 | 285 | TE1    | -832.5  | -361 | 347 | DVDD   | 1957.5 | -361 |
| 224 | C52P   | -3577.5 | -361 | 286 | TE1    | -787.5  | -361 | 348 | VSS    | 2002.5 | -361 |
| 225 | C52P   | -3532.5 | -361 | 287 | TE1    | -742.5  | -361 | 349 | VSS    | 2047.5 | -361 |
| 226 | C52P   | -3487.5 | -361 | 288 | TE1    | -697.5  | -361 | 350 | VSS    | 2092.5 | -361 |
| 227 | C52P   | -3442.5 | -361 | 289 | TE1    | -652.5  | -361 | 351 | VSS    | 2137.5 | -361 |
| 228 | C52N   | -3397.5 | -361 | 290 | RESX   | -607.5  | -361 | 352 | VSS    | 2182.5 | -361 |
| 229 | C52N   | -3352.5 | -361 | 291 | RESX   | -562.5  | -361 | 353 | VSS    | 2227.5 | -361 |
| 230 | C52N   | -3307.5 | -361 | 292 | RESX   | -517.5  | -361 | 354 | VSS    | 2272.5 | -361 |
| 231 | C52N   | -3262.5 | -361 | 293 | RESX   | -472.5  | -361 | 355 | VSS    | 2317.5 | -361 |
| 232 | C52N   | -3217.5 | -361 | 294 | DUMMY  | -427.5  | -361 | 356 | VSS    | 2362.5 | -361 |
| 233 | C52N   | -3172.5 | -361 | 295 | DUMMY  | -382.5  | -361 | 357 | VSS    | 2407.5 | -361 |
| 234 | AVEE   | -3127.5 | -361 | 296 | DUMMY  | -337.5  | -361 | 358 | EXTN   | 2452.5 | -361 |
| 235 | AVEE   | -3082.5 | -361 | 297 | DUMMY  | -292.5  | -361 | 359 | EXTN   | 2497.5 | -361 |
| 236 | AVEE   | -3037.5 | -361 | 298 | DUMMY  | -247.5  | -361 | 360 | EXTN   | 2542.5 | -361 |
| 237 | AVEE   | -2992.5 | -361 | 299 | DUMMY  | -202.5  | -361 | 361 | EXTN   | 2587.5 | -361 |
| 238 | VSS    | -2947.5 | -361 | 300 | VSS    | -157.5  | -361 | 362 | EXTN   | 2632.5 | -361 |
| 239 | VSS    | -2902.5 | -361 | 301 | VSS    | -112.5  | -361 | 363 | EXTN   | 2677.5 | -361 |
| 240 | D7     | -2857.5 | -361 | 302 | PSWAP  | -67.5   | -361 | 364 | EXTN   | 2722.5 | -361 |
| 241 | D6     | -2812.5 | -361 | 303 | PSWAP  | -22.5   | -361 | 365 | EXTN   | 2767.5 | -361 |
| 242 | D5     | -2767.5 | -361 | 304 | VDDI   | 22.5    | -361 | 366 | EXTP   | 2812.5 | -361 |
| 243 | D4     | -2722.5 | -361 | 305 | VDDI   | 67.5    | -361 | 367 | EXTP   | 2857.5 | -361 |
| 244 | D3     | -2677.5 | -361 | 306 | DSWAP0 | 112.5   | -361 | 368 | EXTP   | 2902.5 | -361 |
| 245 | D2     | -2632.5 | -361 | 307 | DSWAP0 | 157.5   | -361 | 369 | EXTP   | 2947.5 | -361 |
| 246 | D1     | -2587.5 | -361 | 308 | VSS    | 202.5   | -361 | 370 | EXTP   | 2992.5 | -361 |

|     |        |         |      |     |        |         |      |     |        |        |      |
|-----|--------|---------|------|-----|--------|---------|------|-----|--------|--------|------|
| 185 | DUMMYN | -5332.5 | -361 | 247 | D0     | -2542.5 | -361 | 309 | VSS    | 247.5  | -361 |
| 186 | DUMMYN | -5287.5 | -361 | 248 | HS     | -2497.5 | -361 | 310 | DUMMY  | 292.5  | -361 |
| 187 | DUMMYN | -5242.5 | -361 | 249 | VS     | -2452.5 | -361 | 311 | DUMMY  | 337.5  | -361 |
| 188 | DUMMYN | -5197.5 | -361 | 250 | DE     | -2407.5 | -361 | 312 | VDDI   | 382.5  | -361 |
| 189 | DUMMYN | -5152.5 | -361 | 251 | PL     | -2362.5 | -361 | 313 | VDDI   | 427.5  | -361 |
| 190 | DUMMYN | -5107.5 | -361 | 252 | DUMMY  | -2317.5 | -361 | 314 | DUMMY  | 472.5  | -361 |
| 191 | DUMMYN | -5062.5 | -361 | 253 | DUMMY  | -2272.5 | -361 | 315 | DUMMY  | 517.5  | -361 |
| 192 | DUMMYN | -5017.5 | -361 | 254 | DUMMY  | -2227.5 | -361 | 316 | VSS    | 562.5  | -361 |
| 193 | AVDD   | -4972.5 | -361 | 255 | DUMMY  | -2182.5 | -361 | 317 | VSS    | 607.5  | -361 |
| 194 | AVDD   | -4927.5 | -361 | 256 | IM0    | -2137.5 | -361 | 318 | DSWAP1 | 652.5  | -361 |
| 195 | AVDD   | -4882.5 | -361 | 257 | IM0    | -2092.5 | -361 | 319 | DSWAP1 | 697.5  | -361 |
| 196 | AVDD   | -4837.5 | -361 | 258 | IM1    | -2047.5 | -361 | 320 | VDDI   | 742.5  | -361 |
| 197 | AVDD   | -4792.5 | -361 | 259 | IM1    | -2002.5 | -361 | 321 | VDDI   | 787.5  | -361 |
| 198 | AVDD   | -4747.5 | -361 | 260 | DUMMY  | -1957.5 | -361 | 322 | BSTM0  | 832.5  | -361 |
| 199 | AVDD   | -4702.5 | -361 | 261 | DUMMY  | -1912.5 | -361 | 323 | BSTM0  | 877.5  | -361 |
| 200 | VSS    | -4657.5 | -361 | 262 | DUMMY  | -1867.5 | -361 | 324 | BSTM1  | 922.5  | -361 |
| 201 | VSS    | -4612.5 | -361 | 263 | DUMMY  | -1822.5 | -361 | 325 | BSTM1  | 967.5  | -361 |
| 202 | VSS    | -4567.5 | -361 | 264 | DCX    | -1777.5 | -361 | 326 | BSTM2  | 1012.5 | -361 |
| 203 | VSS    | -4522.5 | -361 | 265 | DCX    | -1732.5 | -361 | 327 | BSTM2  | 1057.5 | -361 |
| 204 | VSS    | -4477.5 | -361 | 266 | CSX    | -1687.5 | -361 | 328 | VDDA   | 1102.5 | -361 |
| 205 | VSS    | -4432.5 | -361 | 267 | CSX    | -1642.5 | -361 | 329 | VDDA   | 1147.5 | -361 |
| 206 | VSS    | -4387.5 | -361 | 268 | SCL    | -1597.5 | -361 | 330 | VDDA   | 1192.5 | -361 |
| 207 | VSS    | -4342.5 | -361 | 269 | SCL    | -1552.5 | -361 | 331 | VDDA   | 1237.5 | -361 |
| 208 | C51P   | -4297.5 | -361 | 270 | SDI    | -1507.5 | -361 | 332 | VDDA   | 1282.5 | -361 |
| 209 | C51P   | -4252.5 | -361 | 271 | SDI    | -1462.5 | -361 | 333 | VDDA   | 1327.5 | -361 |
| 210 | C51P   | -4207.5 | -361 | 272 | SDO    | -1417.5 | -361 | 334 | VDDA   | 1372.5 | -361 |
| 211 | C51P   | -4162.5 | -361 | 273 | SDO    | -1372.5 | -361 | 335 | VDDA   | 1417.5 | -361 |
| 212 | C51P   | -4117.5 | -361 | 274 | LEDPWM | -1327.5 | -361 | 336 | VDDA   | 1462.5 | -361 |
| 213 | C51P   | -4072.5 | -361 | 275 | LEDPWM | -1282.5 | -361 | 337 | VDDA   | 1507.5 | -361 |
| 214 | C51N   | -4027.5 | -361 | 276 | LEDPWM | -1237.5 | -361 | 338 | DVDD   | 1552.5 | -361 |
| 215 | C51N   | -3982.5 | -361 | 277 | LEDPWM | -1192.5 | -361 | 339 | DVDD   | 1597.5 | -361 |
| 216 | C51N   | -3937.5 | -361 | 278 | TE     | -1147.5 | -361 | 340 | DVDD   | 1642.5 | -361 |
| 217 | C51N   | -3892.5 | -361 | 279 | TE     | -1102.5 | -361 | 341 | DVDD   | 1687.5 | -361 |
| 218 | C51N   | -3847.5 | -361 | 280 | TE     | -1057.5 | -361 | 342 | DVDD   | 1732.5 | -361 |
| 219 | C51N   | -3802.5 | -361 | 281 | TE     | -1012.5 | -361 | 343 | DVDD   | 1777.5 | -361 |
| 220 | VSS    | -3757.5 | -361 | 282 | TE     | -967.5  | -361 | 344 | DVDD   | 1822.5 | -361 |
| 221 | VSS    | -3712.5 | -361 | 283 | TE     | -922.5  | -361 | 345 | DVDD   | 1867.5 | -361 |
| 222 | C52P   | -3667.5 | -361 | 284 | TE1    | -877.5  | -361 | 346 | DVDD   | 1912.5 | -361 |
| 223 | C52P   | -3622.5 | -361 | 285 | TE1    | -832.5  | -361 | 347 | DVDD   | 1957.5 | -361 |
| 224 | C52P   | -3577.5 | -361 | 286 | TE1    | -787.5  | -361 | 348 | VSS    | 2002.5 | -361 |
| 225 | C52P   | -3532.5 | -361 | 287 | TE1    | -742.5  | -361 | 349 | VSS    | 2047.5 | -361 |
| 226 | C52P   | -3487.5 | -361 | 288 | TE1    | -697.5  | -361 | 350 | VSS    | 2092.5 | -361 |
| 227 | C52P   | -3442.5 | -361 | 289 | TE1    | -652.5  | -361 | 351 | VSS    | 2137.5 | -361 |
| 228 | C52N   | -3397.5 | -361 | 290 | RESX   | -607.5  | -361 | 352 | VSS    | 2182.5 | -361 |
| 229 | C52N   | -3352.5 | -361 | 291 | RESX   | -562.5  | -361 | 353 | VSS    | 2227.5 | -361 |
| 230 | C52N   | -3307.5 | -361 | 292 | RESX   | -517.5  | -361 | 354 | VSS    | 2272.5 | -361 |
| 231 | C52N   | -3262.5 | -361 | 293 | RESX   | -472.5  | -361 | 355 | VSS    | 2317.5 | -361 |
| 232 | C52N   | -3217.5 | -361 | 294 | DUMMY  | -427.5  | -361 | 356 | VSS    | 2362.5 | -361 |
| 233 | C52N   | -3172.5 | -361 | 295 | DUMMY  | -382.5  | -361 | 357 | VSS    | 2407.5 | -361 |
| 234 | AVEE   | -3127.5 | -361 | 296 | DUMMY  | -337.5  | -361 | 358 | EXTN   | 2452.5 | -361 |
| 235 | AVEE   | -3082.5 | -361 | 297 | DUMMY  | -292.5  | -361 | 359 | EXTN   | 2497.5 | -361 |
| 236 | AVEE   | -3037.5 | -361 | 298 | DUMMY  | -247.5  | -361 | 360 | EXTN   | 2542.5 | -361 |
| 237 | AVEE   | -2992.5 | -361 | 299 | DUMMY  | -202.5  | -361 | 361 | EXTN   | 2587.5 | -361 |
| 238 | VSS    | -2947.5 | -361 | 300 | VSS    | -157.5  | -361 | 362 | EXTN   | 2632.5 | -361 |
| 239 | VSS    | -2902.5 | -361 | 301 | VSS    | -112.5  | -361 | 363 | EXTN   | 2677.5 | -361 |
| 240 | D7     | -2857.5 | -361 | 302 | PSWAP  | -67.5   | -361 | 364 | EXTN   | 2722.5 | -361 |
| 241 | D6     | -2812.5 | -361 | 303 | PSWAP  | -22.5   | -361 | 365 | EXTN   | 2767.5 | -361 |
| 242 | D5     | -2767.5 | -361 | 304 | VDDI   | 22.5    | -361 | 366 | EXTP   | 2812.5 | -361 |
| 243 | D4     | -2722.5 | -361 | 305 | VDDI   | 67.5    | -361 | 367 | EXTP   | 2857.5 | -361 |
| 244 | D3     | -2677.5 | -361 | 306 | DSWAP0 | 112.5   | -361 | 368 | EXTP   | 2902.5 | -361 |
| 245 | D2     | -2632.5 | -361 | 307 | DSWAP0 | 157.5   | -361 | 369 | EXTP   | 2947.5 | -361 |
| 246 | D1     | -2587.5 | -361 | 308 | VSS    | 202.5   | -361 | 370 | EXTP   | 2992.5 | -361 |

|     |         |        |      |     |      |        |      |     |       |         |      |
|-----|---------|--------|------|-----|------|--------|------|-----|-------|---------|------|
| 371 | EXTP    | 3037.5 | -361 | 433 | C41P | 5827.5 | -361 | 495 | C22P  | 8617.5  | -361 |
| 372 | EXTP    | 3082.5 | -361 | 434 | C41P | 5872.5 | -361 | 496 | C22N  | 8662.5  | -361 |
| 373 | EXTP    | 3127.5 | -361 | 435 | C41P | 5917.5 | -361 | 497 | C22N  | 8707.5  | -361 |
| 374 | MTP_PWR | 3172.5 | -361 | 436 | C41P | 5962.5 | -361 | 498 | C22N  | 8752.5  | -361 |
| 375 | MTP_PWR | 3217.5 | -361 | 437 | C41P | 6007.5 | -361 | 499 | C22N  | 8797.5  | -361 |
| 376 | MTP_PWR | 3262.5 | -361 | 438 | C41P | 6052.5 | -361 | 500 | C22N  | 8842.5  | -361 |
| 377 | MTP_PWR | 3307.5 | -361 | 439 | C41P | 6097.5 | -361 | 501 | C22N  | 8887.5  | -361 |
| 378 | MTP_PWR | 3352.5 | -361 | 440 | C41N | 6142.5 | -361 | 502 | C22N  | 8932.5  | -361 |
| 379 | MTP_PWR | 3397.5 | -361 | 441 | C41N | 6187.5 | -361 | 503 | VGH   | 8977.5  | -361 |
| 380 | VDDR    | 3442.5 | -361 | 442 | C41N | 6232.5 | -361 | 504 | VGH   | 9022.5  | -361 |
| 381 | VDDR    | 3487.5 | -361 | 443 | C41N | 6277.5 | -361 | 505 | VGH   | 9067.5  | -361 |
| 382 | VDDR    | 3532.5 | -361 | 444 | C41N | 6322.5 | -361 | 506 | VGH   | 9112.5  | -361 |
| 383 | VDDR    | 3577.5 | -361 | 445 | C41N | 6367.5 | -361 | 507 | VGH   | 9157.5  | -361 |
| 384 | VDDR    | 3622.5 | -361 | 446 | C41N | 6412.5 | -361 | 508 | VGH   | 9202.5  | -361 |
| 385 | VDDR    | 3667.5 | -361 | 447 | C42P | 6457.5 | -361 | 509 | VDDDB | 9247.5  | -361 |
| 386 | VDDR    | 3712.5 | -361 | 448 | C42P | 6502.5 | -361 | 510 | VDDDB | 9292.5  | -361 |
| 387 | DUMMYN  | 3757.5 | -361 | 449 | C42P | 6547.5 | -361 | 511 | VDDDB | 9337.5  | -361 |
| 388 | DUMMYN  | 3802.5 | -361 | 450 | C42P | 6592.5 | -361 | 512 | VDDDB | 9382.5  | -361 |
| 389 | DUMMYN  | 3847.5 | -361 | 451 | C42P | 6637.5 | -361 | 513 | VDDDB | 9427.5  | -361 |
| 390 | DUMMYN  | 3892.5 | -361 | 452 | C42P | 6682.5 | -361 | 514 | VDDDB | 9472.5  | -361 |
| 391 | DUMMYN  | 3937.5 | -361 | 453 | C42P | 6727.5 | -361 | 515 | VSS   | 9517.5  | -361 |
| 392 | DUMMYN  | 3982.5 | -361 | 454 | C42N | 6772.5 | -361 | 516 | VSS   | 9562.5  | -361 |
| 393 | DUMMYN  | 4027.5 | -361 | 455 | C42N | 6817.5 | -361 | 517 | VSS   | 9607.5  | -361 |
| 394 | DUMMYN  | 4072.5 | -361 | 456 | C42N | 6862.5 | -361 | 518 | VSS   | 9652.5  | -361 |
| 395 | AVSS    | 4117.5 | -361 | 457 | C42N | 6907.5 | -361 | 519 | VSS   | 9697.5  | -361 |
| 396 | AVSS    | 4162.5 | -361 | 458 | C42N | 6952.5 | -361 | 520 | VSS   | 9742.5  | -361 |
| 397 | AVSS    | 4207.5 | -361 | 459 | C42N | 6997.5 | -361 | 521 | VSS   | 9787.5  | -361 |
| 398 | AVSS    | 4252.5 | -361 | 460 | C42N | 7042.5 | -361 | 522 | VSS   | 9832.5  | -361 |
| 399 | AVSS    | 4297.5 | -361 | 461 | CSP  | 7087.5 | -361 | 523 | C23P  | 9877.5  | -361 |
| 400 | AVSS    | 4342.5 | -361 | 462 | CSP  | 7132.5 | -361 | 524 | C23P  | 9922.5  | -361 |
| 401 | AVSS    | 4387.5 | -361 | 463 | AVDD | 7177.5 | -361 | 525 | C23P  | 9967.5  | -361 |
| 402 | AVSS    | 4432.5 | -361 | 464 | AVDD | 7222.5 | -361 | 526 | C23P  | 10012.5 | -361 |
| 403 | AVSS    | 4477.5 | -361 | 465 | AVDD | 7267.5 | -361 | 527 | C23P  | 10057.5 | -361 |
| 404 | AVSS    | 4522.5 | -361 | 466 | AVDD | 7312.5 | -361 | 528 | C23P  | 10102.5 | -361 |
| 405 | AVSS    | 4567.5 | -361 | 467 | AVDD | 7357.5 | -361 | 529 | C23P  | 10147.5 | -361 |
| 406 | VGMP    | 4612.5 | -361 | 468 | AVEE | 7402.5 | -361 | 530 | C23N  | 10192.5 | -361 |
| 407 | VGMP    | 4657.5 | -361 | 469 | AVEE | 7447.5 | -361 | 531 | C23N  | 10237.5 | -361 |
| 408 | VGMP    | 4702.5 | -361 | 470 | AVEE | 7492.5 | -361 | 532 | C23N  | 10282.5 | -361 |
| 409 | VGMN    | 4747.5 | -361 | 471 | AVEE | 7537.5 | -361 | 533 | C23N  | 10327.5 | -361 |
| 410 | VGMN    | 4792.5 | -361 | 472 | AVEE | 7582.5 | -361 | 534 | C23N  | 10372.5 | -361 |
| 411 | VGMN    | 4837.5 | -361 | 473 | CSN  | 7627.5 | -361 | 535 | C23N  | 10417.5 | -361 |
| 412 | VREF    | 4882.5 | -361 | 474 | CSN  | 7672.5 | -361 | 536 | C23N  | 10462.5 | -361 |
| 413 | VREF    | 4927.5 | -361 | 475 | C21P | 7717.5 | -361 | 537 | C24P  | 10507.5 | -361 |
| 414 | VREF    | 4972.5 | -361 | 476 | C21P | 7762.5 | -361 | 538 | C24P  | 10552.5 | -361 |
| 415 | VSS     | 5017.5 | -361 | 477 | C21P | 7807.5 | -361 | 539 | C24P  | 10597.5 | -361 |
| 416 | VSS     | 5062.5 | -361 | 478 | C21P | 7852.5 | -361 | 540 | C24P  | 10642.5 | -361 |
| 417 | VSS     | 5107.5 | -361 | 479 | C21P | 7897.5 | -361 | 541 | C24P  | 10687.5 | -361 |
| 418 | VSS     | 5152.5 | -361 | 480 | C21P | 7942.5 | -361 | 542 | C24P  | 10732.5 | -361 |
| 419 | VSS     | 5197.5 | -361 | 481 | C21P | 7987.5 | -361 | 543 | C24P  | 10777.5 | -361 |
| 420 | VDDDB   | 5242.5 | -361 | 482 | C21N | 8032.5 | -361 | 544 | C24N  | 10822.5 | -361 |
| 421 | VDDDB   | 5287.5 | -361 | 483 | C21N | 8077.5 | -361 | 545 | C24N  | 10867.5 | -361 |
| 422 | VDDDB   | 5332.5 | -361 | 484 | C21N | 8122.5 | -361 | 546 | C24N  | 10912.5 | -361 |
| 423 | VDDDB   | 5377.5 | -361 | 485 | C21N | 8167.5 | -361 | 547 | C24N  | 10957.5 | -361 |
| 424 | VDDDB   | 5422.5 | -361 | 486 | C21N | 8212.5 | -361 | 548 | C24N  | 11002.5 | -361 |
| 425 | VDDDB   | 5467.5 | -361 | 487 | C21N | 8257.5 | -361 | 549 | C24N  | 11047.5 | -361 |
| 426 | VDDDB   | 5512.5 | -361 | 488 | C21N | 8302.5 | -361 | 550 | C24N  | 11092.5 | -361 |
| 427 | VDDDB   | 5557.5 | -361 | 489 | C22P | 8347.5 | -361 | 551 | VGL   | 11137.5 | -361 |
| 428 | VCL     | 5602.5 | -361 | 490 | C22P | 8392.5 | -361 | 552 | VGL   | 11182.5 | -361 |
| 429 | VCL     | 5647.5 | -361 | 491 | C22P | 8437.5 | -361 | 553 | VGL   | 11227.5 | -361 |
| 430 | VCL     | 5692.5 | -361 | 492 | C22P | 8482.5 | -361 | 554 | VGL   | 11272.5 | -361 |
| 431 | VCL     | 5737.5 | -361 | 493 | C22P | 8527.5 | -361 | 555 | VGL   | 11317.5 | -361 |
| 432 | VCL     | 5782.5 | -361 | 494 | C22P | 8572.5 | -361 | 556 | VGL   | 11362.5 | -361 |

|     |          |         |       |     |     |         |       |     |      |         |       |
|-----|----------|---------|-------|-----|-----|---------|-------|-----|------|---------|-------|
| 557 | VRGH     | 11407.5 | -361  | 619 | S4  | 13491.5 | 193.5 | 681 | S66  | 12809.5 | 373.5 |
| 558 | VRGH     | 11452.5 | -361  | 620 | S5  | 13480.5 | 283.5 | 682 | S67  | 12798.5 | 193.5 |
| 559 | VRGH     | 11497.5 | -361  | 621 | S6  | 13469.5 | 373.5 | 683 | S68  | 12787.5 | 283.5 |
| 560 | VRGH     | 11542.5 | -361  | 622 | S7  | 13458.5 | 193.5 | 684 | S69  | 12776.5 | 373.5 |
| 561 | VRGH     | 11587.5 | -361  | 623 | S8  | 13447.5 | 283.5 | 685 | S70  | 12765.5 | 193.5 |
| 562 | VRGH     | 11632.5 | -361  | 624 | S9  | 13436.5 | 373.5 | 686 | S71  | 12754.5 | 283.5 |
| 563 | VRGH     | 11677.5 | -361  | 625 | S10 | 13425.5 | 193.5 | 687 | S72  | 12743.5 | 373.5 |
| 564 | VGLO2    | 11722.5 | -361  | 626 | S11 | 13414.5 | 283.5 | 688 | S73  | 12732.5 | 193.5 |
| 565 | VGLO2    | 11767.5 | -361  | 627 | S12 | 13403.5 | 373.5 | 689 | S74  | 12721.5 | 283.5 |
| 566 | VGLO2    | 11812.5 | -361  | 628 | S13 | 13392.5 | 193.5 | 690 | S75  | 12710.5 | 373.5 |
| 567 | VGLO2    | 11857.5 | -361  | 629 | S14 | 13381.5 | 283.5 | 691 | S76  | 12699.5 | 193.5 |
| 568 | VGLO2    | 11902.5 | -361  | 630 | S15 | 13370.5 | 373.5 | 692 | S77  | 12688.5 | 283.5 |
| 569 | VGLO2    | 11947.5 | -361  | 631 | S16 | 13359.5 | 193.5 | 693 | S78  | 12677.5 | 373.5 |
| 570 | VGLO2    | 11992.5 | -361  | 632 | S17 | 13348.5 | 283.5 | 694 | S79  | 12666.5 | 193.5 |
| 571 | VGLO1    | 12037.5 | -361  | 633 | S18 | 13337.5 | 373.5 | 695 | S80  | 12655.5 | 283.5 |
| 572 | VGLO1    | 12082.5 | -361  | 634 | S19 | 13326.5 | 193.5 | 696 | S81  | 12644.5 | 373.5 |
| 573 | VGLO1    | 12127.5 | -361  | 635 | S20 | 13315.5 | 283.5 | 697 | S82  | 12633.5 | 193.5 |
| 574 | VGLO1    | 12172.5 | -361  | 636 | S21 | 13304.5 | 373.5 | 698 | S83  | 12622.5 | 283.5 |
| 575 | VGLO1    | 12217.5 | -361  | 637 | S22 | 13293.5 | 193.5 | 699 | S84  | 12611.5 | 373.5 |
| 576 | VGLO1    | 12262.5 | -361  | 638 | S23 | 13282.5 | 283.5 | 700 | S85  | 12600.5 | 193.5 |
| 577 | VCOMR    | 12307.5 | -361  | 639 | S24 | 13271.5 | 373.5 | 701 | S86  | 12589.5 | 283.5 |
| 578 | VCOMR    | 12352.5 | -361  | 640 | S25 | 13260.5 | 193.5 | 702 | S87  | 12578.5 | 373.5 |
| 579 | DUMMY    | 12397.5 | -361  | 641 | S26 | 13249.5 | 283.5 | 703 | S88  | 12567.5 | 193.5 |
| 580 | DUMMY    | 12442.5 | -361  | 642 | S27 | 13238.5 | 373.5 | 704 | S89  | 12556.5 | 283.5 |
| 581 | VCOM     | 12497.5 | -361  | 643 | S28 | 13227.5 | 193.5 | 705 | S90  | 12545.5 | 373.5 |
| 582 | VCOM     | 12542.5 | -361  | 644 | S29 | 13216.5 | 283.5 | 706 | S91  | 12534.5 | 193.5 |
| 583 | VCOM     | 12587.5 | -361  | 645 | S30 | 13205.5 | 373.5 | 707 | S92  | 12523.5 | 283.5 |
| 584 | GOUT_R22 | 12632.5 | -361  | 646 | S31 | 13194.5 | 193.5 | 708 | S93  | 12512.5 | 373.5 |
| 585 | GOUT_R21 | 12677.5 | -361  | 647 | S32 | 13183.5 | 283.5 | 709 | S94  | 12501.5 | 193.5 |
| 586 | GOUT_R20 | 12722.5 | -361  | 648 | S33 | 13172.5 | 373.5 | 710 | S95  | 12490.5 | 283.5 |
| 587 | GOUT_R19 | 12767.5 | -361  | 649 | S34 | 13161.5 | 193.5 | 711 | S96  | 12479.5 | 373.5 |
| 588 | GOUT_R18 | 12812.5 | -361  | 650 | S35 | 13150.5 | 283.5 | 712 | S97  | 12468.5 | 193.5 |
| 589 | GOUT_R17 | 12857.5 | -361  | 651 | S36 | 13139.5 | 373.5 | 713 | S98  | 12457.5 | 283.5 |
| 590 | GOUT_R16 | 12902.5 | -361  | 652 | S37 | 13128.5 | 193.5 | 714 | S99  | 12446.5 | 373.5 |
| 591 | GOUT_R15 | 12947.5 | -361  | 653 | S38 | 13117.5 | 283.5 | 715 | S100 | 12435.5 | 193.5 |
| 592 | GOUT_R14 | 12992.5 | -361  | 654 | S39 | 13106.5 | 373.5 | 716 | S101 | 12424.5 | 283.5 |
| 593 | GOUT_R13 | 13037.5 | -361  | 655 | S40 | 13095.5 | 193.5 | 717 | S102 | 12413.5 | 373.5 |
| 594 | GOUT_R12 | 13082.5 | -361  | 656 | S41 | 13084.5 | 283.5 | 718 | S103 | 12402.5 | 193.5 |
| 595 | GOUT_R11 | 13127.5 | -361  | 657 | S42 | 13073.5 | 373.5 | 719 | S104 | 12391.5 | 283.5 |
| 596 | GOUT_R10 | 13172.5 | -361  | 658 | S43 | 13062.5 | 193.5 | 720 | S105 | 12380.5 | 373.5 |
| 597 | GOUT_R9  | 13217.5 | -361  | 659 | S44 | 13051.5 | 283.5 | 721 | S106 | 12369.5 | 193.5 |
| 598 | GOUT_R8  | 13262.5 | -361  | 660 | S45 | 13040.5 | 373.5 | 722 | S107 | 12358.5 | 283.5 |
| 599 | GOUT_R7  | 13307.5 | -361  | 661 | S46 | 13029.5 | 193.5 | 723 | S108 | 12347.5 | 373.5 |
| 600 | GOUT_R6  | 13352.5 | -361  | 662 | S47 | 13018.5 | 283.5 | 724 | S109 | 12336.5 | 193.5 |
| 601 | GOUT_R5  | 13397.5 | -361  | 663 | S48 | 13007.5 | 373.5 | 725 | S110 | 12325.5 | 283.5 |
| 602 | GOUT_R4  | 13442.5 | -361  | 664 | S49 | 12996.5 | 193.5 | 726 | S111 | 12314.5 | 373.5 |
| 603 | GOUT_R3  | 13487.5 | -361  | 665 | S50 | 12985.5 | 283.5 | 727 | S112 | 12303.5 | 193.5 |
| 604 | GOUT_R2  | 13532.5 | -361  | 666 | S51 | 12974.5 | 373.5 | 728 | S113 | 12292.5 | 283.5 |
| 605 | GOUT_R1  | 13577.5 | -361  | 667 | S52 | 12963.5 | 193.5 | 729 | S114 | 12281.5 | 373.5 |
| 606 | DUMMY2   | 13622.5 | -361  | 668 | S53 | 12952.5 | 283.5 | 730 | S115 | 12270.5 | 193.5 |
| 607 | DUMMY3   | 13623.5 | 193.5 | 669 | S54 | 12941.5 | 373.5 | 731 | S116 | 12259.5 | 283.5 |
| 608 | DUMMY4   | 13612.5 | 283.5 | 670 | S55 | 12930.5 | 193.5 | 732 | S117 | 12248.5 | 373.5 |
| 609 | DUMMY5   | 13601.5 | 373.5 | 671 | S56 | 12919.5 | 283.5 | 733 | S118 | 12237.5 | 193.5 |
| 610 | DUMMY6   | 13590.5 | 193.5 | 672 | S57 | 12908.5 | 373.5 | 734 | S119 | 12226.5 | 283.5 |
| 611 | DUMMY7   | 13579.5 | 283.5 | 673 | S58 | 12897.5 | 193.5 | 735 | S120 | 12215.5 | 373.5 |
| 612 | DUMMY8   | 13568.5 | 373.5 | 674 | S59 | 12886.5 | 283.5 | 736 | S121 | 12204.5 | 193.5 |
| 613 | DUMMY9   | 13557.5 | 193.5 | 675 | S60 | 12875.5 | 373.5 | 737 | S122 | 12193.5 | 283.5 |
| 614 | SDUM0    | 13546.5 | 283.5 | 676 | S61 | 12864.5 | 193.5 | 738 | S123 | 12182.5 | 373.5 |
| 615 | SDUM1    | 13535.5 | 373.5 | 677 | S62 | 12853.5 | 283.5 | 739 | S124 | 12171.5 | 193.5 |
| 616 | S1       | 13524.5 | 193.5 | 678 | S63 | 12842.5 | 373.5 | 740 | S125 | 12160.5 | 283.5 |
| 617 | S2       | 13513.5 | 283.5 | 679 | S64 | 12831.5 | 193.5 | 741 | S126 | 12149.5 | 373.5 |
| 618 | S3       | 13502.5 | 373.5 | 680 | S65 | 12820.5 | 283.5 | 742 | S127 | 12138.5 | 193.5 |

|     |      |         |       |     |      |         |       |     |      |         |       |
|-----|------|---------|-------|-----|------|---------|-------|-----|------|---------|-------|
| 743 | S128 | 12127.5 | 283.5 | 805 | S190 | 11445.5 | 193.5 | 867 | S252 | 10763.5 | 373.5 |
| 744 | S129 | 12116.5 | 373.5 | 806 | S191 | 11434.5 | 283.5 | 868 | S253 | 10752.5 | 193.5 |
| 745 | S130 | 12105.5 | 193.5 | 807 | S192 | 11423.5 | 373.5 | 869 | S254 | 10741.5 | 283.5 |
| 746 | S131 | 12094.5 | 283.5 | 808 | S193 | 11412.5 | 193.5 | 870 | S255 | 10730.5 | 373.5 |
| 747 | S132 | 12083.5 | 373.5 | 809 | S194 | 11401.5 | 283.5 | 871 | S256 | 10719.5 | 193.5 |
| 748 | S133 | 12072.5 | 193.5 | 810 | S195 | 11390.5 | 373.5 | 872 | S257 | 10708.5 | 283.5 |
| 749 | S134 | 12061.5 | 283.5 | 811 | S196 | 11379.5 | 193.5 | 873 | S258 | 10697.5 | 373.5 |
| 750 | S135 | 12050.5 | 373.5 | 812 | S197 | 11368.5 | 283.5 | 874 | S259 | 10686.5 | 193.5 |
| 751 | S136 | 12039.5 | 193.5 | 813 | S198 | 11357.5 | 373.5 | 875 | S260 | 10675.5 | 283.5 |
| 752 | S137 | 12028.5 | 283.5 | 814 | S199 | 11346.5 | 193.5 | 876 | S261 | 10664.5 | 373.5 |
| 753 | S138 | 12017.5 | 373.5 | 815 | S200 | 11335.5 | 283.5 | 877 | S262 | 10653.5 | 193.5 |
| 754 | S139 | 12006.5 | 193.5 | 816 | S201 | 11324.5 | 373.5 | 878 | S263 | 10642.5 | 283.5 |
| 755 | S140 | 11995.5 | 283.5 | 817 | S202 | 11313.5 | 193.5 | 879 | S264 | 10631.5 | 373.5 |
| 756 | S141 | 11984.5 | 373.5 | 818 | S203 | 11302.5 | 283.5 | 880 | S265 | 10620.5 | 193.5 |
| 757 | S142 | 11973.5 | 193.5 | 819 | S204 | 11291.5 | 373.5 | 881 | S266 | 10609.5 | 283.5 |
| 758 | S143 | 11962.5 | 283.5 | 820 | S205 | 11280.5 | 193.5 | 882 | S267 | 10598.5 | 373.5 |
| 759 | S144 | 11951.5 | 373.5 | 821 | S206 | 11269.5 | 283.5 | 883 | S268 | 10587.5 | 193.5 |
| 760 | S145 | 11940.5 | 193.5 | 822 | S207 | 11258.5 | 373.5 | 884 | S269 | 10576.5 | 283.5 |
| 761 | S146 | 11929.5 | 283.5 | 823 | S208 | 11247.5 | 193.5 | 885 | S270 | 10565.5 | 373.5 |
| 762 | S147 | 11918.5 | 373.5 | 824 | S209 | 11236.5 | 283.5 | 886 | S271 | 10554.5 | 193.5 |
| 763 | S148 | 11907.5 | 193.5 | 825 | S210 | 11225.5 | 373.5 | 887 | S272 | 10543.5 | 283.5 |
| 764 | S149 | 11896.5 | 283.5 | 826 | S211 | 11214.5 | 193.5 | 888 | S273 | 10532.5 | 373.5 |
| 765 | S150 | 11885.5 | 373.5 | 827 | S212 | 11203.5 | 283.5 | 889 | S274 | 10521.5 | 193.5 |
| 766 | S151 | 11874.5 | 193.5 | 828 | S213 | 11192.5 | 373.5 | 890 | S275 | 10510.5 | 283.5 |
| 767 | S152 | 11863.5 | 283.5 | 829 | S214 | 11181.5 | 193.5 | 891 | S276 | 10499.5 | 373.5 |
| 768 | S153 | 11852.5 | 373.5 | 830 | S215 | 11170.5 | 283.5 | 892 | S277 | 10488.5 | 193.5 |
| 769 | S154 | 11841.5 | 193.5 | 831 | S216 | 11159.5 | 373.5 | 893 | S278 | 10477.5 | 283.5 |
| 770 | S155 | 11830.5 | 283.5 | 832 | S217 | 11148.5 | 193.5 | 894 | S279 | 10466.5 | 373.5 |
| 771 | S156 | 11819.5 | 373.5 | 833 | S218 | 11137.5 | 283.5 | 895 | S280 | 10455.5 | 193.5 |
| 772 | S157 | 11808.5 | 193.5 | 834 | S219 | 11126.5 | 373.5 | 896 | S281 | 10444.5 | 283.5 |
| 773 | S158 | 11797.5 | 283.5 | 835 | S220 | 11115.5 | 193.5 | 897 | S282 | 10433.5 | 373.5 |
| 774 | S159 | 11786.5 | 373.5 | 836 | S221 | 11104.5 | 283.5 | 898 | S283 | 10422.5 | 193.5 |
| 775 | S160 | 11775.5 | 193.5 | 837 | S222 | 11093.5 | 373.5 | 899 | S284 | 10411.5 | 283.5 |
| 776 | S161 | 11764.5 | 283.5 | 838 | S223 | 11082.5 | 193.5 | 900 | S285 | 10400.5 | 373.5 |
| 777 | S162 | 11753.5 | 373.5 | 839 | S224 | 11071.5 | 283.5 | 901 | S286 | 10389.5 | 193.5 |
| 778 | S163 | 11742.5 | 193.5 | 840 | S225 | 11060.5 | 373.5 | 902 | S287 | 10378.5 | 283.5 |
| 779 | S164 | 11731.5 | 283.5 | 841 | S226 | 11049.5 | 193.5 | 903 | S288 | 10367.5 | 373.5 |
| 780 | S165 | 11720.5 | 373.5 | 842 | S227 | 11038.5 | 283.5 | 904 | S289 | 10356.5 | 193.5 |
| 781 | S166 | 11709.5 | 193.5 | 843 | S228 | 11027.5 | 373.5 | 905 | S290 | 10345.5 | 283.5 |
| 782 | S167 | 11698.5 | 283.5 | 844 | S229 | 11016.5 | 193.5 | 906 | S291 | 10334.5 | 373.5 |
| 783 | S168 | 11687.5 | 373.5 | 845 | S230 | 11005.5 | 283.5 | 907 | S292 | 10323.5 | 193.5 |
| 784 | S169 | 11676.5 | 193.5 | 846 | S231 | 10994.5 | 373.5 | 908 | S293 | 10312.5 | 283.5 |
| 785 | S170 | 11665.5 | 283.5 | 847 | S232 | 10983.5 | 193.5 | 909 | S294 | 10301.5 | 373.5 |
| 786 | S171 | 11654.5 | 373.5 | 848 | S233 | 10972.5 | 283.5 | 910 | S295 | 10290.5 | 193.5 |
| 787 | S172 | 11643.5 | 193.5 | 849 | S234 | 10961.5 | 373.5 | 911 | S296 | 10279.5 | 283.5 |
| 788 | S173 | 11632.5 | 283.5 | 850 | S235 | 10950.5 | 193.5 | 912 | S297 | 10268.5 | 373.5 |
| 789 | S174 | 11621.5 | 373.5 | 851 | S236 | 10939.5 | 283.5 | 913 | S298 | 10257.5 | 193.5 |
| 790 | S175 | 11610.5 | 193.5 | 852 | S237 | 10928.5 | 373.5 | 914 | S299 | 10246.5 | 283.5 |
| 791 | S176 | 11599.5 | 283.5 | 853 | S238 | 10917.5 | 193.5 | 915 | S300 | 10235.5 | 373.5 |
| 792 | S177 | 11588.5 | 373.5 | 854 | S239 | 10906.5 | 283.5 | 916 | S301 | 10224.5 | 193.5 |
| 793 | S178 | 11577.5 | 193.5 | 855 | S240 | 10895.5 | 373.5 | 917 | S302 | 10213.5 | 283.5 |
| 794 | S179 | 11566.5 | 283.5 | 856 | S241 | 10884.5 | 193.5 | 918 | S303 | 10202.5 | 373.5 |
| 795 | S180 | 11555.5 | 373.5 | 857 | S242 | 10873.5 | 283.5 | 919 | S304 | 10191.5 | 193.5 |
| 796 | S181 | 11544.5 | 193.5 | 858 | S243 | 10862.5 | 373.5 | 920 | S305 | 10180.5 | 283.5 |
| 797 | S182 | 11533.5 | 283.5 | 859 | S244 | 10851.5 | 193.5 | 921 | S306 | 10169.5 | 373.5 |
| 798 | S183 | 11522.5 | 373.5 | 860 | S245 | 10840.5 | 283.5 | 922 | S307 | 10158.5 | 193.5 |
| 799 | S184 | 11511.5 | 193.5 | 861 | S246 | 10829.5 | 373.5 | 923 | S308 | 10147.5 | 283.5 |
| 800 | S185 | 11500.5 | 283.5 | 862 | S247 | 10818.5 | 193.5 | 924 | S309 | 10136.5 | 373.5 |
| 801 | S186 | 11489.5 | 373.5 | 863 | S248 | 10807.5 | 283.5 | 925 | S310 | 10125.5 | 193.5 |
| 802 | S187 | 11478.5 | 193.5 | 864 | S249 | 10796.5 | 373.5 | 926 | S311 | 10114.5 | 283.5 |
| 803 | S188 | 11467.5 | 283.5 | 865 | S250 | 10785.5 | 193.5 | 927 | S312 | 10103.5 | 373.5 |
| 804 | S189 | 11456.5 | 373.5 | 866 | S251 | 10774.5 | 283.5 | 928 | S313 | 10092.5 | 193.5 |



|     |      |         |       |      |      |        |       |      |      |        |       |
|-----|------|---------|-------|------|------|--------|-------|------|------|--------|-------|
| 929 | S314 | 10081.5 | 283.5 | 991  | S376 | 9399.5 | 193.5 | 1053 | S438 | 8717.5 | 373.5 |
| 930 | S315 | 10070.5 | 373.5 | 992  | S377 | 9388.5 | 283.5 | 1054 | S439 | 8706.5 | 193.5 |
| 931 | S316 | 10059.5 | 193.5 | 993  | S378 | 9377.5 | 373.5 | 1055 | S440 | 8695.5 | 283.5 |
| 932 | S317 | 10048.5 | 283.5 | 994  | S379 | 9366.5 | 193.5 | 1056 | S441 | 8684.5 | 373.5 |
| 933 | S318 | 10037.5 | 373.5 | 995  | S380 | 9355.5 | 283.5 | 1057 | S442 | 8673.5 | 193.5 |
| 934 | S319 | 10026.5 | 193.5 | 996  | S381 | 9344.5 | 373.5 | 1058 | S443 | 8662.5 | 283.5 |
| 935 | S320 | 10015.5 | 283.5 | 997  | S382 | 9333.5 | 193.5 | 1059 | S444 | 8651.5 | 373.5 |
| 936 | S321 | 10004.5 | 373.5 | 998  | S383 | 9322.5 | 283.5 | 1060 | S445 | 8640.5 | 193.5 |
| 937 | S322 | 9993.5  | 193.5 | 999  | S384 | 9311.5 | 373.5 | 1061 | S446 | 8629.5 | 283.5 |
| 938 | S323 | 9982.5  | 283.5 | 1000 | S385 | 9300.5 | 193.5 | 1062 | S447 | 8618.5 | 373.5 |
| 939 | S324 | 9971.5  | 373.5 | 1001 | S386 | 9289.5 | 283.5 | 1063 | S448 | 8607.5 | 193.5 |
| 940 | S325 | 9960.5  | 193.5 | 1002 | S387 | 9278.5 | 373.5 | 1064 | S449 | 8596.5 | 283.5 |
| 941 | S326 | 9949.5  | 283.5 | 1003 | S388 | 9267.5 | 193.5 | 1065 | S450 | 8585.5 | 373.5 |
| 942 | S327 | 9938.5  | 373.5 | 1004 | S389 | 9256.5 | 283.5 | 1066 | S451 | 8574.5 | 193.5 |
| 943 | S328 | 9927.5  | 193.5 | 1005 | S390 | 9245.5 | 373.5 | 1067 | S452 | 8563.5 | 283.5 |
| 944 | S329 | 9916.5  | 283.5 | 1006 | S391 | 9234.5 | 193.5 | 1068 | S453 | 8552.5 | 373.5 |
| 945 | S330 | 9905.5  | 373.5 | 1007 | S392 | 9223.5 | 283.5 | 1069 | S454 | 8541.5 | 193.5 |
| 946 | S331 | 9894.5  | 193.5 | 1008 | S393 | 9212.5 | 373.5 | 1070 | S455 | 8530.5 | 283.5 |
| 947 | S332 | 9883.5  | 283.5 | 1009 | S394 | 9201.5 | 193.5 | 1071 | S456 | 8519.5 | 373.5 |
| 948 | S333 | 9872.5  | 373.5 | 1010 | S395 | 9190.5 | 283.5 | 1072 | S457 | 8508.5 | 193.5 |
| 949 | S334 | 9861.5  | 193.5 | 1011 | S396 | 9179.5 | 373.5 | 1073 | S458 | 8497.5 | 283.5 |
| 950 | S335 | 9850.5  | 283.5 | 1012 | S397 | 9168.5 | 193.5 | 1074 | S459 | 8486.5 | 373.5 |
| 951 | S336 | 9839.5  | 373.5 | 1013 | S398 | 9157.5 | 283.5 | 1075 | S460 | 8475.5 | 193.5 |
| 952 | S337 | 9828.5  | 193.5 | 1014 | S399 | 9146.5 | 373.5 | 1076 | S461 | 8464.5 | 283.5 |
| 953 | S338 | 9817.5  | 283.5 | 1015 | S400 | 9135.5 | 193.5 | 1077 | S462 | 8453.5 | 373.5 |
| 954 | S339 | 9806.5  | 373.5 | 1016 | S401 | 9124.5 | 283.5 | 1078 | S463 | 8442.5 | 193.5 |
| 955 | S340 | 9795.5  | 193.5 | 1017 | S402 | 9113.5 | 373.5 | 1079 | S464 | 8431.5 | 283.5 |
| 956 | S341 | 9784.5  | 283.5 | 1018 | S403 | 9102.5 | 193.5 | 1080 | S465 | 8420.5 | 373.5 |
| 957 | S342 | 9773.5  | 373.5 | 1019 | S404 | 9091.5 | 283.5 | 1081 | S466 | 8409.5 | 193.5 |
| 958 | S343 | 9762.5  | 193.5 | 1020 | S405 | 9080.5 | 373.5 | 1082 | S467 | 8398.5 | 283.5 |
| 959 | S344 | 9751.5  | 283.5 | 1021 | S406 | 9069.5 | 193.5 | 1083 | S468 | 8387.5 | 373.5 |
| 960 | S345 | 9740.5  | 373.5 | 1022 | S407 | 9058.5 | 283.5 | 1084 | S469 | 8376.5 | 193.5 |
| 961 | S346 | 9729.5  | 193.5 | 1023 | S408 | 9047.5 | 373.5 | 1085 | S470 | 8365.5 | 283.5 |
| 962 | S347 | 9718.5  | 283.5 | 1024 | S409 | 9036.5 | 193.5 | 1086 | S471 | 8354.5 | 373.5 |
| 963 | S348 | 9707.5  | 373.5 | 1025 | S410 | 9025.5 | 283.5 | 1087 | S472 | 8343.5 | 193.5 |
| 964 | S349 | 9696.5  | 193.5 | 1026 | S411 | 9014.5 | 373.5 | 1088 | S473 | 8332.5 | 283.5 |
| 965 | S350 | 9685.5  | 283.5 | 1027 | S412 | 9003.5 | 193.5 | 1089 | S474 | 8321.5 | 373.5 |
| 966 | S351 | 9674.5  | 373.5 | 1028 | S413 | 8992.5 | 283.5 | 1090 | S475 | 8310.5 | 193.5 |
| 967 | S352 | 9663.5  | 193.5 | 1029 | S414 | 8981.5 | 373.5 | 1091 | S476 | 8299.5 | 283.5 |
| 968 | S353 | 9652.5  | 283.5 | 1030 | S415 | 8970.5 | 193.5 | 1092 | S477 | 8288.5 | 373.5 |
| 969 | S354 | 9641.5  | 373.5 | 1031 | S416 | 8959.5 | 283.5 | 1093 | S478 | 8277.5 | 193.5 |
| 970 | S355 | 9630.5  | 193.5 | 1032 | S417 | 8948.5 | 373.5 | 1094 | S479 | 8266.5 | 283.5 |
| 971 | S356 | 9619.5  | 283.5 | 1033 | S418 | 8937.5 | 193.5 | 1095 | S480 | 8255.5 | 373.5 |
| 972 | S357 | 9608.5  | 373.5 | 1034 | S419 | 8926.5 | 283.5 | 1096 | S481 | 8244.5 | 193.5 |
| 973 | S358 | 9597.5  | 193.5 | 1035 | S420 | 8915.5 | 373.5 | 1097 | S482 | 8233.5 | 283.5 |
| 974 | S359 | 9586.5  | 283.5 | 1036 | S421 | 8904.5 | 193.5 | 1098 | S483 | 8222.5 | 373.5 |
| 975 | S360 | 9575.5  | 373.5 | 1037 | S422 | 8893.5 | 283.5 | 1099 | S484 | 8211.5 | 193.5 |
| 976 | S361 | 9564.5  | 193.5 | 1038 | S423 | 8882.5 | 373.5 | 1100 | S485 | 8200.5 | 283.5 |
| 977 | S362 | 9553.5  | 283.5 | 1039 | S424 | 8871.5 | 193.5 | 1101 | S486 | 8189.5 | 373.5 |
| 978 | S363 | 9542.5  | 373.5 | 1040 | S425 | 8860.5 | 283.5 | 1102 | S487 | 8178.5 | 193.5 |
| 979 | S364 | 9531.5  | 193.5 | 1041 | S426 | 8849.5 | 373.5 | 1103 | S488 | 8167.5 | 283.5 |
| 980 | S365 | 9520.5  | 283.5 | 1042 | S427 | 8838.5 | 193.5 | 1104 | S489 | 8156.5 | 373.5 |
| 981 | S366 | 9509.5  | 373.5 | 1043 | S428 | 8827.5 | 283.5 | 1105 | S490 | 8145.5 | 193.5 |
| 982 | S367 | 9498.5  | 193.5 | 1044 | S429 | 8816.5 | 373.5 | 1106 | S491 | 8134.5 | 283.5 |
| 983 | S368 | 9487.5  | 283.5 | 1045 | S430 | 8805.5 | 193.5 | 1107 | S492 | 8123.5 | 373.5 |
| 984 | S369 | 9476.5  | 373.5 | 1046 | S431 | 8794.5 | 283.5 | 1108 | S493 | 8112.5 | 193.5 |
| 985 | S370 | 9465.5  | 193.5 | 1047 | S432 | 8783.5 | 373.5 | 1109 | S494 | 8101.5 | 283.5 |
| 986 | S371 | 9454.5  | 283.5 | 1048 | S433 | 8772.5 | 193.5 | 1110 | S495 | 8090.5 | 373.5 |
| 987 | S372 | 9443.5  | 373.5 | 1049 | S434 | 8761.5 | 283.5 | 1111 | S496 | 8079.5 | 193.5 |
| 988 | S373 | 9432.5  | 193.5 | 1050 | S435 | 8750.5 | 373.5 | 1112 | S497 | 8068.5 | 283.5 |
| 989 | S374 | 9421.5  | 283.5 | 1051 | S436 | 8739.5 | 193.5 | 1113 | S498 | 8057.5 | 373.5 |
| 990 | S375 | 9410.5  | 373.5 | 1052 | S437 | 8728.5 | 283.5 | 1114 | S499 | 8046.5 | 193.5 |

|      |      |        |       |      |         |        |       |      |      |        |       |
|------|------|--------|-------|------|---------|--------|-------|------|------|--------|-------|
| 1115 | S500 | 8035.5 | 283.5 | 1177 | S562    | 7353.5 | 193.5 | 1239 | S606 | 6671.5 | 373.5 |
| 1116 | S501 | 8024.5 | 373.5 | 1178 | S563    | 7342.5 | 283.5 | 1240 | S607 | 6660.5 | 193.5 |
| 1117 | S502 | 8013.5 | 193.5 | 1179 | S564    | 7331.5 | 373.5 | 1241 | S608 | 6649.5 | 283.5 |
| 1118 | S503 | 8002.5 | 283.5 | 1180 | S565    | 7320.5 | 193.5 | 1242 | S609 | 6638.5 | 373.5 |
| 1119 | S504 | 7991.5 | 373.5 | 1181 | S566    | 7309.5 | 283.5 | 1243 | S610 | 6627.5 | 193.5 |
| 1120 | S505 | 7980.5 | 193.5 | 1182 | S567    | 7298.5 | 373.5 | 1244 | S611 | 6616.5 | 283.5 |
| 1121 | S506 | 7969.5 | 283.5 | 1183 | S568    | 7287.5 | 193.5 | 1245 | S612 | 6605.5 | 373.5 |
| 1122 | S507 | 7958.5 | 373.5 | 1184 | S569    | 7276.5 | 283.5 | 1246 | S613 | 6594.5 | 193.5 |
| 1123 | S508 | 7947.5 | 193.5 | 1185 | S570    | 7265.5 | 373.5 | 1247 | S614 | 6583.5 | 283.5 |
| 1124 | S509 | 7936.5 | 283.5 | 1186 | S571    | 7254.5 | 193.5 | 1248 | S615 | 6572.5 | 373.5 |
| 1125 | S510 | 7925.5 | 373.5 | 1187 | S572    | 7243.5 | 283.5 | 1249 | S616 | 6561.5 | 193.5 |
| 1126 | S511 | 7914.5 | 193.5 | 1188 | S573    | 7232.5 | 373.5 | 1250 | S617 | 6550.5 | 283.5 |
| 1127 | S512 | 7903.5 | 283.5 | 1189 | S574    | 7221.5 | 193.5 | 1251 | S618 | 6539.5 | 373.5 |
| 1128 | S513 | 7892.5 | 373.5 | 1190 | S575    | 7210.5 | 283.5 | 1252 | S619 | 6528.5 | 193.5 |
| 1129 | S514 | 7881.5 | 193.5 | 1191 | S576    | 7199.5 | 373.5 | 1253 | S620 | 6517.5 | 283.5 |
| 1130 | S515 | 7870.5 | 283.5 | 1192 | S577    | 7188.5 | 193.5 | 1254 | S621 | 6506.5 | 373.5 |
| 1131 | S516 | 7859.5 | 373.5 | 1193 | S578    | 7177.5 | 283.5 | 1255 | S622 | 6495.5 | 193.5 |
| 1132 | S517 | 7848.5 | 193.5 | 1194 | S579    | 7166.5 | 373.5 | 1256 | S623 | 6484.5 | 283.5 |
| 1133 | S518 | 7837.5 | 283.5 | 1195 | S580    | 7155.5 | 193.5 | 1257 | S624 | 6473.5 | 373.5 |
| 1134 | S519 | 7826.5 | 373.5 | 1196 | S581    | 7144.5 | 283.5 | 1258 | S625 | 6462.5 | 193.5 |
| 1135 | S520 | 7815.5 | 193.5 | 1197 | S582    | 7133.5 | 373.5 | 1259 | S626 | 6451.5 | 283.5 |
| 1136 | S521 | 7804.5 | 283.5 | 1198 | S583    | 7122.5 | 193.5 | 1260 | S627 | 6440.5 | 373.5 |
| 1137 | S522 | 7793.5 | 373.5 | 1199 | S584    | 7111.5 | 283.5 | 1261 | S628 | 6429.5 | 193.5 |
| 1138 | S523 | 7782.5 | 193.5 | 1200 | S585    | 7100.5 | 373.5 | 1262 | S629 | 6418.5 | 283.5 |
| 1139 | S524 | 7771.5 | 283.5 | 1201 | S586    | 7089.5 | 193.5 | 1263 | S630 | 6407.5 | 373.5 |
| 1140 | S525 | 7760.5 | 373.5 | 1202 | S587    | 7078.5 | 283.5 | 1264 | S631 | 6396.5 | 193.5 |
| 1141 | S526 | 7749.5 | 193.5 | 1203 | S588    | 7067.5 | 373.5 | 1265 | S632 | 6385.5 | 283.5 |
| 1142 | S527 | 7738.5 | 283.5 | 1204 | S589    | 7056.5 | 193.5 | 1266 | S633 | 6374.5 | 373.5 |
| 1143 | S528 | 7727.5 | 373.5 | 1205 | S590    | 7045.5 | 283.5 | 1267 | S634 | 6363.5 | 193.5 |
| 1144 | S529 | 7716.5 | 193.5 | 1206 | S591    | 7034.5 | 373.5 | 1268 | S635 | 6352.5 | 283.5 |
| 1145 | S530 | 7705.5 | 283.5 | 1207 | S592    | 7023.5 | 193.5 | 1269 | S636 | 6341.5 | 373.5 |
| 1146 | S531 | 7694.5 | 373.5 | 1208 | S593    | 7012.5 | 283.5 | 1270 | S637 | 6330.5 | 193.5 |
| 1147 | S532 | 7683.5 | 193.5 | 1209 | S594    | 7001.5 | 373.5 | 1271 | S638 | 6319.5 | 283.5 |
| 1148 | S533 | 7672.5 | 283.5 | 1210 | S595    | 6990.5 | 193.5 | 1272 | S639 | 6308.5 | 373.5 |
| 1149 | S534 | 7661.5 | 373.5 | 1211 | S596    | 6979.5 | 283.5 | 1273 | S640 | 6297.5 | 193.5 |
| 1150 | S535 | 7650.5 | 193.5 | 1212 | S597    | 6968.5 | 373.5 | 1274 | S641 | 6286.5 | 283.5 |
| 1151 | S536 | 7639.5 | 283.5 | 1213 | S598    | 6957.5 | 193.5 | 1275 | S642 | 6275.5 | 373.5 |
| 1152 | S537 | 7628.5 | 373.5 | 1214 | S599    | 6946.5 | 283.5 | 1276 | S643 | 6264.5 | 193.5 |
| 1153 | S538 | 7617.5 | 193.5 | 1215 | S600    | 6935.5 | 373.5 | 1277 | S644 | 6253.5 | 283.5 |
| 1154 | S539 | 7606.5 | 283.5 | 1216 | DUMMY11 | 6924.5 | 193.5 | 1278 | S645 | 6242.5 | 373.5 |
| 1155 | S540 | 7595.5 | 373.5 | 1217 | DUMMY12 | 6913.5 | 283.5 | 1279 | S646 | 6231.5 | 193.5 |
| 1156 | S541 | 7584.5 | 193.5 | 1218 | DUMMY13 | 6902.5 | 373.5 | 1280 | S647 | 6220.5 | 283.5 |
| 1157 | S542 | 7573.5 | 283.5 | 1219 | DUMMY14 | 6891.5 | 193.5 | 1281 | S648 | 6209.5 | 373.5 |
| 1158 | S543 | 7562.5 | 373.5 | 1220 | DUMMY15 | 6880.5 | 283.5 | 1282 | S649 | 6198.5 | 193.5 |
| 1159 | S544 | 7551.5 | 193.5 | 1221 | DUMMY16 | 6869.5 | 373.5 | 1283 | S650 | 6187.5 | 283.5 |
| 1160 | S545 | 7540.5 | 283.5 | 1222 | DUMMY17 | 6858.5 | 193.5 | 1284 | S651 | 6176.5 | 373.5 |
| 1161 | S546 | 7529.5 | 373.5 | 1223 | DUMMY18 | 6847.5 | 283.5 | 1285 | S652 | 6165.5 | 193.5 |
| 1162 | S547 | 7518.5 | 193.5 | 1224 | DUMMY19 | 6836.5 | 373.5 | 1286 | S653 | 6154.5 | 283.5 |
| 1163 | S548 | 7507.5 | 283.5 | 1225 | DUMMY20 | 6825.5 | 193.5 | 1287 | S654 | 6143.5 | 373.5 |
| 1164 | S549 | 7496.5 | 373.5 | 1226 | DUMMY21 | 6814.5 | 283.5 | 1288 | S655 | 6132.5 | 193.5 |
| 1165 | S550 | 7485.5 | 193.5 | 1227 | DUMMY22 | 6803.5 | 373.5 | 1289 | S656 | 6121.5 | 283.5 |
| 1166 | S551 | 7474.5 | 283.5 | 1228 | DUMMY23 | 6792.5 | 193.5 | 1290 | S657 | 6110.5 | 373.5 |
| 1167 | S552 | 7463.5 | 373.5 | 1229 | DUMMY24 | 6781.5 | 283.5 | 1291 | S658 | 6099.5 | 193.5 |
| 1168 | S553 | 7452.5 | 193.5 | 1230 | DUMMY25 | 6770.5 | 373.5 | 1292 | S659 | 6088.5 | 283.5 |
| 1169 | S554 | 7441.5 | 283.5 | 1231 | DUMMY26 | 6759.5 | 193.5 | 1293 | S660 | 6077.5 | 373.5 |
| 1170 | S555 | 7430.5 | 373.5 | 1232 | DUMMY27 | 6748.5 | 283.5 | 1294 | S661 | 6066.5 | 193.5 |
| 1171 | S556 | 7419.5 | 193.5 | 1233 | DUMMY28 | 6737.5 | 373.5 | 1295 | S662 | 6055.5 | 283.5 |
| 1172 | S557 | 7408.5 | 283.5 | 1234 | S601    | 6726.5 | 193.5 | 1296 | S663 | 6044.5 | 373.5 |
| 1173 | S558 | 7397.5 | 373.5 | 1235 | S602    | 6715.5 | 283.5 | 1297 | S664 | 6033.5 | 193.5 |
| 1174 | S559 | 7386.5 | 193.5 | 1236 | S603    | 6704.5 | 373.5 | 1298 | S665 | 6022.5 | 283.5 |
| 1175 | S560 | 7375.5 | 283.5 | 1237 | S604    | 6693.5 | 193.5 | 1299 | S666 | 6011.5 | 373.5 |
| 1176 | S561 | 7364.5 | 373.5 | 1238 | S605    | 6682.5 | 283.5 | 1300 | S667 | 6000.5 | 193.5 |



|      |      |        |       |      |      |        |       |      |      |        |       |
|------|------|--------|-------|------|------|--------|-------|------|------|--------|-------|
| 1301 | S668 | 5989.5 | 283.5 | 1363 | S730 | 5307.5 | 193.5 | 1425 | S792 | 4625.5 | 373.5 |
| 1302 | S669 | 5978.5 | 373.5 | 1364 | S731 | 5296.5 | 283.5 | 1426 | S793 | 4614.5 | 193.5 |
| 1303 | S670 | 5967.5 | 193.5 | 1365 | S732 | 5285.5 | 373.5 | 1427 | S794 | 4603.5 | 283.5 |
| 1304 | S671 | 5956.5 | 283.5 | 1366 | S733 | 5274.5 | 193.5 | 1428 | S795 | 4592.5 | 373.5 |
| 1305 | S672 | 5945.5 | 373.5 | 1367 | S734 | 5263.5 | 283.5 | 1429 | S796 | 4581.5 | 193.5 |
| 1306 | S673 | 5934.5 | 193.5 | 1368 | S735 | 5252.5 | 373.5 | 1430 | S797 | 4570.5 | 283.5 |
| 1307 | S674 | 5923.5 | 283.5 | 1369 | S736 | 5241.5 | 193.5 | 1431 | S798 | 4559.5 | 373.5 |
| 1308 | S675 | 5912.5 | 373.5 | 1370 | S737 | 5230.5 | 283.5 | 1432 | S799 | 4548.5 | 193.5 |
| 1309 | S676 | 5901.5 | 193.5 | 1371 | S738 | 5219.5 | 373.5 | 1433 | S800 | 4537.5 | 283.5 |
| 1310 | S677 | 5890.5 | 283.5 | 1372 | S739 | 5208.5 | 193.5 | 1434 | S801 | 4526.5 | 373.5 |
| 1311 | S678 | 5879.5 | 373.5 | 1373 | S740 | 5197.5 | 283.5 | 1435 | S802 | 4515.5 | 193.5 |
| 1312 | S679 | 5868.5 | 193.5 | 1374 | S741 | 5186.5 | 373.5 | 1436 | S803 | 4504.5 | 283.5 |
| 1313 | S680 | 5857.5 | 283.5 | 1375 | S742 | 5175.5 | 193.5 | 1437 | S804 | 4493.5 | 373.5 |
| 1314 | S681 | 5846.5 | 373.5 | 1376 | S743 | 5164.5 | 283.5 | 1438 | S805 | 4482.5 | 193.5 |
| 1315 | S682 | 5835.5 | 193.5 | 1377 | S744 | 5153.5 | 373.5 | 1439 | S806 | 4471.5 | 283.5 |
| 1316 | S683 | 5824.5 | 283.5 | 1378 | S745 | 5142.5 | 193.5 | 1440 | S807 | 4460.5 | 373.5 |
| 1317 | S684 | 5813.5 | 373.5 | 1379 | S746 | 5131.5 | 283.5 | 1441 | S808 | 4449.5 | 193.5 |
| 1318 | S685 | 5802.5 | 193.5 | 1380 | S747 | 5120.5 | 373.5 | 1442 | S809 | 4438.5 | 283.5 |
| 1319 | S686 | 5791.5 | 283.5 | 1381 | S748 | 5109.5 | 193.5 | 1443 | S810 | 4427.5 | 373.5 |
| 1320 | S687 | 5780.5 | 373.5 | 1382 | S749 | 5098.5 | 283.5 | 1444 | S811 | 4416.5 | 193.5 |
| 1321 | S688 | 5769.5 | 193.5 | 1383 | S750 | 5087.5 | 373.5 | 1445 | S812 | 4405.5 | 283.5 |
| 1322 | S689 | 5758.5 | 283.5 | 1384 | S751 | 5076.5 | 193.5 | 1446 | S813 | 4394.5 | 373.5 |
| 1323 | S690 | 5747.5 | 373.5 | 1385 | S752 | 5065.5 | 283.5 | 1447 | S814 | 4383.5 | 193.5 |
| 1324 | S691 | 5736.5 | 193.5 | 1386 | S753 | 5054.5 | 373.5 | 1448 | S815 | 4372.5 | 283.5 |
| 1325 | S692 | 5725.5 | 283.5 | 1387 | S754 | 5043.5 | 193.5 | 1449 | S816 | 4361.5 | 373.5 |
| 1326 | S693 | 5714.5 | 373.5 | 1388 | S755 | 5032.5 | 283.5 | 1450 | S817 | 4350.5 | 193.5 |
| 1327 | S694 | 5703.5 | 193.5 | 1389 | S756 | 5021.5 | 373.5 | 1451 | S818 | 4339.5 | 283.5 |
| 1328 | S695 | 5692.5 | 283.5 | 1390 | S757 | 5010.5 | 193.5 | 1452 | S819 | 4328.5 | 373.5 |
| 1329 | S696 | 5681.5 | 373.5 | 1391 | S758 | 4999.5 | 283.5 | 1453 | S820 | 4317.5 | 193.5 |
| 1330 | S697 | 5670.5 | 193.5 | 1392 | S759 | 4988.5 | 373.5 | 1454 | S821 | 4306.5 | 283.5 |
| 1331 | S698 | 5659.5 | 283.5 | 1393 | S760 | 4977.5 | 193.5 | 1455 | S822 | 4295.5 | 373.5 |
| 1332 | S699 | 5648.5 | 373.5 | 1394 | S761 | 4966.5 | 283.5 | 1456 | S823 | 4284.5 | 193.5 |
| 1333 | S700 | 5637.5 | 193.5 | 1395 | S762 | 4955.5 | 373.5 | 1457 | S824 | 4273.5 | 283.5 |
| 1334 | S701 | 5626.5 | 283.5 | 1396 | S763 | 4944.5 | 193.5 | 1458 | S825 | 4262.5 | 373.5 |
| 1335 | S702 | 5615.5 | 373.5 | 1397 | S764 | 4933.5 | 283.5 | 1459 | S826 | 4251.5 | 193.5 |
| 1336 | S703 | 5604.5 | 193.5 | 1398 | S765 | 4922.5 | 373.5 | 1460 | S827 | 4240.5 | 283.5 |
| 1337 | S704 | 5593.5 | 283.5 | 1399 | S766 | 4911.5 | 193.5 | 1461 | S828 | 4229.5 | 373.5 |
| 1338 | S705 | 5582.5 | 373.5 | 1400 | S767 | 4900.5 | 283.5 | 1462 | S829 | 4218.5 | 193.5 |
| 1339 | S706 | 5571.5 | 193.5 | 1401 | S768 | 4889.5 | 373.5 | 1463 | S830 | 4207.5 | 283.5 |
| 1340 | S707 | 5560.5 | 283.5 | 1402 | S769 | 4878.5 | 193.5 | 1464 | S831 | 4196.5 | 373.5 |
| 1341 | S708 | 5549.5 | 373.5 | 1403 | S770 | 4867.5 | 283.5 | 1465 | S832 | 4185.5 | 193.5 |
| 1342 | S709 | 5538.5 | 193.5 | 1404 | S771 | 4856.5 | 373.5 | 1466 | S833 | 4174.5 | 283.5 |
| 1343 | S710 | 5527.5 | 283.5 | 1405 | S772 | 4845.5 | 193.5 | 1467 | S834 | 4163.5 | 373.5 |
| 1344 | S711 | 5516.5 | 373.5 | 1406 | S773 | 4834.5 | 283.5 | 1468 | S835 | 4152.5 | 193.5 |
| 1345 | S712 | 5505.5 | 193.5 | 1407 | S774 | 4823.5 | 373.5 | 1469 | S836 | 4141.5 | 283.5 |
| 1346 | S713 | 5494.5 | 283.5 | 1408 | S775 | 4812.5 | 193.5 | 1470 | S837 | 4130.5 | 373.5 |
| 1347 | S714 | 5483.5 | 373.5 | 1409 | S776 | 4801.5 | 283.5 | 1471 | S838 | 4119.5 | 193.5 |
| 1348 | S715 | 5472.5 | 193.5 | 1410 | S777 | 4790.5 | 373.5 | 1472 | S839 | 4108.5 | 283.5 |
| 1349 | S716 | 5461.5 | 283.5 | 1411 | S778 | 4779.5 | 193.5 | 1473 | S840 | 4097.5 | 373.5 |
| 1350 | S717 | 5450.5 | 373.5 | 1412 | S779 | 4768.5 | 283.5 | 1474 | S841 | 4086.5 | 193.5 |
| 1351 | S718 | 5439.5 | 193.5 | 1413 | S780 | 4757.5 | 373.5 | 1475 | S842 | 4075.5 | 283.5 |
| 1352 | S719 | 5428.5 | 283.5 | 1414 | S781 | 4746.5 | 193.5 | 1476 | S843 | 4064.5 | 373.5 |
| 1353 | S720 | 5417.5 | 373.5 | 1415 | S782 | 4735.5 | 283.5 | 1477 | S844 | 4053.5 | 193.5 |
| 1354 | S721 | 5406.5 | 193.5 | 1416 | S783 | 4724.5 | 373.5 | 1478 | S845 | 4042.5 | 283.5 |
| 1355 | S722 | 5395.5 | 283.5 | 1417 | S784 | 4713.5 | 193.5 | 1479 | S846 | 4031.5 | 373.5 |
| 1356 | S723 | 5384.5 | 373.5 | 1418 | S785 | 4702.5 | 283.5 | 1480 | S847 | 4020.5 | 193.5 |
| 1357 | S724 | 5373.5 | 193.5 | 1419 | S786 | 4691.5 | 373.5 | 1481 | S848 | 4009.5 | 283.5 |
| 1358 | S725 | 5362.5 | 283.5 | 1420 | S787 | 4680.5 | 193.5 | 1482 | S849 | 3998.5 | 373.5 |
| 1359 | S726 | 5351.5 | 373.5 | 1421 | S788 | 4669.5 | 283.5 | 1483 | S850 | 3987.5 | 193.5 |
| 1360 | S727 | 5340.5 | 193.5 | 1422 | S789 | 4658.5 | 373.5 | 1484 | S851 | 3976.5 | 283.5 |
| 1361 | S728 | 5329.5 | 283.5 | 1423 | S790 | 4647.5 | 193.5 | 1485 | S852 | 3965.5 | 373.5 |
| 1362 | S729 | 5318.5 | 373.5 | 1424 | S791 | 4636.5 | 283.5 | 1486 | S853 | 3954.5 | 193.5 |

|      |      |        |       |      |      |        |       |      |       |        |       |
|------|------|--------|-------|------|------|--------|-------|------|-------|--------|-------|
| 1487 | S854 | 3943.5 | 283.5 | 1549 | S916 | 3261.5 | 193.5 | 1611 | S978  | 2579.5 | 373.5 |
| 1488 | S855 | 3932.5 | 373.5 | 1550 | S917 | 3250.5 | 283.5 | 1612 | S979  | 2568.5 | 193.5 |
| 1489 | S856 | 3921.5 | 193.5 | 1551 | S918 | 3239.5 | 373.5 | 1613 | S980  | 2557.5 | 283.5 |
| 1490 | S857 | 3910.5 | 283.5 | 1552 | S919 | 3228.5 | 193.5 | 1614 | S981  | 2546.5 | 373.5 |
| 1491 | S858 | 3899.5 | 373.5 | 1553 | S920 | 3217.5 | 283.5 | 1615 | S982  | 2535.5 | 193.5 |
| 1492 | S859 | 3888.5 | 193.5 | 1554 | S921 | 3206.5 | 373.5 | 1616 | S983  | 2524.5 | 283.5 |
| 1493 | S860 | 3877.5 | 283.5 | 1555 | S922 | 3195.5 | 193.5 | 1617 | S984  | 2513.5 | 373.5 |
| 1494 | S861 | 3866.5 | 373.5 | 1556 | S923 | 3184.5 | 283.5 | 1618 | S985  | 2502.5 | 193.5 |
| 1495 | S862 | 3855.5 | 193.5 | 1557 | S924 | 3173.5 | 373.5 | 1619 | S986  | 2491.5 | 283.5 |
| 1496 | S863 | 3844.5 | 283.5 | 1558 | S925 | 3162.5 | 193.5 | 1620 | S987  | 2480.5 | 373.5 |
| 1497 | S864 | 3833.5 | 373.5 | 1559 | S926 | 3151.5 | 283.5 | 1621 | S988  | 2469.5 | 193.5 |
| 1498 | S865 | 3822.5 | 193.5 | 1560 | S927 | 3140.5 | 373.5 | 1622 | S989  | 2458.5 | 283.5 |
| 1499 | S866 | 3811.5 | 283.5 | 1561 | S928 | 3129.5 | 193.5 | 1623 | S990  | 2447.5 | 373.5 |
| 1500 | S867 | 3800.5 | 373.5 | 1562 | S929 | 3118.5 | 283.5 | 1624 | S991  | 2436.5 | 193.5 |
| 1501 | S868 | 3789.5 | 193.5 | 1563 | S930 | 3107.5 | 373.5 | 1625 | S992  | 2425.5 | 283.5 |
| 1502 | S869 | 3778.5 | 283.5 | 1564 | S931 | 3096.5 | 193.5 | 1626 | S993  | 2414.5 | 373.5 |
| 1503 | S870 | 3767.5 | 373.5 | 1565 | S932 | 3085.5 | 283.5 | 1627 | S994  | 2403.5 | 193.5 |
| 1504 | S871 | 3756.5 | 193.5 | 1566 | S933 | 3074.5 | 373.5 | 1628 | S995  | 2392.5 | 283.5 |
| 1505 | S872 | 3745.5 | 283.5 | 1567 | S934 | 3063.5 | 193.5 | 1629 | S996  | 2381.5 | 373.5 |
| 1506 | S873 | 3734.5 | 373.5 | 1568 | S935 | 3052.5 | 283.5 | 1630 | S997  | 2370.5 | 193.5 |
| 1507 | S874 | 3723.5 | 193.5 | 1569 | S936 | 3041.5 | 373.5 | 1631 | S998  | 2359.5 | 283.5 |
| 1508 | S875 | 3712.5 | 283.5 | 1570 | S937 | 3030.5 | 193.5 | 1632 | S999  | 2348.5 | 373.5 |
| 1509 | S876 | 3701.5 | 373.5 | 1571 | S938 | 3019.5 | 283.5 | 1633 | S1000 | 2337.5 | 193.5 |
| 1510 | S877 | 3690.5 | 193.5 | 1572 | S939 | 3008.5 | 373.5 | 1634 | S1001 | 2326.5 | 283.5 |
| 1511 | S878 | 3679.5 | 283.5 | 1573 | S940 | 2997.5 | 193.5 | 1635 | S1002 | 2315.5 | 373.5 |
| 1512 | S879 | 3668.5 | 373.5 | 1574 | S941 | 2986.5 | 283.5 | 1636 | S1003 | 2304.5 | 193.5 |
| 1513 | S880 | 3657.5 | 193.5 | 1575 | S942 | 2975.5 | 373.5 | 1637 | S1004 | 2293.5 | 283.5 |
| 1514 | S881 | 3646.5 | 283.5 | 1576 | S943 | 2964.5 | 193.5 | 1638 | S1005 | 2282.5 | 373.5 |
| 1515 | S882 | 3635.5 | 373.5 | 1577 | S944 | 2953.5 | 283.5 | 1639 | S1006 | 2271.5 | 193.5 |
| 1516 | S883 | 3624.5 | 193.5 | 1578 | S945 | 2942.5 | 373.5 | 1640 | S1007 | 2260.5 | 283.5 |
| 1517 | S884 | 3613.5 | 283.5 | 1579 | S946 | 2931.5 | 193.5 | 1641 | S1008 | 2249.5 | 373.5 |
| 1518 | S885 | 3602.5 | 373.5 | 1580 | S947 | 2920.5 | 283.5 | 1642 | S1009 | 2238.5 | 193.5 |
| 1519 | S886 | 3591.5 | 193.5 | 1581 | S948 | 2909.5 | 373.5 | 1643 | S1010 | 2227.5 | 283.5 |
| 1520 | S887 | 3580.5 | 283.5 | 1582 | S949 | 2898.5 | 193.5 | 1644 | S1011 | 2216.5 | 373.5 |
| 1521 | S888 | 3569.5 | 373.5 | 1583 | S950 | 2887.5 | 283.5 | 1645 | S1012 | 2205.5 | 193.5 |
| 1522 | S889 | 3558.5 | 193.5 | 1584 | S951 | 2876.5 | 373.5 | 1646 | S1013 | 2194.5 | 283.5 |
| 1523 | S890 | 3547.5 | 283.5 | 1585 | S952 | 2865.5 | 193.5 | 1647 | S1014 | 2183.5 | 373.5 |
| 1524 | S891 | 3536.5 | 373.5 | 1586 | S953 | 2854.5 | 283.5 | 1648 | S1015 | 2172.5 | 193.5 |
| 1525 | S892 | 3525.5 | 193.5 | 1587 | S954 | 2843.5 | 373.5 | 1649 | S1016 | 2161.5 | 283.5 |
| 1526 | S893 | 3514.5 | 283.5 | 1588 | S955 | 2832.5 | 193.5 | 1650 | S1017 | 2150.5 | 373.5 |
| 1527 | S894 | 3503.5 | 373.5 | 1589 | S956 | 2821.5 | 283.5 | 1651 | S1018 | 2139.5 | 193.5 |
| 1528 | S895 | 3492.5 | 193.5 | 1590 | S957 | 2810.5 | 373.5 | 1652 | S1019 | 2128.5 | 283.5 |
| 1529 | S896 | 3481.5 | 283.5 | 1591 | S958 | 2799.5 | 193.5 | 1653 | S1020 | 2117.5 | 373.5 |
| 1530 | S897 | 3470.5 | 373.5 | 1592 | S959 | 2788.5 | 283.5 | 1654 | S1021 | 2106.5 | 193.5 |
| 1531 | S898 | 3459.5 | 193.5 | 1593 | S960 | 2777.5 | 373.5 | 1655 | S1022 | 2095.5 | 283.5 |
| 1532 | S899 | 3448.5 | 283.5 | 1594 | S961 | 2766.5 | 193.5 | 1656 | S1023 | 2084.5 | 373.5 |
| 1533 | S900 | 3437.5 | 373.5 | 1595 | S962 | 2755.5 | 283.5 | 1657 | S1024 | 2073.5 | 193.5 |
| 1534 | S901 | 3426.5 | 193.5 | 1596 | S963 | 2744.5 | 373.5 | 1658 | S1025 | 2062.5 | 283.5 |
| 1535 | S902 | 3415.5 | 283.5 | 1597 | S964 | 2733.5 | 193.5 | 1659 | S1026 | 2051.5 | 373.5 |
| 1536 | S903 | 3404.5 | 373.5 | 1598 | S965 | 2722.5 | 283.5 | 1660 | S1027 | 2040.5 | 193.5 |
| 1537 | S904 | 3393.5 | 193.5 | 1599 | S966 | 2711.5 | 373.5 | 1661 | S1028 | 2029.5 | 283.5 |
| 1538 | S905 | 3382.5 | 283.5 | 1600 | S967 | 2700.5 | 193.5 | 1662 | S1029 | 2018.5 | 373.5 |
| 1539 | S906 | 3371.5 | 373.5 | 1601 | S968 | 2689.5 | 283.5 | 1663 | S1030 | 2007.5 | 193.5 |
| 1540 | S907 | 3360.5 | 193.5 | 1602 | S969 | 2678.5 | 373.5 | 1664 | S1031 | 1996.5 | 283.5 |
| 1541 | S908 | 3349.5 | 283.5 | 1603 | S970 | 2667.5 | 193.5 | 1665 | S1032 | 1985.5 | 373.5 |
| 1542 | S909 | 3338.5 | 373.5 | 1604 | S971 | 2656.5 | 283.5 | 1666 | S1033 | 1974.5 | 193.5 |
| 1543 | S910 | 3327.5 | 193.5 | 1605 | S972 | 2645.5 | 373.5 | 1667 | S1034 | 1963.5 | 283.5 |
| 1544 | S911 | 3316.5 | 283.5 | 1606 | S973 | 2634.5 | 193.5 | 1668 | S1035 | 1952.5 | 373.5 |
| 1545 | S912 | 3305.5 | 373.5 | 1607 | S974 | 2623.5 | 283.5 | 1669 | S1036 | 1941.5 | 193.5 |
| 1546 | S913 | 3294.5 | 193.5 | 1608 | S975 | 2612.5 | 373.5 | 1670 | S1037 | 1930.5 | 283.5 |
| 1547 | S914 | 3283.5 | 283.5 | 1609 | S976 | 2601.5 | 193.5 | 1671 | S1038 | 1919.5 | 373.5 |
| 1548 | S915 | 3272.5 | 373.5 | 1610 | S977 | 2590.5 | 283.5 | 1672 | S1039 | 1908.5 | 193.5 |

|      |       |        |       |      |       |        |       |      |         |        |       |
|------|-------|--------|-------|------|-------|--------|-------|------|---------|--------|-------|
| 1673 | S1040 | 1897.5 | 283.5 | 1735 | S1102 | 1215.5 | 193.5 | 1797 | S1164   | 533.5  | 373.5 |
| 1674 | S1041 | 1886.5 | 373.5 | 1736 | S1103 | 1204.5 | 283.5 | 1798 | S1165   | 522.5  | 193.5 |
| 1675 | S1042 | 1875.5 | 193.5 | 1737 | S1104 | 1193.5 | 373.5 | 1799 | S1166   | 511.5  | 283.5 |
| 1676 | S1043 | 1864.5 | 283.5 | 1738 | S1105 | 1182.5 | 193.5 | 1800 | S1167   | 500.5  | 373.5 |
| 1677 | S1044 | 1853.5 | 373.5 | 1739 | S1106 | 1171.5 | 283.5 | 1801 | S1168   | 489.5  | 193.5 |
| 1678 | S1045 | 1842.5 | 193.5 | 1740 | S1107 | 1160.5 | 373.5 | 1802 | S1169   | 478.5  | 283.5 |
| 1679 | S1046 | 1831.5 | 283.5 | 1741 | S1108 | 1149.5 | 193.5 | 1803 | S1170   | 467.5  | 373.5 |
| 1680 | S1047 | 1820.5 | 373.5 | 1742 | S1109 | 1138.5 | 283.5 | 1804 | S1171   | 456.5  | 193.5 |
| 1681 | S1048 | 1809.5 | 193.5 | 1743 | S1110 | 1127.5 | 373.5 | 1805 | S1172   | 445.5  | 283.5 |
| 1682 | S1049 | 1798.5 | 283.5 | 1744 | S1111 | 1116.5 | 193.5 | 1806 | S1173   | 434.5  | 373.5 |
| 1683 | S1050 | 1787.5 | 373.5 | 1745 | S1112 | 1105.5 | 283.5 | 1807 | S1174   | 423.5  | 193.5 |
| 1684 | S1051 | 1776.5 | 193.5 | 1746 | S1113 | 1094.5 | 373.5 | 1808 | S1175   | 412.5  | 283.5 |
| 1685 | S1052 | 1765.5 | 283.5 | 1747 | S1114 | 1083.5 | 193.5 | 1809 | S1176   | 401.5  | 373.5 |
| 1686 | S1053 | 1754.5 | 373.5 | 1748 | S1115 | 1072.5 | 283.5 | 1810 | S1177   | 390.5  | 193.5 |
| 1687 | S1054 | 1743.5 | 193.5 | 1749 | S1116 | 1061.5 | 373.5 | 1811 | S1178   | 379.5  | 283.5 |
| 1688 | S1055 | 1732.5 | 283.5 | 1750 | S1117 | 1050.5 | 193.5 | 1812 | S1179   | 368.5  | 373.5 |
| 1689 | S1056 | 1721.5 | 373.5 | 1751 | S1118 | 1039.5 | 283.5 | 1813 | S1180   | 357.5  | 193.5 |
| 1690 | S1057 | 1710.5 | 193.5 | 1752 | S1119 | 1028.5 | 373.5 | 1814 | S1181   | 346.5  | 283.5 |
| 1691 | S1058 | 1699.5 | 283.5 | 1753 | S1120 | 1017.5 | 193.5 | 1815 | S1182   | 335.5  | 373.5 |
| 1692 | S1059 | 1688.5 | 373.5 | 1754 | S1121 | 1006.5 | 283.5 | 1816 | S1183   | 324.5  | 193.5 |
| 1693 | S1060 | 1677.5 | 193.5 | 1755 | S1122 | 995.5  | 373.5 | 1817 | S1184   | 313.5  | 283.5 |
| 1694 | S1061 | 1666.5 | 283.5 | 1756 | S1123 | 984.5  | 193.5 | 1818 | S1185   | 302.5  | 373.5 |
| 1695 | S1062 | 1655.5 | 373.5 | 1757 | S1124 | 973.5  | 283.5 | 1819 | S1186   | 291.5  | 193.5 |
| 1696 | S1063 | 1644.5 | 193.5 | 1758 | S1125 | 962.5  | 373.5 | 1820 | S1187   | 280.5  | 283.5 |
| 1697 | S1064 | 1633.5 | 283.5 | 1759 | S1126 | 951.5  | 193.5 | 1821 | S1188   | 269.5  | 373.5 |
| 1698 | S1065 | 1622.5 | 373.5 | 1760 | S1127 | 940.5  | 283.5 | 1822 | S1189   | 258.5  | 193.5 |
| 1699 | S1066 | 1611.5 | 193.5 | 1761 | S1128 | 929.5  | 373.5 | 1823 | S1190   | 247.5  | 283.5 |
| 1700 | S1067 | 1600.5 | 283.5 | 1762 | S1129 | 918.5  | 193.5 | 1824 | S1191   | 236.5  | 373.5 |
| 1701 | S1068 | 1589.5 | 373.5 | 1763 | S1130 | 907.5  | 283.5 | 1825 | S1192   | 225.5  | 193.5 |
| 1702 | S1069 | 1578.5 | 193.5 | 1764 | S1131 | 896.5  | 373.5 | 1826 | S1193   | 214.5  | 283.5 |
| 1703 | S1070 | 1567.5 | 283.5 | 1765 | S1132 | 885.5  | 193.5 | 1827 | S1194   | 203.5  | 373.5 |
| 1704 | S1071 | 1556.5 | 373.5 | 1766 | S1133 | 874.5  | 283.5 | 1828 | S1195   | 192.5  | 193.5 |
| 1705 | S1072 | 1545.5 | 193.5 | 1767 | S1134 | 863.5  | 373.5 | 1829 | S1196   | 181.5  | 283.5 |
| 1706 | S1073 | 1534.5 | 283.5 | 1768 | S1135 | 852.5  | 193.5 | 1830 | S1197   | 170.5  | 373.5 |
| 1707 | S1074 | 1523.5 | 373.5 | 1769 | S1136 | 841.5  | 283.5 | 1831 | S1198   | 159.5  | 193.5 |
| 1708 | S1075 | 1512.5 | 193.5 | 1770 | S1137 | 830.5  | 373.5 | 1832 | S1199   | 148.5  | 283.5 |
| 1709 | S1076 | 1501.5 | 283.5 | 1771 | S1138 | 819.5  | 193.5 | 1833 | S1200   | 137.5  | 373.5 |
| 1710 | S1077 | 1490.5 | 373.5 | 1772 | S1139 | 808.5  | 283.5 | 1834 | DUMMY29 | 126.5  | 193.5 |
| 1711 | S1078 | 1479.5 | 193.5 | 1773 | S1140 | 797.5  | 373.5 | 1835 | DUMMY30 | 115.5  | 283.5 |
| 1712 | S1079 | 1468.5 | 283.5 | 1774 | S1141 | 786.5  | 193.5 | 1836 | DUMMY31 | 104.5  | 373.5 |
| 1713 | S1080 | 1457.5 | 373.5 | 1775 | S1142 | 775.5  | 283.5 | 1837 | DUMMY32 | 93.5   | 193.5 |
| 1714 | S1081 | 1446.5 | 193.5 | 1776 | S1143 | 764.5  | 373.5 | 1838 | DUMMY33 | 82.5   | 283.5 |
| 1715 | S1082 | 1435.5 | 283.5 | 1777 | S1144 | 753.5  | 193.5 | 1839 | DUMMY34 | 71.5   | 373.5 |
| 1716 | S1083 | 1424.5 | 373.5 | 1778 | S1145 | 742.5  | 283.5 | 1840 | DUMMY35 | 60.5   | 193.5 |
| 1717 | S1084 | 1413.5 | 193.5 | 1779 | S1146 | 731.5  | 373.5 | 1841 | DUMMY36 | 49.5   | 283.5 |
| 1718 | S1085 | 1402.5 | 283.5 | 1780 | S1147 | 720.5  | 193.5 | 1842 | DUMMY37 | 38.5   | 373.5 |
| 1719 | S1086 | 1391.5 | 373.5 | 1781 | S1148 | 709.5  | 283.5 | 1843 | DUMMY38 | 27.5   | 193.5 |
| 1720 | S1087 | 1380.5 | 193.5 | 1782 | S1149 | 698.5  | 373.5 | 1844 | DUMMY39 | 16.5   | 283.5 |
| 1721 | S1088 | 1369.5 | 283.5 | 1783 | S1150 | 687.5  | 193.5 | 1845 | DUMMY40 | 5.5    | 373.5 |
| 1722 | S1089 | 1358.5 | 373.5 | 1784 | S1151 | 676.5  | 283.5 | 1846 | DUMMY41 | -5.5   | 193.5 |
| 1723 | S1090 | 1347.5 | 193.5 | 1785 | S1152 | 665.5  | 373.5 | 1847 | DUMMY42 | -16.5  | 283.5 |
| 1724 | S1091 | 1336.5 | 283.5 | 1786 | S1153 | 654.5  | 193.5 | 1848 | DUMMY43 | -27.5  | 373.5 |
| 1725 | S1092 | 1325.5 | 373.5 | 1787 | S1154 | 643.5  | 283.5 | 1849 | DUMMY44 | -38.5  | 193.5 |
| 1726 | S1093 | 1314.5 | 193.5 | 1788 | S1155 | 632.5  | 373.5 | 1850 | DUMMY45 | -49.5  | 283.5 |
| 1727 | S1094 | 1303.5 | 283.5 | 1789 | S1156 | 621.5  | 193.5 | 1851 | DUMMY46 | -60.5  | 373.5 |
| 1728 | S1095 | 1292.5 | 373.5 | 1790 | S1157 | 610.5  | 283.5 | 1852 | DUMMY47 | -71.5  | 193.5 |
| 1729 | S1096 | 1281.5 | 193.5 | 1791 | S1158 | 599.5  | 373.5 | 1853 | DUMMY48 | -82.5  | 283.5 |
| 1730 | S1097 | 1270.5 | 283.5 | 1792 | S1159 | 588.5  | 193.5 | 1854 | DUMMY49 | -93.5  | 373.5 |
| 1731 | S1098 | 1259.5 | 373.5 | 1793 | S1160 | 577.5  | 283.5 | 1855 | DUMMY50 | -104.5 | 193.5 |
| 1732 | S1099 | 1248.5 | 193.5 | 1794 | S1161 | 566.5  | 373.5 | 1856 | DUMMY51 | -115.5 | 283.5 |
| 1733 | S1100 | 1237.5 | 283.5 | 1795 | S1162 | 555.5  | 193.5 | 1857 | DUMMY52 | -126.5 | 373.5 |
| 1734 | S1101 | 1226.5 | 373.5 | 1796 | S1163 | 544.5  | 283.5 | 1858 | S1201   | -137.5 | 193.5 |

|      |       |        |       |      |       |         |       |      |       |         |       |
|------|-------|--------|-------|------|-------|---------|-------|------|-------|---------|-------|
| 1859 | S1202 | -148.5 | 283.5 | 1921 | S1264 | -830.5  | 193.5 | 1983 | S1326 | -1512.5 | 373.5 |
| 1860 | S1203 | -159.5 | 373.5 | 1922 | S1265 | -841.5  | 283.5 | 1984 | S1327 | -1523.5 | 193.5 |
| 1861 | S1204 | -170.5 | 193.5 | 1923 | S1266 | -852.5  | 373.5 | 1985 | S1328 | -1534.5 | 283.5 |
| 1862 | S1205 | -181.5 | 283.5 | 1924 | S1267 | -863.5  | 193.5 | 1986 | S1329 | -1545.5 | 373.5 |
| 1863 | S1206 | -192.5 | 373.5 | 1925 | S1268 | -874.5  | 283.5 | 1987 | S1330 | -1556.5 | 193.5 |
| 1864 | S1207 | -203.5 | 193.5 | 1926 | S1269 | -885.5  | 373.5 | 1988 | S1331 | -1567.5 | 283.5 |
| 1865 | S1208 | -214.5 | 283.5 | 1927 | S1270 | -896.5  | 193.5 | 1989 | S1332 | -1578.5 | 373.5 |
| 1866 | S1209 | -225.5 | 373.5 | 1928 | S1271 | -907.5  | 283.5 | 1990 | S1333 | -1589.5 | 193.5 |
| 1867 | S1210 | -236.5 | 193.5 | 1929 | S1272 | -918.5  | 373.5 | 1991 | S1334 | -1600.5 | 283.5 |
| 1868 | S1211 | -247.5 | 283.5 | 1930 | S1273 | -929.5  | 193.5 | 1992 | S1335 | -1611.5 | 373.5 |
| 1869 | S1212 | -258.5 | 373.5 | 1931 | S1274 | -940.5  | 283.5 | 1993 | S1336 | -1622.5 | 193.5 |
| 1870 | S1213 | -269.5 | 193.5 | 1932 | S1275 | -951.5  | 373.5 | 1994 | S1337 | -1633.5 | 283.5 |
| 1871 | S1214 | -280.5 | 283.5 | 1933 | S1276 | -962.5  | 193.5 | 1995 | S1338 | -1644.5 | 373.5 |
| 1872 | S1215 | -291.5 | 373.5 | 1934 | S1277 | -973.5  | 283.5 | 1996 | S1339 | -1655.5 | 193.5 |
| 1873 | S1216 | -302.5 | 193.5 | 1935 | S1278 | -984.5  | 373.5 | 1997 | S1340 | -1666.5 | 283.5 |
| 1874 | S1217 | -313.5 | 283.5 | 1936 | S1279 | -995.5  | 193.5 | 1998 | S1341 | -1677.5 | 373.5 |
| 1875 | S1218 | -324.5 | 373.5 | 1937 | S1280 | -1006.5 | 283.5 | 1999 | S1342 | -1688.5 | 193.5 |
| 1876 | S1219 | -335.5 | 193.5 | 1938 | S1281 | -1017.5 | 373.5 | 2000 | S1343 | -1699.5 | 283.5 |
| 1877 | S1220 | -346.5 | 283.5 | 1939 | S1282 | -1028.5 | 193.5 | 2001 | S1344 | -1710.5 | 373.5 |
| 1878 | S1221 | -357.5 | 373.5 | 1940 | S1283 | -1039.5 | 283.5 | 2002 | S1345 | -1721.5 | 193.5 |
| 1879 | S1222 | -368.5 | 193.5 | 1941 | S1284 | -1050.5 | 373.5 | 2003 | S1346 | -1732.5 | 283.5 |
| 1880 | S1223 | -379.5 | 283.5 | 1942 | S1285 | -1061.5 | 193.5 | 2004 | S1347 | -1743.5 | 373.5 |
| 1881 | S1224 | -390.5 | 373.5 | 1943 | S1286 | -1072.5 | 283.5 | 2005 | S1348 | -1754.5 | 193.5 |
| 1882 | S1225 | -401.5 | 193.5 | 1944 | S1287 | -1083.5 | 373.5 | 2006 | S1349 | -1765.5 | 283.5 |
| 1883 | S1226 | -412.5 | 283.5 | 1945 | S1288 | -1094.5 | 193.5 | 2007 | S1350 | -1776.5 | 373.5 |
| 1884 | S1227 | -423.5 | 373.5 | 1946 | S1289 | -1105.5 | 283.5 | 2008 | S1351 | -1787.5 | 193.5 |
| 1885 | S1228 | -434.5 | 193.5 | 1947 | S1290 | -1116.5 | 373.5 | 2009 | S1352 | -1798.5 | 283.5 |
| 1886 | S1229 | -445.5 | 283.5 | 1948 | S1291 | -1127.5 | 193.5 | 2010 | S1353 | -1809.5 | 373.5 |
| 1887 | S1230 | -456.5 | 373.5 | 1949 | S1292 | -1138.5 | 283.5 | 2011 | S1354 | -1820.5 | 193.5 |
| 1888 | S1231 | -467.5 | 193.5 | 1950 | S1293 | -1149.5 | 373.5 | 2012 | S1355 | -1831.5 | 283.5 |
| 1889 | S1232 | -478.5 | 283.5 | 1951 | S1294 | -1160.5 | 193.5 | 2013 | S1356 | -1842.5 | 373.5 |
| 1890 | S1233 | -489.5 | 373.5 | 1952 | S1295 | -1171.5 | 283.5 | 2014 | S1357 | -1853.5 | 193.5 |
| 1891 | S1234 | -500.5 | 193.5 | 1953 | S1296 | -1182.5 | 373.5 | 2015 | S1358 | -1864.5 | 283.5 |
| 1892 | S1235 | -511.5 | 283.5 | 1954 | S1297 | -1193.5 | 193.5 | 2016 | S1359 | -1875.5 | 373.5 |
| 1893 | S1236 | -522.5 | 373.5 | 1955 | S1298 | -1204.5 | 283.5 | 2017 | S1360 | -1886.5 | 193.5 |
| 1894 | S1237 | -533.5 | 193.5 | 1956 | S1299 | -1215.5 | 373.5 | 2018 | S1361 | -1897.5 | 283.5 |
| 1895 | S1238 | -544.5 | 283.5 | 1957 | S1300 | -1226.5 | 193.5 | 2019 | S1362 | -1908.5 | 373.5 |
| 1896 | S1239 | -555.5 | 373.5 | 1958 | S1301 | -1237.5 | 283.5 | 2020 | S1363 | -1919.5 | 193.5 |
| 1897 | S1240 | -566.5 | 193.5 | 1959 | S1302 | -1248.5 | 373.5 | 2021 | S1364 | -1930.5 | 283.5 |
| 1898 | S1241 | -577.5 | 283.5 | 1960 | S1303 | -1259.5 | 193.5 | 2022 | S1365 | -1941.5 | 373.5 |
| 1899 | S1242 | -588.5 | 373.5 | 1961 | S1304 | -1270.5 | 283.5 | 2023 | S1366 | -1952.5 | 193.5 |
| 1900 | S1243 | -599.5 | 193.5 | 1962 | S1305 | -1281.5 | 373.5 | 2024 | S1367 | -1963.5 | 283.5 |
| 1901 | S1244 | -610.5 | 283.5 | 1963 | S1306 | -1292.5 | 193.5 | 2025 | S1368 | -1974.5 | 373.5 |
| 1902 | S1245 | -621.5 | 373.5 | 1964 | S1307 | -1303.5 | 283.5 | 2026 | S1369 | -1985.5 | 193.5 |
| 1903 | S1246 | -632.5 | 193.5 | 1965 | S1308 | -1314.5 | 373.5 | 2027 | S1370 | -1996.5 | 283.5 |
| 1904 | S1247 | -643.5 | 283.5 | 1966 | S1309 | -1325.5 | 193.5 | 2028 | S1371 | -2007.5 | 373.5 |
| 1905 | S1248 | -654.5 | 373.5 | 1967 | S1310 | -1336.5 | 283.5 | 2029 | S1372 | -2018.5 | 193.5 |
| 1906 | S1249 | -665.5 | 193.5 | 1968 | S1311 | -1347.5 | 373.5 | 2030 | S1373 | -2029.5 | 283.5 |
| 1907 | S1250 | -676.5 | 283.5 | 1969 | S1312 | -1358.5 | 193.5 | 2031 | S1374 | -2040.5 | 373.5 |
| 1908 | S1251 | -687.5 | 373.5 | 1970 | S1313 | -1369.5 | 283.5 | 2032 | S1375 | -2051.5 | 193.5 |
| 1909 | S1252 | -698.5 | 193.5 | 1971 | S1314 | -1380.5 | 373.5 | 2033 | S1376 | -2062.5 | 283.5 |
| 1910 | S1253 | -709.5 | 283.5 | 1972 | S1315 | -1391.5 | 193.5 | 2034 | S1377 | -2073.5 | 373.5 |
| 1911 | S1254 | -720.5 | 373.5 | 1973 | S1316 | -1402.5 | 283.5 | 2035 | S1378 | -2084.5 | 193.5 |
| 1912 | S1255 | -731.5 | 193.5 | 1974 | S1317 | -1413.5 | 373.5 | 2036 | S1379 | -2095.5 | 283.5 |
| 1913 | S1256 | -742.5 | 283.5 | 1975 | S1318 | -1424.5 | 193.5 | 2037 | S1380 | -2106.5 | 373.5 |
| 1914 | S1257 | -753.5 | 373.5 | 1976 | S1319 | -1435.5 | 283.5 | 2038 | S1381 | -2117.5 | 193.5 |
| 1915 | S1258 | -764.5 | 193.5 | 1977 | S1320 | -1446.5 | 373.5 | 2039 | S1382 | -2128.5 | 283.5 |
| 1916 | S1259 | -775.5 | 283.5 | 1978 | S1321 | -1457.5 | 193.5 | 2040 | S1383 | -2139.5 | 373.5 |
| 1917 | S1260 | -786.5 | 373.5 | 1979 | S1322 | -1468.5 | 283.5 | 2041 | S1384 | -2150.5 | 193.5 |
| 1918 | S1261 | -797.5 | 193.5 | 1980 | S1323 | -1479.5 | 373.5 | 2042 | S1385 | -2161.5 | 283.5 |
| 1919 | S1262 | -808.5 | 283.5 | 1981 | S1324 | -1490.5 | 193.5 | 2043 | S1386 | -2172.5 | 373.5 |
| 1920 | S1263 | -819.5 | 373.5 | 1982 | S1325 | -1501.5 | 283.5 | 2044 | S1387 | -2183.5 | 193.5 |

|      |       |         |       |      |       |         |       |      |       |         |       |
|------|-------|---------|-------|------|-------|---------|-------|------|-------|---------|-------|
| 2045 | S1388 | -2194.5 | 283.5 | 2107 | S1450 | -2876.5 | 193.5 | 2169 | S1512 | -3558.5 | 373.5 |
| 2046 | S1389 | -2205.5 | 373.5 | 2108 | S1451 | -2887.5 | 283.5 | 2170 | S1513 | -3569.5 | 193.5 |
| 2047 | S1390 | -2216.5 | 193.5 | 2109 | S1452 | -2898.5 | 373.5 | 2171 | S1514 | -3580.5 | 283.5 |
| 2048 | S1391 | -2227.5 | 283.5 | 2110 | S1453 | -2909.5 | 193.5 | 2172 | S1515 | -3591.5 | 373.5 |
| 2049 | S1392 | -2238.5 | 373.5 | 2111 | S1454 | -2920.5 | 283.5 | 2173 | S1516 | -3602.5 | 193.5 |
| 2050 | S1393 | -2249.5 | 193.5 | 2112 | S1455 | -2931.5 | 373.5 | 2174 | S1517 | -3613.5 | 283.5 |
| 2051 | S1394 | -2260.5 | 283.5 | 2113 | S1456 | -2942.5 | 193.5 | 2175 | S1518 | -3624.5 | 373.5 |
| 2052 | S1395 | -2271.5 | 373.5 | 2114 | S1457 | -2953.5 | 283.5 | 2176 | S1519 | -3635.5 | 193.5 |
| 2053 | S1396 | -2282.5 | 193.5 | 2115 | S1458 | -2964.5 | 373.5 | 2177 | S1520 | -3646.5 | 283.5 |
| 2054 | S1397 | -2293.5 | 283.5 | 2116 | S1459 | -2975.5 | 193.5 | 2178 | S1521 | -3657.5 | 373.5 |
| 2055 | S1398 | -2304.5 | 373.5 | 2117 | S1460 | -2986.5 | 283.5 | 2179 | S1522 | -3668.5 | 193.5 |
| 2056 | S1399 | -2315.5 | 193.5 | 2118 | S1461 | -2997.5 | 373.5 | 2180 | S1523 | -3679.5 | 283.5 |
| 2057 | S1400 | -2326.5 | 283.5 | 2119 | S1462 | -3008.5 | 193.5 | 2181 | S1524 | -3690.5 | 373.5 |
| 2058 | S1401 | -2337.5 | 373.5 | 2120 | S1463 | -3019.5 | 283.5 | 2182 | S1525 | -3701.5 | 193.5 |
| 2059 | S1402 | -2348.5 | 193.5 | 2121 | S1464 | -3030.5 | 373.5 | 2183 | S1526 | -3712.5 | 283.5 |
| 2060 | S1403 | -2359.5 | 283.5 | 2122 | S1465 | -3041.5 | 193.5 | 2184 | S1527 | -3723.5 | 373.5 |
| 2061 | S1404 | -2370.5 | 373.5 | 2123 | S1466 | -3052.5 | 283.5 | 2185 | S1528 | -3734.5 | 193.5 |
| 2062 | S1405 | -2381.5 | 193.5 | 2124 | S1467 | -3063.5 | 373.5 | 2186 | S1529 | -3745.5 | 283.5 |
| 2063 | S1406 | -2392.5 | 283.5 | 2125 | S1468 | -3074.5 | 193.5 | 2187 | S1530 | -3756.5 | 373.5 |
| 2064 | S1407 | -2403.5 | 373.5 | 2126 | S1469 | -3085.5 | 283.5 | 2188 | S1531 | -3767.5 | 193.5 |
| 2065 | S1408 | -2414.5 | 193.5 | 2127 | S1470 | -3096.5 | 373.5 | 2189 | S1532 | -3778.5 | 283.5 |
| 2066 | S1409 | -2425.5 | 283.5 | 2128 | S1471 | -3107.5 | 193.5 | 2190 | S1533 | -3789.5 | 373.5 |
| 2067 | S1410 | -2436.5 | 373.5 | 2129 | S1472 | -3118.5 | 283.5 | 2191 | S1534 | -3800.5 | 193.5 |
| 2068 | S1411 | -2447.5 | 193.5 | 2130 | S1473 | -3129.5 | 373.5 | 2192 | S1535 | -3811.5 | 283.5 |
| 2069 | S1412 | -2458.5 | 283.5 | 2131 | S1474 | -3140.5 | 193.5 | 2193 | S1536 | -3822.5 | 373.5 |
| 2070 | S1413 | -2469.5 | 373.5 | 2132 | S1475 | -3151.5 | 283.5 | 2194 | S1537 | -3833.5 | 193.5 |
| 2071 | S1414 | -2480.5 | 193.5 | 2133 | S1476 | -3162.5 | 373.5 | 2195 | S1538 | -3844.5 | 283.5 |
| 2072 | S1415 | -2491.5 | 283.5 | 2134 | S1477 | -3173.5 | 193.5 | 2196 | S1539 | -3855.5 | 373.5 |
| 2073 | S1416 | -2502.5 | 373.5 | 2135 | S1478 | -3184.5 | 283.5 | 2197 | S1540 | -3866.5 | 193.5 |
| 2074 | S1417 | -2513.5 | 193.5 | 2136 | S1479 | -3195.5 | 373.5 | 2198 | S1541 | -3877.5 | 283.5 |
| 2075 | S1418 | -2524.5 | 283.5 | 2137 | S1480 | -3206.5 | 193.5 | 2199 | S1542 | -3888.5 | 373.5 |
| 2076 | S1419 | -2535.5 | 373.5 | 2138 | S1481 | -3217.5 | 283.5 | 2200 | S1543 | -3899.5 | 193.5 |
| 2077 | S1420 | -2546.5 | 193.5 | 2139 | S1482 | -3228.5 | 373.5 | 2201 | S1544 | -3910.5 | 283.5 |
| 2078 | S1421 | -2557.5 | 283.5 | 2140 | S1483 | -3239.5 | 193.5 | 2202 | S1545 | -3921.5 | 373.5 |
| 2079 | S1422 | -2568.5 | 373.5 | 2141 | S1484 | -3250.5 | 283.5 | 2203 | S1546 | -3932.5 | 193.5 |
| 2080 | S1423 | -2579.5 | 193.5 | 2142 | S1485 | -3261.5 | 373.5 | 2204 | S1547 | -3943.5 | 283.5 |
| 2081 | S1424 | -2590.5 | 283.5 | 2143 | S1486 | -3272.5 | 193.5 | 2205 | S1548 | -3954.5 | 373.5 |
| 2082 | S1425 | -2601.5 | 373.5 | 2144 | S1487 | -3283.5 | 283.5 | 2206 | S1549 | -3965.5 | 193.5 |
| 2083 | S1426 | -2612.5 | 193.5 | 2145 | S1488 | -3294.5 | 373.5 | 2207 | S1550 | -3976.5 | 283.5 |
| 2084 | S1427 | -2623.5 | 283.5 | 2146 | S1489 | -3305.5 | 193.5 | 2208 | S1551 | -3987.5 | 373.5 |
| 2085 | S1428 | -2634.5 | 373.5 | 2147 | S1490 | -3316.5 | 283.5 | 2209 | S1552 | -3998.5 | 193.5 |
| 2086 | S1429 | -2645.5 | 193.5 | 2148 | S1491 | -3327.5 | 373.5 | 2210 | S1553 | -4009.5 | 283.5 |
| 2087 | S1430 | -2656.5 | 283.5 | 2149 | S1492 | -3338.5 | 193.5 | 2211 | S1554 | -4020.5 | 373.5 |
| 2088 | S1431 | -2667.5 | 373.5 | 2150 | S1493 | -3349.5 | 283.5 | 2212 | S1555 | -4031.5 | 193.5 |
| 2089 | S1432 | -2678.5 | 193.5 | 2151 | S1494 | -3360.5 | 373.5 | 2213 | S1556 | -4042.5 | 283.5 |
| 2090 | S1433 | -2689.5 | 283.5 | 2152 | S1495 | -3371.5 | 193.5 | 2214 | S1557 | -4053.5 | 373.5 |
| 2091 | S1434 | -2700.5 | 373.5 | 2153 | S1496 | -3382.5 | 283.5 | 2215 | S1558 | -4064.5 | 193.5 |
| 2092 | S1435 | -2711.5 | 193.5 | 2154 | S1497 | -3393.5 | 373.5 | 2216 | S1559 | -4075.5 | 283.5 |
| 2093 | S1436 | -2722.5 | 283.5 | 2155 | S1498 | -3404.5 | 193.5 | 2217 | S1560 | -4086.5 | 373.5 |
| 2094 | S1437 | -2733.5 | 373.5 | 2156 | S1499 | -3415.5 | 283.5 | 2218 | S1561 | -4097.5 | 193.5 |
| 2095 | S1438 | -2744.5 | 193.5 | 2157 | S1500 | -3426.5 | 373.5 | 2219 | S1562 | -4108.5 | 283.5 |
| 2096 | S1439 | -2755.5 | 283.5 | 2158 | S1501 | -3437.5 | 193.5 | 2220 | S1563 | -4119.5 | 373.5 |
| 2097 | S1440 | -2766.5 | 373.5 | 2159 | S1502 | -3448.5 | 283.5 | 2221 | S1564 | -4130.5 | 193.5 |
| 2098 | S1441 | -2777.5 | 193.5 | 2160 | S1503 | -3459.5 | 373.5 | 2222 | S1565 | -4141.5 | 283.5 |
| 2099 | S1442 | -2788.5 | 283.5 | 2161 | S1504 | -3470.5 | 193.5 | 2223 | S1566 | -4152.5 | 373.5 |
| 2100 | S1443 | -2799.5 | 373.5 | 2162 | S1505 | -3481.5 | 283.5 | 2224 | S1567 | -4163.5 | 193.5 |
| 2101 | S1444 | -2810.5 | 193.5 | 2163 | S1506 | -3492.5 | 373.5 | 2225 | S1568 | -4174.5 | 283.5 |
| 2102 | S1445 | -2821.5 | 283.5 | 2164 | S1507 | -3503.5 | 193.5 | 2226 | S1569 | -4185.5 | 373.5 |
| 2103 | S1446 | -2832.5 | 373.5 | 2165 | S1508 | -3514.5 | 283.5 | 2227 | S1570 | -4196.5 | 193.5 |
| 2104 | S1447 | -2843.5 | 193.5 | 2166 | S1509 | -3525.5 | 373.5 | 2228 | S1571 | -4207.5 | 283.5 |
| 2105 | S1448 | -2854.5 | 283.5 | 2167 | S1510 | -3536.5 | 193.5 | 2229 | S1572 | -4218.5 | 373.5 |
| 2106 | S1449 | -2865.5 | 373.5 | 2168 | S1511 | -3547.5 | 283.5 | 2230 | S1573 | -4229.5 | 193.5 |

|      |       |         |       |      |       |         |       |      |       |         |       |
|------|-------|---------|-------|------|-------|---------|-------|------|-------|---------|-------|
| 2231 | S1574 | -4240.5 | 283.5 | 2293 | S1636 | -4922.5 | 193.5 | 2355 | S1698 | -5604.5 | 373.5 |
| 2232 | S1575 | -4251.5 | 373.5 | 2294 | S1637 | -4933.5 | 283.5 | 2356 | S1699 | -5615.5 | 193.5 |
| 2233 | S1576 | -4262.5 | 193.5 | 2295 | S1638 | -4944.5 | 373.5 | 2357 | S1700 | -5626.5 | 283.5 |
| 2234 | S1577 | -4273.5 | 283.5 | 2296 | S1639 | -4955.5 | 193.5 | 2358 | S1701 | -5637.5 | 373.5 |
| 2235 | S1578 | -4284.5 | 373.5 | 2297 | S1640 | -4966.5 | 283.5 | 2359 | S1702 | -5648.5 | 193.5 |
| 2236 | S1579 | -4295.5 | 193.5 | 2298 | S1641 | -4977.5 | 373.5 | 2360 | S1703 | -5659.5 | 283.5 |
| 2237 | S1580 | -4306.5 | 283.5 | 2299 | S1642 | -4988.5 | 193.5 | 2361 | S1704 | -5670.5 | 373.5 |
| 2238 | S1581 | -4317.5 | 373.5 | 2300 | S1643 | -4999.5 | 283.5 | 2362 | S1705 | -5681.5 | 193.5 |
| 2239 | S1582 | -4328.5 | 193.5 | 2301 | S1644 | -5010.5 | 373.5 | 2363 | S1706 | -5692.5 | 283.5 |
| 2240 | S1583 | -4339.5 | 283.5 | 2302 | S1645 | -5021.5 | 193.5 | 2364 | S1707 | -5703.5 | 373.5 |
| 2241 | S1584 | -4350.5 | 373.5 | 2303 | S1646 | -5032.5 | 283.5 | 2365 | S1708 | -5714.5 | 193.5 |
| 2242 | S1585 | -4361.5 | 193.5 | 2304 | S1647 | -5043.5 | 373.5 | 2366 | S1709 | -5725.5 | 283.5 |
| 2243 | S1586 | -4372.5 | 283.5 | 2305 | S1648 | -5054.5 | 193.5 | 2367 | S1710 | -5736.5 | 373.5 |
| 2244 | S1587 | -4383.5 | 373.5 | 2306 | S1649 | -5065.5 | 283.5 | 2368 | S1711 | -5747.5 | 193.5 |
| 2245 | S1588 | -4394.5 | 193.5 | 2307 | S1650 | -5076.5 | 373.5 | 2369 | S1712 | -5758.5 | 283.5 |
| 2246 | S1589 | -4405.5 | 283.5 | 2308 | S1651 | -5087.5 | 193.5 | 2370 | S1713 | -5769.5 | 373.5 |
| 2247 | S1590 | -4416.5 | 373.5 | 2309 | S1652 | -5098.5 | 283.5 | 2371 | S1714 | -5780.5 | 193.5 |
| 2248 | S1591 | -4427.5 | 193.5 | 2310 | S1653 | -5109.5 | 373.5 | 2372 | S1715 | -5791.5 | 283.5 |
| 2249 | S1592 | -4438.5 | 283.5 | 2311 | S1654 | -5120.5 | 193.5 | 2373 | S1716 | -5802.5 | 373.5 |
| 2250 | S1593 | -4449.5 | 373.5 | 2312 | S1655 | -5131.5 | 283.5 | 2374 | S1717 | -5813.5 | 193.5 |
| 2251 | S1594 | -4460.5 | 193.5 | 2313 | S1656 | -5142.5 | 373.5 | 2375 | S1718 | -5824.5 | 283.5 |
| 2252 | S1595 | -4471.5 | 283.5 | 2314 | S1657 | -5153.5 | 193.5 | 2376 | S1719 | -5835.5 | 373.5 |
| 2253 | S1596 | -4482.5 | 373.5 | 2315 | S1658 | -5164.5 | 283.5 | 2377 | S1720 | -5846.5 | 193.5 |
| 2254 | S1597 | -4493.5 | 193.5 | 2316 | S1659 | -5175.5 | 373.5 | 2378 | S1721 | -5857.5 | 283.5 |
| 2255 | S1598 | -4504.5 | 283.5 | 2317 | S1660 | -5186.5 | 193.5 | 2379 | S1722 | -5868.5 | 373.5 |
| 2256 | S1599 | -4515.5 | 373.5 | 2318 | S1661 | -5197.5 | 283.5 | 2380 | S1723 | -5879.5 | 193.5 |
| 2257 | S1600 | -4526.5 | 193.5 | 2319 | S1662 | -5208.5 | 373.5 | 2381 | S1724 | -5890.5 | 283.5 |
| 2258 | S1601 | -4537.5 | 283.5 | 2320 | S1663 | -5219.5 | 193.5 | 2382 | S1725 | -5901.5 | 373.5 |
| 2259 | S1602 | -4548.5 | 373.5 | 2321 | S1664 | -5230.5 | 283.5 | 2383 | S1726 | -5912.5 | 193.5 |
| 2260 | S1603 | -4559.5 | 193.5 | 2322 | S1665 | -5241.5 | 373.5 | 2384 | S1727 | -5923.5 | 283.5 |
| 2261 | S1604 | -4570.5 | 283.5 | 2323 | S1666 | -5252.5 | 193.5 | 2385 | S1728 | -5934.5 | 373.5 |
| 2262 | S1605 | -4581.5 | 373.5 | 2324 | S1667 | -5263.5 | 283.5 | 2386 | S1729 | -5945.5 | 193.5 |
| 2263 | S1606 | -4592.5 | 193.5 | 2325 | S1668 | -5274.5 | 373.5 | 2387 | S1730 | -5956.5 | 283.5 |
| 2264 | S1607 | -4603.5 | 283.5 | 2326 | S1669 | -5285.5 | 193.5 | 2388 | S1731 | -5967.5 | 373.5 |
| 2265 | S1608 | -4614.5 | 373.5 | 2327 | S1670 | -5296.5 | 283.5 | 2389 | S1732 | -5978.5 | 193.5 |
| 2266 | S1609 | -4625.5 | 193.5 | 2328 | S1671 | -5307.5 | 373.5 | 2390 | S1733 | -5989.5 | 283.5 |
| 2267 | S1610 | -4636.5 | 283.5 | 2329 | S1672 | -5318.5 | 193.5 | 2391 | S1734 | -6000.5 | 373.5 |
| 2268 | S1611 | -4647.5 | 373.5 | 2330 | S1673 | -5329.5 | 283.5 | 2392 | S1735 | -6011.5 | 193.5 |
| 2269 | S1612 | -4658.5 | 193.5 | 2331 | S1674 | -5340.5 | 373.5 | 2393 | S1736 | -6022.5 | 283.5 |
| 2270 | S1613 | -4669.5 | 283.5 | 2332 | S1675 | -5351.5 | 193.5 | 2394 | S1737 | -6033.5 | 373.5 |
| 2271 | S1614 | -4680.5 | 373.5 | 2333 | S1676 | -5362.5 | 283.5 | 2395 | S1738 | -6044.5 | 193.5 |
| 2272 | S1615 | -4691.5 | 193.5 | 2334 | S1677 | -5373.5 | 373.5 | 2396 | S1739 | -6055.5 | 283.5 |
| 2273 | S1616 | -4702.5 | 283.5 | 2335 | S1678 | -5384.5 | 193.5 | 2397 | S1740 | -6066.5 | 373.5 |
| 2274 | S1617 | -4713.5 | 373.5 | 2336 | S1679 | -5395.5 | 283.5 | 2398 | S1741 | -6077.5 | 193.5 |
| 2275 | S1618 | -4724.5 | 193.5 | 2337 | S1680 | -5406.5 | 373.5 | 2399 | S1742 | -6088.5 | 283.5 |
| 2276 | S1619 | -4735.5 | 283.5 | 2338 | S1681 | -5417.5 | 193.5 | 2400 | S1743 | -6099.5 | 373.5 |
| 2277 | S1620 | -4746.5 | 373.5 | 2339 | S1682 | -5428.5 | 283.5 | 2401 | S1744 | -6110.5 | 193.5 |
| 2278 | S1621 | -4757.5 | 193.5 | 2340 | S1683 | -5439.5 | 373.5 | 2402 | S1745 | -6121.5 | 283.5 |
| 2279 | S1622 | -4768.5 | 283.5 | 2341 | S1684 | -5450.5 | 193.5 | 2403 | S1746 | -6132.5 | 373.5 |
| 2280 | S1623 | -4779.5 | 373.5 | 2342 | S1685 | -5461.5 | 283.5 | 2404 | S1747 | -6143.5 | 193.5 |
| 2281 | S1624 | -4790.5 | 193.5 | 2343 | S1686 | -5472.5 | 373.5 | 2405 | S1748 | -6154.5 | 283.5 |
| 2282 | S1625 | -4801.5 | 283.5 | 2344 | S1687 | -5483.5 | 193.5 | 2406 | S1749 | -6165.5 | 373.5 |
| 2283 | S1626 | -4812.5 | 373.5 | 2345 | S1688 | -5494.5 | 283.5 | 2407 | S1750 | -6176.5 | 193.5 |
| 2284 | S1627 | -4823.5 | 193.5 | 2346 | S1689 | -5505.5 | 373.5 | 2408 | S1751 | -6187.5 | 283.5 |
| 2285 | S1628 | -4834.5 | 283.5 | 2347 | S1690 | -5516.5 | 193.5 | 2409 | S1752 | -6198.5 | 373.5 |
| 2286 | S1629 | -4845.5 | 373.5 | 2348 | S1691 | -5527.5 | 283.5 | 2410 | S1753 | -6209.5 | 193.5 |
| 2287 | S1630 | -4856.5 | 193.5 | 2349 | S1692 | -5538.5 | 373.5 | 2411 | S1754 | -6220.5 | 283.5 |
| 2288 | S1631 | -4867.5 | 283.5 | 2350 | S1693 | -5549.5 | 193.5 | 2412 | S1755 | -6231.5 | 373.5 |
| 2289 | S1632 | -4878.5 | 373.5 | 2351 | S1694 | -5560.5 | 283.5 | 2413 | S1756 | -6242.5 | 193.5 |
| 2290 | S1633 | -4889.5 | 193.5 | 2352 | S1695 | -5571.5 | 373.5 | 2414 | S1757 | -6253.5 | 283.5 |
| 2291 | S1634 | -4900.5 | 283.5 | 2353 | S1696 | -5582.5 | 193.5 | 2415 | S1758 | -6264.5 | 373.5 |
| 2292 | S1635 | -4911.5 | 373.5 | 2354 | S1697 | -5593.5 | 283.5 | 2416 | S1759 | -6275.5 | 193.5 |



|      |         |         |       |      |       |         |       |      |       |         |       |
|------|---------|---------|-------|------|-------|---------|-------|------|-------|---------|-------|
| 2417 | S1760   | -6286.5 | 283.5 | 2479 | S1804 | -6968.5 | 193.5 | 2541 | S1866 | -7650.5 | 373.5 |
| 2418 | S1761   | -6297.5 | 373.5 | 2480 | S1805 | -6979.5 | 283.5 | 2542 | S1867 | -7661.5 | 193.5 |
| 2419 | S1762   | -6308.5 | 193.5 | 2481 | S1806 | -6990.5 | 373.5 | 2543 | S1868 | -7672.5 | 283.5 |
| 2420 | S1763   | -6319.5 | 283.5 | 2482 | S1807 | -7001.5 | 193.5 | 2544 | S1869 | -7683.5 | 373.5 |
| 2421 | S1764   | -6330.5 | 373.5 | 2483 | S1808 | -7012.5 | 283.5 | 2545 | S1870 | -7694.5 | 193.5 |
| 2422 | S1765   | -6341.5 | 193.5 | 2484 | S1809 | -7023.5 | 373.5 | 2546 | S1871 | -7705.5 | 283.5 |
| 2423 | S1766   | -6352.5 | 283.5 | 2485 | S1810 | -7034.5 | 193.5 | 2547 | S1872 | -7716.5 | 373.5 |
| 2424 | S1767   | -6363.5 | 373.5 | 2486 | S1811 | -7045.5 | 283.5 | 2548 | S1873 | -7727.5 | 193.5 |
| 2425 | S1768   | -6374.5 | 193.5 | 2487 | S1812 | -7056.5 | 373.5 | 2549 | S1874 | -7738.5 | 283.5 |
| 2426 | S1769   | -6385.5 | 283.5 | 2488 | S1813 | -7067.5 | 193.5 | 2550 | S1875 | -7749.5 | 373.5 |
| 2427 | S1770   | -6396.5 | 373.5 | 2489 | S1814 | -7078.5 | 283.5 | 2551 | S1876 | -7760.5 | 193.5 |
| 2428 | S1771   | -6407.5 | 193.5 | 2490 | S1815 | -7089.5 | 373.5 | 2552 | S1877 | -7771.5 | 283.5 |
| 2429 | S1772   | -6418.5 | 283.5 | 2491 | S1816 | -7100.5 | 193.5 | 2553 | S1878 | -7782.5 | 373.5 |
| 2430 | S1773   | -6429.5 | 373.5 | 2492 | S1817 | -7111.5 | 283.5 | 2554 | S1879 | -7793.5 | 193.5 |
| 2431 | S1774   | -6440.5 | 193.5 | 2493 | S1818 | -7122.5 | 373.5 | 2555 | S1880 | -7804.5 | 283.5 |
| 2432 | S1775   | -6451.5 | 283.5 | 2494 | S1819 | -7133.5 | 193.5 | 2556 | S1881 | -7815.5 | 373.5 |
| 2433 | S1776   | -6462.5 | 373.5 | 2495 | S1820 | -7144.5 | 283.5 | 2557 | S1882 | -7826.5 | 193.5 |
| 2434 | S1777   | -6473.5 | 193.5 | 2496 | S1821 | -7155.5 | 373.5 | 2558 | S1883 | -7837.5 | 283.5 |
| 2435 | S1778   | -6484.5 | 283.5 | 2497 | S1822 | -7166.5 | 193.5 | 2559 | S1884 | -7848.5 | 373.5 |
| 2436 | S1779   | -6495.5 | 373.5 | 2498 | S1823 | -7177.5 | 283.5 | 2560 | S1885 | -7859.5 | 193.5 |
| 2437 | S1780   | -6506.5 | 193.5 | 2499 | S1824 | -7188.5 | 373.5 | 2561 | S1886 | -7870.5 | 283.5 |
| 2438 | S1781   | -6517.5 | 283.5 | 2500 | S1825 | -7199.5 | 193.5 | 2562 | S1887 | -7881.5 | 373.5 |
| 2439 | S1782   | -6528.5 | 373.5 | 2501 | S1826 | -7210.5 | 283.5 | 2563 | S1888 | -7892.5 | 193.5 |
| 2440 | S1783   | -6539.5 | 193.5 | 2502 | S1827 | -7221.5 | 373.5 | 2564 | S1889 | -7903.5 | 283.5 |
| 2441 | S1784   | -6550.5 | 283.5 | 2503 | S1828 | -7232.5 | 193.5 | 2565 | S1890 | -7914.5 | 373.5 |
| 2442 | S1785   | -6561.5 | 373.5 | 2504 | S1829 | -7243.5 | 283.5 | 2566 | S1891 | -7925.5 | 193.5 |
| 2443 | S1786   | -6572.5 | 193.5 | 2505 | S1830 | -7254.5 | 373.5 | 2567 | S1892 | -7936.5 | 283.5 |
| 2444 | S1787   | -6583.5 | 283.5 | 2506 | S1831 | -7265.5 | 193.5 | 2568 | S1893 | -7947.5 | 373.5 |
| 2445 | S1788   | -6594.5 | 373.5 | 2507 | S1832 | -7276.5 | 283.5 | 2569 | S1894 | -7958.5 | 193.5 |
| 2446 | S1789   | -6605.5 | 193.5 | 2508 | S1833 | -7287.5 | 373.5 | 2570 | S1895 | -7969.5 | 283.5 |
| 2447 | S1790   | -6616.5 | 283.5 | 2509 | S1834 | -7298.5 | 193.5 | 2571 | S1896 | -7980.5 | 373.5 |
| 2448 | S1791   | -6627.5 | 373.5 | 2510 | S1835 | -7309.5 | 283.5 | 2572 | S1897 | -7991.5 | 193.5 |
| 2449 | S1792   | -6638.5 | 193.5 | 2511 | S1836 | -7320.5 | 373.5 | 2573 | S1898 | -8002.5 | 283.5 |
| 2450 | S1793   | -6649.5 | 283.5 | 2512 | S1837 | -7331.5 | 193.5 | 2574 | S1899 | -8013.5 | 373.5 |
| 2451 | S1794   | -6660.5 | 373.5 | 2513 | S1838 | -7342.5 | 283.5 | 2575 | S1900 | -8024.5 | 193.5 |
| 2452 | S1795   | -6671.5 | 193.5 | 2514 | S1839 | -7353.5 | 373.5 | 2576 | S1901 | -8035.5 | 283.5 |
| 2453 | S1796   | -6682.5 | 283.5 | 2515 | S1840 | -7364.5 | 193.5 | 2577 | S1902 | -8046.5 | 373.5 |
| 2454 | S1797   | -6693.5 | 373.5 | 2516 | S1841 | -7375.5 | 283.5 | 2578 | S1903 | -8057.5 | 193.5 |
| 2455 | S1798   | -6704.5 | 193.5 | 2517 | S1842 | -7386.5 | 373.5 | 2579 | S1904 | -8068.5 | 283.5 |
| 2456 | S1799   | -6715.5 | 283.5 | 2518 | S1843 | -7397.5 | 193.5 | 2580 | S1905 | -8079.5 | 373.5 |
| 2457 | S1800   | -6726.5 | 373.5 | 2519 | S1844 | -7408.5 | 283.5 | 2581 | S1906 | -8090.5 | 193.5 |
| 2458 | DUMMY53 | -6737.5 | 193.5 | 2520 | S1845 | -7419.5 | 373.5 | 2582 | S1907 | -8101.5 | 283.5 |
| 2459 | DUMMY54 | -6748.5 | 283.5 | 2521 | S1846 | -7430.5 | 193.5 | 2583 | S1908 | -8112.5 | 373.5 |
| 2460 | DUMMY55 | -6759.5 | 373.5 | 2522 | S1847 | -7441.5 | 283.5 | 2584 | S1909 | -8123.5 | 193.5 |
| 2461 | DUMMY56 | -6770.5 | 193.5 | 2523 | S1848 | -7452.5 | 373.5 | 2585 | S1910 | -8134.5 | 283.5 |
| 2462 | DUMMY57 | -6781.5 | 283.5 | 2524 | S1849 | -7463.5 | 193.5 | 2586 | S1911 | -8145.5 | 373.5 |
| 2463 | DUMMY58 | -6792.5 | 373.5 | 2525 | S1850 | -7474.5 | 283.5 | 2587 | S1912 | -8156.5 | 193.5 |
| 2464 | DUMMY59 | -6803.5 | 193.5 | 2526 | S1851 | -7485.5 | 373.5 | 2588 | S1913 | -8167.5 | 283.5 |
| 2465 | DUMMY60 | -6814.5 | 283.5 | 2527 | S1852 | -7496.5 | 193.5 | 2589 | S1914 | -8178.5 | 373.5 |
| 2466 | DUMMY61 | -6825.5 | 373.5 | 2528 | S1853 | -7507.5 | 283.5 | 2590 | S1915 | -8189.5 | 193.5 |
| 2467 | DUMMY62 | -6836.5 | 193.5 | 2529 | S1854 | -7518.5 | 373.5 | 2591 | S1916 | -8200.5 | 283.5 |
| 2468 | DUMMY63 | -6847.5 | 283.5 | 2530 | S1855 | -7529.5 | 193.5 | 2592 | S1917 | -8211.5 | 373.5 |
| 2469 | DUMMY64 | -6858.5 | 373.5 | 2531 | S1856 | -7540.5 | 283.5 | 2593 | S1918 | -8222.5 | 193.5 |
| 2470 | DUMMY65 | -6869.5 | 193.5 | 2532 | S1857 | -7551.5 | 373.5 | 2594 | S1919 | -8233.5 | 283.5 |
| 2471 | DUMMY66 | -6880.5 | 283.5 | 2533 | S1858 | -7562.5 | 193.5 | 2595 | S1920 | -8244.5 | 373.5 |
| 2472 | DUMMY67 | -6891.5 | 373.5 | 2534 | S1859 | -7573.5 | 283.5 | 2596 | S1921 | -8255.5 | 193.5 |
| 2473 | DUMMY68 | -6902.5 | 193.5 | 2535 | S1860 | -7584.5 | 373.5 | 2597 | S1922 | -8266.5 | 283.5 |
| 2474 | DUMMY69 | -6913.5 | 283.5 | 2536 | S1861 | -7595.5 | 193.5 | 2598 | S1923 | -8277.5 | 373.5 |
| 2475 | DUMMY70 | -6924.5 | 373.5 | 2537 | S1862 | -7606.5 | 283.5 | 2599 | S1924 | -8288.5 | 193.5 |
| 2476 | S1801   | -6935.5 | 193.5 | 2538 | S1863 | -7617.5 | 373.5 | 2600 | S1925 | -8299.5 | 283.5 |
| 2477 | S1802   | -6946.5 | 283.5 | 2539 | S1864 | -7628.5 | 193.5 | 2601 | S1926 | -8310.5 | 373.5 |
| 2478 | S1803   | -6957.5 | 373.5 | 2540 | S1865 | -7639.5 | 283.5 | 2602 | S1927 | -8321.5 | 193.5 |



|      |       |         |       |      |       |         |       |      |       |          |       |
|------|-------|---------|-------|------|-------|---------|-------|------|-------|----------|-------|
| 2603 | S1928 | -8332.5 | 283.5 | 2665 | S1990 | -9014.5 | 193.5 | 2727 | S2052 | -9696.5  | 373.5 |
| 2604 | S1929 | -8343.5 | 373.5 | 2666 | S1991 | -9025.5 | 283.5 | 2728 | S2053 | -9707.5  | 193.5 |
| 2605 | S1930 | -8354.5 | 193.5 | 2667 | S1992 | -9036.5 | 373.5 | 2729 | S2054 | -9718.5  | 283.5 |
| 2606 | S1931 | -8365.5 | 283.5 | 2668 | S1993 | -9047.5 | 193.5 | 2730 | S2055 | -9729.5  | 373.5 |
| 2607 | S1932 | -8376.5 | 373.5 | 2669 | S1994 | -9058.5 | 283.5 | 2731 | S2056 | -9740.5  | 193.5 |
| 2608 | S1933 | -8387.5 | 193.5 | 2670 | S1995 | -9069.5 | 373.5 | 2732 | S2057 | -9751.5  | 283.5 |
| 2609 | S1934 | -8398.5 | 283.5 | 2671 | S1996 | -9080.5 | 193.5 | 2733 | S2058 | -9762.5  | 373.5 |
| 2610 | S1935 | -8409.5 | 373.5 | 2672 | S1997 | -9091.5 | 283.5 | 2734 | S2059 | -9773.5  | 193.5 |
| 2611 | S1936 | -8420.5 | 193.5 | 2673 | S1998 | -9102.5 | 373.5 | 2735 | S2060 | -9784.5  | 283.5 |
| 2612 | S1937 | -8431.5 | 283.5 | 2674 | S1999 | -9113.5 | 193.5 | 2736 | S2061 | -9795.5  | 373.5 |
| 2613 | S1938 | -8442.5 | 373.5 | 2675 | S2000 | -9124.5 | 283.5 | 2737 | S2062 | -9806.5  | 193.5 |
| 2614 | S1939 | -8453.5 | 193.5 | 2676 | S2001 | -9135.5 | 373.5 | 2738 | S2063 | -9817.5  | 283.5 |
| 2615 | S1940 | -8464.5 | 283.5 | 2677 | S2002 | -9146.5 | 193.5 | 2739 | S2064 | -9828.5  | 373.5 |
| 2616 | S1941 | -8475.5 | 373.5 | 2678 | S2003 | -9157.5 | 283.5 | 2740 | S2065 | -9839.5  | 193.5 |
| 2617 | S1942 | -8486.5 | 193.5 | 2679 | S2004 | -9168.5 | 373.5 | 2741 | S2066 | -9850.5  | 283.5 |
| 2618 | S1943 | -8497.5 | 283.5 | 2680 | S2005 | -9179.5 | 193.5 | 2742 | S2067 | -9861.5  | 373.5 |
| 2619 | S1944 | -8508.5 | 373.5 | 2681 | S2006 | -9190.5 | 283.5 | 2743 | S2068 | -9872.5  | 193.5 |
| 2620 | S1945 | -8519.5 | 193.5 | 2682 | S2007 | -9201.5 | 373.5 | 2744 | S2069 | -9883.5  | 283.5 |
| 2621 | S1946 | -8530.5 | 283.5 | 2683 | S2008 | -9212.5 | 193.5 | 2745 | S2070 | -9894.5  | 373.5 |
| 2622 | S1947 | -8541.5 | 373.5 | 2684 | S2009 | -9223.5 | 283.5 | 2746 | S2071 | -9905.5  | 193.5 |
| 2623 | S1948 | -8552.5 | 193.5 | 2685 | S2010 | -9234.5 | 373.5 | 2747 | S2072 | -9916.5  | 283.5 |
| 2624 | S1949 | -8563.5 | 283.5 | 2686 | S2011 | -9245.5 | 193.5 | 2748 | S2073 | -9927.5  | 373.5 |
| 2625 | S1950 | -8574.5 | 373.5 | 2687 | S2012 | -9256.5 | 283.5 | 2749 | S2074 | -9938.5  | 193.5 |
| 2626 | S1951 | -8585.5 | 193.5 | 2688 | S2013 | -9267.5 | 373.5 | 2750 | S2075 | -9949.5  | 283.5 |
| 2627 | S1952 | -8596.5 | 283.5 | 2689 | S2014 | -9278.5 | 193.5 | 2751 | S2076 | -9960.5  | 373.5 |
| 2628 | S1953 | -8607.5 | 373.5 | 2690 | S2015 | -9289.5 | 283.5 | 2752 | S2077 | -9971.5  | 193.5 |
| 2629 | S1954 | -8618.5 | 193.5 | 2691 | S2016 | -9300.5 | 373.5 | 2753 | S2078 | -9982.5  | 283.5 |
| 2630 | S1955 | -8629.5 | 283.5 | 2692 | S2017 | -9311.5 | 193.5 | 2754 | S2079 | -9993.5  | 373.5 |
| 2631 | S1956 | -8640.5 | 373.5 | 2693 | S2018 | -9322.5 | 283.5 | 2755 | S2080 | -10004.5 | 193.5 |
| 2632 | S1957 | -8651.5 | 193.5 | 2694 | S2019 | -9333.5 | 373.5 | 2756 | S2081 | -10015.5 | 283.5 |
| 2633 | S1958 | -8662.5 | 283.5 | 2695 | S2020 | -9344.5 | 193.5 | 2757 | S2082 | -10026.5 | 373.5 |
| 2634 | S1959 | -8673.5 | 373.5 | 2696 | S2021 | -9355.5 | 283.5 | 2758 | S2083 | -10037.5 | 193.5 |
| 2635 | S1960 | -8684.5 | 193.5 | 2697 | S2022 | -9366.5 | 373.5 | 2759 | S2084 | -10048.5 | 283.5 |
| 2636 | S1961 | -8695.5 | 283.5 | 2698 | S2023 | -9377.5 | 193.5 | 2760 | S2085 | -10059.5 | 373.5 |
| 2637 | S1962 | -8706.5 | 373.5 | 2699 | S2024 | -9388.5 | 283.5 | 2761 | S2086 | -10070.5 | 193.5 |
| 2638 | S1963 | -8717.5 | 193.5 | 2700 | S2025 | -9399.5 | 373.5 | 2762 | S2087 | -10081.5 | 283.5 |
| 2639 | S1964 | -8728.5 | 283.5 | 2701 | S2026 | -9410.5 | 193.5 | 2763 | S2088 | -10092.5 | 373.5 |
| 2640 | S1965 | -8739.5 | 373.5 | 2702 | S2027 | -9421.5 | 283.5 | 2764 | S2089 | -10103.5 | 193.5 |
| 2641 | S1966 | -8750.5 | 193.5 | 2703 | S2028 | -9432.5 | 373.5 | 2765 | S2090 | -10114.5 | 283.5 |
| 2642 | S1967 | -8761.5 | 283.5 | 2704 | S2029 | -9443.5 | 193.5 | 2766 | S2091 | -10125.5 | 373.5 |
| 2643 | S1968 | -8772.5 | 373.5 | 2705 | S2030 | -9454.5 | 283.5 | 2767 | S2092 | -10136.5 | 193.5 |
| 2644 | S1969 | -8783.5 | 193.5 | 2706 | S2031 | -9465.5 | 373.5 | 2768 | S2093 | -10147.5 | 283.5 |
| 2645 | S1970 | -8794.5 | 283.5 | 2707 | S2032 | -9476.5 | 193.5 | 2769 | S2094 | -10158.5 | 373.5 |
| 2646 | S1971 | -8805.5 | 373.5 | 2708 | S2033 | -9487.5 | 283.5 | 2770 | S2095 | -10169.5 | 193.5 |
| 2647 | S1972 | -8816.5 | 193.5 | 2709 | S2034 | -9498.5 | 373.5 | 2771 | S2096 | -10180.5 | 283.5 |
| 2648 | S1973 | -8827.5 | 283.5 | 2710 | S2035 | -9509.5 | 193.5 | 2772 | S2097 | -10191.5 | 373.5 |
| 2649 | S1974 | -8838.5 | 373.5 | 2711 | S2036 | -9520.5 | 283.5 | 2773 | S2098 | -10202.5 | 193.5 |
| 2650 | S1975 | -8849.5 | 193.5 | 2712 | S2037 | -9531.5 | 373.5 | 2774 | S2099 | -10213.5 | 283.5 |
| 2651 | S1976 | -8860.5 | 283.5 | 2713 | S2038 | -9542.5 | 193.5 | 2775 | S2100 | -10224.5 | 373.5 |
| 2652 | S1977 | -8871.5 | 373.5 | 2714 | S2039 | -9553.5 | 283.5 | 2776 | S2101 | -10235.5 | 193.5 |
| 2653 | S1978 | -8882.5 | 193.5 | 2715 | S2040 | -9564.5 | 373.5 | 2777 | S2102 | -10246.5 | 283.5 |
| 2654 | S1979 | -8893.5 | 283.5 | 2716 | S2041 | -9575.5 | 193.5 | 2778 | S2103 | -10257.5 | 373.5 |
| 2655 | S1980 | -8904.5 | 373.5 | 2717 | S2042 | -9586.5 | 283.5 | 2779 | S2104 | -10268.5 | 193.5 |
| 2656 | S1981 | -8915.5 | 193.5 | 2718 | S2043 | -9597.5 | 373.5 | 2780 | S2105 | -10279.5 | 283.5 |
| 2657 | S1982 | -8926.5 | 283.5 | 2719 | S2044 | -9608.5 | 193.5 | 2781 | S2106 | -10290.5 | 373.5 |
| 2658 | S1983 | -8937.5 | 373.5 | 2720 | S2045 | -9619.5 | 283.5 | 2782 | S2107 | -10301.5 | 193.5 |
| 2659 | S1984 | -8948.5 | 193.5 | 2721 | S2046 | -9630.5 | 373.5 | 2783 | S2108 | -10312.5 | 283.5 |
| 2660 | S1985 | -8959.5 | 283.5 | 2722 | S2047 | -9641.5 | 193.5 | 2784 | S2109 | -10323.5 | 373.5 |
| 2661 | S1986 | -8970.5 | 373.5 | 2723 | S2048 | -9652.5 | 283.5 | 2785 | S2110 | -10334.5 | 193.5 |
| 2662 | S1987 | -8981.5 | 193.5 | 2724 | S2049 | -9663.5 | 373.5 | 2786 | S2111 | -10345.5 | 283.5 |
| 2663 | S1988 | -8992.5 | 283.5 | 2725 | S2050 | -9674.5 | 193.5 | 2787 | S2112 | -10356.5 | 373.5 |
| 2664 | S1989 | -9003.5 | 373.5 | 2726 | S2051 | -9685.5 | 283.5 | 2788 | S2113 | -10367.5 | 193.5 |

|      |       |          |       |      |       |          |       |      |       |          |       |
|------|-------|----------|-------|------|-------|----------|-------|------|-------|----------|-------|
| 2789 | S2114 | -10378.5 | 283.5 | 2851 | S2176 | -11060.5 | 193.5 | 2913 | S2238 | -11742.5 | 373.5 |
| 2790 | S2115 | -10389.5 | 373.5 | 2852 | S2177 | -11071.5 | 283.5 | 2914 | S2239 | -11753.5 | 193.5 |
| 2791 | S2116 | -10400.5 | 193.5 | 2853 | S2178 | -11082.5 | 373.5 | 2915 | S2240 | -11764.5 | 283.5 |
| 2792 | S2117 | -10411.5 | 283.5 | 2854 | S2179 | -11093.5 | 193.5 | 2916 | S2241 | -11775.5 | 373.5 |
| 2793 | S2118 | -10422.5 | 373.5 | 2855 | S2180 | -11104.5 | 283.5 | 2917 | S2242 | -11786.5 | 193.5 |
| 2794 | S2119 | -10433.5 | 193.5 | 2856 | S2181 | -11115.5 | 373.5 | 2918 | S2243 | -11797.5 | 283.5 |
| 2795 | S2120 | -10444.5 | 283.5 | 2857 | S2182 | -11126.5 | 193.5 | 2919 | S2244 | -11808.5 | 373.5 |
| 2796 | S2121 | -10455.5 | 373.5 | 2858 | S2183 | -11137.5 | 283.5 | 2920 | S2245 | -11819.5 | 193.5 |
| 2797 | S2122 | -10466.5 | 193.5 | 2859 | S2184 | -11148.5 | 373.5 | 2921 | S2246 | -11830.5 | 283.5 |
| 2798 | S2123 | -10477.5 | 283.5 | 2860 | S2185 | -11159.5 | 193.5 | 2922 | S2247 | -11841.5 | 373.5 |
| 2799 | S2124 | -10488.5 | 373.5 | 2861 | S2186 | -11170.5 | 283.5 | 2923 | S2248 | -11852.5 | 193.5 |
| 2800 | S2125 | -10499.5 | 193.5 | 2862 | S2187 | -11181.5 | 373.5 | 2924 | S2249 | -11863.5 | 283.5 |
| 2801 | S2126 | -10510.5 | 283.5 | 2863 | S2188 | -11192.5 | 193.5 | 2925 | S2250 | -11874.5 | 373.5 |
| 2802 | S2127 | -10521.5 | 373.5 | 2864 | S2189 | -11203.5 | 283.5 | 2926 | S2251 | -11885.5 | 193.5 |
| 2803 | S2128 | -10532.5 | 193.5 | 2865 | S2190 | -11214.5 | 373.5 | 2927 | S2252 | -11896.5 | 283.5 |
| 2804 | S2129 | -10543.5 | 283.5 | 2866 | S2191 | -11225.5 | 193.5 | 2928 | S2253 | -11907.5 | 373.5 |
| 2805 | S2130 | -10554.5 | 373.5 | 2867 | S2192 | -11236.5 | 283.5 | 2929 | S2254 | -11918.5 | 193.5 |
| 2806 | S2131 | -10565.5 | 193.5 | 2868 | S2193 | -11247.5 | 373.5 | 2930 | S2255 | -11929.5 | 283.5 |
| 2807 | S2132 | -10576.5 | 283.5 | 2869 | S2194 | -11258.5 | 193.5 | 2931 | S2256 | -11940.5 | 373.5 |
| 2808 | S2133 | -10587.5 | 373.5 | 2870 | S2195 | -11269.5 | 283.5 | 2932 | S2257 | -11951.5 | 193.5 |
| 2809 | S2134 | -10598.5 | 193.5 | 2871 | S2196 | -11280.5 | 373.5 | 2933 | S2258 | -11962.5 | 283.5 |
| 2810 | S2135 | -10609.5 | 283.5 | 2872 | S2197 | -11291.5 | 193.5 | 2934 | S2259 | -11973.5 | 373.5 |
| 2811 | S2136 | -10620.5 | 373.5 | 2873 | S2198 | -11302.5 | 283.5 | 2935 | S2260 | -11984.5 | 193.5 |
| 2812 | S2137 | -10631.5 | 193.5 | 2874 | S2199 | -11313.5 | 373.5 | 2936 | S2261 | -11995.5 | 283.5 |
| 2813 | S2138 | -10642.5 | 283.5 | 2875 | S2200 | -11324.5 | 193.5 | 2937 | S2262 | -12006.5 | 373.5 |
| 2814 | S2139 | -10653.5 | 373.5 | 2876 | S2201 | -11335.5 | 283.5 | 2938 | S2263 | -12017.5 | 193.5 |
| 2815 | S2140 | -10664.5 | 193.5 | 2877 | S2202 | -11346.5 | 373.5 | 2939 | S2264 | -12028.5 | 283.5 |
| 2816 | S2141 | -10675.5 | 283.5 | 2878 | S2203 | -11357.5 | 193.5 | 2940 | S2265 | -12039.5 | 373.5 |
| 2817 | S2142 | -10686.5 | 373.5 | 2879 | S2204 | -11368.5 | 283.5 | 2941 | S2266 | -12050.5 | 193.5 |
| 2818 | S2143 | -10697.5 | 193.5 | 2880 | S2205 | -11379.5 | 373.5 | 2942 | S2267 | -12061.5 | 283.5 |
| 2819 | S2144 | -10708.5 | 283.5 | 2881 | S2206 | -11390.5 | 193.5 | 2943 | S2268 | -12072.5 | 373.5 |
| 2820 | S2145 | -10719.5 | 373.5 | 2882 | S2207 | -11401.5 | 283.5 | 2944 | S2269 | -12083.5 | 193.5 |
| 2821 | S2146 | -10730.5 | 193.5 | 2883 | S2208 | -11412.5 | 373.5 | 2945 | S2270 | -12094.5 | 283.5 |
| 2822 | S2147 | -10741.5 | 283.5 | 2884 | S2209 | -11423.5 | 193.5 | 2946 | S2271 | -12105.5 | 373.5 |
| 2823 | S2148 | -10752.5 | 373.5 | 2885 | S2210 | -11434.5 | 283.5 | 2947 | S2272 | -12116.5 | 193.5 |
| 2824 | S2149 | -10763.5 | 193.5 | 2886 | S2211 | -11445.5 | 373.5 | 2948 | S2273 | -12127.5 | 283.5 |
| 2825 | S2150 | -10774.5 | 283.5 | 2887 | S2212 | -11456.5 | 193.5 | 2949 | S2274 | -12138.5 | 373.5 |
| 2826 | S2151 | -10785.5 | 373.5 | 2888 | S2213 | -11467.5 | 283.5 | 2950 | S2275 | -12149.5 | 193.5 |
| 2827 | S2152 | -10796.5 | 193.5 | 2889 | S2214 | -11478.5 | 373.5 | 2951 | S2276 | -12160.5 | 283.5 |
| 2828 | S2153 | -10807.5 | 283.5 | 2890 | S2215 | -11489.5 | 193.5 | 2952 | S2277 | -12171.5 | 373.5 |
| 2829 | S2154 | -10818.5 | 373.5 | 2891 | S2216 | -11500.5 | 283.5 | 2953 | S2278 | -12182.5 | 193.5 |
| 2830 | S2155 | -10829.5 | 193.5 | 2892 | S2217 | -11511.5 | 373.5 | 2954 | S2279 | -12193.5 | 283.5 |
| 2831 | S2156 | -10840.5 | 283.5 | 2893 | S2218 | -11522.5 | 193.5 | 2955 | S2280 | -12204.5 | 373.5 |
| 2832 | S2157 | -10851.5 | 373.5 | 2894 | S2219 | -11533.5 | 283.5 | 2956 | S2281 | -12215.5 | 193.5 |
| 2833 | S2158 | -10862.5 | 193.5 | 2895 | S2220 | -11544.5 | 373.5 | 2957 | S2282 | -12226.5 | 283.5 |
| 2834 | S2159 | -10873.5 | 283.5 | 2896 | S2221 | -11555.5 | 193.5 | 2958 | S2283 | -12237.5 | 373.5 |
| 2835 | S2160 | -10884.5 | 373.5 | 2897 | S2222 | -11566.5 | 283.5 | 2959 | S2284 | -12248.5 | 193.5 |
| 2836 | S2161 | -10895.5 | 193.5 | 2898 | S2223 | -11577.5 | 373.5 | 2960 | S2285 | -12259.5 | 283.5 |
| 2837 | S2162 | -10906.5 | 283.5 | 2899 | S2224 | -11588.5 | 193.5 | 2961 | S2286 | -12270.5 | 373.5 |
| 2838 | S2163 | -10917.5 | 373.5 | 2900 | S2225 | -11599.5 | 283.5 | 2962 | S2287 | -12281.5 | 193.5 |
| 2839 | S2164 | -10928.5 | 193.5 | 2901 | S2226 | -11610.5 | 373.5 | 2963 | S2288 | -12292.5 | 283.5 |
| 2840 | S2165 | -10939.5 | 283.5 | 2902 | S2227 | -11621.5 | 193.5 | 2964 | S2289 | -12303.5 | 373.5 |
| 2841 | S2166 | -10950.5 | 373.5 | 2903 | S2228 | -11632.5 | 283.5 | 2965 | S2290 | -12314.5 | 193.5 |
| 2842 | S2167 | -10961.5 | 193.5 | 2904 | S2229 | -11643.5 | 373.5 | 2966 | S2291 | -12325.5 | 283.5 |
| 2843 | S2168 | -10972.5 | 283.5 | 2905 | S2230 | -11654.5 | 193.5 | 2967 | S2292 | -12336.5 | 373.5 |
| 2844 | S2169 | -10983.5 | 373.5 | 2906 | S2231 | -11665.5 | 283.5 | 2968 | S2293 | -12347.5 | 193.5 |
| 2845 | S2170 | -10994.5 | 193.5 | 2907 | S2232 | -11676.5 | 373.5 | 2969 | S2294 | -12358.5 | 283.5 |
| 2846 | S2171 | -11005.5 | 283.5 | 2908 | S2233 | -11687.5 | 193.5 | 2970 | S2295 | -12369.5 | 373.5 |
| 2847 | S2172 | -11016.5 | 373.5 | 2909 | S2234 | -11698.5 | 283.5 | 2971 | S2296 | -12380.5 | 193.5 |
| 2848 | S2173 | -11027.5 | 193.5 | 2910 | S2235 | -11709.5 | 373.5 | 2972 | S2297 | -12391.5 | 283.5 |
| 2849 | S2174 | -11038.5 | 283.5 | 2911 | S2236 | -11720.5 | 193.5 | 2973 | S2298 | -12402.5 | 373.5 |
| 2850 | S2175 | -11049.5 | 373.5 | 2912 | S2237 | -11731.5 | 283.5 | 2974 | S2299 | -12413.5 | 193.5 |

|      |       |          |       |
|------|-------|----------|-------|
| 2975 | S2300 | -12424.5 | 283.5 |
| 2976 | S2301 | -12435.5 | 373.5 |
| 2977 | S2302 | -12446.5 | 193.5 |
| 2978 | S2303 | -12457.5 | 283.5 |
| 2979 | S2304 | -12468.5 | 373.5 |
| 2980 | S2305 | -12479.5 | 193.5 |
| 2981 | S2306 | -12490.5 | 283.5 |
| 2982 | S2307 | -12501.5 | 373.5 |
| 2983 | S2308 | -12512.5 | 193.5 |
| 2984 | S2309 | -12523.5 | 283.5 |
| 2985 | S2310 | -12534.5 | 373.5 |
| 2986 | S2311 | -12545.5 | 193.5 |
| 2987 | S2312 | -12556.5 | 283.5 |
| 2988 | S2313 | -12567.5 | 373.5 |
| 2989 | S2314 | -12578.5 | 193.5 |
| 2990 | S2315 | -12589.5 | 283.5 |
| 2991 | S2316 | -12600.5 | 373.5 |
| 2992 | S2317 | -12611.5 | 193.5 |
| 2993 | S2318 | -12622.5 | 283.5 |
| 2994 | S2319 | -12633.5 | 373.5 |
| 2995 | S2320 | -12644.5 | 193.5 |
| 2996 | S2321 | -12655.5 | 283.5 |
| 2997 | S2322 | -12666.5 | 373.5 |
| 2998 | S2323 | -12677.5 | 193.5 |
| 2999 | S2324 | -12688.5 | 283.5 |
| 3000 | S2325 | -12699.5 | 373.5 |
| 3001 | S2326 | -12710.5 | 193.5 |
| 3002 | S2327 | -12721.5 | 283.5 |
| 3003 | S2328 | -12732.5 | 373.5 |
| 3004 | S2329 | -12743.5 | 193.5 |
| 3005 | S2330 | -12754.5 | 283.5 |
| 3006 | S2331 | -12765.5 | 373.5 |
| 3007 | S2332 | -12776.5 | 193.5 |
| 3008 | S2333 | -12787.5 | 283.5 |
| 3009 | S2334 | -12798.5 | 373.5 |
| 3010 | S2335 | -12809.5 | 193.5 |
| 3011 | S2336 | -12820.5 | 283.5 |
| 3012 | S2337 | -12831.5 | 373.5 |
| 3013 | S2338 | -12842.5 | 193.5 |
| 3014 | S2339 | -12853.5 | 283.5 |
| 3015 | S2340 | -12864.5 | 373.5 |
| 3016 | S2341 | -12875.5 | 193.5 |
| 3017 | S2342 | -12886.5 | 283.5 |
| 3018 | S2343 | -12897.5 | 373.5 |
| 3019 | S2344 | -12908.5 | 193.5 |
| 3020 | S2345 | -12919.5 | 283.5 |
| 3021 | S2346 | -12930.5 | 373.5 |
| 3022 | S2347 | -12941.5 | 193.5 |
| 3023 | S2348 | -12952.5 | 283.5 |
| 3024 | S2349 | -12963.5 | 373.5 |
| 3025 | S2350 | -12974.5 | 193.5 |
| 3026 | S2351 | -12985.5 | 283.5 |
| 3027 | S2352 | -12996.5 | 373.5 |
| 3028 | S2353 | -13007.5 | 193.5 |
| 3029 | S2354 | -13018.5 | 283.5 |
| 3030 | S2355 | -13029.5 | 373.5 |
| 3031 | S2356 | -13040.5 | 193.5 |
| 3032 | S2357 | -13051.5 | 283.5 |
| 3033 | S2358 | -13062.5 | 373.5 |
| 3034 | S2359 | -13073.5 | 193.5 |
| 3035 | S2360 | -13084.5 | 283.5 |
| 3036 | S2361 | -13095.5 | 373.5 |

|      |         |          |       |
|------|---------|----------|-------|
| 3037 | S2362   | -13106.5 | 193.5 |
| 3038 | S2363   | -13117.5 | 283.5 |
| 3039 | S2364   | -13128.5 | 373.5 |
| 3040 | S2365   | -13139.5 | 193.5 |
| 3041 | S2366   | -13150.5 | 283.5 |
| 3042 | S2367   | -13161.5 | 373.5 |
| 3043 | S2368   | -13172.5 | 193.5 |
| 3044 | S2369   | -13183.5 | 283.5 |
| 3045 | S2370   | -13194.5 | 373.5 |
| 3046 | S2371   | -13205.5 | 193.5 |
| 3047 | S2372   | -13216.5 | 283.5 |
| 3048 | S2373   | -13227.5 | 373.5 |
| 3049 | S2374   | -13238.5 | 193.5 |
| 3050 | S2375   | -13249.5 | 283.5 |
| 3051 | S2376   | -13260.5 | 373.5 |
| 3052 | S2377   | -13271.5 | 193.5 |
| 3053 | S2378   | -13282.5 | 283.5 |
| 3054 | S2379   | -13293.5 | 373.5 |
| 3055 | S2380   | -13304.5 | 193.5 |
| 3056 | S2381   | -13315.5 | 283.5 |
| 3057 | S2382   | -13326.5 | 373.5 |
| 3058 | S2383   | -13337.5 | 193.5 |
| 3059 | S2384   | -13348.5 | 283.5 |
| 3060 | S2385   | -13359.5 | 373.5 |
| 3061 | S2386   | -13370.5 | 193.5 |
| 3062 | S2387   | -13381.5 | 283.5 |
| 3063 | S2388   | -13392.5 | 373.5 |
| 3064 | S2389   | -13403.5 | 193.5 |
| 3065 | S2390   | -13414.5 | 283.5 |
| 3066 | S2391   | -13425.5 | 373.5 |
| 3067 | S2392   | -13436.5 | 193.5 |
| 3068 | S2393   | -13447.5 | 283.5 |
| 3069 | S2394   | -13458.5 | 373.5 |
| 3070 | S2395   | -13469.5 | 193.5 |
| 3071 | S2396   | -13480.5 | 283.5 |
| 3072 | S2397   | -13491.5 | 373.5 |
| 3073 | S2398   | -13502.5 | 193.5 |
| 3074 | S2399   | -13513.5 | 283.5 |
| 3075 | S2400   | -13524.5 | 373.5 |
| 3076 | SDUM2   | -13535.5 | 193.5 |
| 3077 | SDUM3   | -13546.5 | 283.5 |
| 3078 | DUMMY73 | -13557.5 | 373.5 |
| 3079 | DUMMY74 | -13568.5 | 193.5 |
| 3080 | DUMMY75 | -13579.5 | 283.5 |
| 3081 | DUMMY76 | -13590.5 | 373.5 |
| 3082 | DUMMY77 | -13601.5 | 193.5 |
| 3083 | DUMMY78 | -13612.5 | 283.5 |
| 3084 | DUMMY79 | -13623.5 | 373.5 |

| Coordinate           | X-axis | Y-axis |
|----------------------|--------|--------|
| Alignment Left Up    | -13706 | 381    |
| Alignment Left Down  | -13706 | 294    |
| Alignment Right Up   | 13706  | 381    |
| Alignment Right Down | 13706  | 294    |

## 6. Function Description

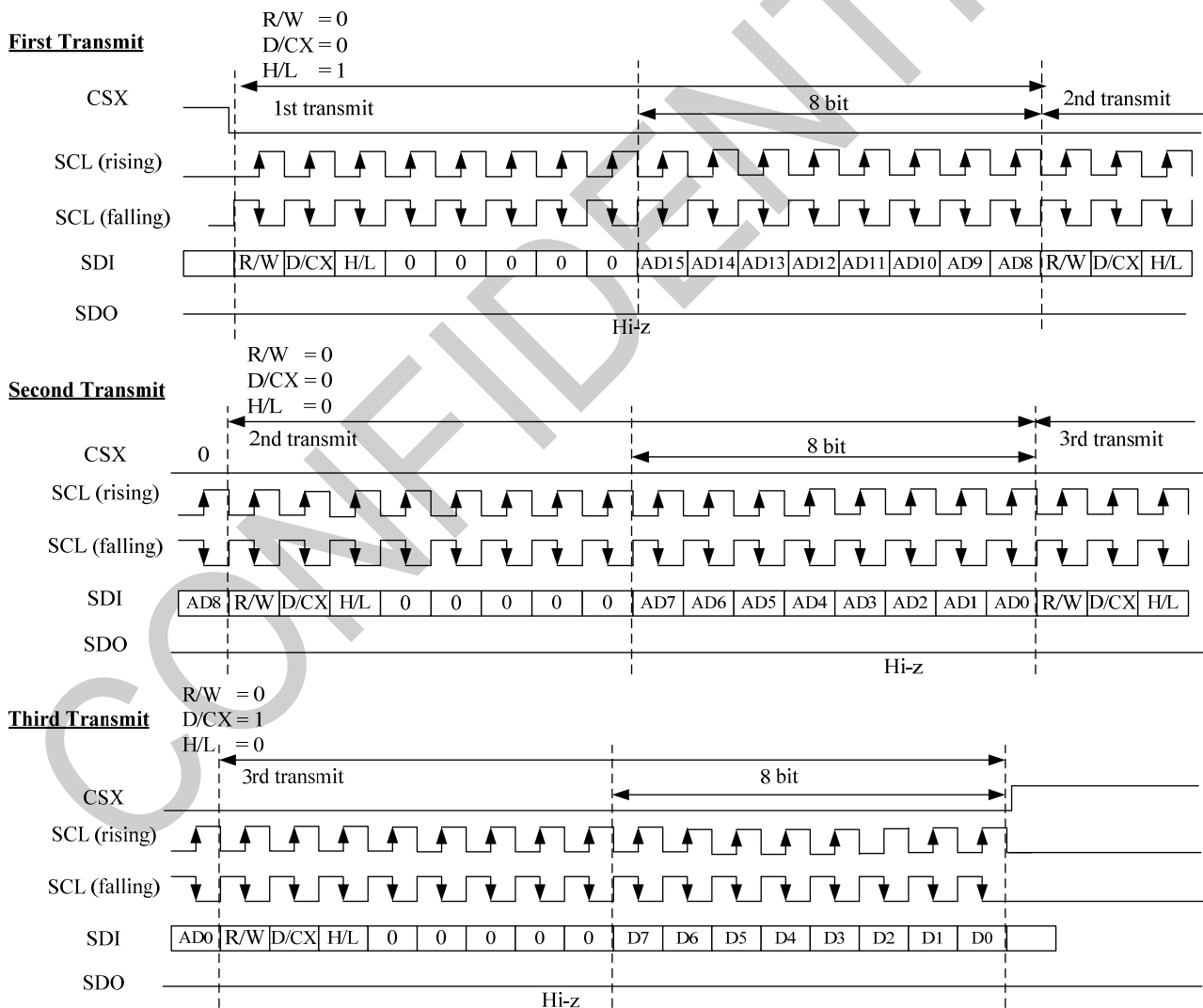
### 6.1 Serial-Parallel Interface

#### 6.1.1 Write Cycle and Sequence

During a write cycle, host processor sends a single bit of data to the display module via the interface. The SPI interface utilizes CSX, SCL, SDI and SDO signals. SCL is driven from high to low and then pulled back to be high for a complete write cycle. The host processor provides information during the write cycle. On the other hand, display module reads the information at the rising edge of SCL.

During a write sequence, host processor writes one or more bytes of information to display module via the interface. A write sequence is initiated when CSX is driven from high to low and ended once CSX is pulled to high. Each byte is either nine or sixteen write cycles in length. If the optional DCX signal is used, a byte is eight write cycles long. DCX is driven low while command information is on the interface and is pulled to be high when data is present.

The SPI interface write command sequences are described in the following figures.

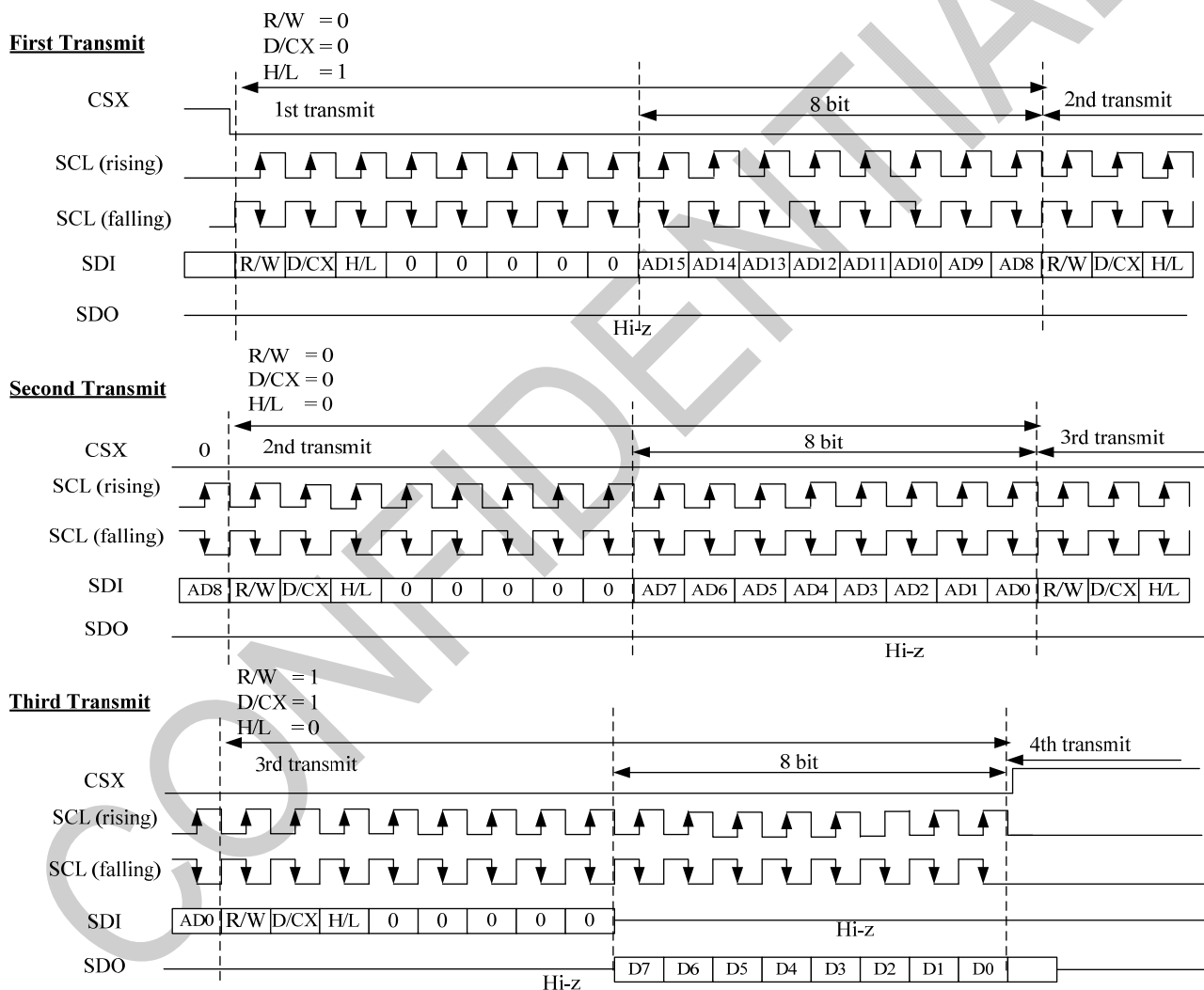


## 6.1.2 Read Cycle and Sequence

During a read cycle, host processor reads a single bit of data from display module via the interface. The SPI interface utilizes CSX, SCL and SDO signals. SCL is driven from high to low and then pulled back to high for a complete read cycle. The display module provides information during a read cycle and the host processor reads the display module information at the rising edge of SCL.

During a read sequence, host processor reads one or more bytes of information from the display module via the interface. A read sequence is initiated when CSX is driven from high to low and ended when CSX is pulled high. Each byte is either nine or sixteen write cycles in length. If the optional DCX signal is used, a byte is eight read cycles long. DCX is driven low while command information is on the interface and pulled high when data is present.

The SPI interface read command sequences are described in the following figures.

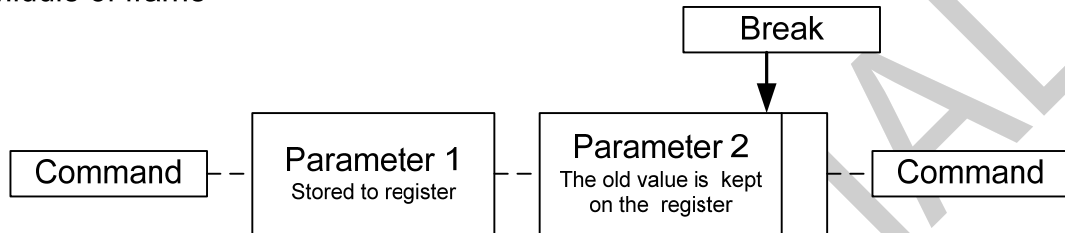


## 6.1.3 Break and Pause Sequence

The host processor can break a read or write sequence by pulling CSX high during processing a command or data byte. And, display module shall reset its interface so that it can receive the same byte when CSX is driven low again.

The host processor can postpone a read or write sequence by pulling CSX high between command or data bytes. The display module must wait until the host processor drives CSX low to continue the read or write sequence at the point where it was postponed.

### 1. Middle of frame

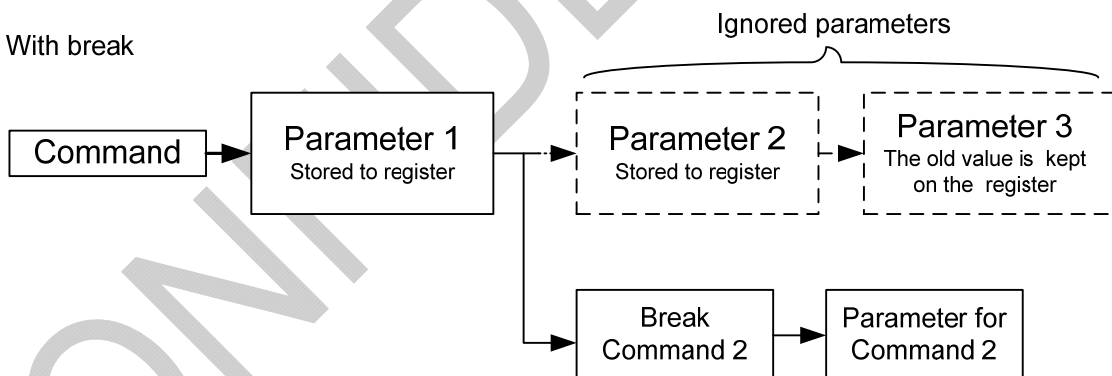


### 2. Between frames

Without break



With break



## 6.2 Display Serial Interface (DSI)

DSI-compliant peripherals support both of two basic modes of operation, *Command Mode* and *Video Mode*. The mode definitions reflect the intended use of DSI for display interconnect but not intend to restrict DSI from operating in other applications.

*Command Mode* refers to the operation in which transactions primarily take the form of sending commands to a display module, which incorporates a display controller including local registers. It requires a bidirectional interface so that systems can utilize this mode to write data onto or to read data from those local registers. The host processor indirectly controls activity of the peripheral by sending commands and parameters to the display controller. Moreover, the host processor can also read status information of a display module via it.

*Video Mode* refers to operation in which a real-time pixel stream data send from the host processor to the peripheral. Under normal operation, display module relies on host processor to provide image data with sufficient bandwidth to avoid flicker or other visible artifacts shown in the displayed image. Video information should only be transmitted by using High Speed Mode.

Configuration between MCU(Master) and RM68200(Slave)

| Lane Pair   | Available Operation Mode                                                                       |
|-------------|------------------------------------------------------------------------------------------------|
| Clock Lane  | Unidirectional Lane<br>Clock only                                                              |
| Data Lane 0 | Bi-directional Lane<br>Forward High-speed<br>Bi-directional Escape Mode<br>Bi-directional LPDT |
| Data Lane 1 | Unidirectional Lane<br>Forward High-Speed<br>Escape Mode<br>No LPDT                            |
| Data Lane 2 | Unidirectional Lane<br>Forward High-speed<br>Escape Mode<br>No LPDT                            |
| Data Lane 3 | Unidirectional Lane<br>Forward High-speed<br>Escape Mode<br>No LPDT                            |

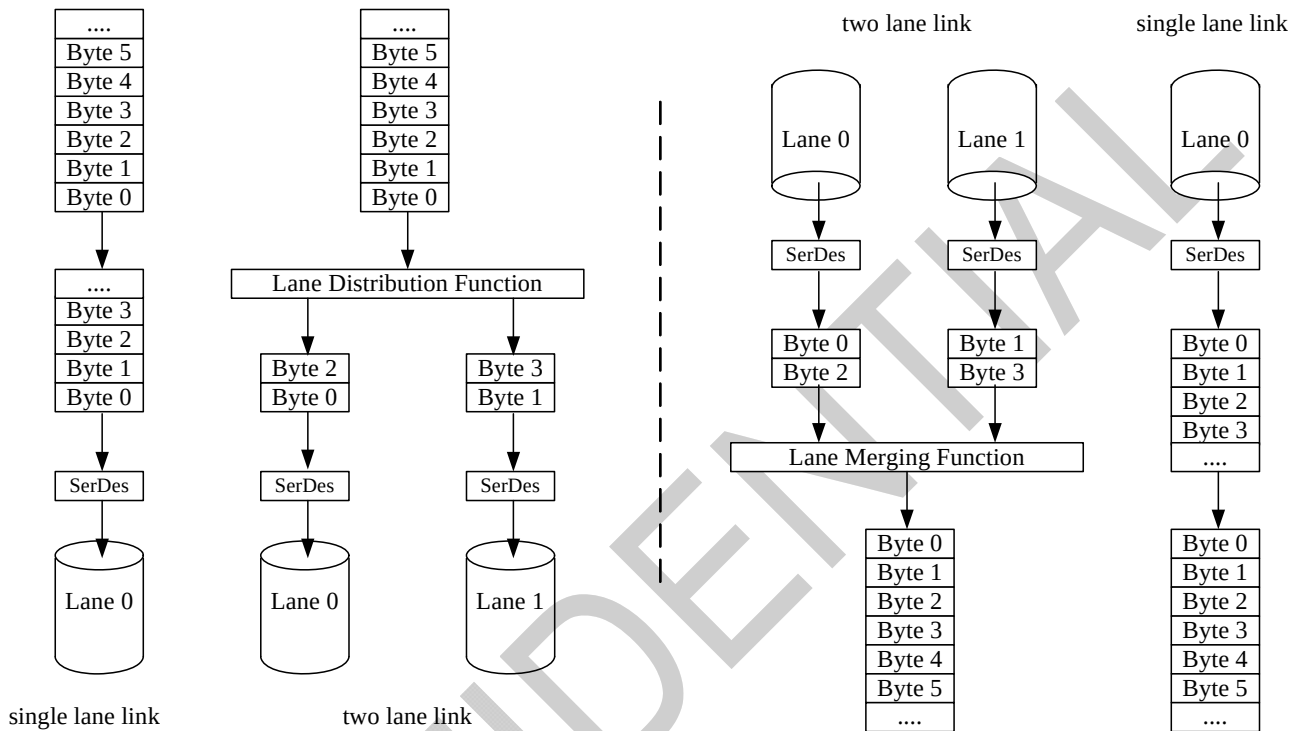
Notes:

1. We recommend users to stay in STOP state for 500ns when switching from LPDT to HSMT.
2. We recommend users to adopt EoTp to enhance overall robustness of the system during HSMT.



## 6.2.1 Protocol

On the transmitter side of a *DSI Link*, parallel data, signal events, and commands are converted to packets and sent across the serial Link. On the other hand, the receiver side of a *DSI Link* performs the data conversion, decomposing received packets into parallel data, signal events and commands. The data processing flow is shown in the following.



There are two kinds of transmitted packets. That is, short packet and long packet. Some of the key items comprised in the packet are commonly used for both of packet types:

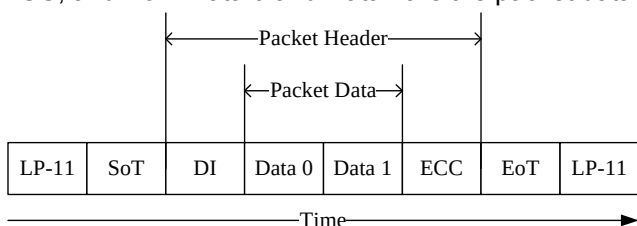
LP-11 : Low-Power mode  
SoT : Start-of-Transmission  
EoT : End-of-Transmission  
ECC : Error-Correction-Code  
DI : Data Identification

| Virtual Channel (VC) |       | Data Type (DT) |       |       |       |       |       |
|----------------------|-------|----------------|-------|-------|-------|-------|-------|
| Bit 7                | Bit 6 | Bit 5          | Bit 4 | Bit 3 | Bit 2 | Bit 1 | Bit 0 |

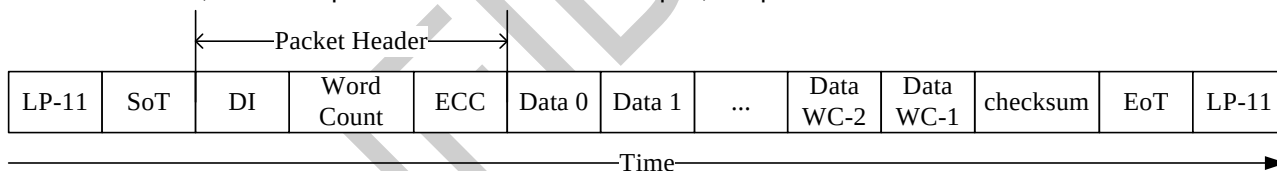
**\*\* Virtual Channel:** these two bits identify the data as directed to one of four virtual channels

**\*\* Data Type:** It specifies the packet structure and packet format

For the short packet as shown in the following, its data structure comprises of LP-11, SoT, DI, Data-0/1, ECC, and EoT. Data 0 and Data1 are the packet data



For the long packet as shown in the following, its data structure comprises of LP-11, SoT, DI, Word Count, ECC, Data(0) ~ Data(WC-1), checksum, and EoT. Where *Word Count* is the number of total data bytes of this long packet, *Checksum* is the 16-bit CRC generator to check packet data. If the calculated checksum of receiver is equal to the one contained in the packet data, then those packet data are regard as correct. However, if the comparison result shows non-equal, the packet data are considered to be incorrect.



**6.2.2 Processor-to-Peripheral Transactions**

Packet Data Types, from Processor to Peripheral

| Data Type | Data Type (Binary) | Description                                           | Packet Size |
|-----------|--------------------|-------------------------------------------------------|-------------|
| 01h       | 00 0001            | Sync Event, V Sync Start                              | Short       |
| 11h       | 01 0001            | Sync Event, V Sync End                                | Short       |
| 21h       | 10 0001            | Sync Event, H Sync Start                              | Short       |
| 31h       | 11 0001            | Sync Event, H Sync End                                | Short       |
| 08h       | 00 1000            | End of Transmission packet (EoTp)                     | Short       |
| 02h       | 00 0010            | Color Mode (CM) Off Command                           | Short       |
| 12h       | 01 0010            | Color Mode (CM) On Command                            | Short       |
| 22h       | 10 0010            | reserved                                              | Short       |
| 32h       | 11 0010            | reserved                                              | Short       |
| 03h       | 00 0011            | reserved                                              | Short       |
| 13h       | 01 0011            | Generic Short WRITE, 1 parameter                      | Short       |
| 23h       | 10 0011            | Generic Short WRITE, 2 parameters                     | Short       |
| 04h       | 00 0100            | reserved                                              | Short       |
| 14h       | 01 0100            | Generic READ, 1 parameter                             | Short       |
| 24h       | 10 0100            | Generic READ, 2 parameters                            | Short       |
| 05h       | 00 0101            | DCS Short WRITE, no parameters                        | Short       |
| 15h       | 01 0101            | DCS Short WRITE, 1 parameter                          | Short       |
| 06h       | 00 0110            | DCS READ, no parameters                               | Short       |
| 37h       | 11 0111            | Set Maximum Return Packet Size                        | Short       |
| 09h       | 00 1001            | Null Packet, no data                                  | Long        |
| 19h       | 01 1001            | Blanking Packet, no data                              | Long        |
| 29h       | 10 1001            | Generic Long Write                                    | Long        |
| 39h       | 11 1001            | DCS Long Write/write_LUT Command Packet               | Long        |
| 0Eh       | 00 1110            | Packed Pixel Stream, 16-bit RGB, 5-6-5 Format         | Long        |
| 1Eh       | 01 1110            | Packed Pixel Stream, 18-bit RGB, 6-6-6 Format         | Long        |
| 2Eh       | 10 1110            | Loosely Packed Pixel Stream, 18-bit RGB, 6-6-6 Format | Long        |
| 3Eh       | 11 1110            | Packed Pixel Stream, 24-bit RGB, 8-8-8 Format         | Long        |

**Sync Event, Data Type = xx 0001:**

Sync Events are all short packets and time-accurately. They can perform like the start and end of sync pulses. To represent timing information as accurately as possible, a V Sync Start event represents the start of the VSA and also implies an H Sync Start event for the first line of the VSA. Hence, a V Sync End event implies an H Sync Start event for the last line of the VSA. Sync events may be concatenated with blanking packets to convey inter-line timing accurately and avoid the overhead of switching between LPS and HS for every event. Note there is a power penalty for keeping the data line in HS mode.

**EoT packet:**

This short packet is used to indicate the end of a high speed (HS) transmission. This packet will enhance overall system reliability. Although the main objective of the EoTp is to enhance robustness during HS transmission mode, RM68200 can detect and interpret arriving EoTps regardless of transmission mode (HS or LP modes)

**Color Mode Off / On Command:**

They are short packet commands to switch video display module between normal display mode and low-color mode for power saving.

**Generic short write / read packet:**

Generic Short WRITE command is a short packet for sending generic data to the peripheral. Generic READ request is a short packet for acquiring data from the peripheral.

**DCS short write and read commands:**

DCS short write command is used to write a single data byte command to display module. If there is a valid parameter byte, data type bit 4 shall be set to 1. If there is no valid parameter byte, data type bit 4 shall be set to 0 and the parameter byte shall be 00h. On the other hand, DCS read command is used to request data from display module.

**DCS Long Write / write\_LUT command:**

The commands are used to send larger blocks of data to a display module.

**Maximum return packet size:**

This command specifies the maximum size of the payload in a long packet transmission from a display module to host processor.

**Null Packet:**

This is a mechanism for keeping the data lane(s) in high speed mode while sending dummy data.

**Blanking Packet:**

A Blanking packet is used to convey blanking timing information in a Long packet. The packet represents a period between active scan lines of a Video Mode display, where traditional display timing is provided from the host processor to the display module. The blanking period may have Sync Event packets interspersed between blanking segments. Blanking packets may contain arbitrary data as payload.

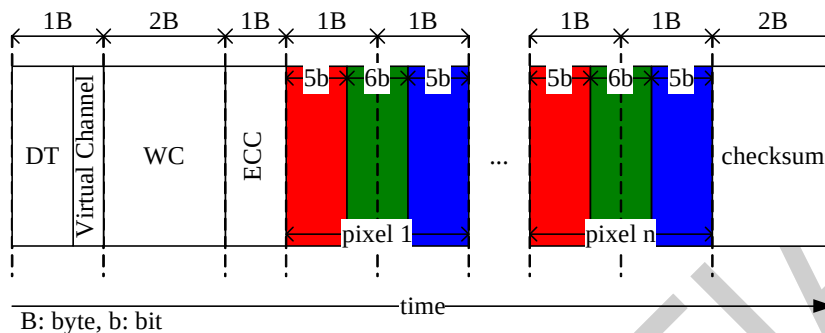
**Generic Long Write:**

This is used to transmit arbitrary blocks of data from a host processor to a peripheral.

## Packed Pixel Stream:

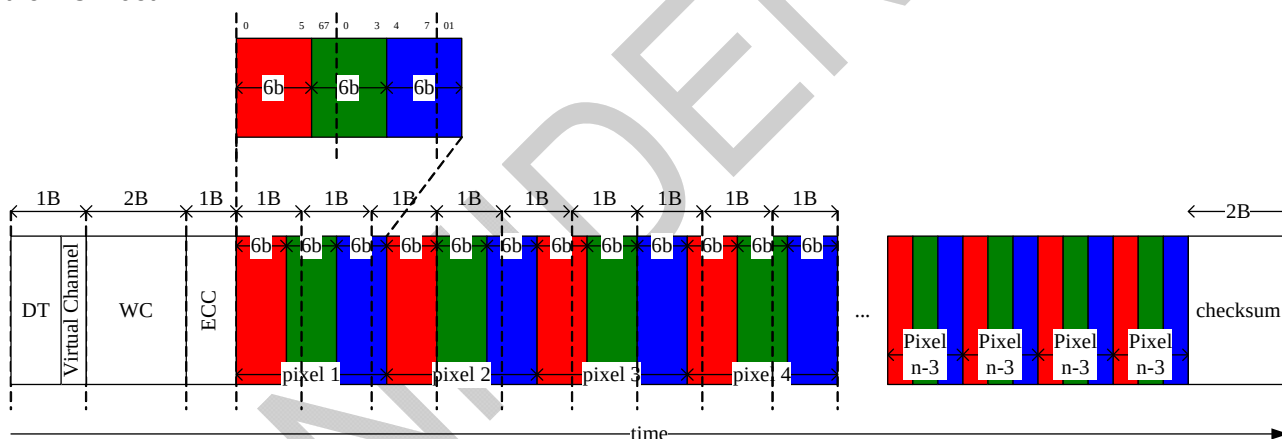
### 16-bit Format, Data Type: 0x0Eh

The pixel format is five bits red, six bits green and five bits blue. The green component is split across two bytes. Within a color component, the LSB is sent first, the MSB last.



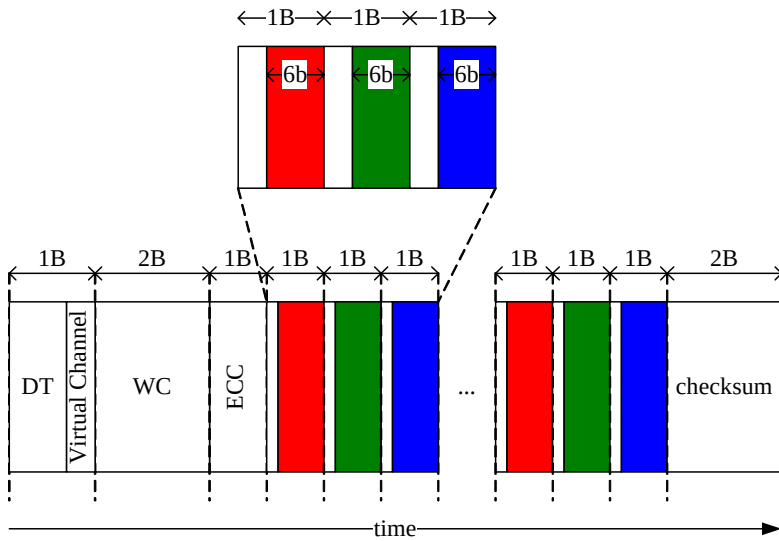
### 18-bit format, Data Type: 0x1Eh

The pixel format is six bits red, six bits green, and six bits blue. Within a color component, the LSB is sent first, the MSB last.



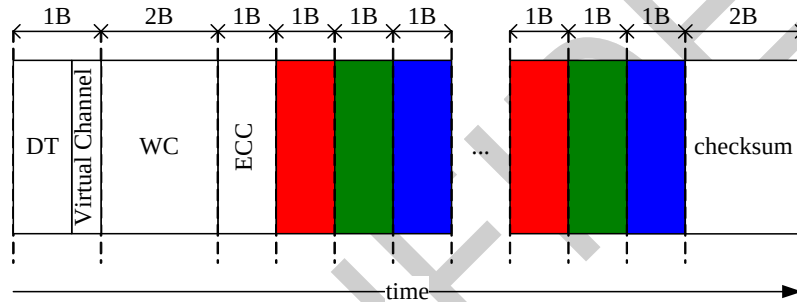
■ 18-bit format in three bytes, Data Type: 0x2Eh

This is 18-bit pixel loosely packed format, each R, G or B color component is six bits but shifted to the upper bits of byte.



Packet pixel stream, 24-bit format, Data Type: 0x3Eh

The pixel format is eight bits red, eight bits green and eight bits blue.



### 6.2.3 Peripheral-to-Processor LP Transmission

All Command Mode systems require bidirectional capability for returning READ data, acknowledge, or error information to the host processor. Multi-Lane systems shall use Lane 0 for all peripheral-to-processor transmission. Reverse-direction signaling shall only use low power mode transmission.

Packet structure for peripheral-to-processor transaction is the same as for the processor-to-peripheral direction. For the processor-to-peripheral direction, two basic packet formats are the same as the peripheral-to-processor direction: Short and Long packet structure. BTA shall take place after every peripheral-to-processor transaction. This bus turn-around control between host processor and the peripheral follows completion of the LP transmission from the peripheral.

There are four basic types of peripheral-to-processor transactions:

1. *Tearing Effect:*

It is a Trigger message sent to convey display timing information to the host processor.

2. *Acknowledge:*

It is a Trigger Message sent when the current transmission, as well as all preceding transmissions since the last peripheral to host communication, either triggers or packets, is received by the peripheral with no errors.

3. *Acknowledge and Error Report:*

It is a Short packet sent if any errors were detected in preceding transmissions from the host processor.

4. *Response to Read Request:*

It may be a Short or Long packet that returns data requested by the preceding READ command from the processor.



#### 6.2.4 Appropriate Response to Commands and ACK Requests

In general, if the host processor completes a transmission to the peripheral with BTA asserted, the peripheral shall respond with one or more appropriate packets, and then return bus control to the host processor. Interpretation of processor-to-peripheral transactions with BTA asserted, and the expected responses, are as follows:

- *Following a non-Read command*, the peripheral shall respond with Acknowledge if no errors were detected or stored since the last peripheral to host communication.
- *Following a Read request*, the peripheral shall send the requested READ data if no errors were detected and stored since the last peripheral to host communication.
- *Following a Read request*, if only a single-bit ECC error was detected and corrected, the peripheral shall send the requested READ data in a Long or Short packet, followed by a 4-byte Acknowledge and Error Report packet in the same LP transmission.
- *Following a non-Read command*, if only a single-bit ECC error was detected and corrected, the peripheral shall proceed to execute the command, and shall respond to BTA by sending a 4-byte Acknowledge and Error Report packet.
- *Following a Read request*, if multi-bit ECC errors were detected and not corrected, the peripheral shall send a 4-byte Acknowledge and Error Report packet without sending Read data.
- *Following a non-Read command*, if multi-bit ECC errors were detected and not corrected, the peripheral shall not execute the command, and shall send a 4-byte Acknowledge and Error Report packet.
- *Following any command*, if SoT Error, SoT Sync Error, or DSI VC ID invalid or DSI protocol violation was detected, or the DSI command was not recognized, the peripheral shall send a 4-byte Acknowledge and Error Report response, with the appropriate error flags set, as well as any error bits from any preceding transmissions stored since the last peripheral to host communication, in the two-byte error field.
- *Following any command*, if EoT Sync Error or LP Transmit Sync Error is detected, or a checksum error is detected in the payload, the peripheral shall send a 4-byte Acknowledge and Error Report packet with appropriate error flags set, as well as any error bits from any preceding transmissions stored since the last peripheral to host communication.

### 6.2.5 Error Reporting Format

Acknowledge and Error Report confirms that the preceding command or data sent from the host processor to a peripheral was received and indicates what type of error detected on the transmission and preceding transmission. An error report is a Short packet comprised of two bytes(Byte-1 and Byte-2) following the Data Identifier byte(Byte-0), with an ECC byte(Byte-3) following the Error Report bytes. By convention, detection and reporting of each error type is signified by setting the corresponding bit to "1". The following table shows the bit assignment for all error report:

| Bit | Description                                    |
|-----|------------------------------------------------|
| 0   | SoT Error                                      |
| 1   | SoT Sync Error                                 |
| 2   | EoT Sync Error                                 |
| 3   | Escape Mode Entry Command Error                |
| 4   | Low-Power Transmit Sync Error                  |
| 5   | HS Receive Timeout Error                       |
| 6   | False Control Error                            |
| 7   | Reserved                                       |
| 8   | ECC Error, single-bit (detected and corrected) |
| 9   | ECC Error, multi-bit (detected, not corrected) |
| 10  | Checksum Error (Long packet only)              |
| 11  | DSI Data Type Not Recognized                   |
| 12  | DSI VC ID Invalid                              |
| 13  | reserved                                       |
| 14  | reserved                                       |
| 15  | reserved                                       |

### 6.2.6 Peripheral-to-Processor Transaction – Detail Format Description

#### Data Types for Peripheral-to-Processor Packets

| Data type, hex | Data type , binary   | Description                                   | Packet size |
|----------------|----------------------|-----------------------------------------------|-------------|
| 0x00 – 0x01    | 00 000X              | Reserved                                      | short       |
| 0x02           | 00 0010              | Acknowledge and error report                  | short       |
| 0x03 – 0x07    | 00 0011-<br>00 011   | reserved                                      |             |
| 0x08           | 00 1000              | End of Transmission packet (EoTp)             | short       |
| 0x09 – 0x10    | 00 1001 –<br>01 0000 | Reserved                                      |             |
| 11h            | 01 0001              | Generic Short Read Response, 1 byte returned  | short       |
| 12h            | 01 0010              | Generic Short Read Response, 2 bytes returned | short       |
| 0x13 – 0x19    | 01 0011 –<br>01 1001 | Reserved                                      |             |
| 1Ah            | 01 1010              | Generic Long Read response                    | long        |
| 1Ch            | 01 1100              | DCS Long Read response                        | long        |
| 0x1D – 0x20    | 01 1101 –<br>10 0000 | Reserved                                      |             |
| 21h            | 10 0001              | DCS Short Read Response, 1 byte returned      | short       |
| 22h            | 10 0010              | DCS Short Read Response, 2 bytes returned     | short       |
| 0x23 – 0x3F    | 10 0011 –            | Reserved                                      |             |

**Acknowledge and error report**, it is sent in response to any command, or read, request, with BTA asserted when a reportable error is detected in the preceding, or earlier, transmission from the host processor.

**Generic Short Read Response**, this is the short-packet response to *Generic READ Request*. Packet composition is the Data Identifier (DI) byte, two bytes of payload data and an ECC byte. If the command itself is possibly corrupt, due to an uncorrectable ECC error, SoT or SoT Sync error, the requested READ data packet shall not be sent and only the *Acknowledge and Error Report* packet shall be sent.

**Generic Long Read Response**, this is the long-packet response to *Generic READ Request*. Packet composition is DI followed by a two-byte Word Count, an ECC byte, N bytes of payload, and a two-byte Checksum. If the command itself is possibly corrupt, due to an uncorrectable ECC error, SoT or SoT Sync error, the requested READ data packet shall not be sent and only the *Acknowledge and Error Report* packet shall be sent.

**DCS Long Read Response**, This is a long-packet response to *DCS Read Request*. Packet composition is DI followed by a two-byte Word Count, an ECC byte, N bytes of payload, and a two-byte Checksum. If the DCS command itself is possibly corrupt, due to uncorrectable ECC error, SoT or SoT Sync error, the requested READ data packet shall not be sent and only the *Acknowledge and Error Report* packet shall be sent.

**DCS Short Read Response**, this is the short-packet response to *DCS Read Request*. Packet composition is DI, two bytes of payload data and an ECC byte. If the command itself is possibly corrupt, due to an uncorrectable ECC error, SoT or SoT Sync error, the requested READ data packet shall not be sent and only the *Acknowledge and Error Report* packet shall be sent.

### 6.2.7 Error Correction Code (ECC)

ECC shall always be generated and appended in the Packet Header from the host processor. Peripherals with Bidirectional Links shall also generate and send ECC.

The number of parity or error check bits required is given by the Hamming rule, which uses parity to correct a single-bit error or detect a two-bit error, but are not capable of doing both simultaneously. DSI uses Hamming-modified codes where an extra parity bit is used to support both single error correction as well as two-bit error detection.

Since Packet Headers are fixed at four bytes (twenty-four data bits and eight ECC bits), P6 and P7 of the ECC byte are unused and shall be set to zero by the transmitter. The receiver shall ignore P6 and P7 and set both bits to zero before processing ECC.

The parity bits of ECC are defined as below:

P7 = 0

P6 = 0

P5 =  $D_{10} \oplus D_{11} \oplus D_{12} \oplus D_{13} \oplus D_{14} \oplus D_{15} \oplus D_{16} \oplus D_{17} \oplus D_{18} \oplus D_{19} \oplus D_{21} \oplus D_{22} \oplus D_{23}$

P4 =  $D_4 \oplus D_5 \oplus D_6 \oplus D_7 \oplus D_8 \oplus D_9 \oplus D_{16} \oplus D_{17} \oplus D_{18} \oplus D_{19} \oplus D_{20} \oplus D_{22} \oplus D_{23}$

P3 =  $D_1 \oplus D_2 \oplus D_3 \oplus D_7 \oplus D_8 \oplus D_9 \oplus D_{13} \oplus D_{14} \oplus D_{15} \oplus D_{19} \oplus D_{20} \oplus D_{21} \oplus D_{23}$

P2 =  $D_0 \oplus D_2 \oplus D_3 \oplus D_5 \oplus D_6 \oplus D_9 \oplus D_{11} \oplus D_{12} \oplus D_{15} \oplus D_{18} \oplus D_{20} \oplus D_{21} \oplus D_{22}$

P1 =  $D_0 \oplus D_1 \oplus D_3 \oplus D_4 \oplus D_6 \oplus D_8 \oplus D_{10} \oplus D_{12} \oplus D_{14} \oplus D_{17} \oplus D_{20} \oplus D_{21} \oplus D_{22} \oplus D_{23}$

P0 =  $D_0 \oplus D_1 \oplus D_2 \oplus D_4 \oplus D_5 \oplus D_7 \oplus D_{10} \oplus D_{11} \oplus D_{13} \oplus D_{16} \oplus D_{20} \oplus D_{21} \oplus D_{22} \oplus D_{23}$

ECC Parity Generation Rules, the encoding of parity and decoding of syndromes:

| Data Bit | P7 | P6 | P5 | P4 | P3 | P2 | P1 | P0 | Hex  |
|----------|----|----|----|----|----|----|----|----|------|
| 0        | 0  | 0  | 0  | 0  | 0  | 1  | 1  | 1  | 0x07 |
| 1        | 0  | 0  | 0  | 0  | 1  | 0  | 1  | 1  | 0x0B |
| 2        | 0  | 0  | 0  | 0  | 1  | 1  | 0  | 1  | 0x0D |
| 3        | 0  | 0  | 0  | 0  | 1  | 1  | 1  | 0  | 0x0E |
| 4        | 0  | 0  | 0  | 1  | 0  | 0  | 1  | 1  | 0x13 |
| 5        | 0  | 0  | 0  | 1  | 0  | 1  | 0  | 1  | 0x15 |
| 6        | 0  | 0  | 0  | 1  | 0  | 1  | 1  | 0  | 0x16 |
| 7        | 0  | 0  | 0  | 1  | 1  | 0  | 0  | 1  | 0x19 |
| 8        | 0  | 0  | 0  | 1  | 1  | 0  | 1  | 0  | 0x1A |
| 9        | 0  | 0  | 0  | 1  | 1  | 1  | 0  | 0  | 0x1C |
| 10       | 0  | 0  | 1  | 0  | 0  | 0  | 1  | 1  | 0x23 |
| 11       | 0  | 0  | 1  | 0  | 0  | 1  | 0  | 1  | 0x25 |
| 12       | 0  | 0  | 1  | 0  | 0  | 1  | 1  | 0  | 0x26 |
| 13       | 0  | 0  | 1  | 0  | 1  | 0  | 0  | 1  | 0x29 |
| 14       | 0  | 0  | 1  | 0  | 1  | 0  | 1  | 0  | 0x2A |
| 15       | 0  | 0  | 1  | 0  | 1  | 1  | 0  | 0  | 0x2C |
| 16       | 0  | 0  | 1  | 1  | 0  | 0  | 0  | 1  | 0x31 |
| 17       | 0  | 0  | 1  | 1  | 0  | 0  | 1  | 0  | 0x32 |
| 18       | 0  | 0  | 1  | 1  | 0  | 1  | 0  | 0  | 0x34 |
| 19       | 0  | 0  | 1  | 1  | 1  | 0  | 0  | 0  | 0x38 |
| 20       | 0  | 0  | 0  | 1  | 1  | 1  | 1  | 1  | 0x1F |
| 21       | 0  | 0  | 1  | 0  | 1  | 1  | 1  | 1  | 0x2F |
| 22       | 0  | 0  | 1  | 1  | 0  | 1  | 1  | 1  | 0x37 |
| 23       | 0  | 0  | 1  | 1  | 1  | 0  | 1  | 1  | 0x3B |

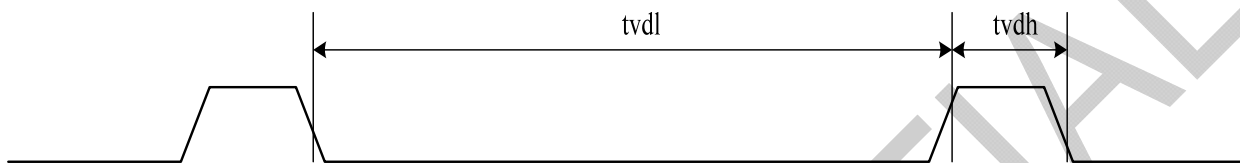
## 6.3 Tearing Effect Output

The tearing effect output from the peripheral to the HOST is a synchronization signal for panel display. Functionally, this output is used by the HOST to synchronize internal VSYNC when displaying video images.

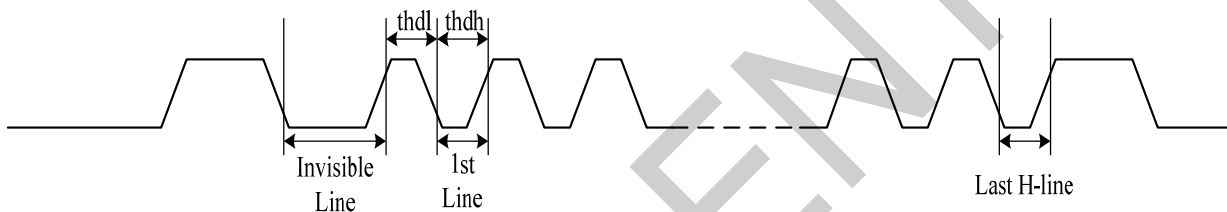
This signal can be enabled or disabled by the `set_tear_off(0x34h)` and `set_tear_on(0x35h)` commands. The selection of tearing effect output model is defined by the parameter of command `set_tear_on(0x35h)`. The number of display lines used as TE output's stop trigger in every frame is set by `set_tear_scanline(0x44h)`.

### 6.3.1 Tearing Effect Mode

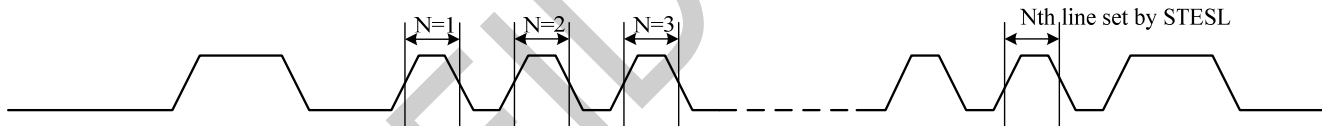
Mode 1, the tearing effect output signal consists of V-sync information only:



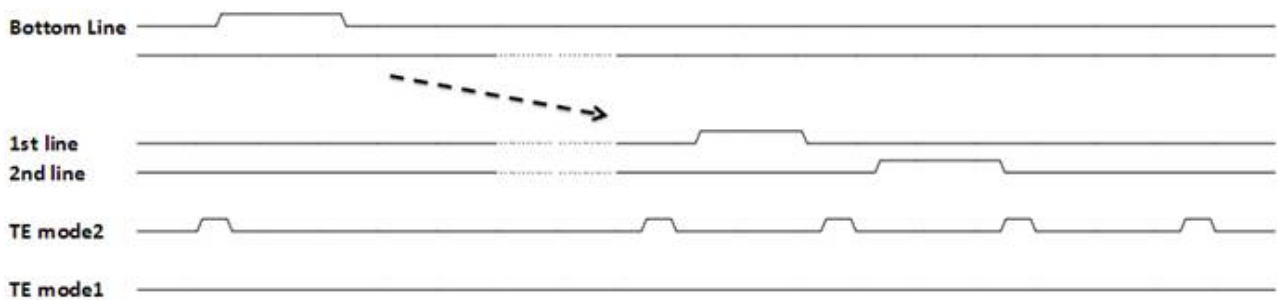
Mode 2, the tearing effect output signal consists of both V-sync and H-sync information:



Mode 3, the tearing effect output signal stops toggling when vertical scanning reaches line N in every frame:



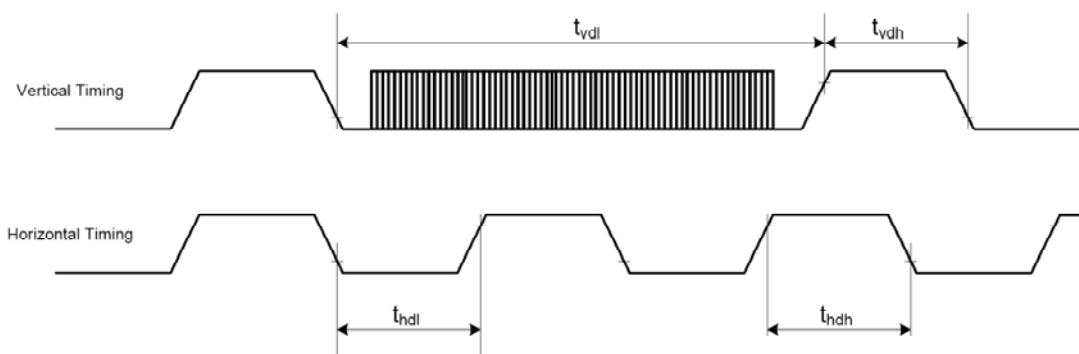
N = The N-th scanning line which set by register N[15:0] of command STESL(0x44h).



Note. During Sleep In mode, the tearing effect output signal is active low.

### 6.3.2 Timing of Tearing Effect

The timing definitions of tearing effect signal are described as below:



AC characteristics of Tearing Effect Signal (Frame Rate = 60.5Hz)

| Symbol | Parameter                       | Min. | Max. | Unit | Description |
|--------|---------------------------------|------|------|------|-------------|
| tvdl   | Vertical timing low duration    | TBD  |      | ms   |             |
| tvdh   | Vertical timing high duration   | TBD  |      | us   |             |
| thdl   | Horizontal timing low duration  | TBD  |      | us   |             |
| thdh   | Horizontal timing high duration | TBD  |      | us   |             |

\*\* The signal's rise and fall times ( $t_f$ ,  $t_r$ ) are limited to be equal to or less than 15ns.

Mode selection for TE output

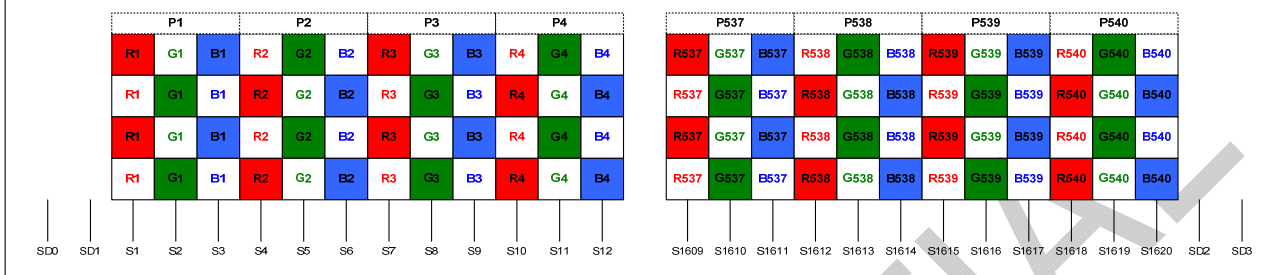
| TEON (0x35h)       | TELOM (0x35h[0]) | TE Output   |
|--------------------|------------------|-------------|
| Command is not set | X                | GND         |
| Command is set     | 0                | TE (Mode 1) |
| Command is set     | 1                | TE (Mode 2) |

## 6.4 Panel Type

### 6.4.1 Normal Type

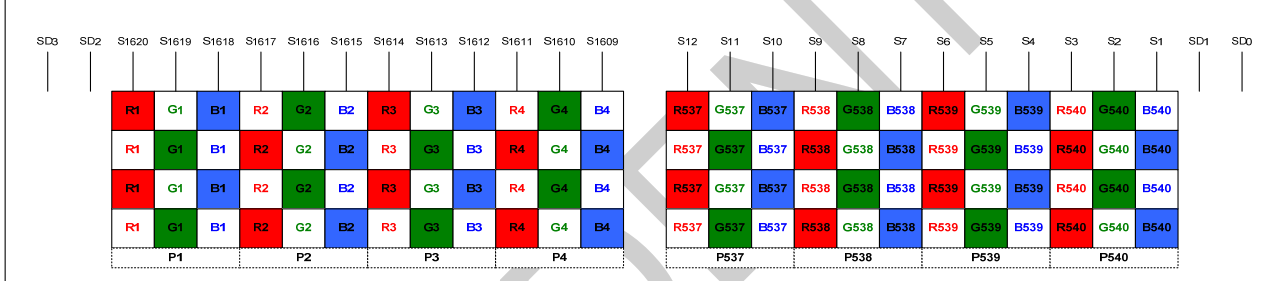
Bottom Side

normal (CTB=0, CRL=0, CRGB=0)



Upper Side

normal (CTB=0, CRL=1, CRGB=1)



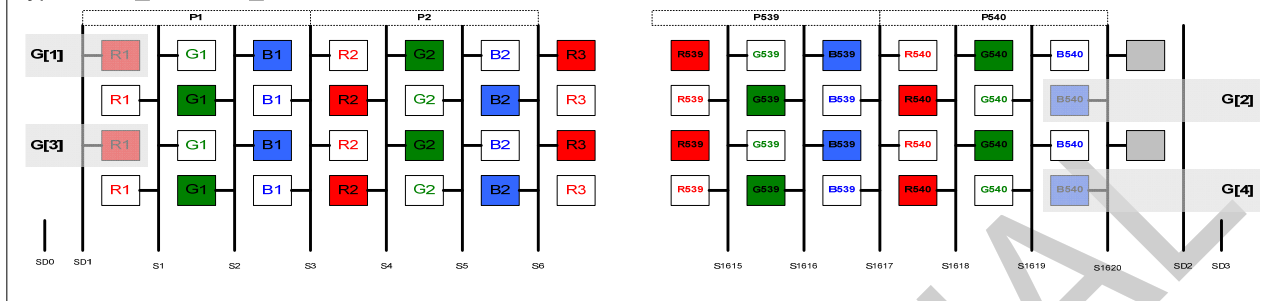


## 6.4.2 Zig-Zag Type

### ● Zig-Zag Type 1

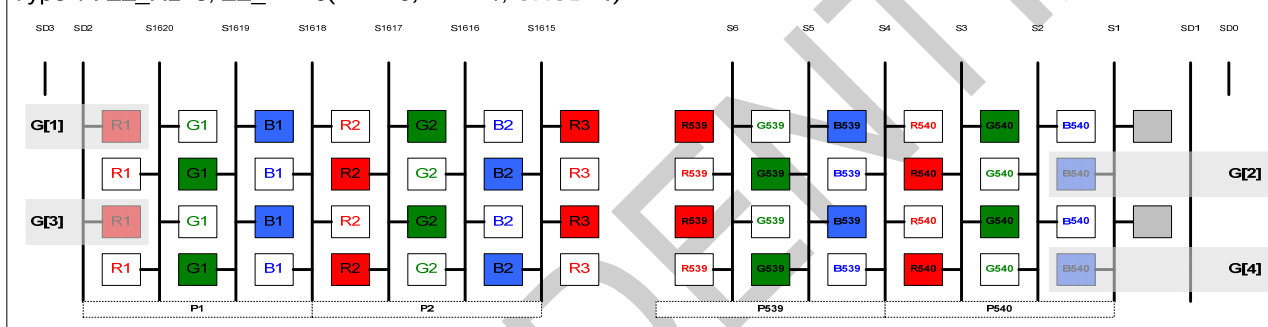
Bottom Side

Type 1 : ZZ\_RL=1, ZZ\_EO=0(CTB=0, CRL=0, CRGB=0)



Upper Side

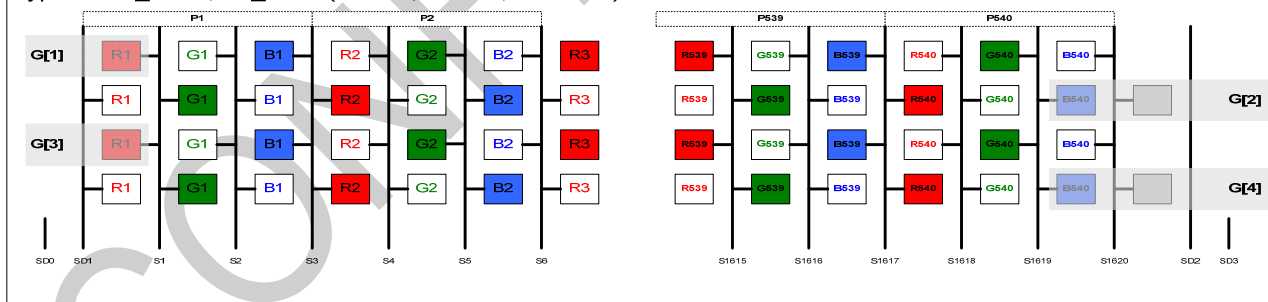
Type 1 : ZZ\_RL=0, ZZ\_EO=0(CTB=0, CRL=1, CRGB=1)



### ● Zig-Zag Type 2

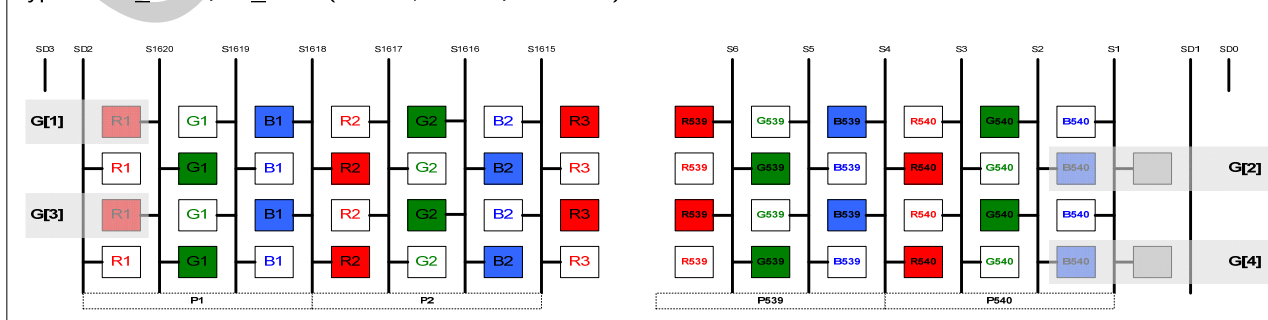
Bottom Side

Type 2 : ZZ\_RL=1, ZZ\_EO=1(CTB=0, CRL=0, CRGB=0)



Upper Side

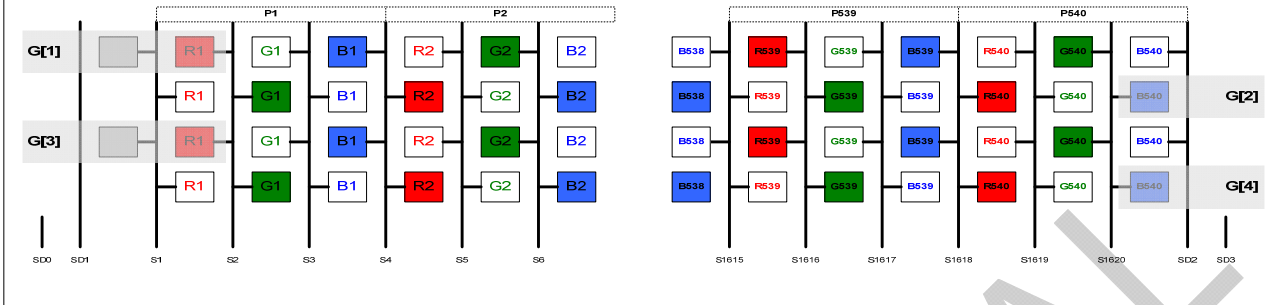
Type 2 : ZZ\_RL=0, ZZ\_EO=1(CTB=0, CRL=1, CRGB=1)



## ● Zig-Zag Type 3

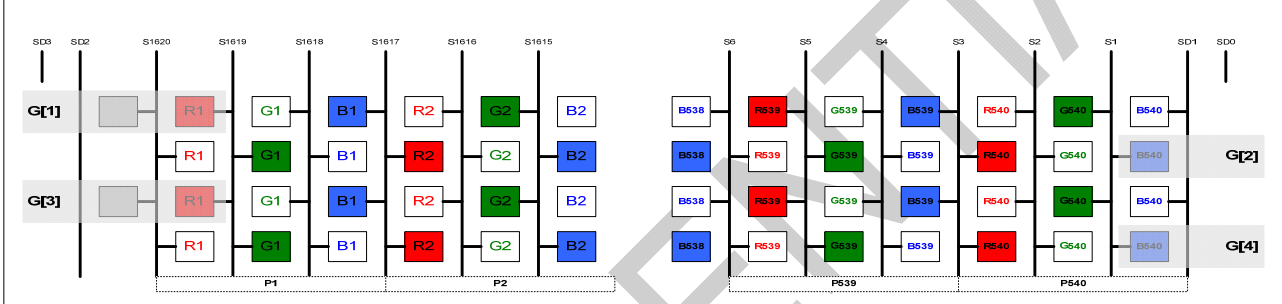
### Bottom Side

Type 3 : ZZ\_RL=0, ZZ\_EO=0(CTB=0, CRL=0, CRGB=0)



### Upper Side

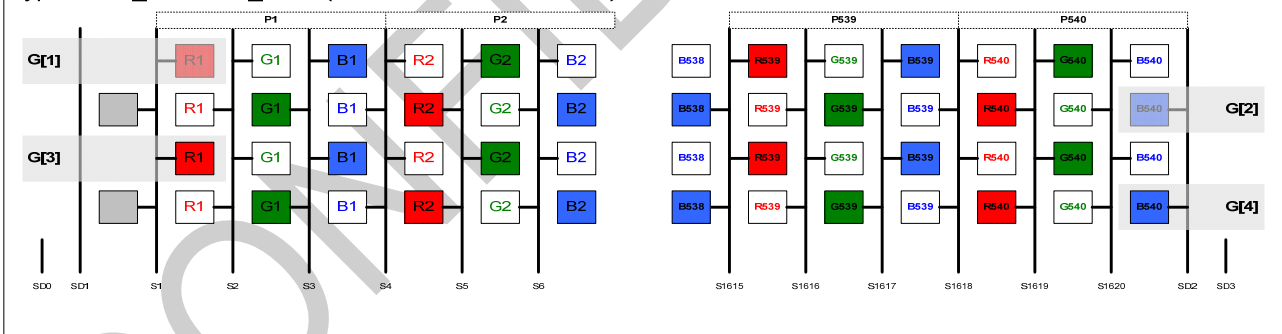
Type 3 : ZZ\_RL=1, ZZ\_EO=0(CTB=0, CRL=1, CRGB=1)



## ● Zig-Zag Type 4

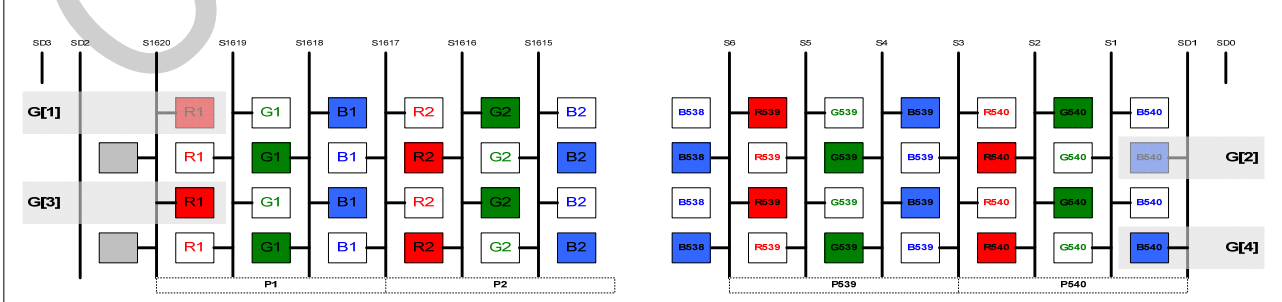
### Bottom Side

Type 4 : ZZ\_RL=0, ZZ\_EO=1(CTB=0, CRL=0, CRGB=0)



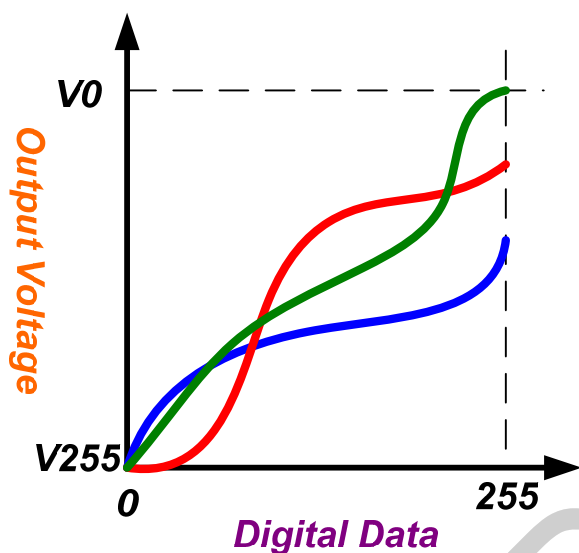
### Upper Side

Type 4 : ZZ\_RL=1, ZZ\_EO=1(CTB=0, CRL=1, CRGB=1)



## 6.5 Independent Gamma Correction Function

RM68200 supports independent gamma adjustment function for R, G and B dots separately to display 16.7M colors. This function is implemented by selecting different grayscale levels set by gamma adjustment registers. This LSI also provides gradient control registers for gamma curve adjustments. The driver's source outputs corresponding to different color dots are briefly depicted in the following figure. The relation between source outputs and color dots is set by the selection of panel type.



## 6.6 Dynamic Backlight Control

The function of Dynamic Display Backlight Control is used to reduce the power consumption of display backlight without sacrificing too much display quality. The function of Content Adaptive Brightness Control (CABC) is included for the calculation of target backlight brightness and consequently to generate a proper PWM signal to control other lighting device(s) (such as power IC or LED driver IC). For the registers described in this section are all located in CMD2 P3. If users want to use these functions, please contact Raydium for detailed information and register setting.

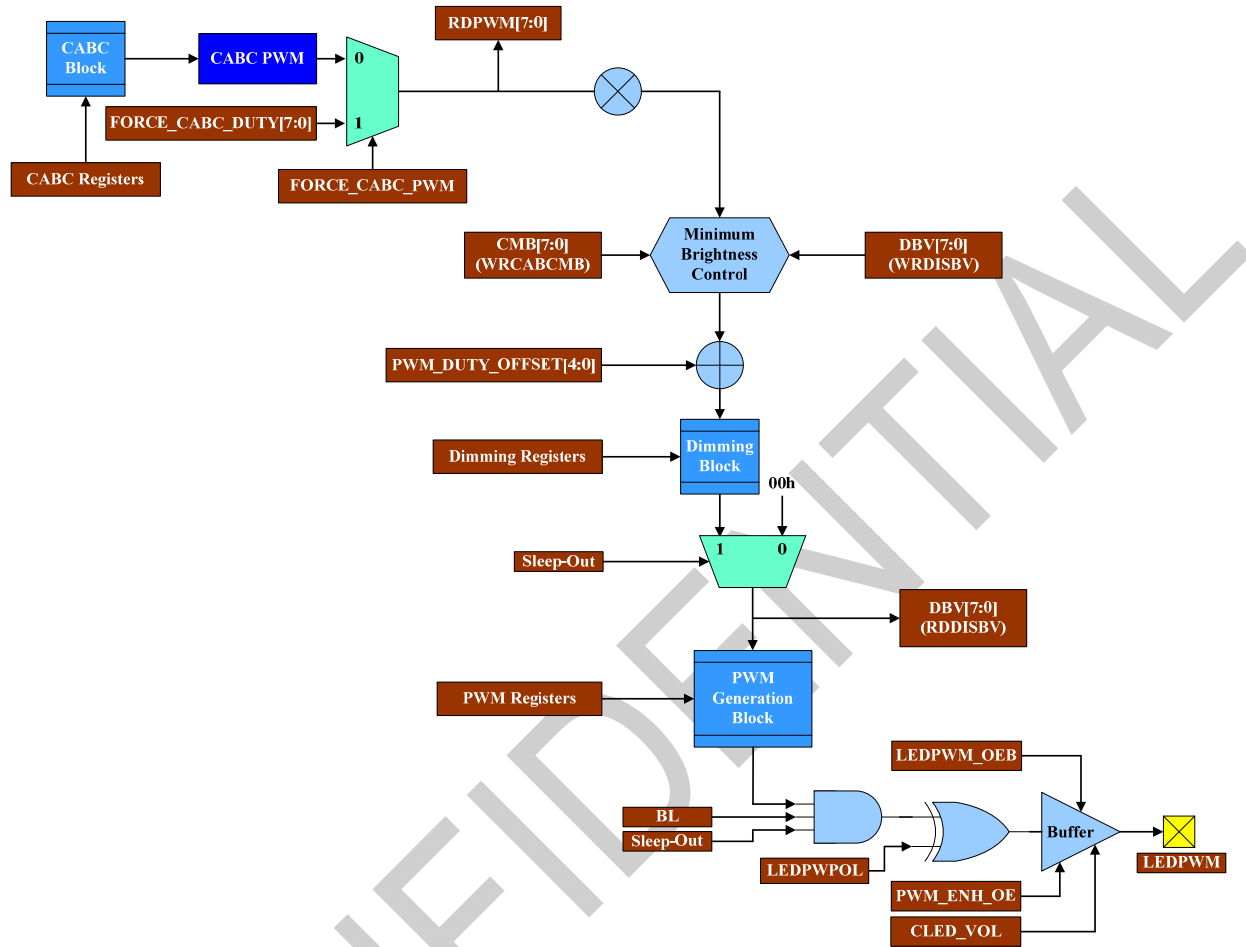
### 6.6.1 Content Adaptive Brightness Control (CABC)

Contents adaptation means that the grayscale level of image contents is changed simultaneously corresponding to the change of backlight's luminance so as to keep the same perceived brightness. The adjustment of grayscale levels and brightness reduction, or say backlight power consumption, is based on the contents of whole display data. And, the display image and brightness are dynamically controlled by CABC block. In order to reduce power consumption of backlight element and maintain an acceptable display quality at the same time, 4 different schemes are given for users to select. They are off mode, user interface image mode, still picture mode, and moving image mode, respectively. Descriptions of these four modes are listed below:

1. Off mode: Content Adaptive Brightness Control functionality is totally off.
2. UI [User interface] image mode: Optimized for UI image. It is kept image quality as much as possible. Target power consumption reduction ratio: 10% or less. User can achieve prefer brightness for UI-Mode by setting the registers "ui\_idx1" ~ "ui\_idx12" in CMD2 P3.
3. Still picture mode: Optimized for still picture. Some image quality degradation would be acceptable. Target power consumption reduction ratio: more than 30%. The RM68200 will automatically estimate a better gamma setting based on image contents. User can achieve prefer brightness for still picture mode by setting the registers "st\_idx1" ~ "st\_idx12" in CMD2 P3..
4. Moving image mode: Optimized for moving image e.g. Video clip. It is focused on the biggest power reduction with image quality degradation. Target power consumption reduction ratio: more than 30%. The RM68200 will automatically estimate a better gamma setting based on image contents. User can achieve prefer brightness for moving image mode by setting the registers "mv\_idx1" ~ "mv\_idx12" in CMD2 P3.

## 6.6.2 PWM Control Architecture

The following diagram is the brief flow chart of internal display backlight control.



The register bit “BL” is used to turn on or turn off backlight control function. Normally, if user want to disable the display backlight completely and immediately, user can set “BL = 0”. Relations between “BL” and LEDPWM output are list below:

| BL | LEDPWPOL | Status of LEDPWM                            |
|----|----------|---------------------------------------------|
| 0  | 0        | 0                                           |
|    | 1        | 1                                           |
| 1  | 0        | Toggle with original polarity of PWM signal |
|    | 1        | Toggle with inversed polarity of PWM signal |

The register “BCTRL” is used to turn on or turn off backlight control block. Display brightness which is controlled by LEDPWM will be turn off when BCTRL is set to be 0.

| BCTRL | Value of DBV[7:0] (RDDISBV) | Display Backlight Status |
|-------|-----------------------------|--------------------------|
| 0     | 0x00h                       | Off                      |
| 1     | Determined by CABC          | On                       |

The data writing to DBV[7:0] (WRDISBV) is used to adjust the backlight brightness manually. On the other hand, the data read from register DBV[7:0] (RDDISBV) is used to indicate the output of PWM signal duty. That is, DBV[7:0](RDDISBV) is also affected by CABC function when CABC function is enabled.

The register setting CMB[7:0](WRCABCCMB) is used to limit the minimum PWM duty in order to prevent the backlight brightness from being too dark.

If the register FORCE\_CABC\_PWM is set to be 1, register data “FORCE\_CABC\_DUTY[7:0]\*DBV[7:0]” are used to perform a fixed PWM duty of CABC output.

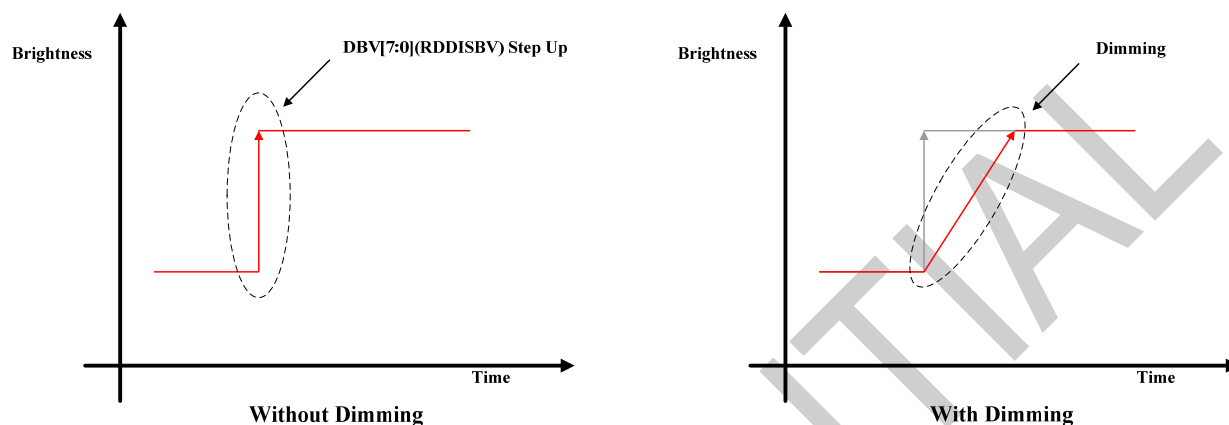
| Sleep-Out Mode,<br>BL=1, and BCTRL=1, FORCE_CABC_PWM=0,<br>CMB[7:0](WRCABCCMB) = 0x00h, and PWM_DUTY_OFFSET[4:0]=0x00h |                  |                                    |
|------------------------------------------------------------------------------------------------------------------------|------------------|------------------------------------|
| CABC Status                                                                                                            | RDPWM[7:0]       | Display Backlight Brightness       |
| Off Mode                                                                                                               | 0xFFh            | DBV[7:0] (WRDISBV)                 |
| UI-Mode /<br>Still-Mode /<br>Moving-Mode                                                                               | By CABC Function | DBV[7:0] (WRDISBV) x CABC Function |

The register data “DD” is used to enable or disable the dimming function for Manual Brightness Control. Smooth transition of PWM duty is performed when “DD=1”. Dimming function is applied only when driver IC is in sleep-out status. CABC function is available only when driver IC is under sleep-out. Availability of functions is listed in below table.

| Driver IC Status | CABC Function | Dimming Function | Display Backlight Status |
|------------------|---------------|------------------|--------------------------|
| Sleep-In         | Not Available | Not Available    | Turn-Off                 |
| Sleep-Out        | Available     | Available        | Controllable             |

### 6.6.3 Dimming Function for Manual Brightness Control

A dimming function is implemented to determine how fast the brightness changes from old level to new one and what are brightness levels during the transition. The curves of brightness dimming for *Manual Brightness Control* can be configured either the same or not the same in increment and decrement directions. The basic concept of dimming function is described below.



There are two types of PWM duty dimming mechanism are provided, and they are “Fixed-Time Dimming” and “Fixed-Slope Dimming”. The settings of dimming type for rising dimming and falling dimming are independent. The register “SEL\_IN” is used for rising dimming (increment dimming), and the register “SEL\_DE” is used for falling dimming (decrement dimming).

| SEL_IN | SEL_DE | Rising Dimming Type | Falling Dimming Type |
|--------|--------|---------------------|----------------------|
| 0      | 0      | Fixed-Time Dimming  | Fixed-Time Dimming   |
| 0      | 1      | Fixed-Time Dimming  | Fixed-Slope Dimming  |
| 1      | 0      | Fixed-Slope Dimming | Fixed-Time Dimming   |
| 1      | 1      | Fixed-Slope Dimming | Fixed-Slope Dimming  |



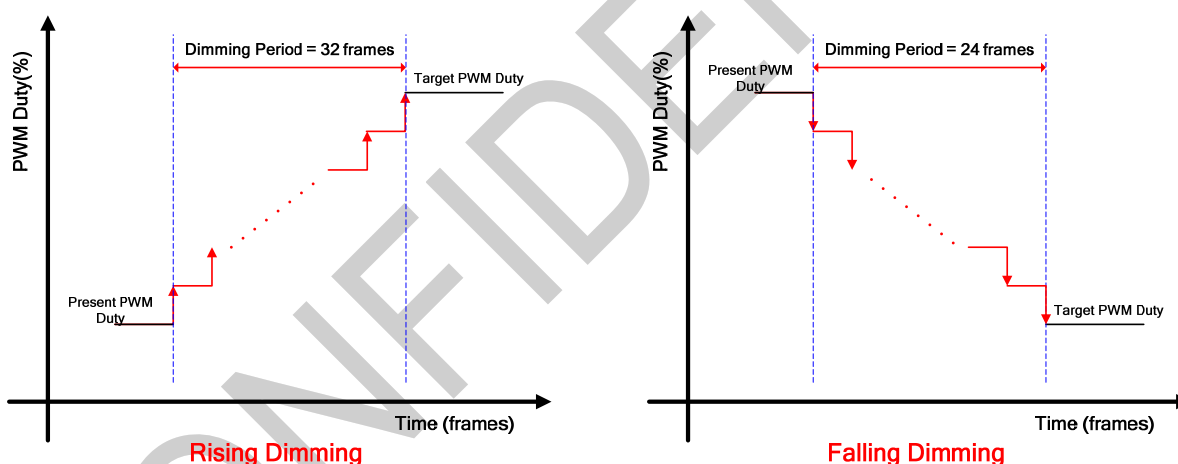
## 1. Fixed-Time Dimming Type

The total dimming steps can be set by registers “DMSTP\_L[2:0]”, “DM\_IN[3:0]”, and “DM\_DE[3:0]”. These three registers can also used to determine the total dimming time. The registers “DM\_IN[3:0]” and “DM\_DE[3:0]” are frame-based and the unit for them is “frame(s) per step”. On the other hand, the unit of registers dim\_step\_off[2:0], dim\_step\_still[2:0], and dim\_step\_mov[2:0] are all in “step(s)”. The following table is an example for these parameters:

| Register Name                                                 | Value | Description                                                                                        |
|---------------------------------------------------------------|-------|----------------------------------------------------------------------------------------------------|
| SEL_IN                                                        | 0     | Fixed-Time dimming for rising dimming                                                              |
| SEL_DE                                                        | 0     | Fixed-Time dimming for falling dimming                                                             |
| DM_IN[3:0]                                                    | 7h    | The number of frames needed per for an increase of step duty is 8                                  |
| DM_DE[3:0]                                                    | 5h    | The number of frames needed per for an decrease of step duty is 6                                  |
| dim_step_off[2:0]<br>dim_step_still[2:0]<br>dim_step_mov[2:0] | 1h    | 1. dimming steps is 4 steps<br>2. The setting can be set individually for different operation mode |

From the register data described above, the brightness dimming characteristic is shown below:

- (1) Total dimming time of “rising dimming” is 32-frames time length (8 frames x 4).
- (2) Total dimming time of “falling dimming” is 24-frames time length (6 frames x 4).



## 2. Fixed-Slope Dimming Type

The increasing/decreasing PWM duty during a time period can be set by register “STEP\_IN[3:0]”, “STEP\_DE[3:0]”, “DM\_IN[3:0]”, and “DM\_DE[3:0]”, respectively. These registers can also be used to determine some characteristics of dimming curves. The unit of registers STEP\_IN[3:0] and STEP\_DE [3:0] is “duty ratio” (0xFFh is 100%, and 0x00h is 0%). The following table is an example for these parameters:

| Register Name | Value | Description                                                       |
|---------------|-------|-------------------------------------------------------------------|
| SEL_IN        | 1     | Fixed-Slope dimming for rising dimming                            |
| SEL_DE        | 1     | Fixed-Slope dimming for falling dimming                           |
| STEP_IN[3:0]  | 8h    | PWM increment is 8 for each step                                  |
| STEP_DN[3:0]  | 5h    | PWM decrement is 5 for each step                                  |
| DM_IN[3:0]    | 3h    | The number of frames needed per for an increase of step duty is 8 |
| DM_DE[3:0]    | 4h    | The number of frames needed per for a decrease of step duty is 6  |

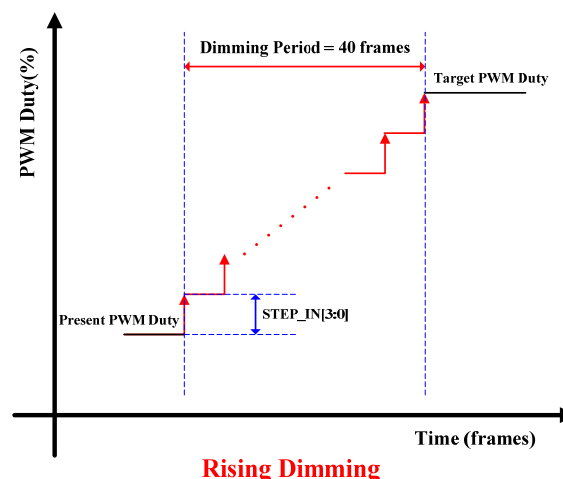
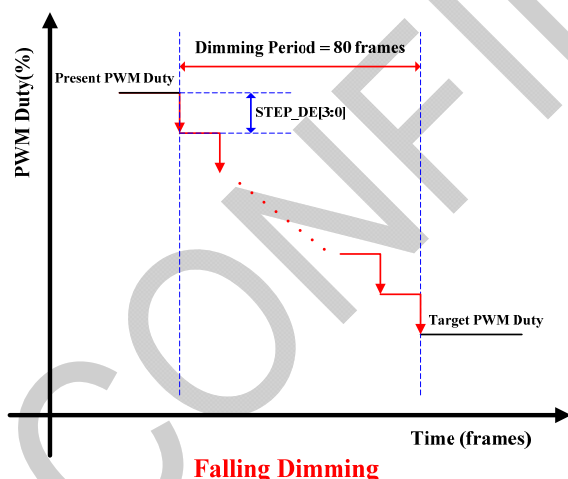
From the register data described above, the brightness dimming characteristic is shown below:

1. When the present PWM duty is 0x64h(100 in decimal) and target PWM duty is 0x14h(20 in decimal),  
Total dimming steps = (Present PWM Duty - Target PWM duty) / (PWM decrement) = (100-20)/5=16 steps

So total dimming time for falling dimming is 80 frames (16 steps x 5 frames/step)

2. When the present PWM duty is 0x14h(20 in decimal) and new PWM duty is 0x64h(100 in decimal),  
Total dimming steps = (Target PWM Duty - Present PWM duty) / (PWM increment)=(100 - 20)/8=10 steps

So total dimming time for rising dimming is 40 frames (10 steps x 4 frames/step)



## 6.6.4 PWM Signal Setting for CABC

The register data PWMDIV[7:0] can change the frequency of PWM signal (LEDPWM) and the register "PWM\_DUTY\_OFFSET[4:0]" can perform a duty compensation of the PWM output signal. The " $F_{OSC}$ " is used to provide clock source for the internal PWM circuit. Three PWM operation frequency can be chosen by setting register "PWMF[1:0]", and PWM frequency can be quickly estimated by the below formula:

| PWMF[1:0] | PWM Operation Frequency ( $F_{osc}$ ) |
|-----------|---------------------------------------|
| 00        | 8.3 MHz                               |
| 01        | 12.5 MHz                              |
| 10        | 25 MHz                                |
| 11        | (not used)                            |

$$PWM \text{ Frequency} = \frac{F_{osc}}{256 \times PWMDIV[7:0]}$$

$$PWM \text{ Duty} = \frac{DBV[7:0](RDISBV) + 1}{256}$$

**For example:**

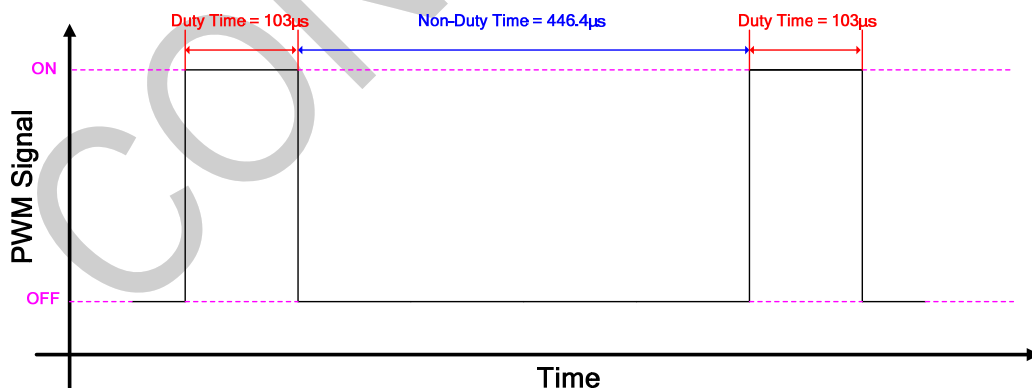
If PWMDIV[7:0]=0x0Fh, PWMF[1:0]=0x0h, and DBV[7:0]=0x2Fh(RDISBV)

$PWM \text{ Frequency} = 8.3\text{MHz} / (256 \times 15) = 2.161\text{kHz}$ .

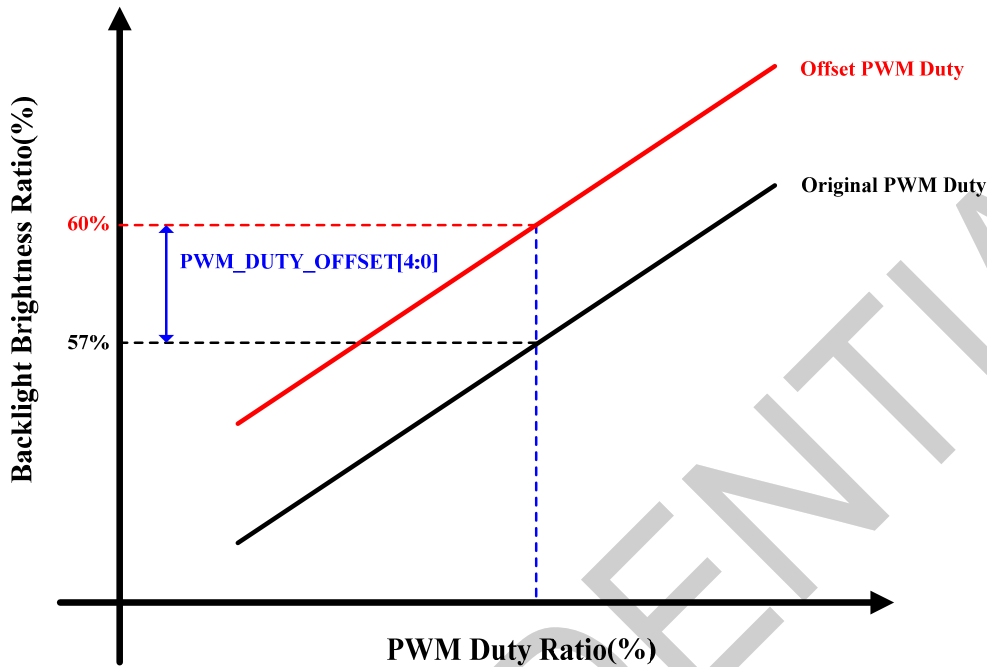
DBV[7:0]=0x2Fh means PWM duty ratio is about 18.75%

$PWM \text{ duty time} = 18.75\% \times 1/2.161\text{kHz} = 86.76\mu\text{s}$

$PWM \text{ non-duty time} = 81.25\% \times 1/2.161\text{kHz} = 375.9\mu\text{s}$



Since the external LED driver needs some stable time to drive the LED backlight, this necessary stable time will reduce the effective PWM duty period. The PWM\_DUTY\_OFFSET[4:0] is simply used to compensate the loss of PWM duty period. For example, assume original PWM duty of LEDPWM signal is 60%, but the actual backlight brightness driven by LED driver is 57%. User can set PWM\_DUTY\_OFFSET[4:0] to achieve 60% backlight brightness.



## 6.7 MTP Programming

### 6.7.1 Available Pages for MTP Write

Once the data in these pages are programmed, the page index is recorded to be 1'b1. User can read this index to check whether the corresponding page is programmed or not. The pages available for MTP programming are list as below for reference.

For CMD2 Page1 and Page3 are display algorithm-dependent, so the registers are not released to customers through this datasheet.

#### CMD2 MTP Summary Table Programmed by Customer

##### **CMD2 Page0:** Display Option, Power etc.

Register address: 0x0040h ~ 0xFF40h

Total Program Count = **1**

Enable Program => OTP\_cmd2\_P0\_PGM\_EN = D[5] of 0xF040h

Program Done Flag => CMD2\_P0\_Done = D[5] of 0xEE40h

##### **CMD2 Page1:** 3 Gamma etc.

Register address: 0x0050h ~ 0xFF50h

Total Program Count = **1**

Enable Program => OTP\_cmd2\_P1\_PGM\_EN = D[4] of 0xF040h

Program Done Flag => CMD2\_P1\_Done = D[4] of 0xEE40h,

##### **CMD2 Page2:** Display GOA Timing etc.

Register address: 0x0070h ~ 0xFF70h

Total Program Count = **1**

Enable Program => OTP\_cmd2\_P2\_PGM\_EN = D[3] of 0xF040h

Program Done Flag => CMD2\_P2\_Done = D[3] of 0xEE40h

##### **CMD2 Page3:** CABC

Register address: 0x0080h ~ 0x5F80h & 0x7180h ~ 0xFF80h

Total Program Count = **1**

Enable Program => OTP\_cmd2\_P3\_PGM\_EN = D[2] of 0xF040h

Program Done Flag => CMD2\_P3\_Done = D[2] of 0xEE40h

##### **CMD2 Page3:** Gamma2.2 + CABC Gamma Offset(11 sets).

Register address: 0x6080h ~ 0x7F80h

Total Program Count = **1**

Enable Program => OTP\_cmd2\_gma\_PGM\_EN = D[0] of 0xF040h

Program Done Flag => CMD2\_gma\_Done = D[0] of 0xEE40h

##### **CMD2 Page0:** ID Code, VCOM

Register address: 0x4540h ~ 0x4940h & 0xE140h ~ 0xE740h

Total Program Count = **2**

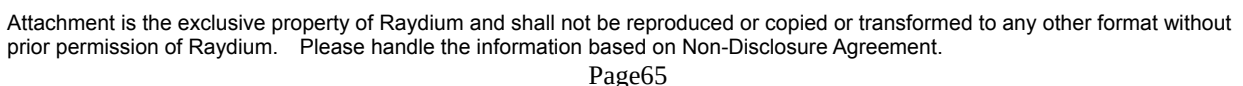
1<sup>st</sup> Enable Program => OTP\_CMD2\_2ndEN = D[6] of 0xF040h

1<sup>st</sup> Program Done Flag => CMD2\_2nd\_Done = D[6] of 0xEE40h

2<sup>nd</sup> Enable Program => OTP\_CMD2\_3rdEN = D[7] of 0xF040h

2<sup>nd</sup> Program Done Flag => CMD2\_3rd\_Done = D[7] of 0xEE40h

The MTP trim flow is depicted in the following. Schemes using both external power, MTP\_PWR, and internal power, VGH, are included. The program verify after data write needs either hardware reset or software reset for MTP data reload.

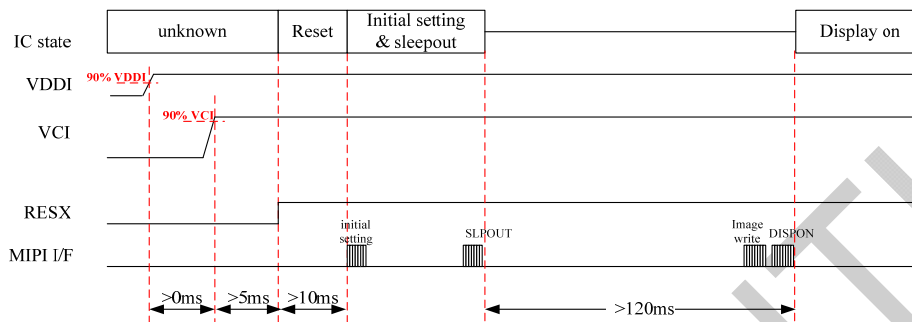


## 6.8 Power On/Off Sequence

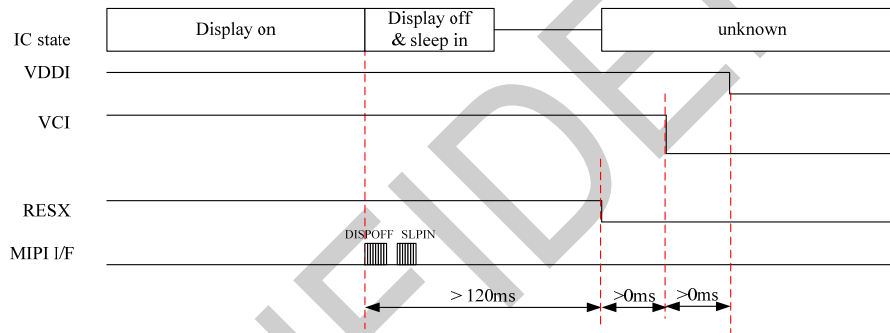
### 6.8.1 2-Power mode with Power IC or PFM mode

The power on/off sequence for 2-power mode, in which input powers are VCI and VDDI, is depicted in the following. Please follow the power input sequence to avoid triggering any abnormal state.

#### Power On sequence



#### Power Off sequence

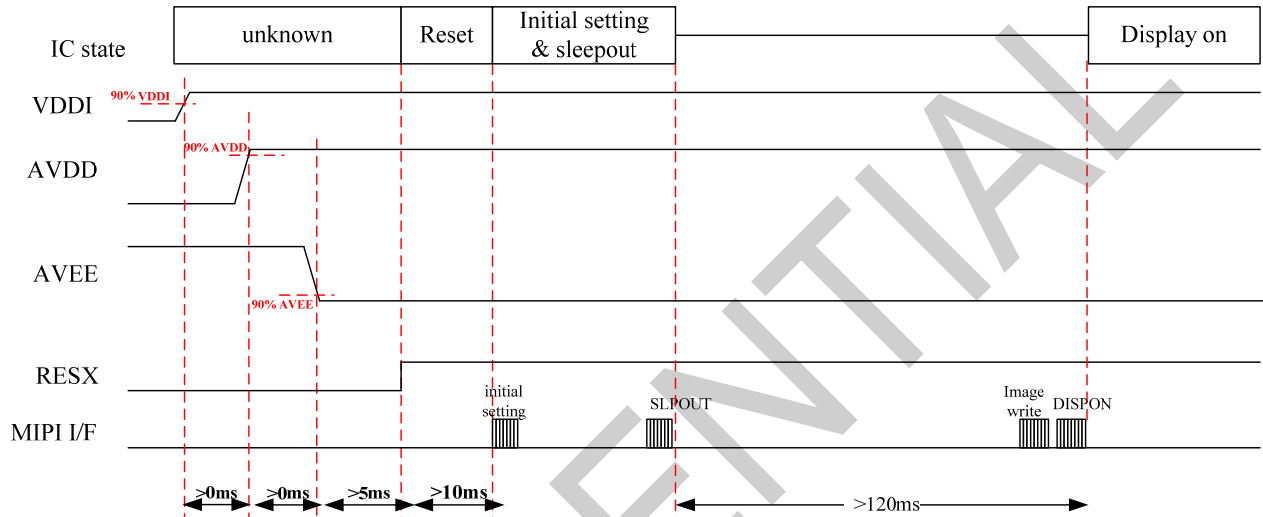




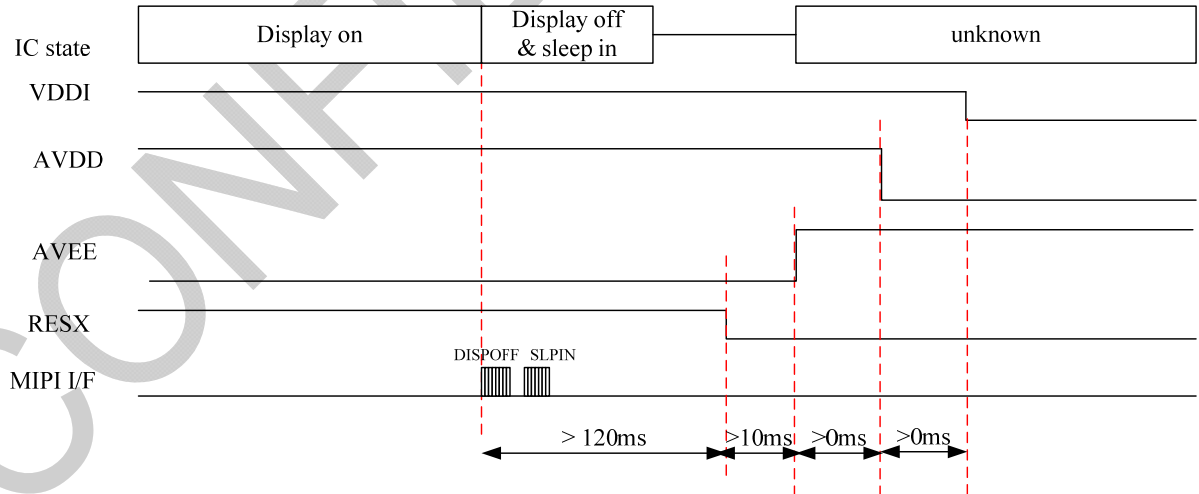
## 6.8.2 3-Power IC mode

The power on/off sequence for 3-power mode, in which input powers are VDDI, AVDD and AVEE, is depicted in the following. Please follow the power input sequence to avoid triggering any abnormal state.

### Power On sequence



### Power Off sequence



### **6.8.3 Uncontrolled Power Off**

The uncontrolled power off means a situation when e.g. there is removed a battery without the controlled power off sequence. There will not be any damage for the display module or the display module will not cause any damage to the host or lines of the interface. At an uncontrolled power off the display will go blank and there will not be any visible effects and remains blank until "Power-On Sequence" Powers it up.

CONFIDENTIAL

## 7. DCS Command

### 7.1 DCS Command List

| Command |      |       | W/R | Function                                  | D7              | D6               | D5     | D4            | D3            | D2 | D1 | D0    | Default (hex) | MTP |   |
|---------|------|-------|-----|-------------------------------------------|-----------------|------------------|--------|---------------|---------------|----|----|-------|---------------|-----|---|
| Page    | Add. | Para. |     |                                           |                 |                  |        |               |               |    |    |       |               |     |   |
| CMD1    | 00h  | -     | W   | NOP                                       | No Argument     |                  |        |               |               |    |    |       | -             | -   |   |
| CMD1    | 01h  | -     | W   | Software reset                            | No Argument     |                  |        |               |               |    |    |       | -             | -   |   |
| CMD1    | 04h  | 1st   | R   | Read display identification information   | ID1[7:0]        |                  |        |               |               |    |    |       | 00h           | -   |   |
| CMD1    | 04h  | 2nd   |     |                                           | ID2[7:0]        |                  |        |               |               |    |    |       | 80h           | -   |   |
| CMD1    | 04h  | 3rd   |     |                                           | ID3[7:0]        |                  |        |               |               |    |    |       | 00h           | -   |   |
| CMD1    | 05h  | -     | R   | Read number of the errors on DSI          | P[7:0]          |                  |        |               |               |    |    |       | 00h           | -   |   |
| CMD1    | 0Ah  | 1st   | R   | Read display power mode                   | RDDPM[7:0]      |                  |        |               |               |    |    |       | 08h           | -   |   |
| CMD1    | 0Bh  | 1st   | R   | Read display MADCTR                       | RDDMADCTR[7:0]  |                  |        |               |               |    |    |       | 00h           | -   |   |
| CMD1    | 0Ch  | 1st   | R   | Read display pixel format                 | RDDCOLMOD[7:0]  |                  |        |               |               |    |    |       | 70h           | -   |   |
| CMD1    | 0Dh  | 1st   | R   | Read display image mode                   | RDDIM[7:0]      |                  |        |               |               |    |    |       | 00h           | -   |   |
| CMD1    | 0Eh  | 1st   | R   | Read display signal mode                  | RDDSM[7:0]      |                  |        |               |               |    |    |       | 00h           | -   |   |
| CMD1    | 0Fh  | 1st   | R   | Read display self-diagnostic result       | RDDSDR[7:0]     |                  |        |               |               |    |    |       | 00h           | -   |   |
| CMD1    | 10h  | -     | W   | Sleep-in                                  | No Argument     |                  |        |               |               |    |    |       | -             | -   |   |
| CMD1    | 11h  | -     | W   | Sleep-out                                 | No Argument     |                  |        |               |               |    |    |       | -             | -   |   |
| CMD1    | 12h  | -     | W   | Partial display mode on                   | No Argument     |                  |        |               |               |    |    |       | -             | -   |   |
| CMD1    | 13h  | -     | W   | Normal display mode on                    | No Argument     |                  |        |               |               |    |    |       | -             | -   |   |
| CMD1    | 20h  | -     | W   | Display inversion off                     | No Argument     |                  |        |               |               |    |    |       | -             | -   |   |
| CMD1    | 21h  | -     | W   | Display inversion on                      | No Argument     |                  |        |               |               |    |    |       | -             | -   |   |
| CMD1    | 22h  | -     | W   | All pixel off                             | No Argument     |                  |        |               |               |    |    |       | -             | -   |   |
| CMD1    | 23h  | -     | W   | All pixel on                              | No Argument     |                  |        |               |               |    |    |       | -             | -   |   |
| CMD1    | 28h  | -     | W   | Display off                               | No Argument     |                  |        |               |               |    |    |       | -             | -   |   |
| CMD1    | 29h  | -     | W   | Display on                                | No Argument     |                  |        |               |               |    |    |       | -             | -   |   |
| CMD1    | 30h  | 1st   | W   | Partial area                              | SR[15:8]        |                  |        |               |               |    |    |       | 00h           | -   |   |
| CMD1    |      | 2nd   | W   |                                           | SR[7:0]         |                  |        |               |               |    |    |       | 00h           | -   |   |
| CMD1    |      | 3rd   | W   |                                           | ER[15:8]        |                  |        |               |               |    |    |       | 04h           | -   |   |
| CMD1    |      | 4th   | W   |                                           | ER[7:0]         |                  |        |               |               |    |    |       | FFh           | -   |   |
| CMD1    | 34h  | -     | W   | Tearing effect line off                   | No Argument     |                  |        |               |               |    |    |       | -             | -   |   |
| CMD1    | 35h  | -     | W   | Tearing effect line on                    |                 |                  |        |               |               |    |    | TELOM | 00h           | -   |   |
| CMD1    | 36h  | -     | W   | Scan direction control                    | MADCTR[7:0]     |                  |        |               |               |    |    |       | 00h           | -   |   |
| CMD1    | 38h  | -     | W   | Idle mode off                             | No Argument     |                  |        |               |               |    |    |       | -             | -   |   |
| CMD1    | 39h  | -     | W   | Enter idle mode                           | No Argument     |                  |        |               |               |    |    |       | -             | -   |   |
| CMD1    | 44h  | 1st   | W   | Set tear scan-line                        | STS[15:8]       |                  |        |               |               |    |    |       | 00h           | -   |   |
| CMD1    |      | 2nd   | W   |                                           | STS[7:0]        |                  |        |               |               |    |    |       | 00h           | -   |   |
| CMD1    | 45h  | 1st   | R   | Get scan line                             | GTS[15:8]       |                  |        |               |               |    |    |       | 00h           | -   |   |
| CMD1    |      | 2nd   | R   |                                           | GTS[7:0]        |                  |        |               |               |    |    |       | 00h           | -   |   |
| CMD1    | 4Fh  | -     | W   |                                           |                 |                  |        |               |               |    |    | DSTB  | 00h           | -   |   |
| CMD1    | 51h  | -     | W   | Write display brightness                  | DBV[7:0]        |                  |        |               |               |    |    |       | 00h           | -   |   |
| CMD1    | 52h  | -     | R   | Read display brightness                   | DBV[7:0]        |                  |        |               |               |    |    |       | 00h           | -   |   |
| CMD1    | 53h  | -     | W   | Write CTRL display                        | SLR_LEVEL [1:0] | BCTRL            | 0      | DD            | BL            | 0  | 0  |       | 00h           | -   |   |
| CMD1    | 54h  | -     | R   | Read CTRL display                         | SLR_LEVEL [1:0] | BCTRL            | 0      | DD            | BL            | 0  | 0  |       | 00h           | -   |   |
| CMD1    | 55h  | -     | W   | Write Content Adaptive Brightness Control | CTE_EN          | CTE_LEVEL[3:0]   |        |               |               | 0  | C1 | C0    |               | 48h | - |
| CMD1    | 56h  | -     | R   | Read Content Adaptive Brightness Control  | CTE_EN          | CTE_LEVEL[3:0]   |        |               |               | 0  | C1 | C0    |               | 48h | - |
| CMD1    | 58h  | -     | W   | Set color enhancement                     | EN_EDGE E_ENH   | EDGE_LEVE L[1:0] | SLR_EN | EN_VIVI D_ENH | CE_LEVEL[2:0] |    |    |       | 21h           | -   |   |
| CMD1    | 59h  | -     | R   | Read color enhancement                    | EN_EDGE E_ENH   | EDGE_LEVE L[1:0] | SLR_EN | EN_VIVI D_ENH | CE_LEVEL[2:0] |    |    |       | 00h           | -   |   |
| CMD1    | 5Eh  | -     | W   | Write CABG Minimum Brightness             | CMB[7:0]        |                  |        |               |               |    |    |       | 00h           | -   |   |
| CMD1    | 5Fh  | -     | W   | Read CABG Minimum Brightness              | CMB[7:0]        |                  |        |               |               |    |    |       | 00h           | -   |   |
| CMD1    | A1h  | 1st   | R   | Read DDB                                  | SID[7:0]        |                  |        |               |               |    |    |       | 00h           | -   |   |

|      |     |     |   |                          |           |   |   |   |                 |   |   |   |     |   |
|------|-----|-----|---|--------------------------|-----------|---|---|---|-----------------|---|---|---|-----|---|
| CMD1 |     | 2nd | R |                          | SID[15:8] |   |   |   |                 |   |   |   | 00h | - |
| CMD1 |     | 3rd | R |                          | MID[7:0]  |   |   |   |                 |   |   |   | 00h | - |
| CMD1 |     | 4th | R |                          | MID[15:8] |   |   |   |                 |   |   |   | 00h | - |
| CMD1 |     | 5th | R |                          | 1         | 1 | 1 | 1 | 1               | 1 | 1 | 1 | FFh | - |
| CMD1 | A8h | 1st | R | Read DDB Continuous      | SID[7:0]  |   |   |   |                 |   |   |   | 00h | - |
| CMD1 |     | 2nd | R |                          | SID[15:8] |   |   |   |                 |   |   |   | 00h | - |
| CMD1 |     | 3rd | R |                          | MID[7:0]  |   |   |   |                 |   |   |   | 00h | - |
| CMD1 |     | 4th | R |                          | MID[15:8] |   |   |   |                 |   |   |   | 00h | - |
| CMD1 |     | 5th | R |                          | 1         | 1 | 1 | 1 | 1               | 1 | 1 | 1 | FFh | - |
| CMD1 | AAh | -   | R | Read first checksum      | FCS[7:0]  |   |   |   |                 |   |   |   | 00h | - |
| CMD1 | AFh | -   | R | Read continuous checksum | CCS[7:0]  |   |   |   |                 |   |   |   | 00h | - |
| CMD1 | DAh | -   | R | Read ID1                 | ID1[7:0]  |   |   |   |                 |   |   |   | 00h | - |
| CMD1 | DBh | -   | R | Read ID2                 | ID2[7:0]  |   |   |   |                 |   |   |   | 80h | - |
| CMD1 | DCh | -   | R | Read ID3                 | ID3[7:0]  |   |   |   |                 |   |   |   | 00h | - |
| CMD1 | FEh | -   | W | Write CMD mode page      | 0         | 0 | 0 | 0 | CMD_Page[3:0]   |   |   |   | 00h | - |
| CMD1 | FFh | -   | R | Read CMD page Status     | 0         | 0 | 0 | 0 | CMD_Status[3:0] |   |   |   | 00h | - |

## 7.2 DCS Command Description

### NOP (0000h)

| 0000H                 | NOP (No Operation)                                                                                                                                                                                                                                                                                                                                                                         |               |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------|-------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|-----|
| Instruction           | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address       |       | D15-8       | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI          | Other |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| NOP                   | W                                                                                                                                                                                                                                                                                                                                                                                          | 00h           | 0000h | No Argument |    |    |    |    |    |    |    |    | -   |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Description           | This command is an empty command; it does not have any effect on the display module.                                                                                                                                                                                                                                                                                                       |               |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Restriction           | None                                                                                                                                                                                                                                                                                                                                                                                       |               |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Register Availability | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |               |       |             |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                     | Availability  |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                   | Yes           |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                    | Yes           |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                  | Yes           |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                   | Yes           |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Sleep In              | Yes                                                                                                                                                                                                                                                                                                                                                                                        |               |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Default               | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>N/A</td></tr><tr><td>SW Reset</td><td>N/A</td></tr><tr><td>HW Reset</td><td>N/A</td></tr></table>                                                                                                                                                                                                   |               |       |             |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | N/A | SW Reset                                | N/A | HW Reset                                  | N/A |                                          |     |          |     |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                     | Default Value |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Power On Sequence                                                                                                                                                                                                                                                                                                                                                                          | N/A           |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | SW Reset                                                                                                                                                                                                                                                                                                                                                                                   | N/A           |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| HW Reset              | N/A                                                                                                                                                                                                                                                                                                                                                                                        |               |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Flow Chart            | None                                                                                                                                                                                                                                                                                                                                                                                       |               |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |

**SWRESET(0100h): Software Reset**

| 0100H                                     | SWRESET(Software Reset)                                                                                                                                                                                                                                                                                                                                                                    |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|-------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address |       | D15-8       | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI    | Other |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| SWRESET                                   | W                                                                                                                                                                                                                                                                                                                                                                                          | 01h     | 0100h | No Argument |    |    |    |    |    |    |    |    | -   |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Description                               | When the Software Reset command is executed, it resets the commands and parameters to their S/W Reset default values.                                                                                                                                                                                                                                                                      |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Restriction                               | Software Reset Command cannot be sent during Sleep Out sequence. Any new command cannot be sent within 10-frame period until the device enters Sleep-In mode.                                                                                                                                                                                                                              |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |         |       |             |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                               |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>N/A</td></tr><tr><td>SW Reset</td><td>N/A</td></tr><tr><td>HW Reset</td><td>N/A</td></tr></table>                                                                                                                                                                                                   |         |       |             |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | N/A | SW Reset                                | N/A | HW Reset                                  | N/A |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                              |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Power On Sequence                         | N/A                                                                                                                                                                                                                                                                                                                                                                                        |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| SW Reset                                  | N/A                                                                                                                                                                                                                                                                                                                                                                                        |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| HW Reset                                  | N/A                                                                                                                                                                                                                                                                                                                                                                                        |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Flow Chart                                | <div><div><div>SWRESET (01h)</div><div>↓</div><div>Display whole blank screen</div><div>↓</div><div>Set Commands to S/W Default Value</div><div>↓</div><div>Sleep In Mode</div></div><div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div></div>                                   |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |

**RDDID(0400h): Read Display ID**

| 0400H                                     |                                                                                                                                                                                                                                                                                                                                                                                              |                             |       | RDDID |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |                             |                                          |           |                             |          |           |                             |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------|-------|-------|------|------|------|------|------|------|------|------|-----|--------|---------------|------------------------------------------|-----------|-----------------------------------------|-------------------|-------------------------------------------|-----------------------------|------------------------------------------|-----------|-----------------------------|----------|-----------|-----------------------------|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                          | Address                     |       | D15-8 | D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   | HEX |        |               |                                          |           |                                         |                   |                                           |                             |                                          |           |                             |          |           |                             |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                              | MIPI                        | Other |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |                             |                                          |           |                             |          |           |                             |
| RDDID                                     | R                                                                                                                                                                                                                                                                                                                                                                                            | 04h                         | 0400h | x     | ID17 | ID16 | ID15 | ID14 | ID13 | ID12 | ID11 | ID10 | 00  |        |               |                                          |           |                                         |                   |                                           |                             |                                          |           |                             |          |           |                             |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                              |                             | 0401h | x     | ID27 | ID26 | ID25 | ID24 | ID23 | ID22 | ID21 | ID20 | 80  |        |               |                                          |           |                                         |                   |                                           |                             |                                          |           |                             |          |           |                             |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                              |                             | 0402h | x     | ID37 | ID36 | ID35 | ID34 | ID33 | ID32 | ID31 | ID30 | 00  |        |               |                                          |           |                                         |                   |                                           |                             |                                          |           |                             |          |           |                             |
| Description                               | The 1 <sup>st</sup> parameter (ID1): Module's manufacture ID<br>The 2 <sup>nd</sup> parameter (ID2): Module's driver version ID<br>The 3 <sup>rd</sup> parameter (ID3): Module/driver ID<br>Note: Commands RDDID (0xDAh/0xDBh/0xDC <sub>h</sub> ) read data corresponding to the parameters 1, 2, 3 of command 0x04h, respectively.                                                          |                             |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |                             |                                          |           |                             |          |           |                             |
| Restriction                               |                                                                                                                                                                                                                                                                                                                                                                                              |                             |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |                             |                                          |           |                             |          |           |                             |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>   |                             |       |       |      |      |      |      |      |      |      |      |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes       | Normal Mode On, Idle Mode On, Sleep Out | Yes               | Partial Mode On, Idle Mode Off, Sleep Out | Yes                         | Partial Mode On, Idle Mode On, Sleep Out | Yes       | Sleep In                    | Yes      |           |                             |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                 |                             |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |                             |                                          |           |                             |          |           |                             |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                          |                             |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |                             |                                          |           |                             |          |           |                             |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                          |                             |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |                             |                                          |           |                             |          |           |                             |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                          |                             |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |                             |                                          |           |                             |          |           |                             |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                          |                             |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |                             |                                          |           |                             |          |           |                             |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                          |                             |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |                             |                                          |           |                             |          |           |                             |
| Default                                   | <table><tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr><tr><th>After MTP</th><th>Before MTP</th></tr><tr><td>Power On Sequence</td><td>MTP value</td><td>ID1=00h / ID2=80h / ID3=00h</td></tr><tr><td>SW Reset</td><td>MTP value</td><td>ID1=00h / ID2=80h / ID3=00h</td></tr><tr><td>HW Reset</td><td>MTP value</td><td>ID1=00h / ID2=80h / ID3=00h</td></tr></table> |                             |       |       |      |      |      |      |      |      |      |      |     | Status | Default Value |                                          | After MTP | Before MTP                              | Power On Sequence | MTP value                                 | ID1=00h / ID2=80h / ID3=00h | SW Reset                                 | MTP value | ID1=00h / ID2=80h / ID3=00h | HW Reset | MTP value | ID1=00h / ID2=80h / ID3=00h |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                |                             |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |                             |                                          |           |                             |          |           |                             |
|                                           | After MTP                                                                                                                                                                                                                                                                                                                                                                                    | Before MTP                  |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |                             |                                          |           |                             |          |           |                             |
| Power On Sequence                         | MTP value                                                                                                                                                                                                                                                                                                                                                                                    | ID1=00h / ID2=80h / ID3=00h |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |                             |                                          |           |                             |          |           |                             |
| SW Reset                                  | MTP value                                                                                                                                                                                                                                                                                                                                                                                    | ID1=00h / ID2=80h / ID3=00h |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |                             |                                          |           |                             |          |           |                             |
| HW Reset                                  | MTP value                                                                                                                                                                                                                                                                                                                                                                                    | ID1=00h / ID2=80h / ID3=00h |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |                             |                                          |           |                             |          |           |                             |
| Flow Chart                                | <div><div><div>RDDID (04h)</div><div>Send 1st parameter ID1[7:0]</div><div>Send 2nd parameter ID2[7:0]</div><div>Send 3rd parameter ID3[7:0]</div></div><div><div>Legend</div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>                                                                             |                             |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |                             |                                          |           |                             |          |           |                             |

**RDNUMED(0500h): Read Number of Errors on DSI**

| 0500H                                     |                                                                                                                                                                                                                                                                                                                                                                                            |         |       | RDNUMED |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|---------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address |       | D15-8   | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI    | Other |         |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| RDNUMED                                   | R                                                                                                                                                                                                                                                                                                                                                                                          | 05h     | 0500h | x       | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | 00  |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Description                               | 1. D[7] is set to “1” if there is overflow of D[6..0], which store a number of the parity errors.<br>2. D[7..0] bits are set to “0”s after the read function is completed.<br>3. This command is used for MIPI DSI only. It is of no use for others interface operation.                                                                                                                   |         |       |         |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Restriction                               | -                                                                                                                                                                                                                                                                                                                                                                                          |         |       |         |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |         |       |         |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                               |         |       |         |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |         |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |         |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |         |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |         |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |         |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>SW Reset</td><td>00h</td></tr><tr><td>HW Reset</td><td>00h</td></tr></table>                                                                                                                                                                                                   |         |       |         |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | 00h | SW Reset                                | 00h | HW Reset                                  | 00h |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                              |         |       |         |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Power On Sequence                         | 00h                                                                                                                                                                                                                                                                                                                                                                                        |         |       |         |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| SW Reset                                  | 00h                                                                                                                                                                                                                                                                                                                                                                                        |         |       |         |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| HW Reset                                  | 00h                                                                                                                                                                                                                                                                                                                                                                                        |         |       |         |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Flow Chart                                | <div><div><div>RDDID (05h)</div><div>↓</div><div>Send 1st parameter</div><div>↓</div><div>P[7:0]=00h<br/>RDDSM(0Eh)' s D0 = '0'</div></div><div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div></div>                                                                             |         |       |         |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |



**RDDPM (0A00h): Read Display Power Mode**

| 0A00H                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                    | RDDPM (Read Display Power Mode) |                            |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|----------------------------|-------|--------------------------------------------------|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|-----|
| Instruction           | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                | Address                         |                            | D15-8 | D7                                               | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                    | MIPI                            | Other                      |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| RDDPM                 | R                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0Ah                             | 0A00h                      | x     | D7                                               | D6 | D5 | D4 | D3 | D2 | 0  | 0  | 08  |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Description           | This command indicates the current status of this device as described in the table below:                                                                                                                                                                                                                                                                                                                                                          |                                 |                            |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Bit                                                                                                                                                                                                                                                                                                                                                                                                                                                | Symbol                          | Description                |       | Comment                                          |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | D7                                                                                                                                                                                                                                                                                                                                                                                                                                                 | BSTON                           | Booster Voltage Status     |       | '1'=Booster on,<br>'0'=Booster off               |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | D6                                                                                                                                                                                                                                                                                                                                                                                                                                                 | IDMON                           | Idle Mode On/Off           |       | '1' = Idle Mode On,<br>'0' = Idle Mode Off       |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | D5                                                                                                                                                                                                                                                                                                                                                                                                                                                 | PTLON                           | Partial Mode On/Off        |       | '1' = Partial Mode On,<br>'0' = Partial Mode Off |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | D4                                                                                                                                                                                                                                                                                                                                                                                                                                                 | SLPON                           | Sleep In/Out               |       | '1' = Sleep Out,<br>'0' = Sleep In               |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | D3                                                                                                                                                                                                                                                                                                                                                                                                                                                 | NORON                           | Display Normal Mode On/Off |       | '1' = Normal Display,<br>'0' = Partial Display   |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | D2                                                                                                                                                                                                                                                                                                                                                                                                                                                 | DISON                           | Display On/Off             |       | '1' = Display On,<br>'0' = Display Off           |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Register Availability | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>                                                         |                                 |                            |       |                                                  |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                                                                             | Availability                    |                            |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                                           | Yes                             |                            |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                                            | Yes                             |                            |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                                          | Yes                             |                            |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                                           | Yes                             |                            |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Sleep In              | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                 |                            |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Default               | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>08h</td></tr><tr><td>SW Reset</td><td>08h</td></tr><tr><td>HW Reset</td><td>08h</td></tr></table>                                                                                                                                                                                                                                                           |                                 |                            |       |                                                  |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | 08h | SW Reset                                | 08h | HW Reset                                  | 08h |                                          |     |          |     |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                                                                             | Default Value                   |                            |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Power On Sequence                                                                                                                                                                                                                                                                                                                                                                                                                                  | 08h                             |                            |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | SW Reset                                                                                                                                                                                                                                                                                                                                                                                                                                           | 08h                             |                            |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| HW Reset              | 08h                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                 |                            |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Flow Chart            | <div><div><div>Serial I/F Mode</div><div><div>RDDPM (0Ah)</div><div>↓</div><div>Send D[7:0]</div></div></div><div><div>Parallel I/F Mode</div><div><div>RDDPM (0Ah)</div><div>↓</div><div>Dummy Read</div><div>↓</div><div>Send D[7:0]</div></div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                                 |                            |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |

# RDDMADCTR (0B00h): Read Display MADCTR

| 0B00H                 |                                                                                                                                                                                                                                                                                                                                                                                                                        | RDDMADCTR (Read Display MADCTR) |                 |                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-----------------|-----------------------------------------------------------------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|-----|
| Instruction           | R/W                                                                                                                                                                                                                                                                                                                                                                                                                    | Address                         |                 | D15-8                                                                 | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                                        | MIPI                            | Other           |                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| RDDMADCTR             | R                                                                                                                                                                                                                                                                                                                                                                                                                      | 0Bh                             | 0B00h           | x                                                                     | 0  | 0  | 0  | 0  | D3 | D2 | D1 | D0 | 00  |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Description           | This command indicates the current display status as described in the table below:                                                                                                                                                                                                                                                                                                                                     |                                 |                 |                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Bit                                                                                                                                                                                                                                                                                                                                                                                                                    | Symbol                          | Description     | Comment                                                               |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | D3                                                                                                                                                                                                                                                                                                                                                                                                                     | RGB                             | RGB/BGR Order   | '1' =BGR, "0"=RGB                                                     |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | D2                                                                                                                                                                                                                                                                                                                                                                                                                     | Reserved                        | -               | 0                                                                     |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | D1                                                                                                                                                                                                                                                                                                                                                                                                                     | RSMX                            | Horizontal Flip | '0' = Normal display(36H-D1='1')<br>'1' = Flipped display(36H-D1='1') |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | D0                                                                                                                                                                                                                                                                                                                                                                                                                     | RSMY                            | Vertical Flip   | '0' = Normal display(36H-D0='1')<br>'1' = Flipped display(36H-D0='1') |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Register Availability | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>                             |                                 |                 |                                                                       |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                                                 | Availability                    |                 |                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                                               | Yes                             |                 |                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                | Yes                             |                 |                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                                              | Yes                             |                 |                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                                               | Yes                             |                 |                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Sleep In                                                                                                                                                                                                                                                                                                                                                                                                               | Yes                             |                 |                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Default               | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>SW Reset</td><td>00h</td></tr><tr><td>HW Reset</td><td>00h</td></tr></table>                                                                                                                                                                                                                               |                                 |                 |                                                                       |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | 00h | SW Reset                                | 00h | HW Reset                                  | 00h |                                          |     |          |     |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                                                 | Default Value                   |                 |                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Power On Sequence                                                                                                                                                                                                                                                                                                                                                                                                      | 00h                             |                 |                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | SW Reset                                                                                                                                                                                                                                                                                                                                                                                                               | 00h                             |                 |                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | HW Reset                                                                                                                                                                                                                                                                                                                                                                                                               | 00h                             |                 |                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Flow Chart            | <div><div><div>Serial I/F Mode</div><div><div>RDDMADCTR (0Bh)</div><div>Send D[7:0]</div></div></div><div><div>Parallel I/F Mode</div><div><div>RDDMADCTR (0Bh)</div><div>Dummy Read</div><div>Send D[7:0]</div></div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                                 |                 |                                                                       |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |

**RDDCOLMOD (0C00h): Read Display Pixel Format**

| 0C00H                 |                                                                                                                                                                                                                                                                                                                                                                                                  | RDDCOLMOD (Read Display Pixel Format)         |                  |                          |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|------------------|--------------------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|-----|
| Instruction           | R/W                                                                                                                                                                                                                                                                                                                                                                                              | Address                                       |                  | D15-8                    | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                  | MIPI                                          | Other            |                          |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| RDDCOLMOD             | R                                                                                                                                                                                                                                                                                                                                                                                                | 0Ch                                           | 0C00h            | x                        | D7 | D6 | D5 | D4 | 0  | 0  | 0  | 0  | 70  |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Description           | This command indicates the current status of the display as described in the table below:                                                                                                                                                                                                                                                                                                        |                                               |                  |                          |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Bit                                                                                                                                                                                                                                                                                                                                                                                              | Symbol                                        | Description      | Comment                  |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | D7                                                                                                                                                                                                                                                                                                                                                                                               | VIPF[3]                                       | -                | '0'                      |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | D6                                                                                                                                                                                                                                                                                                                                                                                               | VIPF[2]                                       | DPI Pixel format | '101' = 16-bits / pixel, |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | D5                                                                                                                                                                                                                                                                                                                                                                                               | VIPF[1]                                       |                  | '110' = 18-bits / pixel, |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| D4                    | VIPF[0]                                                                                                                                                                                                                                                                                                                                                                                          | '111' = 24-bits / pixel, others are no define |                  |                          |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Register Availability | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>       |                                               |                  |                          |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                           | Availability                                  |                  |                          |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                         | Yes                                           |                  |                          |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                          | Yes                                           |                  |                          |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                        | Yes                                           |                  |                          |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                         | Yes                                           |                  |                          |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Sleep In              | Yes                                                                                                                                                                                                                                                                                                                                                                                              |                                               |                  |                          |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Default               | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>70h</td></tr><tr><td>SW Reset</td><td>70h</td></tr><tr><td>HW Reset</td><td>70h</td></tr></table>                                                                                                                                                                                                         |                                               |                  |                          |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | 70h | SW Reset                                | 70h | HW Reset                                  | 70h |                                          |     |          |     |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                           | Default Value                                 |                  |                          |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Power On Sequence                                                                                                                                                                                                                                                                                                                                                                                | 70h                                           |                  |                          |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | SW Reset                                                                                                                                                                                                                                                                                                                                                                                         | 70h                                           |                  |                          |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| HW Reset              | 70h                                                                                                                                                                                                                                                                                                                                                                                              |                                               |                  |                          |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Flow Chart            | <div><div><div>Serial I/F Mode</div><div>RDDCOLMOD (0Ch)</div><div>Send D[7:0]</div></div><div><div>Parallel I/F Mode</div><div>RDDCOLMOD (0Ch)</div><div>Dummy Read</div><div>Send D[7:0]</div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                                               |                  |                          |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |

**RDDIM (0D00h): Read Display Image Mode**

| 0D00H                 |                                                                                                                                                                                                                                                                                                                                                                                            | RDDIM (Read Display Image Mode) |                  |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|------------------|-------|--------------------------------------------------|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|-----|
| Instruction           | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address                         |                  | D15-8 | D7                                               | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI                            | Other            |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| RDDIM                 | R                                                                                                                                                                                                                                                                                                                                                                                          | 0Dh                             | 0D00h            | x     | D7                                               | D6 | D5 | D4 | D3 | 0  | 0  | 0  | 00  |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Description           | The display module returns the display image mode status.                                                                                                                                                                                                                                                                                                                                  |                                 |                  |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Bit                                                                                                                                                                                                                                                                                                                                                                                        | Symbol                          | Description      |       | Comment                                          |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | D7                                                                                                                                                                                                                                                                                                                                                                                         | Reserved                        |                  |       | '0'                                              |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | D6                                                                                                                                                                                                                                                                                                                                                                                         | Reserved                        |                  |       | '0'                                              |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | D5                                                                                                                                                                                                                                                                                                                                                                                         | INVON                           | Inversion On/Off |       | "1" = Inversion is On,<br>"0" = Inversion is Off |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | D4                                                                                                                                                                                                                                                                                                                                                                                         | ALLON                           | All Pixel On     |       | '0' = Normal display<br>'1' = White display      |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | D3                                                                                                                                                                                                                                                                                                                                                                                         | ALLOFF                          | All Pixel Off    |       | '0' = Normal display<br>'1' = Black display      |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Register Availability | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |                                 |                  |       |                                                  |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                     | Availability                    |                  |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                   | Yes                             |                  |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                    | Yes                             |                  |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                  | Yes                             |                  |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                   | Yes                             |                  |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Sleep In                                                                                                                                                                                                                                                                                                                                                                                   | Yes                             |                  |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Default               | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>SW Reset</td><td>00h</td></tr><tr><td>HW Reset</td><td>00h</td></tr></table>                                                                                                                                                                                                   |                                 |                  |       |                                                  |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | 00h | SW Reset                                | 00h | HW Reset                                  | 00h |                                          |     |          |     |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                     | Default Value                   |                  |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Power On Sequence                                                                                                                                                                                                                                                                                                                                                                          | 00h                             |                  |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | SW Reset                                                                                                                                                                                                                                                                                                                                                                                   | 00h                             |                  |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | HW Reset                                                                                                                                                                                                                                                                                                                                                                                   | 00h                             |                  |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Flow Chart            | <div><div><div>Serial I/F Mode</div><div><div>RDDIM (0Dh)</div><div>Send D[7:0]</div></div></div><div><div>Parallel I/F Mode</div><div><div>RDDIM (0Dh)</div><div>Dummy Read</div><div>Send D[7:0]</div></div></div><div>Host Driver</div></div>                                                                                                                                           |                                 |                  |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>                                                                                                                                                                                                                              |                                 |                  |       |                                                  |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |

**RDDSM (0E00h): Read Display Signal Mode**

| 0E00H                 |                                                     |              |       | RDDSM (Read Display Signal Mode) |    |    |    |                             |    |    |    |    |     |
|-----------------------|-----------------------------------------------------|--------------|-------|----------------------------------|----|----|----|-----------------------------|----|----|----|----|-----|
| Instruction           | R/W                                                 | Address      |       | D15-8                            | D7 | D6 | D5 | D4                          | D3 | D2 | D1 | D0 | HEX |
|                       |                                                     | MIPI         | Other |                                  |    |    |    |                             |    |    |    |    |     |
| RDDSM                 | R                                                   | 0Eh          | 0E00h | x                                | D7 | D6 | 0  | 0                           | 0  | 0  | 0  | D0 | 00  |
| Description           | The display module returns the Display Signal Mode. |              |       |                                  |    |    |    |                             |    |    |    |    |     |
|                       | Bit                                                 | Symbol       |       | Description                      |    |    |    | Comment                     |    |    |    |    |     |
|                       | D7                                                  | TEON         |       | Tearing Effect Line On/Off       |    |    |    | “1” = On, “0” = Off         |    |    |    |    |     |
|                       | D6                                                  | TELOM        |       | Tearing effect line mode         |    |    |    | “0” = mode1, “1” = mode2    |    |    |    |    |     |
|                       | D0                                                  | Error on DSI |       | Error on DSI                     |    |    |    | ‘0’ = No Error, ‘1’ = Error |    |    |    |    |     |
| Register Availability |                                                     |              |       |                                  |    |    |    |                             |    |    |    |    |     |
|                       |                                                     |              |       |                                  |    |    |    |                             |    |    |    |    |     |
|                       |                                                     |              |       |                                  |    |    |    |                             |    |    |    |    |     |
|                       |                                                     |              |       |                                  |    |    |    |                             |    |    |    |    |     |
|                       |                                                     |              |       |                                  |    |    |    |                             |    |    |    |    |     |
|                       |                                                     |              |       |                                  |    |    |    |                             |    |    |    |    |     |
|                       |                                                     |              |       |                                  |    |    |    |                             |    |    |    |    |     |
| Default               |                                                     |              |       |                                  |    |    |    |                             |    |    |    |    |     |
|                       |                                                     |              |       |                                  |    |    |    |                             |    |    |    |    |     |
|                       |                                                     |              |       |                                  |    |    |    |                             |    |    |    |    |     |
|                       |                                                     |              |       |                                  |    |    |    |                             |    |    |    |    |     |
|                       |                                                     |              |       |                                  |    |    |    |                             |    |    |    |    |     |
| Flow Chart            |                                                     |              |       |                                  |    |    |    |                             |    |    |    |    |     |
|                       |                                                     |              |       |                                  |    |    |    |                             |    |    |    |    |     |
|                       |                                                     |              |       |                                  |    |    |    |                             |    |    |    |    |     |
|                       |                                                     |              |       |                                  |    |    |    |                             |    |    |    |    |     |
|                       |                                                     |              |       |                                  |    |    |    |                             |    |    |    |    |     |

**RDDSDR (0F00h): Read Display Self-Diagnostic Result**

| 0F00H                 |                                                                                                                                                                                                                                                                                                                                                                                                                          |               |                      | RDDSDR (Read Display Self-Diagnostic Result) |    |    |    |                            |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|----------------------|----------------------------------------------|----|----|----|----------------------------|----|----|----|----|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|-----|
| Instruction           | R/W                                                                                                                                                                                                                                                                                                                                                                                                                      | Address       |                      | D15-8                                        | D7 | D6 | D5 | D4                         | D3 | D2 | D1 | D0 | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                                          | MIPI          | Other                |                                              |    |    |    |                            |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| RDDSDR                | R                                                                                                                                                                                                                                                                                                                                                                                                                        | 0Fh           | 0F00h                | x                                            | 0  | 0  | 0  | 0                          | 0  | 0  | 0  | D0 | 00  |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Description           | The display module returns the self-diagnostic results following a Sleep Out command.                                                                                                                                                                                                                                                                                                                                    |               |                      |                                              |    |    |    |                            |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Bit                                                                                                                                                                                                                                                                                                                                                                                                                      | Symbol        | Description          |                                              |    |    |    | Comment                    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | D0                                                                                                                                                                                                                                                                                                                                                                                                                       | CSC           | Checksums Comparison |                                              |    |    |    | '0' = Same, '1' = Not Same |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Register Availability | <table><thead><tr><th>Status</th><th>Availability</th></tr></thead><tbody><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></tbody></table> |               |                      |                                              |    |    |    |                            |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                                                   | Availability  |                      |                                              |    |    |    |                            |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                 | Yes           |                      |                                              |    |    |    |                            |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                  | Yes           |                      |                                              |    |    |    |                            |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                | Yes           |                      |                                              |    |    |    |                            |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                 | Yes           |                      |                                              |    |    |    |                            |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Sleep In              | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |               |                      |                                              |    |    |    |                            |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Default               | <table><thead><tr><th>Status</th><th>Default Value</th></tr></thead><tbody><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>SW Reset</td><td>00h</td></tr><tr><td>HW Reset</td><td>00h</td></tr></tbody></table>                                                                                                                                                                                                   |               |                      |                                              |    |    |    |                            |    |    |    |    |     | Status | Default Value | Power On Sequence                        | 00h | SW Reset                                | 00h | HW Reset                                  | 00h |                                          |     |          |     |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                                                   | Default Value |                      |                                              |    |    |    |                            |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Power On Sequence                                                                                                                                                                                                                                                                                                                                                                                                        | 00h           |                      |                                              |    |    |    |                            |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | SW Reset                                                                                                                                                                                                                                                                                                                                                                                                                 | 00h           |                      |                                              |    |    |    |                            |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| HW Reset              | 00h                                                                                                                                                                                                                                                                                                                                                                                                                      |               |                      |                                              |    |    |    |                            |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Flow Chart            | <div><div><div>Serial I/F Mode</div><div>RDDSDR (0Fh)</div><div>Send D[7:0]</div></div><div><div>Parallel I/F Mode</div><div>RDDSTR (0Fh)</div><div>Dummy Read</div><div>Send D[7:0]</div></div><div>Host Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>                               |               |                      |                                              |    |    |    |                            |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |

**SLPIN (1000h): Sleep In**

| 1000H                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                     | SLPIN (Sleep In) |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------|-------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|---------------|-----------------------------------------|---------------|-------------------------------------------|---------------|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                 | Address          |       | D15-8       | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                     | MIPI             | Other |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| SLPIN                                     | W                                                                                                                                                                                                                                                                                                                                                                                                                                   | 10h              | 1000h | No Argument |    |    |    |    |    |    |    |    | -   |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Description                               | This command causes the display module to enter sleep-in mode. Under this mode the DC/DC converter is stopped, internal display oscillator is stopped, and panel scanning is stopped, either. The control interface and related registers are still working and keeps its values.                                                                                                                                                   |                  |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Restriction                               | 1. This command has no effect when the display module is already in Sleep mode. Sleep Mode can only be exit by the Sleep Out Command (11h).<br>2. It must wait 5msec before sending next command for the supply voltages and clock circuits to stabilize.<br>3. And, it must wait 120msec after sending Sleep Out command (when in Sleep Mode) for the next Sleep In command.                                                       |                  |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>                                          |                  |       |             |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes           | Normal Mode On, Idle Mode On, Sleep Out | Yes           | Partial Mode On, Idle Mode Off, Sleep Out | Yes           | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                                        |                  |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                 |                  |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                 |                  |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                 |                  |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                 |                  |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                 |                  |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Sleep In Mode</td></tr><tr><td>SW Reset</td><td>Sleep In Mode</td></tr><tr><td>HW Reset</td><td>Sleep In Mode</td></tr></table>                                                                                                                                                                                                              |                  |       |             |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | Sleep In Mode | SW Reset                                | Sleep In Mode | HW Reset                                  | Sleep In Mode |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                                       |                  |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Power On Sequence                         | Sleep In Mode                                                                                                                                                                                                                                                                                                                                                                                                                       |                  |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| SW Reset                                  | Sleep In Mode                                                                                                                                                                                                                                                                                                                                                                                                                       |                  |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| HW Reset                                  | Sleep In Mode                                                                                                                                                                                                                                                                                                                                                                                                                       |                  |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Flow Chart                                | <div><div><div>SLPIN (10h)</div><div>Display whole blank screen (Automatic No effect to DISP ON/OFF Command)</div><div>Drain charge from panel</div></div><div><div>Stop DC/DC Converter</div><div>Stop Internal Oscillator</div><div>Sleep In Mode</div></div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div> |                  |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |

**SLPOUT (1100h): Sleep Out**

| 1100H                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |         |       | SLPOUT (Sleep Out) |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|--------------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|---------------|-----------------------------------------|---------------|-------------------------------------------|---------------|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Address |       | D15-8              | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | MIPI    | Other |                    |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| SLPOUT                                    | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 11h     | 1100h | No Argument        |    |    |    |    |    |    |    |    | -   |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Description                               | This command causes the display module to exit Sleep-in mode. All blocks inside display module are enabled. The host processor sends PCLK, HS and VS information to display modules two frames before this command is sent when the display module is in Normal Mode.                                                                                                                                                                                                                                                                                                                                                                                                                            |         |       |                    |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Restriction                               | <div><div>1.</div><div>This command shall not cause any visible effect on the display device when the display module is not in Sleep mode. The host processor must wait five milliseconds after sending this command before sending another command. This delay allows the supply voltages and clock circuits to stabilize.</div><div>2.</div><div>After sending a Sleep-Out command, the host processor must wait 120 ms before sending a Sleep-In command.</div><div>3.</div><div>The display module loads the display module's default values to the registers when exiting the Sleep mode. The display module runs the self-diagnostic functions after this command is received.</div></div> |         |       |                    |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Register Availability                     | <table><thead><tr><th>Status</th><th>Availability</th></tr></thead><tbody><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></tbody></table>                                                                                                                                                                                                                                                                         |         |       |                    |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes           | Normal Mode On, Idle Mode On, Sleep Out | Yes           | Partial Mode On, Idle Mode Off, Sleep Out | Yes           | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |       |                    |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |       |                    |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |       |                    |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |       |                    |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |       |                    |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |       |                    |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Default                                   | <table><thead><tr><th>Status</th><th>Default Value</th></tr></thead><tbody><tr><td>Power On Sequence</td><td>Sleep In Mode</td></tr><tr><td>SW Reset</td><td>Sleep In Mode</td></tr><tr><td>HW Reset</td><td>Sleep In Mode</td></tr></tbody></table>                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |       |                    |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | Sleep In Mode | SW Reset                                | Sleep In Mode | HW Reset                                  | Sleep In Mode |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |       |                    |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Power On Sequence                         | Sleep In Mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |       |                    |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| SW Reset                                  | Sleep In Mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |       |                    |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| HW Reset                                  | Sleep In Mode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |       |                    |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Flow chart                                | <div><div>SLPOUT (11h)</div><div>Start Internal Oscillator</div><div>Start DC-DC Converter</div><div>Charge Offset voltage for LCD Panel</div><div>Display whole blank screen for 2 frames (Automatic No effect to DISP ON/OFF)</div><div>Display Image contents in accordance with the current command table settings</div><div>Sleep Out</div></div> <div><div>Legend</div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div>                                                                                                                                                                                        |         |       |                    |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |



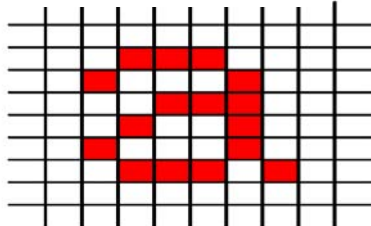
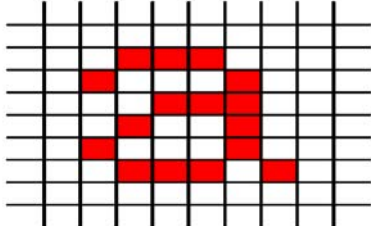
**PTLON (1200h): Partial Display Mode On**

| 1200H                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |       | PTLON (Partial Display Mode On) |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|---------------------------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|------------------------|-----------------------------------------|------------------------|-------------------------------------------|------------------------|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                   | Address |       | D15-8                           | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                       | MIPI    | Other |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| PTLON                                     | W                                                                                                                                                                                                                                                                                                                                                                                                                                     | 12h     | 1200h | No Argument                     |    |    |    |    |    |    |    |    | -   |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Description                               | This command causes the display module to enter the Partial Display Mode. The Partial Display Mode window is described by the Partial Area (30h) command. To leave Partial Display Mode, the Normal Display Mode On (13h) command should be written. The host processor continues to send PCLK, HS and VS information to display modules for two frames after this command is sent when the display module is in Normal Display Mode. |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Restriction                               | This command has no effect when Partial Display Mode is already active.                                                                                                                                                                                                                                                                                                                                                               |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>                                            |         |       |                                 |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes                    | Normal Mode On, Idle Mode On, Sleep Out | Yes                    | Partial Mode On, Idle Mode Off, Sleep Out | Yes                    | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                                          |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Normal display mode On</td></tr><tr><td>SW Reset</td><td>Normal display mode On</td></tr><tr><td>HW Reset</td><td>Normal display mode On</td></tr></table>                                                                                                                                                                                     |         |       |                                 |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | Normal display mode On | SW Reset                                | Normal display mode On | HW Reset                                  | Normal display mode On |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                                         |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Power On Sequence                         | Normal display mode On                                                                                                                                                                                                                                                                                                                                                                                                                |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| SW Reset                                  | Normal display mode On                                                                                                                                                                                                                                                                                                                                                                                                                |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| HW Reset                                  | Normal display mode On                                                                                                                                                                                                                                                                                                                                                                                                                |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Flow Chart                                | Refer to Partial Area (30h)                                                                                                                                                                                                                                                                                                                                                                                                           |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |

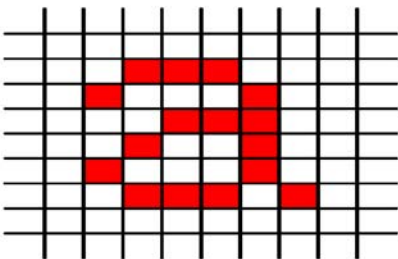
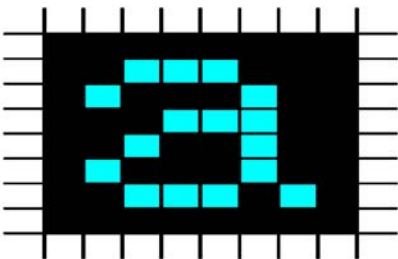
**NORON (1300h): Normal Display Mode On**

| 1300H                                     |                                                                                                                                                                                                                                                                                                                                                                                            | NORON (Normal Display Mode On) |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------|-------|-------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|------------------------|-----------------------------------------|------------------------|-------------------------------------------|------------------------|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address                        |       | D15-8       | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI                           | Other |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| NORON                                     | W                                                                                                                                                                                                                                                                                                                                                                                          | 13h                            | 1300h | No Argument |    |    |    |    |    |    |    |    | -   |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Description                               | This command causes the display module to enter the Normal mode. The host processor sends PCLK, HS and VS information to Type 2 display modules two frames before this command is sent if the display module is in Partial Display Mode.                                                                                                                                                   |                                |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Restriction                               | This command has no effect when Normal Display mode is already active.                                                                                                                                                                                                                                                                                                                     |                                |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |                                |       |             |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes                    | Normal Mode On, Idle Mode On, Sleep Out | Yes                    | Partial Mode On, Idle Mode Off, Sleep Out | Yes                    | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                               |                                |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Normal Display Mode On</td></tr><tr><td>SW Reset</td><td>Normal Display Mode On</td></tr><tr><td>HW Reset</td><td>Normal Display Mode On</td></tr></table>                                                                                                                                          |                                |       |             |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | Normal Display Mode On | SW Reset                                | Normal Display Mode On | HW Reset                                  | Normal Display Mode On |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                              |                                |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Power On Sequence                         | Normal Display Mode On                                                                                                                                                                                                                                                                                                                                                                     |                                |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| SW Reset                                  | Normal Display Mode On                                                                                                                                                                                                                                                                                                                                                                     |                                |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| HW Reset                                  | Normal Display Mode On                                                                                                                                                                                                                                                                                                                                                                     |                                |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |
| Flow Chart                                | Refer to the description of Partial Area (3000h)                                                                                                                                                                                                                                                                                                                                           |                                |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                        |                                         |                        |                                           |                        |                                          |     |          |     |

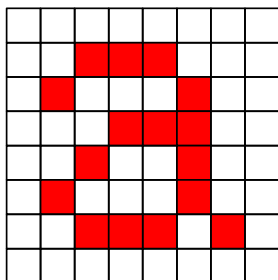
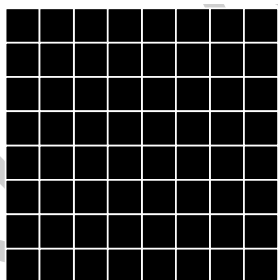
**INVOFF (2000H): Display Inversion Off**

| 2000H                                     |                                                                                                                                                                                                                                                                                                                                                                                                                |         |       | INVOFF (Display Inversion Off) |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|--------------------------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-----------------------|-----------------------------------------|-----------------------|-------------------------------------------|-----------------------|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                                            | Address |       | D15-8                          | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                | MIPI    | Other |                                |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| INVOFF                                    | W                                                                                                                                                                                                                                                                                                                                                                                                              | 20h     | 2000h | No Argument                    |    |    |    |    |    |    |    |    | -   |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Description                               | <div><div>This command causes the display module to stop inverting the image data on the display device. No status bits are changed.</div><div><div><div>Input Image</div></div><div>→</div><div><div>Display Panel</div></div></div></div> |         |       |                                |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Restriction                               | This command has no effect when the display module is not inverting the display image.                                                                                                                                                                                                                                                                                                                         |         |       |                                |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>                     |         |       |                                |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes                   | Normal Mode On, Idle Mode On, Sleep Out | Yes                   | Partial Mode On, Idle Mode Off, Sleep Out | Yes                   | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                   |         |       |                                |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                            |         |       |                                |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                            |         |       |                                |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                            |         |       |                                |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                            |         |       |                                |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                                            |         |       |                                |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Display Inversion off</td></tr><tr><td>SW Reset</td><td>Display Inversion off</td></tr><tr><td>HW Reset</td><td>Display Inversion off</td></tr></table>                                                                                                                                                                 |         |       |                                |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | Display Inversion off | SW Reset                                | Display Inversion off | HW Reset                                  | Display Inversion off |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                  |         |       |                                |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Power On Sequence                         | Display Inversion off                                                                                                                                                                                                                                                                                                                                                                                          |         |       |                                |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| SW Reset                                  | Display Inversion off                                                                                                                                                                                                                                                                                                                                                                                          |         |       |                                |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| HW Reset                                  | Display Inversion off                                                                                                                                                                                                                                                                                                                                                                                          |         |       |                                |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Flow Chart                                | <div><div><div>Display Inversion On Mode</div><div>↓</div><div>INVOFF (20h)</div><div>↓</div><div>Display Inversion OFF Mode</div></div><div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div></div>                                                                                                    |         |       |                                |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |

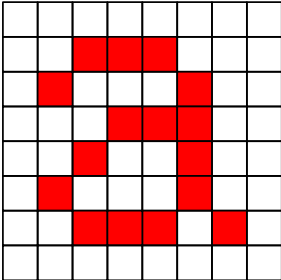
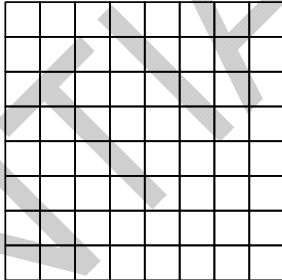
**INVON (2100H): Display Inversion On**

| 2100H                                     |                                                                                                                                                                                                                                                                                                                                                                                            | INVON (Display Inversion On) |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------|-------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-----------------------|-----------------------------------------|-----------------------|-------------------------------------------|-----------------------|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address                      |       | D15-8       | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI                         | Other |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| INVON                                     | W                                                                                                                                                                                                                                                                                                                                                                                          | 21h                          | 2100h | No Argument |    |    |    |    |    |    |    |    | -   |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Description                               | This command causes the display module to invert the image data only on the display device. No status bits are changed.                                                                                                                                                                                                                                                                    |                              |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
|                                           | <div><div><div>Input Image</div></div><div>→</div><div><div>Display Panel</div></div></div>                                                                                                                            |                              |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Restriction                               | This command has no effect when module is already in inversion on mode.                                                                                                                                                                                                                                                                                                                    |                              |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |                              |       |             |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes                   | Normal Mode On, Idle Mode On, Sleep Out | Yes                   | Partial Mode On, Idle Mode Off, Sleep Out | Yes                   | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                               |                              |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                              |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                              |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                              |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                              |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                              |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Display Inversion off</td></tr><tr><td>SW Reset</td><td>Display Inversion off</td></tr><tr><td>HW Reset</td><td>Display Inversion off</td></tr></table>                                                                                                                                             |                              |       |             |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | Display Inversion off | SW Reset                                | Display Inversion off | HW Reset                                  | Display Inversion off |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                              |                              |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Power On Sequence                         | Display Inversion off                                                                                                                                                                                                                                                                                                                                                                      |                              |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| SW Reset                                  | Display Inversion off                                                                                                                                                                                                                                                                                                                                                                      |                              |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| HW Reset                                  | Display Inversion off                                                                                                                                                                                                                                                                                                                                                                      |                              |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Flow Chart                                | <div><div><div>Display Inversion OFF Mode</div><div>↓</div><div>INVON (21h)</div><div>↓</div><div>Display Inversion ON Mode</div></div><div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div></div>                                                                                 |                              |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |

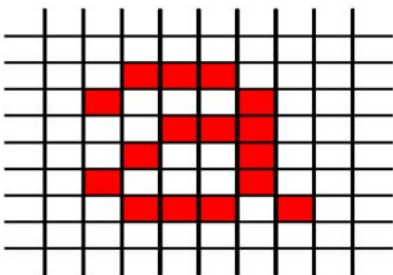
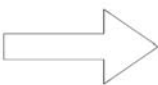
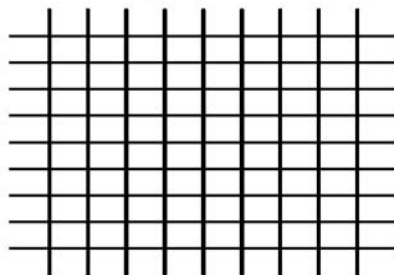
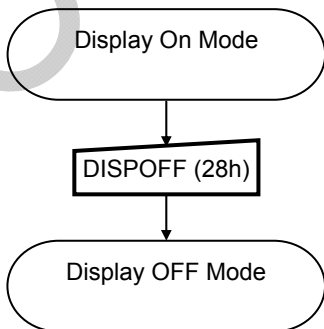
**ALLPOFF (2200H): All Pixel Off**

| 2200H                                     |                                                                                                                                                                                                                                                                                                                                                                                                             |         |       | ALLPOFF     |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|-------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-----------------------|-----------------------------------------|-----------------------|-------------------------------------------|-----------------------|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                                         | Address |       | D15-8       | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                                             | MIPI    | Other |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| ALLPOFF                                   | W                                                                                                                                                                                                                                                                                                                                                                                                           | 22h     | 2200h | No Argument |    |    |    |    |    |    |    |    | -   |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Description                               | <p>This command turns the display panel black in Sleep Out mode and a status of the Display On/Off register can be on or off. This command does not change any other status. “All Pixels On”, “Normal Display Mode On” or “Partial Mode On” commands are used to leave this mode. The display panel is showing the content of the Input Image after “Normal Display On” and “Partial Mode On” commands.</p> |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
|                                           | <div><div><p>Input Image</p></div><div><p>Display Panel</p></div></div>                                                                                                                                                                  |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Restriction                               | -                                                                                                                                                                                                                                                                                                                                                                                                           |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>                  |         |       |             |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes                   | Normal Mode On, Idle Mode On, Sleep Out | Yes                   | Partial Mode On, Idle Mode Off, Sleep Out | Yes                   | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                         |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                         |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                         |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                         |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                                         |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Display Inversion off</td></tr><tr><td>SW Reset</td><td>Display Inversion off</td></tr><tr><td>HW Reset</td><td>Display Inversion off</td></tr></table>                                                                                                                                                              |         |       |             |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | Display Inversion off | SW Reset                                | Display Inversion off | HW Reset                                  | Display Inversion off |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                               |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Power On Sequence                         | Display Inversion off                                                                                                                                                                                                                                                                                                                                                                                       |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| SW Reset                                  | Display Inversion off                                                                                                                                                                                                                                                                                                                                                                                       |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| HW Reset                                  | Display Inversion off                                                                                                                                                                                                                                                                                                                                                                                       |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Flow Chart                                | <div><div><p>Normal Display ON Mode</p><p>ALLPOFF (22h)</p><p>Black Display</p></div><div><p>Legend</p><ul style="list-style-type: none"><li>Command</li><li>Parameter</li><li>Display</li><li>Action</li><li>Mode</li><li>Sequential transfer</li></ul></div></div>                                                                                                                                        |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |

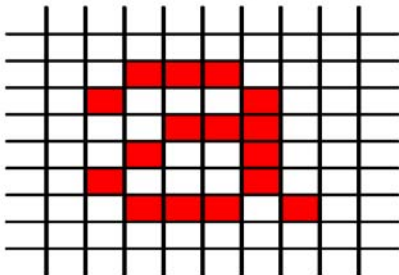
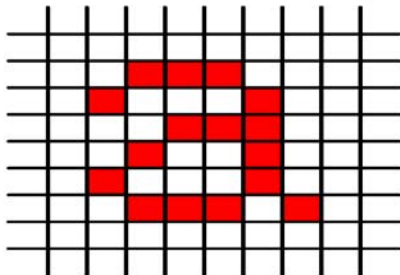
**ALLPON (2300H): All Pixel On**

| 2300H                                     |                                                                                                                                                                                                                                                                                                                                                                                                              |         |       | ALLPON      |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|-------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-----------------------|-----------------------------------------|-----------------------|-------------------------------------------|-----------------------|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                                          | Address |       | D15-8       | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                                              | MIPI    | Other |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| ALLPON                                    | W                                                                                                                                                                                                                                                                                                                                                                                                            | 23h     | 2300h | No Argument |    |    |    |    |    |    |    |    | -   |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Description                               | <p>This command turns the display panel white in Sleep Out mode and a status of the Display On/Off register can be on or off. This command does not change any other status. “All Pixels Off”, “Normal Display Mode On” or “Partial Mode On” commands are used to leave this mode. The display panel is showing the content of the Input Image after “Normal Display On” and “Partial Mode On” commands.</p> |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
|                                           | <div><div><p>Input Image</p></div><div><p>Display Panel</p></div></div>                                                                                                                                                                   |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Restriction                               | -                                                                                                                                                                                                                                                                                                                                                                                                            |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>                   |         |       |             |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes                   | Normal Mode On, Idle Mode On, Sleep Out | Yes                   | Partial Mode On, Idle Mode Off, Sleep Out | Yes                   | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                 |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                          |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                          |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                          |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                          |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                                          |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Display Inversion off</td></tr><tr><td>SW Reset</td><td>Display Inversion off</td></tr><tr><td>HW Reset</td><td>Display Inversion off</td></tr></table>                                                                                                                                                               |         |       |             |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | Display Inversion off | SW Reset                                | Display Inversion off | HW Reset                                  | Display Inversion off |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Power On Sequence                         | Display Inversion off                                                                                                                                                                                                                                                                                                                                                                                        |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| SW Reset                                  | Display Inversion off                                                                                                                                                                                                                                                                                                                                                                                        |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| HW Reset                                  | Display Inversion off                                                                                                                                                                                                                                                                                                                                                                                        |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |
| Flow Chart                                | <div><div><p>Normal Display ON Mode</p><p>↓</p><p>ALLPON (23h)</p><p>↓</p><p>White Display</p></div><div><p>Legend</p><ul style="list-style-type: none"><li>Command</li><li>Parameter</li><li>Display</li><li>Action</li><li>Mode</li><li>Sequential transfer</li></ul></div></div>                                                                                                                          |         |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |                       |                                         |                       |                                           |                       |                                          |     |          |     |

**DISPOFF (2800h): Display Off**

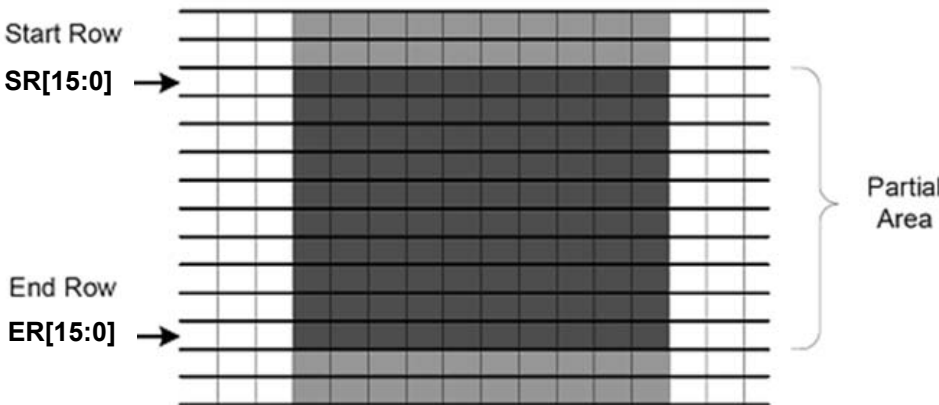
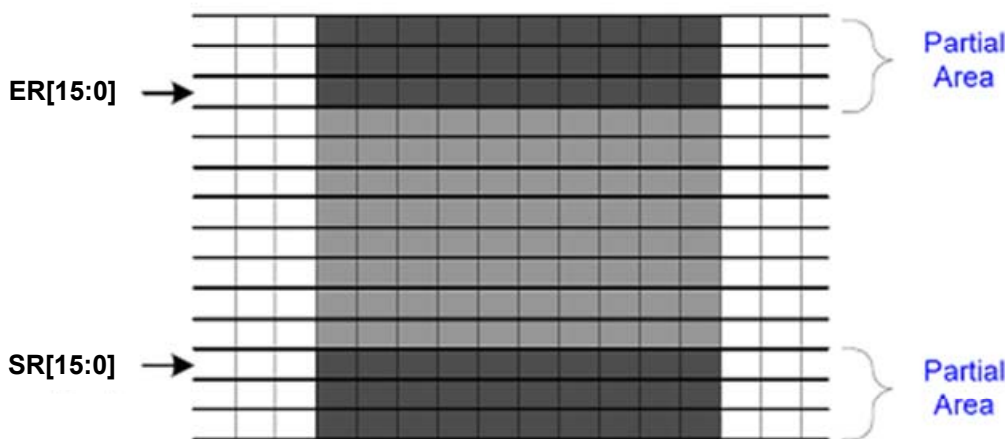
| 2800H                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |       | DISPOFF (Display Off) |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|-----------------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-------------|-----------------------------------------|-------------|-------------------------------------------|-------------|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Address |       | D15-8                 | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | MIPI    | Other |                       |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| DISPOFF                                   | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 28h     | 2800h | No Argument           |    |    |    |    |    |    |    |    | -   |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Description                               | <p>This command causes the display module to stop displaying the image data on the display device. No status bits are changed.</p> <div><div><p>Input Image</p></div><div></div><div><p>Display Panel</p></div></div> |         |       |                       |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Restriction                               | This command has no effect when module is already in display off mode.                                                                                                                                                                                                                                                                                                                                                                                                     |         |       |                       |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>                                                                                 |         |       |                       |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes         | Normal Mode On, Idle Mode On, Sleep Out | Yes         | Partial Mode On, Idle Mode Off, Sleep Out | Yes         | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                                                                               |         |       |                       |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |         |       |                       |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |         |       |                       |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |         |       |                       |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |         |       |                       |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |         |       |                       |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Display Off</td></tr><tr><td>SW Reset</td><td>Display Off</td></tr><tr><td>HW Reset</td><td>Display Off</td></tr></table>                                                                                                                                                                                                                                                           |         |       |                       |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | Display Off | SW Reset                                | Display Off | HW Reset                                  | Display Off |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                                                                              |         |       |                       |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Power On Sequence                         | Display Off                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |       |                       |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| SW Reset                                  | Display Off                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |       |                       |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| HW Reset                                  | Display Off                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |       |                       |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Flow Chart                                | <div><div><pre>graph TD; A([Display On Mode]) --&gt; B[DISPOFF (28h)]; B --&gt; C([Display OFF Mode]);</pre></div><div><p>Legend</p><ul style="list-style-type: none"><li>Command</li><li>Parameter</li><li>Display</li><li>Action</li><li>Mode</li><li>Sequential transfer</li></ul></div></div>                                                                                       |         |       |                       |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |

**DISPON (2900h): Display On**

| 2900H                                     |                                                                                                                                                                                                                                                                                                                                                                                            |         |       | DISPON (Display On) |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|---------------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-------------|-----------------------------------------|-------------|-------------------------------------------|-------------|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address |       | D15-8               | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI    | Other |                     |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| DISPON                                    | W                                                                                                                                                                                                                                                                                                                                                                                          | 29h     | 2900h | No Argument         |    |    |    |    |    |    |    |    | -   |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Description                               | <p>This command causes the display module to start displaying the image data on the display device. No status bits are changed.</p> <div><div><p>Input Image</p></div><div><p>Display Panel</p></div></div>            |         |       |                     |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Restriction                               | This command has no effect when module is already in display on mode.                                                                                                                                                                                                                                                                                                                      |         |       |                     |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |         |       |                     |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes         | Normal Mode On, Idle Mode On, Sleep Out | Yes         | Partial Mode On, Idle Mode Off, Sleep Out | Yes         | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                               |         |       |                     |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |                     |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |                     |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |                     |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |                     |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |                     |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Display Off</td></tr><tr><td>SW Reset</td><td>Display Off</td></tr><tr><td>HW Reset</td><td>Display Off</td></tr></table>                                                                                                                                                                           |         |       |                     |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | Display Off | SW Reset                                | Display Off | HW Reset                                  | Display Off |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                              |         |       |                     |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Power On Sequence                         | Display Off                                                                                                                                                                                                                                                                                                                                                                                |         |       |                     |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| SW Reset                                  | Display Off                                                                                                                                                                                                                                                                                                                                                                                |         |       |                     |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| HW Reset                                  | Display Off                                                                                                                                                                                                                                                                                                                                                                                |         |       |                     |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |
| Flow Chart                                | <div><div><p>Display OFF Mode</p><p>↓</p><p>DISPON (29h)</p><p>↓</p><p>Display ON Mode</p></div><div><p>Legend</p><ul style="list-style-type: none"><li>Command</li><li>Parameter</li><li>Display</li><li>Action</li><li>Mode</li><li>Sequential transfer</li></ul></div></div>                                                                                                            |         |       |                     |    |    |    |    |    |    |    |    |     |        |               |                                          |             |                                         |             |                                           |             |                                          |     |          |     |



**PTLAR (3000h): Partial Area**

| 3000H       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                             |       | PTLAR (Partial Area) |      |      |      |      |      |      |     |     |     |  |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-------|----------------------|------|------|------|------|------|------|-----|-----|-----|--|
| Instruction | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Address                                                                     |       | D15-8                | D7   | D6   | D5   | D4   | D3   | D2   | D1  | D0  | HEX |  |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | MIPI                                                                        | Other |                      |      |      |      |      |      |      |     |     |     |  |
| PTLAR       | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 30h                                                                         | 3000h | x                    | SR15 | SR14 | SR13 | SR12 | SR11 | SR10 | SR9 | SR8 | 00  |  |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                             | 3001h | x                    | SR7  | SR6  | SR5  | SR4  | SR3  | SR2  | SR1 | SR0 | 00  |  |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                             | 3002h | x                    | ER15 | ER14 | ER13 | ER12 | ER11 | ER10 | ER9 | ER8 | 04  |  |
|             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                             | 3003h | x                    | ER7  | ER6  | ER5  | ER4  | ER3  | ER2  | ER1 | ER0 | FF  |  |
| Description | <p>This command defines the Partial Display mode's display area. There are two parameters associated with this command, the first defines the Start Row (SR) and the second the End Row (ER), as illustrated in the following figure.</p> <p>If End Row &gt; Start Row</p>  <p>If End Row &lt; Start Row</p>  <p>If End Row = Start Row then the Partial Area will be one row deep.</p> |                                                                             |       |                      |      |      |      |      |      |      |     |     |     |  |
|             | Restriction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | SR[15:0] and ER[15:0] settings should be within max available Display Area. |       |                      |      |      |      |      |      |      |     |     |     |  |

| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Status  | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes     | Normal Mode On, Idle Mode On, Sleep Out | Yes               | Partial Mode On, Idle Mode Off, Sleep Out | Yes   | Partial Mode On, Idle Mode On, Sleep Out | Yes   | Sleep In | Yes      |       |       |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|---------------|------------------------------------------|---------|-----------------------------------------|-------------------|-------------------------------------------|-------|------------------------------------------|-------|----------|----------|-------|-------|
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |               |                                          |         |                                         |                   |                                           |       |                                          |       |          |          |       |       |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |               |                                          |         |                                         |                   |                                           |       |                                          |       |          |          |       |       |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |               |                                          |         |                                         |                   |                                           |       |                                          |       |          |          |       |       |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |               |                                          |         |                                         |                   |                                           |       |                                          |       |          |          |       |       |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |               |                                          |         |                                         |                   |                                           |       |                                          |       |          |          |       |       |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |               |                                          |         |                                         |                   |                                           |       |                                          |       |          |          |       |       |
| Default                                   | <table><tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr><tr><th>SR[8:0]</th><th>ER[8:0]</th></tr><tr><td>Power On Sequence</td><td>0000h</td><td>04ffh</td></tr><tr><td>SW Reset</td><td>0000h</td><td>04ffh</td></tr><tr><td>HW Reset</td><td>0000h</td><td>04ffh</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Status  | Default Value |                                          | SR[8:0] | ER[8:0]                                 | Power On Sequence | 0000h                                     | 04ffh | SW Reset                                 | 0000h | 04ffh    | HW Reset | 0000h | 04ffh |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |               |                                          |         |                                         |                   |                                           |       |                                          |       |          |          |       |       |
|                                           | SR[8:0]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | ER[8:0] |               |                                          |         |                                         |                   |                                           |       |                                          |       |          |          |       |       |
| Power On Sequence                         | 0000h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 04ffh   |               |                                          |         |                                         |                   |                                           |       |                                          |       |          |          |       |       |
| SW Reset                                  | 0000h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 04ffh   |               |                                          |         |                                         |                   |                                           |       |                                          |       |          |          |       |       |
| HW Reset                                  | 0000h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 04ffh   |               |                                          |         |                                         |                   |                                           |       |                                          |       |          |          |       |       |
| Flow chart                                | <div><div><p>1. To Enter Partial Mode</p><pre>graph TD; PTLAR[PTLAR 30h] --&gt; SR[/1st &amp; 2nd Parameter: SR[9:0]/]; SR --&gt; ER[/3rd &amp; 4th Parameter: ER[9:0]/]; ER --&gt; PTLON[PTLON 12h]; PTLON --&gt; PM([Partial Mode]);</pre></div><div><p>2. To Exit Partial Mode</p><pre>graph TD; PM([Partial Mode]) --&gt; DISPOFF[DISPOFF 28h]; DISPOFF --&gt; NORON[NORON 13h]; NORON --&gt; PMOFF([Partial Mode OFF]); PMOFF --&gt; ID([Image Data D1[B:0], D2[B:0]... Dn[B:0]]); ID --&gt; DISON[DISON 29h];</pre></div><div><p>Optional to prevent tearing effect image display</p><pre>graph LR; DISPOFF -.-&gt; Note[Optional to prevent tearing effect image display];</pre></div><div><p>Legend</p><ul style="list-style-type: none"><li>Command</li><li>Parameter</li><li>Display</li><li>Action</li><li>Mode</li><li>Sequential transfer</li></ul></div></div> <p>Note : B=23</p> |         |               |                                          |         |                                         |                   |                                           |       |                                          |       |          |          |       |       |

**TEOFF (3400h): Tearing Effect Line OFF**

| 3400H                                     |                                                                                                                                                                                                                                                                                                                                                                                            |         |       | TEOFF (Tearing Effect Line OFF) |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|---------------------------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address |       | D15-8                           | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI    | Other |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| TEOFF                                     | W                                                                                                                                                                                                                                                                                                                                                                                          | 34h     | 3400h | No Argument                     |    |    |    |    |    |    |    |    | -   |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Description                               | This command turns off outputting display module's tearing effect output signal to TE pad.                                                                                                                                                                                                                                                                                                 |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Restriction                               | This command has no effect when the Tearing Effect output is already off.                                                                                                                                                                                                                                                                                                                  |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |         |       |                                 |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                               |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>OFF</td></tr><tr><td>SW Reset</td><td>OFF</td></tr><tr><td>HW Reset</td><td>OFF</td></tr></table>                                                                                                                                                                                                   |         |       |                                 |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | OFF | SW Reset                                | OFF | HW Reset                                  | OFF |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                              |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Power On Sequence                         | OFF                                                                                                                                                                                                                                                                                                                                                                                        |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| SW Reset                                  | OFF                                                                                                                                                                                                                                                                                                                                                                                        |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| HW Reset                                  | OFF                                                                                                                                                                                                                                                                                                                                                                                        |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Flow Chart                                | <div><div><div>TE Line Output ON</div><div>TEOFF (34h)</div><div>TE Line Output OFF</div></div><div><div>Legend</div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>                                                                                                                                    |         |       |                                 |    |    |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |

**TEON (3500h): Tearing Effect Line ON**

| 3500H                                     |                                                                                                                                                                                                                                                                                                                                                                                                                          |         |       | TEON (Tearing Effect Line ON) |    |    |    |    |    |    |    |       |     |        |               |                                          |        |                                         |        |                                           |        |                                          |     |          |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|-------------------------------|----|----|----|----|----|----|----|-------|-----|--------|---------------|------------------------------------------|--------|-----------------------------------------|--------|-------------------------------------------|--------|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                                                      | Address |       | D15-8                         | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0    | HEX |        |               |                                          |        |                                         |        |                                           |        |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                          | MIPI    | Other |                               |    |    |    |    |    |    |    |       |     |        |               |                                          |        |                                         |        |                                           |        |                                          |     |          |     |
| TEON                                      | R/W                                                                                                                                                                                                                                                                                                                                                                                                                      | 35h     | 3500h | x                             | 0  | 0  | 0  | 0  | 0  | 0  | 0  | TELOM | 00  |        |               |                                          |        |                                         |        |                                           |        |                                          |     |          |     |
| Description                               | This command turns on the tearing effect output signal to the TE pad. The “Tearing Effect Line On” has one parameter that selects its output mode.                                                                                                                                                                                                                                                                       |         |       |                               |    |    |    |    |    |    |    |       |     |        |               |                                          |        |                                         |        |                                           |        |                                          |     |          |     |
| Restriction                               | The tearing effect output is active low if display module is under sleep mode.                                                                                                                                                                                                                                                                                                                                           |         |       |                               |    |    |    |    |    |    |    |       |     |        |               |                                          |        |                                         |        |                                           |        |                                          |     |          |     |
| Register Availability                     | <table><thead><tr><th>Status</th><th>Availability</th></tr></thead><tbody><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></tbody></table> |         |       |                               |    |    |    |    |    |    |    |       |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes    | Normal Mode On, Idle Mode On, Sleep Out | Yes    | Partial Mode On, Idle Mode Off, Sleep Out | Yes    | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                             |         |       |                               |    |    |    |    |    |    |    |       |     |        |               |                                          |        |                                         |        |                                           |        |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |                               |    |    |    |    |    |    |    |       |     |        |               |                                          |        |                                         |        |                                           |        |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |                               |    |    |    |    |    |    |    |       |     |        |               |                                          |        |                                         |        |                                           |        |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |                               |    |    |    |    |    |    |    |       |     |        |               |                                          |        |                                         |        |                                           |        |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |                               |    |    |    |    |    |    |    |       |     |        |               |                                          |        |                                         |        |                                           |        |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |                               |    |    |    |    |    |    |    |       |     |        |               |                                          |        |                                         |        |                                           |        |                                          |     |          |     |
| Default                                   | <table><thead><tr><th>Status</th><th>Default Value</th></tr></thead><tbody><tr><td>Power On Sequence</td><td>TE OFF</td></tr><tr><td>SW Reset</td><td>TE OFF</td></tr><tr><td>HW Reset</td><td>TE OFF</td></tr></tbody></table>                                                                                                                                                                                          |         |       |                               |    |    |    |    |    |    |    |       |     | Status | Default Value | Power On Sequence                        | TE OFF | SW Reset                                | TE OFF | HW Reset                                  | TE OFF |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                            |         |       |                               |    |    |    |    |    |    |    |       |     |        |               |                                          |        |                                         |        |                                           |        |                                          |     |          |     |
| Power On Sequence                         | TE OFF                                                                                                                                                                                                                                                                                                                                                                                                                   |         |       |                               |    |    |    |    |    |    |    |       |     |        |               |                                          |        |                                         |        |                                           |        |                                          |     |          |     |
| SW Reset                                  | TE OFF                                                                                                                                                                                                                                                                                                                                                                                                                   |         |       |                               |    |    |    |    |    |    |    |       |     |        |               |                                          |        |                                         |        |                                           |        |                                          |     |          |     |
| HW Reset                                  | TE OFF                                                                                                                                                                                                                                                                                                                                                                                                                   |         |       |                               |    |    |    |    |    |    |    |       |     |        |               |                                          |        |                                         |        |                                           |        |                                          |     |          |     |
| Flow Chart                                | <div><div><div>TE Line Output OFF</div><div>TEON (35h)</div><div>1<sup>st</sup> Parameter: TELOM</div><div>TE Line Output ON</div></div><div><div>Legend</div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>                                                                                                                         |         |       |                               |    |    |    |    |    |    |    |       |     |        |               |                                          |        |                                         |        |                                           |        |                                          |     |          |     |

**MADCTR (3600h): Scan Direction Control**

| 3600H                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 | MADCTR (Scan Direction Control)               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------|-----------------------------------------------|----|----|----|----|----|----|----|----|-------------|--------|--------|---------------|----------|----|-------------|---------------|-------------------|---------------|----------|---|---|----|------|-----------------|-----------------------------------------------|----|------|---------------|-----------------------------------------------|
| Instruction                                                                                                                                                                                                                                                                                             | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Address  |                 | D15-8                                         | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX         |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               | MIPI     | Other           |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
| MADCTR                                                                                                                                                                                                                                                                                                  | W                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 36h      | 3600h           | x                                             | 0  | 0  | 0  | 0  | D3 | D2 | D1 | D0 | 00          |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
| Description                                                                                                                                                                                                                                                                                             | This command defines the scan direction of Source and Gate Driver. This command makes no change on the other driver status.                                                                                                                                                                                                                                                                                                                                   |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         | <table><tr><th>Bit</th><th>Symbol</th><th>Description</th><th>Comment</th></tr><tr><td>D3</td><td>RGB</td><td>RGB/BGR Order</td><td>'1' =BGR, "0"=RGB</td></tr><tr><td>D2</td><td>Reserved</td><td></td><td>0</td></tr><tr><td>D1</td><td>RSMX</td><td>Horizontal Flip</td><td>'0' = Normal display<br/>'1' = Flipped display</td></tr><tr><td>D0</td><td>RSMY</td><td>Vertical Flip</td><td>'0' = Normal display<br/>'1' = Flipped display</td></tr></table> |          |                 |                                               |    |    |    |    |    |    |    |    |             | Bit    | Symbol | Description   | Comment  | D3 | RGB         | RGB/BGR Order | '1' =BGR, "0"=RGB | D2            | Reserved |   | 0 | D1 | RSMX | Horizontal Flip | '0' = Normal display<br>'1' = Flipped display | D0 | RSMY | Vertical Flip | '0' = Normal display<br>'1' = Flipped display |
|                                                                                                                                                                                                                                                                                                         | Bit                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Symbol   | Description     | Comment                                       |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         | D3                                                                                                                                                                                                                                                                                                                                                                                                                                                            | RGB      | RGB/BGR Order   | '1' =BGR, "0"=RGB                             |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         | D2                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Reserved |                 | 0                                             |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         | D1                                                                                                                                                                                                                                                                                                                                                                                                                                                            | RSMX     | Horizontal Flip | '0' = Normal display<br>'1' = Flipped display |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         | D0                                                                                                                                                                                                                                                                                                                                                                                                                                                            | RSMY     | Vertical Flip   | '0' = Normal display<br>'1' = Flipped display |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         | <table><tr><th>D1</th><th>D0</th><th>Display Panel</th></tr><tr><td>0</td><td>0</td><td></td></tr><tr><td>0</td><td>1</td><td></td></tr><tr><td>1</td><td>0</td><td></td></tr><tr><td>1</td><td>1</td><td></td></tr></table>                                                                                                                                                                                                                                  |          |                 |                                               |    |    |    |    |    |    |    |    |             | D1     | D0     | Display Panel | 0        | 0  |             | 0             | 1                 |               | 1        | 0 |   | 1  | 1    |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         | D1                                                                                                                                                                                                                                                                                                                                                                                                                                                            | D0       | Display Panel   |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 0        |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
| 0                                                                                                                                                                                                                                                                                                       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
| 1                                                                                                                                                                                                                                                                                                       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
| 1                                                                                                                                                                                                                                                                                                       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
| <table><tr><td rowspan="2">Input Image</td><td colspan="2">D3 = 0</td><td rowspan="2">Display Panel</td></tr><tr><td colspan="2">Sent RGB</td></tr><tr><td rowspan="2">Input Image</td><td colspan="2">D3 = 1</td><td rowspan="2">Display Panel</td></tr><tr><td colspan="2">Sent BGR</td></tr></table> |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    | Input Image | D3 = 0 |        | Display Panel | Sent RGB |    | Input Image | D3 = 1        |                   | Display Panel | Sent BGR |   |   |    |      |                 |                                               |    |      |               |                                               |
| Input Image                                                                                                                                                                                                                                                                                             | D3 = 0                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          | Display Panel   |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         | Sent RGB                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
| Input Image                                                                                                                                                                                                                                                                                             | D3 = 1                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          | Display Panel   |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         | Sent BGR                                                                                                                                                                                                                                                                                                                                                                                                                                                      |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |
|                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                               |          |                 |                                               |    |    |    |    |    |    |    |    |             |        |        |               |          |    |             |               |                   |               |          |   |   |    |      |                 |                                               |    |      |               |                                               |

| Register Availability                     | <table> <tr> <th>Status</th><th>Availability</th></tr> <tr> <td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr> <tr> <td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr> <tr> <td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr> <tr> <td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr> <tr> <td>Sleep In</td><td>Yes</td></tr> </table>                                        | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|-----|
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Default                                   | <table> <tr> <th>Status</th><th>Default Value</th></tr> <tr> <td>Power On Sequence</td><td>00h</td></tr> <tr> <td>SW Reset</td><td>00h</td></tr> <tr> <td>HW Reset</td><td>00h</td></tr> </table>                                                                                                                                                                                                                                              | Status | Default Value | Power On Sequence                        | 00h | SW Reset                                | 00h | HW Reset                                  | 00h |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Power On Sequence                         | 00h                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| SW Reset                                  | 00h                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| HW Reset                                  | 00h                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Flow chart                                | <p>The flow chart shows a rectangular box labeled 'MADCTR (36h)' with an arrow pointing down to a parallelogram box labeled '1<sup>st</sup> Parameter'. To the right is a dashed box labeled 'Legend' containing six symbols: a rectangle for 'Command', a parallelogram for 'Parameter', a horizontal oval for 'Display', a horizontal arrow for 'Action', a rounded rectangle for 'Mode', and a speech bubble for 'Sequential transfer'.</p> |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |

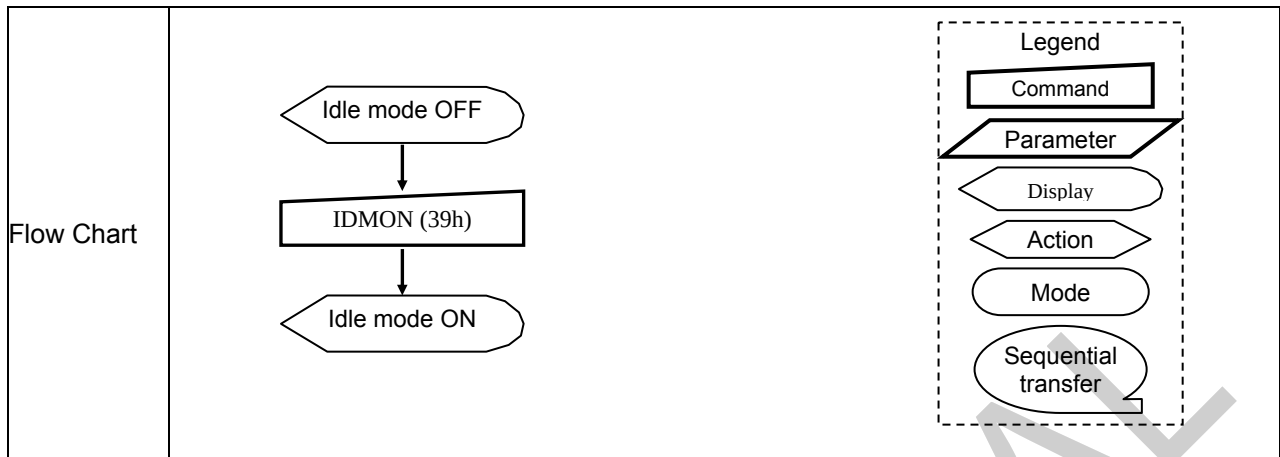
**IDMOFF (3800h): Idle Mode Off**

| 3800H                                     |                                                                                                                                                                                                                                                                                                                                                                                            | IDMOFF (Idle Mode Off) |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------|-------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|---------------|-----------------------------------------|---------------|-------------------------------------------|---------------|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address                |       | D15-8       | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI                   | Other |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| IDMOFF                                    | W                                                                                                                                                                                                                                                                                                                                                                                          | 38h                    | 3800h | No Argument |    |    |    |    |    |    |    |    | -   |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Description                               | This command causes the display module to exit Idle mode.                                                                                                                                                                                                                                                                                                                                  |                        |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Restriction                               | This command has no effect when the display module is not in Idle mode.                                                                                                                                                                                                                                                                                                                    |                        |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |                        |       |             |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes           | Normal Mode On, Idle Mode On, Sleep Out | Yes           | Partial Mode On, Idle Mode Off, Sleep Out | Yes           | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                               |                        |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                        |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                        |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                        |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                        |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                        |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Idle Mode Off</td></tr><tr><td>SW Reset</td><td>Idle Mode Off</td></tr><tr><td>HW Reset</td><td>Idle Mode Off</td></tr></table>                                                                                                                                                                     |                        |       |             |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | Idle Mode Off | SW Reset                                | Idle Mode Off | HW Reset                                  | Idle Mode Off |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                              |                        |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Power On Sequence                         | Idle Mode Off                                                                                                                                                                                                                                                                                                                                                                              |                        |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| SW Reset                                  | Idle Mode Off                                                                                                                                                                                                                                                                                                                                                                              |                        |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| HW Reset                                  | Idle Mode Off                                                                                                                                                                                                                                                                                                                                                                              |                        |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |
| Flow Chart                                | <div><div><div>Idle mode ON</div><div>IDMOFF (38h)</div><div>Idle mode OFF</div></div><div><div>Legend</div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>                                                                                                                                             |                        |       |             |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |

**IDMON (3900h): Enter\_idle\_mode**

| 3900H                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |         |       | Enter_idle_mode |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|-----------------|----|----|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|---------------|-----------------------------------------|---------------|-------------------------------------------|---------------|------------------------------------------|-----|----------|-----|----|----|----|----|----|----|----|----|----|----|----|----|----|-------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|-----|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|-------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|-------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Address |       | D15-8           | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | MIPI    | Other |                 |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| IDMON                                     | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 39h     | 3900h | No Argument     |    |    |    |    |    |    |    |    | -   |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Description                               | <p>This command causes the display module to enter Idle Mode.<br/>In Idle Mode, color expression is reduced. Colors are shown on the display device using the MSB of each of the R, G and B color components in the Input Image.</p> <div><div><p>Input Image</p></div><div><p>Panel Display</p></div></div> <table><tr><td>Color</td><td>R7</td><td>R6</td><td>R5</td><td>R4</td><td>R3</td><td>R2</td><td>R1</td><td>R0</td><td>G7</td><td>G6</td><td>G5</td><td>G4</td><td>G3</td><td>G2</td><td>G1</td><td>G0</td><td>B7</td><td>B6</td><td>B5</td><td>B4</td><td>B3</td><td>B2</td><td>B1</td><td>B0</td></tr><tr><td>Black</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>Blue</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>Red</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>Magenta</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>Green</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>Cyan</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr><tr><td>Yellow</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>White</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td></tr></table> |         |       |                 |    |    |    |    |    |    |    |    |     | Color  | R7            | R6                                       | R5            | R4                                      | R3            | R2                                        | R1            | R0                                       | G7  | G6       | G5  | G4 | G3 | G2 | G1 | G0 | B7 | B6 | B5 | B4 | B3 | B2 | B1 | B0 | Black | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | Blue | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | Red | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | Magenta | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | Green | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | Cyan | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | Yellow | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | White | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
|                                           | Color                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | R7      | R6    | R5              | R4 | R3 | R2 | R1 | R0 | G7 | G6 | G5 | G4  | G3     | G2            | G1                                       | G0            | B7                                      | B6            | B5                                        | B4            | B3                                       | B2  | B1       | B0  |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|                                           | Black                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0       | 0     | 0               | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 0      | 0             | 0                                        | 0             | 0                                       | 0             | 0                                         | 0             | 0                                        | 0   | 0        | 0   |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Blue                                      | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0       | 0     | 0               | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0  | 0   | 0      | 0             | 0                                        | 1             | 1                                       | 1             | 1                                         | 1             | 1                                        | 1   | 1        |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Red                                       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1       | 1     | 1               | 1  | 1  | 1  | 1  | 0  | 0  | 0  | 0  | 0   | 0      | 0             | 0                                        | 0             | 0                                       | 0             | 0                                         | 0             | 0                                        | 0   | 0        |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Magenta                                   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1       | 1     | 1               | 1  | 1  | 1  | 1  | 0  | 0  | 0  | 0  | 0   | 0      | 0             | 0                                        | 1             | 1                                       | 1             | 1                                         | 1             | 1                                        | 1   | 1        |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Green                                     | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0       | 0     | 0               | 0  | 0  | 0  | 0  | 1  | 1  | 1  | 1  | 1   | 1      | 1             | 1                                        | 0             | 0                                       | 0             | 0                                         | 0             | 0                                        | 0   | 0        |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Cyan                                      | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 0       | 0     | 0               | 0  | 0  | 0  | 0  | 1  | 1  | 1  | 1  | 1   | 1      | 1             | 1                                        | 1             | 1                                       | 1             | 1                                         | 1             | 1                                        | 1   | 1        |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Yellow                                    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1       | 1     | 1               | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1   | 1      | 1             | 1                                        | 0             | 0                                       | 0             | 0                                         | 0             | 0                                        | 0   | 0        |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| White                                     | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1       | 1     | 1               | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1  | 1   | 1      | 1             | 1                                        | 1             | 1                                       | 1             | 1                                         | 1             | 1                                        | 1   | 1        |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Restriction                               | This command has no effect when module is already in idle on mode.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |       |                 |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |         |       |                 |    |    |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes           | Normal Mode On, Idle Mode On, Sleep Out | Yes           | Partial Mode On, Idle Mode Off, Sleep Out | Yes           | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |       |                 |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |       |                 |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |       |                 |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |       |                 |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |       |                 |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |       |                 |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>Idle Mode Off</td></tr><tr><td>SW Reset</td><td>Idle Mode Off</td></tr><tr><td>HW Reset</td><td>Idle Mode Off</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |       |                 |    |    |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | Idle Mode Off | SW Reset                                | Idle Mode Off | HW Reset                                  | Idle Mode Off |                                          |     |          |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |                 |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Power On Sequence                         | Idle Mode Off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |                 |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| SW Reset                                  | Idle Mode Off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |                 |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| HW Reset                                  | Idle Mode Off                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |                 |    |    |    |    |    |    |    |    |     |        |               |                                          |               |                                         |               |                                           |               |                                          |     |          |     |    |    |    |    |    |    |    |    |    |    |    |    |    |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |     |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |         |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |





CONFIDENTIAL

**STESL(4400h): Set Tear Scan line**

| 4400H                                     |                                                                                                                                                                                                                                                                                                                                                                                                                          | STESL(Set_Tear_Scanline) |       |       |         |         |         |         |         |         |        |        |     |        |               |                                          |                    |                                         |                    |                                           |                    |                                          |     |          |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-------|-------|---------|---------|---------|---------|---------|---------|--------|--------|-----|--------|---------------|------------------------------------------|--------------------|-----------------------------------------|--------------------|-------------------------------------------|--------------------|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                                                      | Address                  |       | D15-8 | D7      | D6      | D5      | D4      | D3      | D2      | D1     | D0     | HEX |        |               |                                          |                    |                                         |                    |                                           |                    |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                          | MIPI                     | Other |       |         |         |         |         |         |         |        |        |     |        |               |                                          |                    |                                         |                    |                                           |                    |                                          |     |          |     |
| STESL                                     | W                                                                                                                                                                                                                                                                                                                                                                                                                        | 44h                      | 4400h | x     | STS[15] | STS[14] | STS[13] | STS[12] | STS[11] | STS[10] | STS[9] | STS[8] | 00  |        |               |                                          |                    |                                         |                    |                                           |                    |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                          |                          | 4401h | x     | STS[7]  | STS[6]  | STS[5]  | STS[4]  | STS[3]  | STS[2]  | STS[1] | STS[0] | 00  |        |               |                                          |                    |                                         |                    |                                           |                    |                                          |     |          |     |
| Description                               | This command turns on the display Tearing Effect output to the TE pad when the display reaches line N, set by STS[15:0].                                                                                                                                                                                                                                                                                                 |                          |       |       |         |         |         |         |         |         |        |        |     |        |               |                                          |                    |                                         |                    |                                           |                    |                                          |     |          |     |
| Restriction                               | The tearing effect output is active low when the display module is under sleep mode                                                                                                                                                                                                                                                                                                                                      |                          |       |       |         |         |         |         |         |         |        |        |     |        |               |                                          |                    |                                         |                    |                                           |                    |                                          |     |          |     |
| Register Availability                     | <table><thead><tr><th>Status</th><th>Availability</th></tr></thead><tbody><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></tbody></table> |                          |       |       |         |         |         |         |         |         |        |        |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes                | Normal Mode On, Idle Mode On, Sleep Out | Yes                | Partial Mode On, Idle Mode Off, Sleep Out | Yes                | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                             |                          |       |       |         |         |         |         |         |         |        |        |     |        |               |                                          |                    |                                         |                    |                                           |                    |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |                          |       |       |         |         |         |         |         |         |        |        |     |        |               |                                          |                    |                                         |                    |                                           |                    |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |                          |       |       |         |         |         |         |         |         |        |        |     |        |               |                                          |                    |                                         |                    |                                           |                    |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |                          |       |       |         |         |         |         |         |         |        |        |     |        |               |                                          |                    |                                         |                    |                                           |                    |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |                          |       |       |         |         |         |         |         |         |        |        |     |        |               |                                          |                    |                                         |                    |                                           |                    |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |                          |       |       |         |         |         |         |         |         |        |        |     |        |               |                                          |                    |                                         |                    |                                           |                    |                                          |     |          |     |
| Default                                   | <table><thead><tr><th>Status</th><th>Default Value</th></tr></thead><tbody><tr><td>Power On Sequence</td><td>STS[15:0]=16'h0000</td></tr><tr><td>SW Reset</td><td>STS[15:0]=16'h0000</td></tr><tr><td>HW Reset</td><td>STS[15:0]=16'h0000</td></tr></tbody></table>                                                                                                                                                      |                          |       |       |         |         |         |         |         |         |        |        |     | Status | Default Value | Power On Sequence                        | STS[15:0]=16'h0000 | SW Reset                                | STS[15:0]=16'h0000 | HW Reset                                  | STS[15:0]=16'h0000 |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                            |                          |       |       |         |         |         |         |         |         |        |        |     |        |               |                                          |                    |                                         |                    |                                           |                    |                                          |     |          |     |
| Power On Sequence                         | STS[15:0]=16'h0000                                                                                                                                                                                                                                                                                                                                                                                                       |                          |       |       |         |         |         |         |         |         |        |        |     |        |               |                                          |                    |                                         |                    |                                           |                    |                                          |     |          |     |
| SW Reset                                  | STS[15:0]=16'h0000                                                                                                                                                                                                                                                                                                                                                                                                       |                          |       |       |         |         |         |         |         |         |        |        |     |        |               |                                          |                    |                                         |                    |                                           |                    |                                          |     |          |     |
| HW Reset                                  | STS[15:0]=16'h0000                                                                                                                                                                                                                                                                                                                                                                                                       |                          |       |       |         |         |         |         |         |         |        |        |     |        |               |                                          |                    |                                         |                    |                                           |                    |                                          |     |          |     |
| Flow Chart                                | <div><div><div>TE Output On or Off</div><div>↓</div><div>set_tear_scanline</div><div>↓</div><div>Send 1st parameter STS[8]</div><div>↓</div><div>Send 2nd parameter STS[7:0]</div><div>↓</div><div>TE Output On the Nth line</div></div><div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div></div>              |                          |       |       |         |         |         |         |         |         |        |        |     |        |               |                                          |                    |                                         |                    |                                           |                    |                                          |     |          |     |

**GSL (4500h): Get Scan Line**

| 4500H                                     | GSL(Get_Scanline)                                                                                                                                                                                                                                                                                                                                                                                                        |         |       |       |         |         |         |         |         |         |        |        |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|-------|---------|---------|---------|---------|---------|---------|--------|--------|-----|--------|--------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                                                      | Address |       | D15-8 | D7      | D6      | D5      | D4      | D3      | D2      | D1     | D0     | HEX |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                          | MIPI    | Other |       |         |         |         |         |         |         |        |        |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| GSL                                       | R                                                                                                                                                                                                                                                                                                                                                                                                                        | 45h     | 4500h | x     | GTS[15] | GTS[14] | GTS[13] | GTS[12] | GTS[11] | GTS[10] | GTS[9] | GTS[8] | 00  |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                          |         | 4501h | x     | GTS[7]  | GTS[6]  | GTS[5]  | GTS[4]  | GTS[3]  | GTS[2]  | GTS[1] | GTS[0] | 00  |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Description                               | The display returns the current scan line, N, used to update the display device. The total number of scan lines on a display device is defined as VSYNC + VBP + VACT + VFP. The first scan line is defined as the first line of V-Sync and is denoted as Line 0. When in Sleep Mode, the value returned by get_scanline is undefined.                                                                                    |         |       |       |         |         |         |         |         |         |        |        |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Restriction                               | -                                                                                                                                                                                                                                                                                                                                                                                                                        |         |       |       |         |         |         |         |         |         |        |        |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Register Availability                     | <table><thead><tr><th>Status</th><th>Availability</th></tr></thead><tbody><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></tbody></table> |         |       |       |         |         |         |         |         |         |        |        |     | Status | Availability | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                             |         |       |       |         |         |         |         |         |         |        |        |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |       |         |         |         |         |         |         |        |        |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |       |         |         |         |         |         |         |        |        |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |       |         |         |         |         |         |         |        |        |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |       |         |         |         |         |         |         |        |        |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |       |         |         |         |         |         |         |        |        |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Flow Chart                                | <div><div><div>get_scanline</div><div>Wait 3us</div><div>Dummy Read</div><div>Send 1st parameter GTS[9:8]</div><div>Send 2nd parameter GTS[7:0]</div></div><div><div>Legend</div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>                                                                                                      |         |       |       |         |         |         |         |         |         |        |        |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |

**DSTBON (4F00h): Deep Standby Mode On**

| 4F00H                                     |                                                                                                                                                                                                                                                                                                                                                                                                                          |         |       | DSTBON(Deep Standby Mode On) |    |    |    |    |    |    |    |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|------------------------------|----|----|----|----|----|----|----|------|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                                                      | Address |       | D15-8                        | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0   | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                          | MIPI    | Other |                              |    |    |    |    |    |    |    |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| DSTBON                                    | W                                                                                                                                                                                                                                                                                                                                                                                                                        | 4Fh     | 4F00h | x                            | 0  | 0  | 0  | 0  | 0  | 0  | 0  | DSTB | 00  |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Description                               | This command is used to enter deep standby mode.<br>DSTB="1", enter deep standby mode.                                                                                                                                                                                                                                                                                                                                   |         |       |                              |    |    |    |    |    |    |    |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Restriction                               | To exit Deep Standby Mode, input low pulse more than 3 ms to pin RESX.                                                                                                                                                                                                                                                                                                                                                   |         |       |                              |    |    |    |    |    |    |    |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Register Availability                     | <table><thead><tr><th>Status</th><th>Availability</th></tr></thead><tbody><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></tbody></table> |         |       |                              |    |    |    |    |    |    |    |      |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                             |         |       |                              |    |    |    |    |    |    |    |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |                              |    |    |    |    |    |    |    |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |                              |    |    |    |    |    |    |    |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |                              |    |    |    |    |    |    |    |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |                              |    |    |    |    |    |    |    |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |                              |    |    |    |    |    |    |    |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Default                                   | <table><thead><tr><th>Status</th><th>Default Value</th></tr></thead><tbody><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>SW Reset</td><td>00h</td></tr><tr><td>HW Reset</td><td>00h</td></tr></tbody></table>                                                                                                                                                                                                   |         |       |                              |    |    |    |    |    |    |    |      |     | Status | Default Value | Power On Sequence                        | 00h | SW Reset                                | 00h | HW Reset                                  | 00h |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                            |         |       |                              |    |    |    |    |    |    |    |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Power On Sequence                         | 00h                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |                              |    |    |    |    |    |    |    |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| SW Reset                                  | 00h                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |                              |    |    |    |    |    |    |    |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| HW Reset                                  | 00h                                                                                                                                                                                                                                                                                                                                                                                                                      |         |       |                              |    |    |    |    |    |    |    |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Flow chart                                | <div><div><div>DSTBON (4Fh)</div><div>Parameter DSTB=1</div><div>Deep Standby Mode</div></div><div><div>Legend</div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>                                                                                                                                                                   |         |       |                              |    |    |    |    |    |    |    |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |

**WRDISBV (5100h): Write Display Brightness**

| 5100H                                     |                                                                                                                                                                                                                                                                                                                                                                                            |         |       | WRDISBV |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|---------|------|------|------|------|------|------|------|------|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address |       | D15-8   | D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI    | Other |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| WRDISBV                                   | W                                                                                                                                                                                                                                                                                                                                                                                          | 51h     | 5100h | x       | DBV7 | DBV6 | DBV5 | DBV4 | DBV3 | DBV2 | DBV1 | DBV0 | 00  |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Description                               | This command is used to adjust brightness value. In principle relationship is that 00h value means the lowest brightness and 0xFFh value means the highest brightness.                                                                                                                                                                                                                     |         |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Restriction                               | The display supplier cannot use this command for tuning                                                                                                                                                                                                                                                                                                                                    |         |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |         |       |         |      |      |      |      |      |      |      |      |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                               |         |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>SW Reset</td><td>00h</td></tr><tr><td>HW Reset</td><td>00h</td></tr></table>                                                                                                                                                                                                   |         |       |         |      |      |      |      |      |      |      |      |     | Status | Default Value | Power On Sequence                        | 00h | SW Reset                                | 00h | HW Reset                                  | 00h |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                              |         |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Power On Sequence                         | 00h                                                                                                                                                                                                                                                                                                                                                                                        |         |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| SW Reset                                  | 00h                                                                                                                                                                                                                                                                                                                                                                                        |         |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| HW Reset                                  | 00h                                                                                                                                                                                                                                                                                                                                                                                        |         |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Flow chart                                | <div><div><div>WRDISBV (51h)</div><div>Parameter DBV[7:0]</div><div>New Brightness Loaded</div></div><div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div></div>                                                                                                                   |         |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |

**RDDISBV (5200h): Read Display Brightness**

| 5200H                 |                                                                                                                                                                                                                                                                                                                                                                                            |               |       | RDDISBV |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------|---------|------|------|------|------|------|------|------|------|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|-----|
| Instruction           | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address       |       | D15-8   | D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI          | Other |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| RDDISBV               | R                                                                                                                                                                                                                                                                                                                                                                                          | 52h           | 5200h | x       | DBV7 | DBV6 | DBV5 | DBV4 | DBV3 | DBV2 | DBV1 | DBV0 | 00  |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Description           | This command returns brightness value. In principle relationship is that 00h value means the lowest brightness and FFh value means the highest brightness.                                                                                                                                                                                                                                 |               |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Restriction           | -                                                                                                                                                                                                                                                                                                                                                                                          |               |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Register Availability | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |               |       |         |      |      |      |      |      |      |      |      |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                     | Availability  |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                   | Yes           |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                    | Yes           |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                  | Yes           |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                   | Yes           |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Sleep In              | Yes                                                                                                                                                                                                                                                                                                                                                                                        |               |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Default               | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>SW Reset</td><td>00h</td></tr><tr><td>HW Reset</td><td>00h</td></tr></table>                                                                                                                                                                                                   |               |       |         |      |      |      |      |      |      |      |      |     | Status | Default Value | Power On Sequence                        | 00h | SW Reset                                | 00h | HW Reset                                  | 00h |                                          |     |          |     |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                     | Default Value |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Power On Sequence                                                                                                                                                                                                                                                                                                                                                                          | 00h           |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | SW Reset                                                                                                                                                                                                                                                                                                                                                                                   | 00h           |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| HW Reset              | 00h                                                                                                                                                                                                                                                                                                                                                                                        |               |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Flow Chart            | <div><div><div>RDDISBV (52hH)</div><div>Send parameter DBV[7:0]</div></div><div>Host<br/>Driver</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>                                                                                                                  |               |       |         |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |

**WRCTRLD (5300h): Write CTRL Display**

| 5300H                                                                                                                                                |                                                                              |                            | WRCTRLD(Write CTRL Display) |                                                            |                                                                                                 |                                                                                                 |    |    |    |    |    |     |              |
|------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------|-----------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|----|----|----|----|----|-----|--------------|
| Instruction                                                                                                                                          | R/W                                                                          | Address                    |                             | Parameter                                                  |                                                                                                 |                                                                                                 |    |    |    |    |    |     |              |
|                                                                                                                                                      |                                                                              | MIPI                       | Others                      | D15-D8                                                     | D7                                                                                              | D6                                                                                              | D5 | D4 | D3 | D2 | D1 | D0  | HEX          |
| WRCTRLD                                                                                                                                              | W                                                                            | 53h                        | 5300h                       | 00h                                                        | SLR_LEVEL-                                                                                      | BCTRL                                                                                           | -  | DD | BL | -  | -  | 00  |              |
| Description                                                                                                                                          | This command is used to control ambient light, brightness and gamma setting. |                            |                             |                                                            |                                                                                                 |                                                                                                 |    |    |    |    |    |     |              |
|                                                                                                                                                      | BCTRL, Brightness Control Block On/Off,                                      |                            |                             |                                                            |                                                                                                 |                                                                                                 |    |    |    |    |    |     |              |
|                                                                                                                                                      | BCTRL                                                                        | DESCRIPTION                |                             |                                                            |                                                                                                 | DESCRIPTION of LEDPWM Pin                                                                       |    |    |    |    |    |     |              |
|                                                                                                                                                      | 0                                                                            | Off,<br>DBV[7:0] are 00h.  |                             |                                                            |                                                                                                 | LEDPWMPOL="0": keep low (0%)<br>LEDPWMPOL="1": keep high (0%)                                   |    |    |    |    |    |     |              |
|                                                                                                                                                      | 1                                                                            | On,<br>DBV[7:0] are active |                             |                                                            |                                                                                                 | LEDPWMPOL="0": PWM output (high level is duty)<br>LEDPWMPOL="1": PWM output (low level is duty) |    |    |    |    |    |     |              |
|                                                                                                                                                      | ** It is always used to switch brightness for display with dimming effect.   |                            |                             |                                                            |                                                                                                 |                                                                                                 |    |    |    |    |    |     |              |
|                                                                                                                                                      | DD: Display Dimming Control On/Off                                           |                            |                             |                                                            |                                                                                                 |                                                                                                 |    |    |    |    |    |     |              |
|                                                                                                                                                      | DD                                                                           | DESCRIPTION                |                             |                                                            |                                                                                                 |                                                                                                 |    |    |    |    |    |     |              |
|                                                                                                                                                      | 0                                                                            | Display dimming is off     |                             |                                                            |                                                                                                 |                                                                                                 |    |    |    |    |    |     |              |
|                                                                                                                                                      | 1                                                                            | Display dimming is on      |                             |                                                            |                                                                                                 |                                                                                                 |    |    |    |    |    |     |              |
|                                                                                                                                                      | BL: Backlight Control On/Off without Dimming Effect                          |                            |                             |                                                            |                                                                                                 |                                                                                                 |    |    |    |    |    |     |              |
|                                                                                                                                                      | BL                                                                           | DESCRIPTION                |                             |                                                            |                                                                                                 | DESCRIPTION of LEDPWM Pin                                                                       |    |    |    |    |    |     |              |
|                                                                                                                                                      | 0                                                                            | Off                        |                             |                                                            |                                                                                                 | LEDPWMPOL="0": keep low (non-lit)<br>LEDPWMPOL="1": keep high (non-lit)                         |    |    |    |    |    |     |              |
| 1                                                                                                                                                    | On                                                                           |                            |                             |                                                            | LEDPWMPOL="0": PWM output (high level is duty)<br>LEDPWMPOL="1": PWM output (low level is duty) |                                                                                                 |    |    |    |    |    |     |              |
| ** When BL bit change from "On" to "Off", display brightness control is turned off without gradual dimming, even if dimming on (DD="1") is selected. |                                                                              |                            |                             |                                                            |                                                                                                 |                                                                                                 |    |    |    |    |    |     |              |
| SLR_LEVEL : SLR Setting                                                                                                                              |                                                                              |                            |                             |                                                            |                                                                                                 |                                                                                                 |    |    |    |    |    |     |              |
| SLR_LEVEL[1:0]                                                                                                                                       |                                                                              |                            |                             | DESCRIPTION                                                |                                                                                                 |                                                                                                 |    |    |    |    |    |     |              |
|                                                                                                                                                      |                                                                              |                            |                             | SLR level<br>0 : SLR low<br>1 : SLR medium<br>2 : SLR high |                                                                                                 |                                                                                                 |    |    |    |    |    |     |              |
| Restriction                                                                                                                                          | The display supplier cannot use this command for tuning                      |                            |                             |                                                            |                                                                                                 |                                                                                                 |    |    |    |    |    |     |              |
| Register Availability                                                                                                                                |                                                                              |                            |                             |                                                            |                                                                                                 |                                                                                                 |    |    |    |    |    |     |              |
|                                                                                                                                                      | Status                                                                       |                            |                             |                                                            |                                                                                                 |                                                                                                 |    |    |    |    |    |     | Availability |
|                                                                                                                                                      | Normal Mode On, Idle Mode Off, Sleep Out                                     |                            |                             |                                                            |                                                                                                 |                                                                                                 |    |    |    |    |    |     | Yes          |
|                                                                                                                                                      | Normal Mode On, Idle Mode On, Sleep Out                                      |                            |                             |                                                            |                                                                                                 |                                                                                                 |    |    |    |    |    |     | Yes          |
|                                                                                                                                                      | Partial Mode On, Idle Mode Off, Sleep Out                                    |                            |                             |                                                            |                                                                                                 |                                                                                                 |    |    |    |    |    |     | Yes          |
|                                                                                                                                                      | Partial Mode On, Idle Mode On, Sleep Out                                     |                            |                             |                                                            |                                                                                                 |                                                                                                 |    |    |    |    |    |     | Yes          |
| Sleep In                                                                                                                                             |                                                                              |                            |                             |                                                            |                                                                                                 |                                                                                                 |    |    |    |    |    | Yes |              |

| Default           | <table border="1"> <thead> <tr> <th>Status</th><th>Default Value</th></tr> </thead> <tbody> <tr> <td>Power On Sequence</td><td>00h</td></tr> <tr> <td>SW Reset</td><td>00h</td></tr> <tr> <td>HW Reset</td><td>00h</td></tr> </tbody> </table>                                                    | Status | Default Value | Power On Sequence | 00h | SW Reset | 00h | HW Reset | 00h |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------------|-------------------|-----|----------|-----|----------|-----|
| Status            | Default Value                                                                                                                                                                                                                                                                                     |        |               |                   |     |          |     |          |     |
| Power On Sequence | 00h                                                                                                                                                                                                                                                                                               |        |               |                   |     |          |     |          |     |
| SW Reset          | 00h                                                                                                                                                                                                                                                                                               |        |               |                   |     |          |     |          |     |
| HW Reset          | 00h                                                                                                                                                                                                                                                                                               |        |               |                   |     |          |     |          |     |
| Flow chart        | <pre> graph TD     A[WRCTRLD (53h)] --&gt; B[/BCTRL, A, DD, BL, DB/]     B --&gt; C([New Control Value])     </pre> <p><b>Legend</b></p> <ul style="list-style-type: none"> <li>Command</li> <li>Parameter</li> <li>Display</li> <li>Action</li> <li>Mode</li> <li>Sequential transfer</li> </ul> |        |               |                   |     |          |     |          |     |



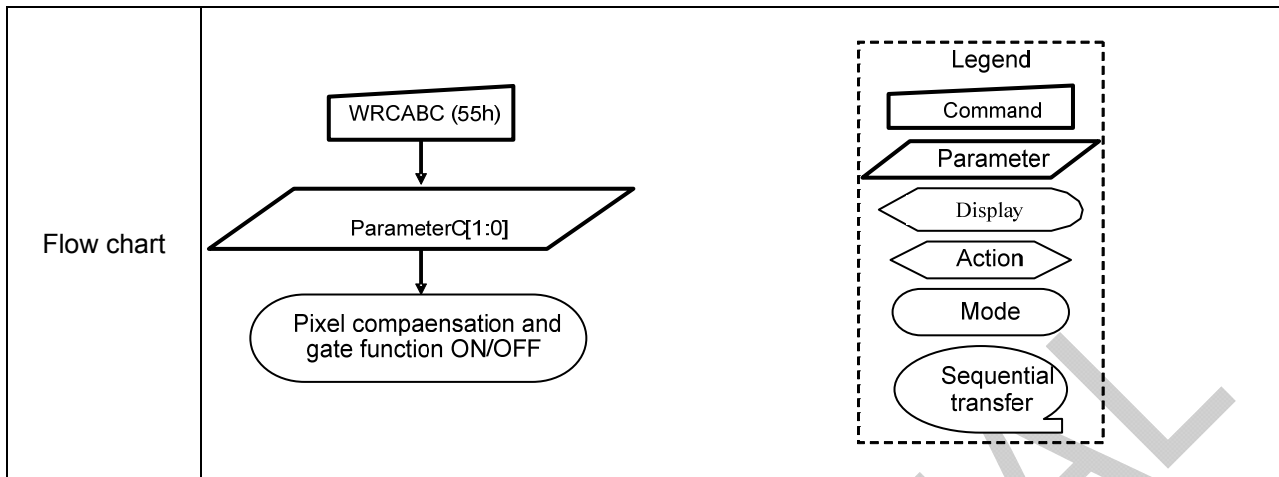
**RDCTRLD (5400h): Read CTRL Display Value**

| 5400H                                                                                                                                        |                                                                              | RDCTRLD(Read CTRL Display Value)                                    |        |           |                                                                                                 |                                                                                                 |    |    |    |     |              |    |     |
|----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------|--------|-----------|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|----|----|----|-----|--------------|----|-----|
| Instruction                                                                                                                                  | R/W                                                                          | Address                                                             |        | Parameter |                                                                                                 |                                                                                                 |    |    |    |     |              |    |     |
|                                                                                                                                              |                                                                              | MIPI                                                                | Others | D15-D8    | D7                                                                                              | D6                                                                                              | D5 | D4 | D3 | D2  | D1           | D0 | HEX |
| RDCTRLD                                                                                                                                      | R                                                                            | 54h                                                                 | 5400h  | 00h       | SLR_LEVEL-                                                                                      | BCTRL                                                                                           | -  | DD | BL | -   | -            | -  | 00  |
| Description                                                                                                                                  | This command is used to control ambient light, brightness and gamma setting. |                                                                     |        |           |                                                                                                 |                                                                                                 |    |    |    |     |              |    |     |
|                                                                                                                                              | BCTRL: Brightness Control Block On/Off                                       |                                                                     |        |           |                                                                                                 |                                                                                                 |    |    |    |     |              |    |     |
|                                                                                                                                              | BCTRL                                                                        | DESCRIPTION LEDPWM Pin                                              |        |           |                                                                                                 | DESCRIPTION LEDPWM Pin                                                                          |    |    |    |     |              |    |     |
|                                                                                                                                              | 0                                                                            | Off,<br>DBV[7:0] are 00h.                                           |        |           |                                                                                                 | LEDPWMPOL="0": keep low (0%)<br>LEDPWMPOL="1": keep high (0%)                                   |    |    |    |     |              |    |     |
|                                                                                                                                              | 1                                                                            | On,<br>DBV[7:0] are active                                          |        |           |                                                                                                 | LEDPWMPOL="0": PWM output (high level is duty)<br>LEDPWMPOL="1": PWM output (low level is duty) |    |    |    |     |              |    |     |
|                                                                                                                                              | ** It is always used to switch brightness for display with dimming effect.   |                                                                     |        |           |                                                                                                 |                                                                                                 |    |    |    |     |              |    |     |
|                                                                                                                                              | DD: Display Dimming Control On/Off                                           |                                                                     |        |           |                                                                                                 |                                                                                                 |    |    |    |     |              |    |     |
|                                                                                                                                              | DD                                                                           | DESCRIPTION                                                         |        |           |                                                                                                 |                                                                                                 |    |    |    |     |              |    |     |
|                                                                                                                                              | 0                                                                            | Display dimming is off                                              |        |           |                                                                                                 |                                                                                                 |    |    |    |     |              |    |     |
|                                                                                                                                              | 1                                                                            | Display dimming is on                                               |        |           |                                                                                                 |                                                                                                 |    |    |    |     |              |    |     |
|                                                                                                                                              | BL: Backlight Control On/Off without Dimming Effect                          |                                                                     |        |           |                                                                                                 |                                                                                                 |    |    |    |     |              |    |     |
|                                                                                                                                              | BL                                                                           | DESCRIPTION                                                         |        |           |                                                                                                 | LEDPWM Pin                                                                                      |    |    |    |     |              |    |     |
| 0                                                                                                                                            | Off                                                                          |                                                                     |        |           | LEDPWMPOL="0": keep low (non-lit)<br>LEDPWMPOL="1": keep high (non-lit)                         |                                                                                                 |    |    |    |     |              |    |     |
| 1                                                                                                                                            | On                                                                           |                                                                     |        |           | LEDPWMPOL="0": PWM output (high level is duty)<br>LEDPWMPOL="1": PWM output (low level is duty) |                                                                                                 |    |    |    |     |              |    |     |
| ** When BL bit change from "On" to "Off", display brightness is turned off without gradual dimming, even if dimming on (DD="1") is selected. |                                                                              |                                                                     |        |           |                                                                                                 |                                                                                                 |    |    |    |     |              |    |     |
| SLR_LEVEL : SLR function level setting                                                                                                       |                                                                              |                                                                     |        |           |                                                                                                 |                                                                                                 |    |    |    |     |              |    |     |
|                                                                                                                                              |                                                                              | DESCRIPTION                                                         |        |           |                                                                                                 |                                                                                                 |    |    |    |     |              |    |     |
| SLR_LEVEL[1:0]                                                                                                                               |                                                                              | SLR function level<br>0 : SLR low<br>1 : SLR medium<br>2 : SLR high |        |           |                                                                                                 |                                                                                                 |    |    |    |     |              |    |     |
| Restriction                                                                                                                                  | The display supplier cannot use this command for tuning                      |                                                                     |        |           |                                                                                                 |                                                                                                 |    |    |    |     |              |    |     |
| Register Availability                                                                                                                        |                                                                              |                                                                     |        |           |                                                                                                 |                                                                                                 |    |    |    |     |              |    |     |
|                                                                                                                                              | Status                                                                       |                                                                     |        |           |                                                                                                 |                                                                                                 |    |    |    |     | Availability |    |     |
|                                                                                                                                              | Normal Mode On, Idle Mode Off, Sleep Out                                     |                                                                     |        |           |                                                                                                 |                                                                                                 |    |    |    |     | Yes          |    |     |
|                                                                                                                                              | Normal Mode On, Idle Mode On, Sleep Out                                      |                                                                     |        |           |                                                                                                 |                                                                                                 |    |    |    |     | Yes          |    |     |
|                                                                                                                                              | Partial Mode On, Idle Mode Off, Sleep Out                                    |                                                                     |        |           |                                                                                                 |                                                                                                 |    |    |    |     | Yes          |    |     |
|                                                                                                                                              | Partial Mode On, Idle Mode On, Sleep Out                                     |                                                                     |        |           |                                                                                                 |                                                                                                 |    |    |    |     | Yes          |    |     |
| Sleep In                                                                                                                                     |                                                                              |                                                                     |        |           |                                                                                                 |                                                                                                 |    |    |    | Yes |              |    |     |

| Default           | <table border="1"> <thead> <tr> <th>Status</th><th>Default Value</th></tr> </thead> <tbody> <tr> <td>Power On Sequence</td><td>00h</td></tr> <tr> <td>SW Reset</td><td>00h</td></tr> <tr> <td>HW Reset</td><td>00h</td></tr> </tbody> </table>                                                                                                                                                                                                                 | Status | Default Value | Power On Sequence | 00h | SW Reset | 00h | HW Reset | 00h |
|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------------|-------------------|-----|----------|-----|----------|-----|
| Status            | Default Value                                                                                                                                                                                                                                                                                                                                                                                                                                                  |        |               |                   |     |          |     |          |     |
| Power On Sequence | 00h                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |               |                   |     |          |     |          |     |
| SW Reset          | 00h                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |               |                   |     |          |     |          |     |
| HW Reset          | 00h                                                                                                                                                                                                                                                                                                                                                                                                                                                            |        |               |                   |     |          |     |          |     |
| Flow Chart        | <p>The flow chart illustrates the communication between a Host Driver and a device. The Host Driver sends the command RDCTRLD (54hH). In response, the device sends parameters BCTRL, A, DD, BL, and DB. A legend on the right defines the symbols used in the flow chart: a rectangle for Command, a parallelogram for Parameter, a hexagon for Display, a chevron for Action, a rounded rectangle for Mode, and a speech bubble for Sequential transfer.</p> |        |               |                   |     |          |     |          |     |

**WRCABC (5500h): Write Content Adaptive Brightness Control**

| 5500H                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                            | WRCABC(Write Content Adaptive Brightness Control) |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|----------|-----------|--------|--------------|--------------|--------------|--------------|----|----|----|-----|-------------|-----------------|------------------------------------------|-------------------|-----------------------------------------|----------------------|-------------------------------------------|--------------------|------------------------------------------|-----------------------|----------|-------------------------|---|-----------------------|-------------|
| Instruction                                                                                                                                                                                                                                                                                                                                                                                                                                                                | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address                                           |          | Parameter |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI                                              | Others   | D15-D8    | D7     | D6           | D5           | D4           | D3           | D2 | D1 | D0 | HEX |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
| WRCABC                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | W                                                                                                                                                                                                                                                                                                                                                                                          | 55h                                               | 5500h    | 00h       | CTE_EN | CTE_LEVEL[3] | CTE_LEVEL[2] | CTE_LEVEL[1] | CTE_LEVEL[0] | 0  | C1 | C0 | 48  |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
| Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                | This command is used to set parameters for image content based adaptive brightness control functionality. There is possible to use 4 different modes for content adaptive image functionality, which are defined on a table below.                                                                                                                                                         |                                                   |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <table><tr><td>C1</td><td>C0</td><td>Function</td></tr><tr><td>0</td><td>0</td><td>Off</td></tr><tr><td>0</td><td>1</td><td>UI Mode</td></tr><tr><td>1</td><td>0</td><td>Still Mode</td></tr><tr><td>1</td><td>1</td><td>Moving Mode</td></tr></table>                                                                                                                                     |                                                   |          |           |        |              |              |              |              |    |    |    |     | C1          | C0              | Function                                 | 0                 | 0                                       | Off                  | 0                                         | 1                  | UI Mode                                  | 1                     | 0        | Still Mode              | 1 | 1                     | Moving Mode |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | C1                                                                                                                                                                                                                                                                                                                                                                                         | C0                                                | Function |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0                                                                                                                                                                                                                                                                                                                                                                                          | 0                                                 | Off      |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
| 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1                                                                                                                                                                                                                                                                                                                                                                                          | UI Mode                                           |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0                                                                                                                                                                                                                                                                                                                                                                                          | Still Mode                                        |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
| 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1                                                                                                                                                                                                                                                                                                                                                                                          | Moving Mode                                       |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
| CTE_LEVEL[3:0]: CTE function level setting                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                            |                                                   |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
| <table><tr><td></td><td>DESCRIPTION</td></tr><tr><td rowspan="6">CTE_LEVEL [3:0]</td><td>CTE function level</td></tr><tr><td>0 : CTE mode1 low</td><td>4 : CTE mode2 low</td></tr><tr><td>1 : CTE mode1 medium</td><td>5 : CTE mode2 low</td></tr><tr><td>2 : CTE mode1 high</td><td>6 : CTE mode2 low</td></tr><tr><td>8 : CTE auto mode low</td><td></td></tr><tr><td>9: CTE auto mode medium</td><td></td></tr><tr><td>A: CTE auto mode high</td><td></td></tr></table> |                                                                                                                                                                                                                                                                                                                                                                                            |                                                   |          |           |        |              |              |              |              |    |    |    |     | DESCRIPTION | CTE_LEVEL [3:0] | CTE function level                       | 0 : CTE mode1 low | 4 : CTE mode2 low                       | 1 : CTE mode1 medium | 5 : CTE mode2 low                         | 2 : CTE mode1 high | 6 : CTE mode2 low                        | 8 : CTE auto mode low |          | 9: CTE auto mode medium |   | A: CTE auto mode high |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                |                                                   |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
| CTE_LEVEL [3:0]                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CTE function level                                                                                                                                                                                                                                                                                                                                                                         |                                                   |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 0 : CTE mode1 low                                                                                                                                                                                                                                                                                                                                                                          | 4 : CTE mode2 low                                 |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1 : CTE mode1 medium                                                                                                                                                                                                                                                                                                                                                                       | 5 : CTE mode2 low                                 |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2 : CTE mode1 high                                                                                                                                                                                                                                                                                                                                                                         | 6 : CTE mode2 low                                 |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 8 : CTE auto mode low                                                                                                                                                                                                                                                                                                                                                                      |                                                   |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 9: CTE auto mode medium                                                                                                                                                                                                                                                                                                                                                                    |                                                   |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
| A: CTE auto mode high                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                            |                                                   |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
| Restriction                                                                                                                                                                                                                                                                                                                                                                                                                                                                | This register is synchronized with V-sync by internal circuit.                                                                                                                                                                                                                                                                                                                             |                                                   |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
| Register Availability                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <table><tr><td>Status</td><td>Availability</td></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |                                                   |          |           |        |              |              |              |              |    |    |    |     | Status      | Availability    | Normal Mode On, Idle Mode Off, Sleep Out | Yes               | Normal Mode On, Idle Mode On, Sleep Out | Yes                  | Partial Mode On, Idle Mode Off, Sleep Out | Yes                | Partial Mode On, Idle Mode On, Sleep Out | Yes                   | Sleep In | Yes                     |   |                       |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Status                                                                                                                                                                                                                                                                                                                                                                                     | Availability                                      |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                   | Yes                                               |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                    | Yes                                               |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                  | Yes                                               |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                   | Yes                                               |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
| Sleep In                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                                   |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
| Default                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <table><tr><td>Status</td><td>Default Value</td></tr><tr><td>Power On Sequence</td><td>48h</td></tr><tr><td>SW Reset</td><td>48h</td></tr><tr><td>HW Reset</td><td>48h</td></tr></table>                                                                                                                                                                                                   |                                                   |          |           |        |              |              |              |              |    |    |    |     | Status      | Default Value   | Power On Sequence                        | 48h               | SW Reset                                | 48h                  | HW Reset                                  | 48h                |                                          |                       |          |                         |   |                       |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Status                                                                                                                                                                                                                                                                                                                                                                                     | Default Value                                     |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Power On Sequence                                                                                                                                                                                                                                                                                                                                                                          | 48h                                               |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SW Reset                                                                                                                                                                                                                                                                                                                                                                                   | 48h                                               |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |
| HW Reset                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 48h                                                                                                                                                                                                                                                                                                                                                                                        |                                                   |          |           |        |              |              |              |              |    |    |    |     |             |                 |                                          |                   |                                         |                      |                                           |                    |                                          |                       |          |                         |   |                       |             |



CONFIDENTIAL

**RDCABC (5600h): Read Content Adaptive Brightness Control**

| 5600H                                     |                                                                                                                                                                                                                                                                                                                                                                                            | RDCABC(Read Content Adaptive Brightness Control) |        |           |        |                |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|--------|-----------|--------|----------------|----|----|----|----|----|----|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|------------|---|---|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address                                          |        | Parameter |        |                |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI                                             | Others | D15-D8    | D7     | D6             | D5 | D4 | D3 | D2 | D1 | D0 | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
| RDCABC                                    | R                                                                                                                                                                                                                                                                                                                                                                                          | 56h                                              | 5600h  | 00h       | CTE_EN | CTE_LEVEL[3:0] |    |    | 0  | C1 | C0 | 48 |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
| Description                               | This command is used to set parameters for image content based adaptive brightness control functionality. There is possible to use 4 different modes for content adaptive image functionality, which are defined on a table below.                                                                                                                                                         |                                                  |        |           |        |                |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
|                                           | <table><tr><td>C1</td><td>C0</td><td>Function</td></tr><tr><td>0</td><td>0</td><td>Off</td></tr><tr><td>0</td><td>1</td><td>UI Mode</td></tr><tr><td>1</td><td>0</td><td>Still Mode</td></tr><tr><td>1</td><td>1</td><td>Moving Mode</td></tr></table> <p>CTE_EN: Contrast Function Enable control.<br/>CTE_LEVEL[3:0]: Fast Mode setting for CTE.</p>                                     |                                                  |        |           |        |                |    |    |    |    |    |    |     | C1     | C0            | Function                                 | 0   | 0                                       | Off | 0                                         | 1   | UI Mode                                  | 1   | 0        | Still Mode | 1 | 1 |
| C1                                        | C0                                                                                                                                                                                                                                                                                                                                                                                         | Function                                         |        |           |        |                |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
| 0                                         | 0                                                                                                                                                                                                                                                                                                                                                                                          | Off                                              |        |           |        |                |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
| 0                                         | 1                                                                                                                                                                                                                                                                                                                                                                                          | UI Mode                                          |        |           |        |                |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
| 1                                         | 0                                                                                                                                                                                                                                                                                                                                                                                          | Still Mode                                       |        |           |        |                |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
| 1                                         | 1                                                                                                                                                                                                                                                                                                                                                                                          | Moving Mode                                      |        |           |        |                |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
| Restriction                               |                                                                                                                                                                                                                                                                                                                                                                                            |                                                  |        |           |        |                |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |                                                  |        |           |        |                |    |    |    |    |    |    |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes        |   |   |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                               |                                                  |        |           |        |                |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |        |           |        |                |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |        |           |        |                |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |        |           |        |                |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |        |           |        |                |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |        |           |        |                |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>48h</td></tr><tr><td>SW Reset</td><td>48h</td></tr><tr><td>HW Reset</td><td>48h</td></tr></table>                                                                                                                                                                                                   |                                                  |        |           |        |                |    |    |    |    |    |    |     | Status | Default Value | Power On Sequence                        | 48h | SW Reset                                | 48h | HW Reset                                  | 48h |                                          |     |          |            |   |   |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                              |                                                  |        |           |        |                |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
| Power On Sequence                         | 48h                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |        |           |        |                |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
| SW Reset                                  | 48h                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |        |           |        |                |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
| HW Reset                                  | 48h                                                                                                                                                                                                                                                                                                                                                                                        |                                                  |        |           |        |                |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |
| Flow Chart                                | <div><div><div>RDCABC (56h)</div><div>↓</div><div>ParameterC[1:0]</div></div><div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div></div>                                                                                                                                           |                                                  |        |           |        |                |    |    |    |    |    |    |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |            |   |   |

**WRCE (5800h): Set color enhance**

| 5800H                 | WRCE (set_color_enhance)                                                                                                                                                                                                                                                                                                                                                                   |              |                                     |           |               |                 |    |        |                                                                                                                                       |               |    |    |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|-------------------------------------|-----------|---------------|-----------------|----|--------|---------------------------------------------------------------------------------------------------------------------------------------|---------------|----|----|-----|--------|--------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|-----|
| Instruction           | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address      |                                     | Parameter |               |                 |    |        |                                                                                                                                       |               |    |    |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI         | Others                              | D15-D8    | D7            | D6              | D5 | D4     | D3                                                                                                                                    | D2            | D1 | D0 | HEX |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| WRCE                  | W                                                                                                                                                                                                                                                                                                                                                                                          | 58h          | 5800h                               | 00h       | EN_EDGE_E_ENH | EDGE_LEVEL[1:0] |    | SLR_EN | EN_VIVID_ENH                                                                                                                          | CE_LEVEL[2:0] |    |    | 21  |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Description           | Bit                                                                                                                                                                                                                                                                                                                                                                                        |              | Description                         |           |               |                 |    |        | Value                                                                                                                                 |               |    |    |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | EN_EDGE_ENH                                                                                                                                                                                                                                                                                                                                                                                |              | EDGE Enhancement function control   |           |               |                 |    |        | '0' : disable;<br>'1': enable                                                                                                         |               |    |    |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | EDGE_LEVEL[1:0]                                                                                                                                                                                                                                                                                                                                                                            |              | Edge enhancement fast mode setting  |           |               |                 |    |        | 0 :edge level low<br>1: edge level mediotm<br>2: edge levell high                                                                     |               |    |    |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | SLR_EN                                                                                                                                                                                                                                                                                                                                                                                     |              | Sun Light Readable Function control |           |               |                 |    |        | '0' : disable;<br>'1': enable                                                                                                         |               |    |    |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | EN_VIVID_ENH                                                                                                                                                                                                                                                                                                                                                                               |              | Vivid Color function control        |           |               |                 |    |        | '0' : disable;<br>'1': enable                                                                                                         |               |    |    |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | CE_LEVEL[2:0]                                                                                                                                                                                                                                                                                                                                                                              |              | Color Enhancement fast mode setting |           |               |                 |    |        | 0: CE level low<br>1: CE level medium<br>2: CE level high<br>4: CE+Skin level low<br>5: CE+Skin level medium<br>6: CE+Skin level high |               |    |    |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Restriction           | -                                                                                                                                                                                                                                                                                                                                                                                          |              |                                     |           |               |                 |    |        |                                                                                                                                       |               |    |    |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Register Availability | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |              |                                     |           |               |                 |    |        |                                                                                                                                       |               |    |    |     | Status | Availability | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                     | Availability |                                     |           |               |                 |    |        |                                                                                                                                       |               |    |    |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                   | Yes          |                                     |           |               |                 |    |        |                                                                                                                                       |               |    |    |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                    | Yes          |                                     |           |               |                 |    |        |                                                                                                                                       |               |    |    |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                  | Yes          |                                     |           |               |                 |    |        |                                                                                                                                       |               |    |    |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                   | Yes          |                                     |           |               |                 |    |        |                                                                                                                                       |               |    |    |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Sleep In              | Yes                                                                                                                                                                                                                                                                                                                                                                                        |              |                                     |           |               |                 |    |        |                                                                                                                                       |               |    |    |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Flow Chart            | <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>                                                                                                                                                                                                                              |              |                                     |           |               |                 |    |        |                                                                                                                                       |               |    |    |     |        |              |                                          |     |                                         |     |                                           |     |                                          |     |          |     |

**RDCE (5900h): Read color enhance**

| 5900H                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | RDCE (read_color_enhance)                                                                                                             |                                                                |           |             |                 |    |        |              |               |    |    |     |        |              |                                          |             |                                         |                              |                                           |                                    |                                                                |        |                                     |                              |              |                              |                              |               |                                     |                                                                                                                                       |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|-----------|-------------|-----------------|----|--------|--------------|---------------|----|----|-----|--------|--------------|------------------------------------------|-------------|-----------------------------------------|------------------------------|-------------------------------------------|------------------------------------|----------------------------------------------------------------|--------|-------------------------------------|------------------------------|--------------|------------------------------|------------------------------|---------------|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Instruction           | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Address                                                                                                                               |                                                                | Parameter |             |                 |    |        |              |               |    |    |     |        |              |                                          |             |                                         |                              |                                           |                                    |                                                                |        |                                     |                              |              |                              |                              |               |                                     |                                                                                                                                       |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | MIPI                                                                                                                                  | Others                                                         | D15-D8    | D7          | D6              | D5 | D4     | D3           | D2            | D1 | D0 | HEX |        |              |                                          |             |                                         |                              |                                           |                                    |                                                                |        |                                     |                              |              |                              |                              |               |                                     |                                                                                                                                       |
| RDCE                  | R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 59h                                                                                                                                   | 5900h                                                          | 00h       | EN_EDGE_ENH | EDGE_LEVEL[1:0] |    | SLR_EN | EN_VIVID_ENH | CE_LEVEL[2:0] |    | 00 |     |        |              |                                          |             |                                         |                              |                                           |                                    |                                                                |        |                                     |                              |              |                              |                              |               |                                     |                                                                                                                                       |
| Description           | <table><tr><th>Bit</th><th>Description</th><th>Value</th></tr><tr><td>EN_EDGE_ENH</td><td>EDGE Enhancement function control</td><td>'0': disable;<br/>'1': enable</td></tr><tr><td>EDGE_LEVEL[1:0]</td><td>Edge enhancement fast mode setting</td><td>0 :edge level low<br/>1: edge level mediu<br/>2: edge level high</td></tr><tr><td>SLR_EN</td><td>Sun Light Readable Function control</td><td>'0': disable;<br/>'1': enable</td></tr><tr><td>EN_VIVID_ENH</td><td>Vivid Color function control</td><td>'0': disable;<br/>'1': enable</td></tr><tr><td>CE_LEVEL[2:0]</td><td>Color Enhancement fast mode setting</td><td>0: CE level low<br/>1: CE level medium<br/>2: CE level high<br/>4: CE+Skin level low<br/>5: CE+Skin level medium<br/>6: CE+Skin level high</td></tr></table> |                                                                                                                                       |                                                                |           |             |                 |    |        |              |               |    |    |     | Bit    | Description  | Value                                    | EN_EDGE_ENH | EDGE Enhancement function control       | '0': disable;<br>'1': enable | EDGE_LEVEL[1:0]                           | Edge enhancement fast mode setting | 0 :edge level low<br>1: edge level mediu<br>2: edge level high | SLR_EN | Sun Light Readable Function control | '0': disable;<br>'1': enable | EN_VIVID_ENH | Vivid Color function control | '0': disable;<br>'1': enable | CE_LEVEL[2:0] | Color Enhancement fast mode setting | 0: CE level low<br>1: CE level medium<br>2: CE level high<br>4: CE+Skin level low<br>5: CE+Skin level medium<br>6: CE+Skin level high |
|                       | Bit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Description                                                                                                                           | Value                                                          |           |             |                 |    |        |              |               |    |    |     |        |              |                                          |             |                                         |                              |                                           |                                    |                                                                |        |                                     |                              |              |                              |                              |               |                                     |                                                                                                                                       |
|                       | EN_EDGE_ENH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | EDGE Enhancement function control                                                                                                     | '0': disable;<br>'1': enable                                   |           |             |                 |    |        |              |               |    |    |     |        |              |                                          |             |                                         |                              |                                           |                                    |                                                                |        |                                     |                              |              |                              |                              |               |                                     |                                                                                                                                       |
|                       | EDGE_LEVEL[1:0]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Edge enhancement fast mode setting                                                                                                    | 0 :edge level low<br>1: edge level mediu<br>2: edge level high |           |             |                 |    |        |              |               |    |    |     |        |              |                                          |             |                                         |                              |                                           |                                    |                                                                |        |                                     |                              |              |                              |                              |               |                                     |                                                                                                                                       |
|                       | SLR_EN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Sun Light Readable Function control                                                                                                   | '0': disable;<br>'1': enable                                   |           |             |                 |    |        |              |               |    |    |     |        |              |                                          |             |                                         |                              |                                           |                                    |                                                                |        |                                     |                              |              |                              |                              |               |                                     |                                                                                                                                       |
|                       | EN_VIVID_ENH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Vivid Color function control                                                                                                          | '0': disable;<br>'1': enable                                   |           |             |                 |    |        |              |               |    |    |     |        |              |                                          |             |                                         |                              |                                           |                                    |                                                                |        |                                     |                              |              |                              |                              |               |                                     |                                                                                                                                       |
| CE_LEVEL[2:0]         | Color Enhancement fast mode setting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0: CE level low<br>1: CE level medium<br>2: CE level high<br>4: CE+Skin level low<br>5: CE+Skin level medium<br>6: CE+Skin level high |                                                                |           |             |                 |    |        |              |               |    |    |     |        |              |                                          |             |                                         |                              |                                           |                                    |                                                                |        |                                     |                              |              |                              |                              |               |                                     |                                                                                                                                       |
| Restriction           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                       |                                                                |           |             |                 |    |        |              |               |    |    |     |        |              |                                          |             |                                         |                              |                                           |                                    |                                                                |        |                                     |                              |              |                              |                              |               |                                     |                                                                                                                                       |
| Register Availability | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                       |                                                                |           |             |                 |    |        |              |               |    |    |     | Status | Availability | Normal Mode On, Idle Mode Off, Sleep Out | Yes         | Normal Mode On, Idle Mode On, Sleep Out | Yes                          | Partial Mode On, Idle Mode Off, Sleep Out | Yes                                | Partial Mode On, Idle Mode On, Sleep Out                       | Yes    | Sleep In                            | Yes                          |              |                              |                              |               |                                     |                                                                                                                                       |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Availability                                                                                                                          |                                                                |           |             |                 |    |        |              |               |    |    |     |        |              |                                          |             |                                         |                              |                                           |                                    |                                                                |        |                                     |                              |              |                              |                              |               |                                     |                                                                                                                                       |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Yes                                                                                                                                   |                                                                |           |             |                 |    |        |              |               |    |    |     |        |              |                                          |             |                                         |                              |                                           |                                    |                                                                |        |                                     |                              |              |                              |                              |               |                                     |                                                                                                                                       |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Yes                                                                                                                                   |                                                                |           |             |                 |    |        |              |               |    |    |     |        |              |                                          |             |                                         |                              |                                           |                                    |                                                                |        |                                     |                              |              |                              |                              |               |                                     |                                                                                                                                       |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Yes                                                                                                                                   |                                                                |           |             |                 |    |        |              |               |    |    |     |        |              |                                          |             |                                         |                              |                                           |                                    |                                                                |        |                                     |                              |              |                              |                              |               |                                     |                                                                                                                                       |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Yes                                                                                                                                   |                                                                |           |             |                 |    |        |              |               |    |    |     |        |              |                                          |             |                                         |                              |                                           |                                    |                                                                |        |                                     |                              |              |                              |                              |               |                                     |                                                                                                                                       |
| Sleep In              | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                       |                                                                |           |             |                 |    |        |              |               |    |    |     |        |              |                                          |             |                                         |                              |                                           |                                    |                                                                |        |                                     |                              |              |                              |                              |               |                                     |                                                                                                                                       |
| Flow Chart            | <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                                       |                                                                |           |             |                 |    |        |              |               |    |    |     |        |              |                                          |             |                                         |                              |                                           |                                    |                                                                |        |                                     |                              |              |                              |                              |               |                                     |                                                                                                                                       |

**WRCABCMB (5E00h): Write CABC Minimum Brightness**

| 5E00H                                     |                                                                                                                                                                                                                                                                                                                                                                                            | WRCABCMB(Write CABC Minimum Brightness) |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|--------|-----------|------|------|------|------|------|------|------|------|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address                                 |        | Parameter |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI                                    | Others | D15-D8    | D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| WRCABCMB                                  | W                                                                                                                                                                                                                                                                                                                                                                                          | 5Eh                                     | 5E00h  | 00h       | CMB7 | CMB6 | CMB5 | CMB4 | CMB3 | CMB2 | CMB1 | CMB0 | 00  |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Description                               | This command is used to set the minimum brightness value of the display for CABC function. In principle relationship is that 00h value means the lowest brightness for CABC and 0xFFh value means the highest brightness for CABC.                                                                                                                                                         |                                         |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Restriction                               | -                                                                                                                                                                                                                                                                                                                                                                                          |                                         |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |                                         |        |           |      |      |      |      |      |      |      |      |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                               |                                         |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                         |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                         |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                         |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                         |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                         |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>SW Reset</td><td>00h</td></tr><tr><td>HW Reset</td><td>00h</td></tr></table>                                                                                                                                                                                                   |                                         |        |           |      |      |      |      |      |      |      |      |     | Status | Default Value | Power On Sequence                        | 00h | SW Reset                                | 00h | HW Reset                                  | 00h |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                              |                                         |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Power On Sequence                         | 00h                                                                                                                                                                                                                                                                                                                                                                                        |                                         |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| SW Reset                                  | 00h                                                                                                                                                                                                                                                                                                                                                                                        |                                         |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| HW Reset                                  | 00h                                                                                                                                                                                                                                                                                                                                                                                        |                                         |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Flow chart                                | <div><div><div>WRCABCMB(5Eh)</div><div>Parameter CMB[7:0]</div><div>New Display Luminance Value Loaded</div></div><div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div></div>                                                                                                      |                                         |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |

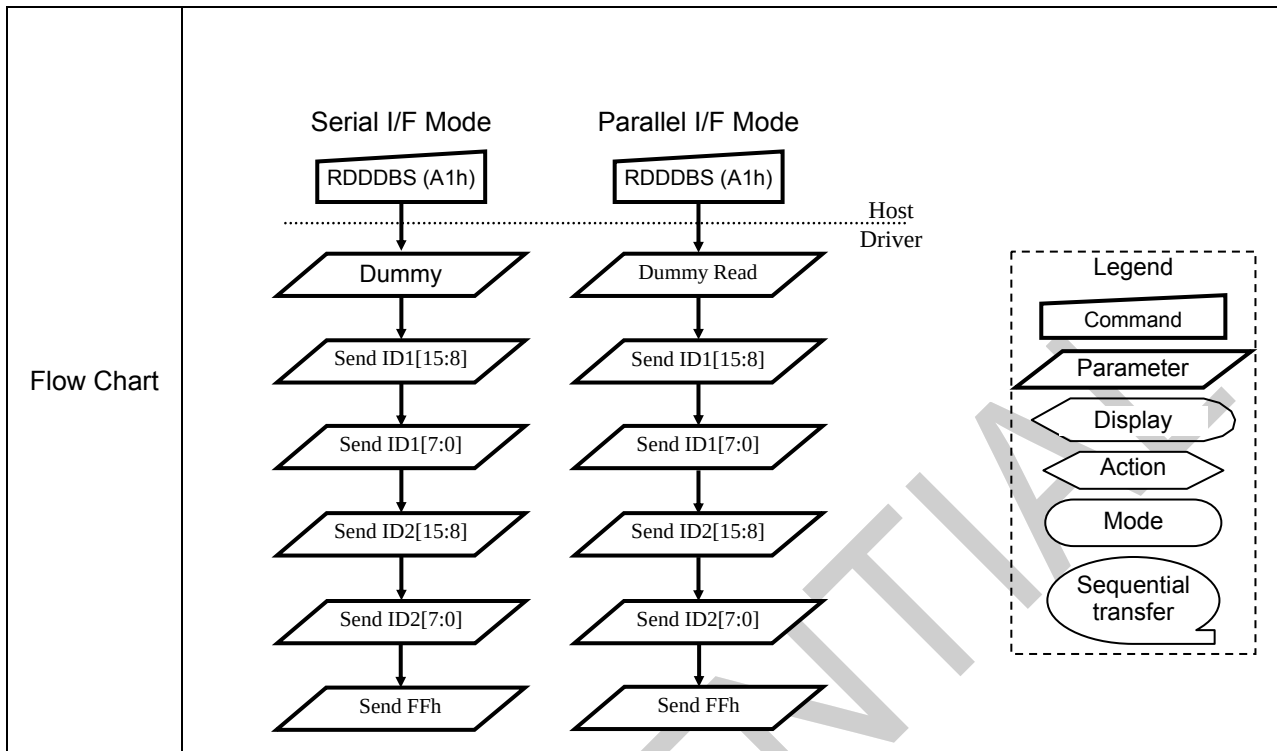


**RDCABCMB (5F00h): Read CABC Minimum Brightness**

| 5F00H                                     |                                                                                                                                                                                                                                                                                                                                                                                                                          | RDCABCMB(Read CABC Minimum Brightness) |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------|--------|-----------|------|------|------|------|------|------|------|------|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                                                      | Address                                |        | Parameter |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                          | MIPI                                   | Others | D15-D8    | D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| RDCABCMB                                  | R                                                                                                                                                                                                                                                                                                                                                                                                                        | 5Fh                                    | 5F00h  | 00h       | CMB7 | CMB6 | CMB5 | CMB4 | CMB3 | CMB2 | CMB1 | CMB0 | 00  |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Description                               | This command returns the minimum brightness value of CABC function. In principle relationship is that 00h value means the lowest brightness for CABC and FFh value means the highest brightness for CABC. CMB[7:0] is minimum brightness for CABC specified with "WRCABCMB Write CABC minimum brightness (0x5Eh)" command.                                                                                               |                                        |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Restriction                               | -                                                                                                                                                                                                                                                                                                                                                                                                                        |                                        |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Register Availability                     | <table><thead><tr><th>Status</th><th>Availability</th></tr></thead><tbody><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></tbody></table> |                                        |        |           |      |      |      |      |      |      |      |      |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                             |                                        |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |                                        |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |                                        |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |                                        |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |                                        |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                      |                                        |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Default                                   | <table><thead><tr><th>Status</th><th>Default Value</th></tr></thead><tbody><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>SW Reset</td><td>00h</td></tr><tr><td>HW Reset</td><td>00h</td></tr></tbody></table>                                                                                                                                                                                                   |                                        |        |           |      |      |      |      |      |      |      |      |     | Status | Default Value | Power On Sequence                        | 00h | SW Reset                                | 00h | HW Reset                                  | 00h |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                            |                                        |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Power On Sequence                         | 00h                                                                                                                                                                                                                                                                                                                                                                                                                      |                                        |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| SW Reset                                  | 00h                                                                                                                                                                                                                                                                                                                                                                                                                      |                                        |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| HW Reset                                  | 00h                                                                                                                                                                                                                                                                                                                                                                                                                      |                                        |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Flow chart                                | <div><div><div>RDCABCMB(5Fh)</div><div>↓</div><div>Send Parameter CMB[7:0]</div></div><div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div></div>                                                                                                                                                                |                                        |        |           |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |

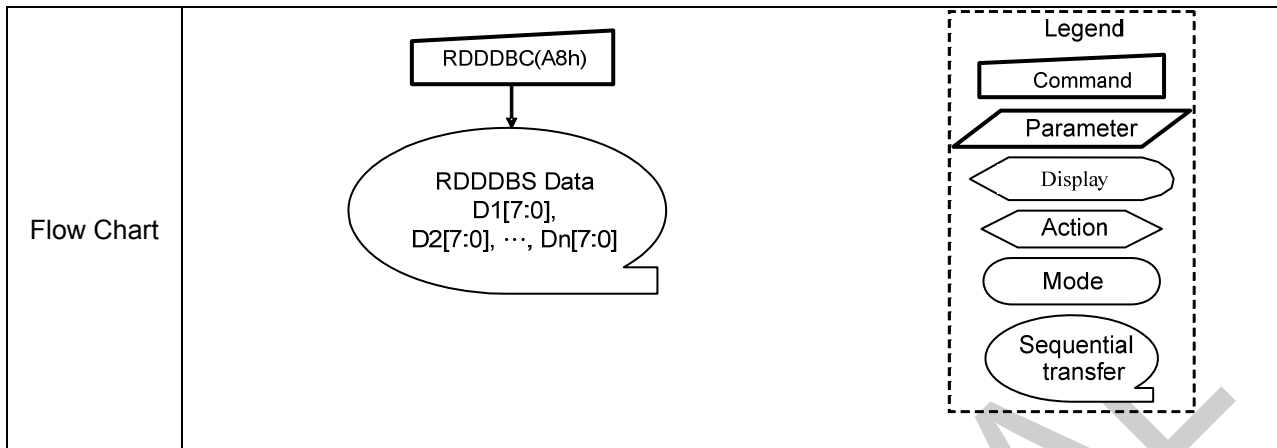
**RDDDBS(A100h) : Read\_DDB\_Start**

| A100H                                     |                                                                                                                                                                                                                                                                                                                                                                                            | RDDDBS(Read_DDB_Start)                                                                                                         |       |       |         |         |         |         |         |         |         |         |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|-------|-------|---------|---------|---------|---------|---------|---------|---------|---------|-----|--------|---------------|------------------------------------------|-----------|-----------------------------------------|-------------------|-------------------------------------------|-----|------------------------------------------|-----------|----------|----------|-----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address                                                                                                                        |       | D15-8 | D7      | D6      | D5      | D4      | D3      | D2      | D1      | D0      | HEX |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI                                                                                                                           | Other |       |         |         |         |         |         |         |         |         |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| RDDDBS                                    | R                                                                                                                                                                                                                                                                                                                                                                                          | A1h                                                                                                                            | A100h | x     | SID[7]  | SID [6] | SID [5] | SID [4] | SID [3] | SID [2] | SID [1] | SID [0] | 00  |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                | A101h | x     | SID[15] | SID[14] | SID[13] | SID[12] | SID[11] | SID[10] | SID[9]  | SID[8]  | 00  |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                | A102h | x     | MID[7]  | MID[6]  | MID[5]  | MID[4]  | MID[3]  | MID[2]  | MID[1]  | MID[0]  | 00  |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                | A103h | x     | MID[15] | MID[14] | MID[13] | MID[12] | MID[11] | MID[10] | MID[9]  | MID[8]  | 00  |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                | A104h | x     | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1   | FF     |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Description                               | 1 <sup>st</sup><br>2 <sup>nd</sup><br>3 <sup>rd</sup><br>4 <sup>th</sup><br>5 <sup>th</sup>                                                                                                                                                                                                                                                                                                | parameter: Supplier ID code<br>parameter: Supplier ID code<br>parameter: Module ID<br>parameter: Module ID<br>Exit code (FFh). |       |       |         |         |         |         |         |         |         |         |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Restriction                               |                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                |       |       |         |         |         |         |         |         |         |         |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |                                                                                                                                |       |       |         |         |         |         |         |         |         |         |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes       | Normal Mode On, Idle Mode On, Sleep Out | Yes               | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes       | Sleep In | Yes      |           |     |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                |       |       |         |         |         |         |         |         |         |         |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                |       |       |         |         |         |         |         |         |         |         |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                |       |       |         |         |         |         |         |         |         |         |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                |       |       |         |         |         |         |         |         |         |         |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                |       |       |         |         |         |         |         |         |         |         |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                |       |       |         |         |         |         |         |         |         |         |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Default                                   | <table><tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr><tr><th>After MTP</th><th>Before MTP</th></tr><tr><td>Power On Sequence</td><td>MTP Value</td><td>00h</td></tr><tr><td>SW Reset</td><td>MTP Value</td><td>00h</td></tr><tr><td>HW Reset</td><td>MTP Value</td><td>00h</td></tr></table>                                                                       |                                                                                                                                |       |       |         |         |         |         |         |         |         |         |     | Status | Default Value |                                          | After MTP | Before MTP                              | Power On Sequence | MTP Value                                 | 00h | SW Reset                                 | MTP Value | 00h      | HW Reset | MTP Value | 00h |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                |       |       |         |         |         |         |         |         |         |         |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                                           | After MTP                                                                                                                                                                                                                                                                                                                                                                                  | Before MTP                                                                                                                     |       |       |         |         |         |         |         |         |         |         |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Power On Sequence                         | MTP Value                                                                                                                                                                                                                                                                                                                                                                                  | 00h                                                                                                                            |       |       |         |         |         |         |         |         |         |         |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| SW Reset                                  | MTP Value                                                                                                                                                                                                                                                                                                                                                                                  | 00h                                                                                                                            |       |       |         |         |         |         |         |         |         |         |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| HW Reset                                  | MTP Value                                                                                                                                                                                                                                                                                                                                                                                  | 00h                                                                                                                            |       |       |         |         |         |         |         |         |         |         |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |



**RDDDBC(A800h): Read DDB Continuous**

| A800H                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            | RDDDBC |       |    |         |         |         |         |         |         |         |         |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------|-------|----|---------|---------|---------|---------|---------|---------|---------|---------|--------|---------------|------------------------------------------|-----------|-----------------------------------------|-------------------|-------------------------------------------|-----|------------------------------------------|-----------|----------|----------|-----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Address    |        | D15-8 | D7 | D6      | D5      | D4      | D3      | D2      | D1      | D0      | HEX     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | MIPI       | Other  |       |    |         |         |         |         |         |         |         |         |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| RDDDBC                                    | R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | A8h        | A800h  | x     | x  | SID[7]  | SID [6] | SID [5] | SID [4] | SID [3] | SID [2] | SID [1] | SID [0] |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            | A801h  | x     | x  | SID[15] | SID[14] | SID[13] | SID[12] | SID[11] | SID[10] | SID[9]  | SID[8]  |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            | A802h  | x     | x  | MID[7]  | MID[6]  | MID[5]  | MID[4]  | MID[3]  | MID[2]  | MID[1]  | MID[0]  |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            | A803h  | x     | x  | MID[15] | MID[14] | MID[13] | MID[12] | MID[11] | MID[10] | MID[9]  | MID[8]  |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |            | A804h  | x     | x  | 1       | 1       | 1       | 1       | 1       | 1       | 1       | 1       |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Description                               | <p>This command returns the supplier identification and display module mode/revision information from the point where RDDDBS command was interrupted by another command.<br/><i>Note: Parameter 0xFF is an “Exit Code”, this means that there is no more data in the DDB block.</i><br/><i>Note: For use example,</i><br/>1. Set maximum return packet size=3<br/>2. Read 0xA1, return 3 bytes SID[7:0], SID[15:8], MID[7:0]<br/>3. Read 0xA8, return 2 bytes MID[15:8],RID[7:0], RID[15:8] and 0xFF</p> |            |        |       |    |         |         |         |         |         |         |         |         |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Restriction                               | A Read DDB Start command (RDDDBS) should be executed at least once before a Read DDB Continue command (RDDDBC) to define the read location. Otherwise, data read with a Read DDB Continue command is undefined.                                                                                                                                                                                                                                                                                          |            |        |       |    |         |         |         |         |         |         |         |         |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table>                                                                                                               |            |        |       |    |         |         |         |         |         |         |         |         | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes       | Normal Mode On, Idle Mode On, Sleep Out | Yes               | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes       | Sleep In | Yes      |           |     |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |            |        |       |    |         |         |         |         |         |         |         |         |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |        |       |    |         |         |         |         |         |         |         |         |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |        |       |    |         |         |         |         |         |         |         |         |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |        |       |    |         |         |         |         |         |         |         |         |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |        |       |    |         |         |         |         |         |         |         |         |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |            |        |       |    |         |         |         |         |         |         |         |         |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Default                                   | <table><tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr><tr><th>After MTP</th><th>Before MTP</th></tr><tr><td>Power On Sequence</td><td>MTP Value</td><td>00h</td></tr><tr><td>SW Reset</td><td>MTP Value</td><td>00h</td></tr><tr><td>HW Reset</td><td>MTP Value</td><td>00h</td></tr></table>                                                                                                                                                                                     |            |        |       |    |         |         |         |         |         |         |         |         | Status | Default Value |                                          | After MTP | Before MTP                              | Power On Sequence | MTP Value                                 | 00h | SW Reset                                 | MTP Value | 00h      | HW Reset | MTP Value | 00h |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |        |       |    |         |         |         |         |         |         |         |         |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                                           | After MTP                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Before MTP |        |       |    |         |         |         |         |         |         |         |         |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Power On Sequence                         | MTP Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 00h        |        |       |    |         |         |         |         |         |         |         |         |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| SW Reset                                  | MTP Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 00h        |        |       |    |         |         |         |         |         |         |         |         |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| HW Reset                                  | MTP Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 00h        |        |       |    |         |         |         |         |         |         |         |         |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |



CONFIDENTIAL

**RDFCS(AA00h): Read First Checksum**

| AA00H                                     |                                                                                                                                                                                                                                                                                                                                                                                            |         |       | RDFCS |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|-------|------|------|------|------|------|------|------|------|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address |       | D15-8 | D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI    | Other |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| RDFCS                                     | R                                                                                                                                                                                                                                                                                                                                                                                          | AAh     | AA00h | x     | FCS7 | FCS6 | FCS5 | FCS4 | FCS3 | FCS2 | FCS1 | FCS0 | 00  |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Description                               | This command returns the first checksum what has been calculated from “User Command Set” area registers (not include “Manufacture Command Set) and the frame memory after the write access to those registers and/or frame memory has been done.                                                                                                                                           |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Restriction                               | It will be necessary to wait 150ms after there is the last write access on “User Command Set” area registers before there can read this checksum value.                                                                                                                                                                                                                                    |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |         |       |       |      |      |      |      |      |      |      |      |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                               |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>S/W Reset</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table>                                                                                                                                                                                                 |         |       |       |      |      |      |      |      |      |      |      |     | Status | Default Value | Power On Sequence                        | 00h | S/W Reset                               | 00h | H/W Reset                                 | 00h |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                              |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Power On Sequence                         | 00h                                                                                                                                                                                                                                                                                                                                                                                        |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| S/W Reset                                 | 00h                                                                                                                                                                                                                                                                                                                                                                                        |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| H/W Reset                                 | 00h                                                                                                                                                                                                                                                                                                                                                                                        |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Flow Chart                                | <div><div><div>RDFCS(AAh)</div><div>Send Parameter FCS[7:0]</div></div><div><div>Legend</div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>                                                                                                                                                            |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |

**RDCCS(AF00h): Read Continue Checksum**

| AF00H                                     |                                                                                                                                                                                                                                                                                                                                                                                            |         |       | RDCCS |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-------|-------|------|------|------|------|------|------|------|------|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address |       | D15-8 | D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI    | Other |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| RDCCS                                     | R                                                                                                                                                                                                                                                                                                                                                                                          | AFh     | AF00h | x     | CCS7 | CCS6 | CCS5 | CCS4 | CCS3 | CCS2 | CCS1 | CCS0 | 00  |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Description                               | This command returns the continue checksum what has been calculated continuously after the first checksum has calculated from “User Command Set” area registers and the frame memory after the write access to those registers and/or frame memory has been done.                                                                                                                          |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Restriction                               | It will be necessary to wait 300ms after there is the last write access on “User Command Set” area registers before there can read this checksum value in the first time.                                                                                                                                                                                                                  |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |         |       |       |      |      |      |      |      |      |      |      |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                               |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Default                                   | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>S/W Reset</td><td>00h</td></tr><tr><td>H/W Reset</td><td>00h</td></tr></table>                                                                                                                                                                                                 |         |       |       |      |      |      |      |      |      |      |      |     | Status | Default Value | Power On Sequence                        | 00h | S/W Reset                               | 00h | H/W Reset                                 | 00h |                                          |     |          |     |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                              |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Power On Sequence                         | 00h                                                                                                                                                                                                                                                                                                                                                                                        |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| S/W Reset                                 | 00h                                                                                                                                                                                                                                                                                                                                                                                        |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| H/W Reset                                 | 00h                                                                                                                                                                                                                                                                                                                                                                                        |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Flow Chart                                | <div><div><div>RDCCS(AFh)</div><div>Send Parameter CCS[7:0]</div></div><div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div></div>                                                                                                                                                 |         |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |

**RDID1(DA00h) : Read ID1**

| DA00H                 |                                                                                                                                                                                                                                                                                                                                                                                            |               |       | RDID1 |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------|-------|------|------|------|------|------|------|------|------|-----|--------|---------------|------------------------------------------|-----|-----------------------------------------|-----|-------------------------------------------|-----|------------------------------------------|-----|----------|-----|
| Instruction           | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address       |       | D15-8 | D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   | HEX |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI          | Other |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| RDID1                 | R                                                                                                                                                                                                                                                                                                                                                                                          | DAh           | DA00h | x     | ID17 | ID16 | ID15 | ID14 | ID13 | ID12 | ID11 | ID10 | 00  |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Description           | This read byte identifies Module's manufacture ID.                                                                                                                                                                                                                                                                                                                                         |               |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Restriction           | -                                                                                                                                                                                                                                                                                                                                                                                          |               |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Register Availability | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |               |       |       |      |      |      |      |      |      |      |      |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes | Normal Mode On, Idle Mode On, Sleep Out | Yes | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes | Sleep In | Yes |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                     | Availability  |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                   | Yes           |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                    | Yes           |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                  | Yes           |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                   | Yes           |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Sleep In              | Yes                                                                                                                                                                                                                                                                                                                                                                                        |               |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Default               | <table><tr><th>Status</th><th>Default Value</th></tr><tr><td>Power On Sequence</td><td>00h</td></tr><tr><td>SW Reset</td><td>00h</td></tr><tr><td>HW Reset</td><td>00h</td></tr></table>                                                                                                                                                                                                   |               |       |       |      |      |      |      |      |      |      |      |     | Status | Default Value | Power On Sequence                        | 00h | SW Reset                                | 00h | HW Reset                                  | 00h |                                          |     |          |     |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                     | Default Value |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | Power On Sequence                                                                                                                                                                                                                                                                                                                                                                          | 00h           |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
|                       | SW Reset                                                                                                                                                                                                                                                                                                                                                                                   | 00h           |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| HW Reset              | 00h                                                                                                                                                                                                                                                                                                                                                                                        |               |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |
| Flow Chart            | <div><div>RDID1(DAh)</div><div>Send Parameter ID1[7:0]</div></div> <div><div>Legend</div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div>                                                                                                                                                                      |               |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |     |                                         |     |                                           |     |                                          |     |          |     |



**RDID2(DB00h) : Read ID2**

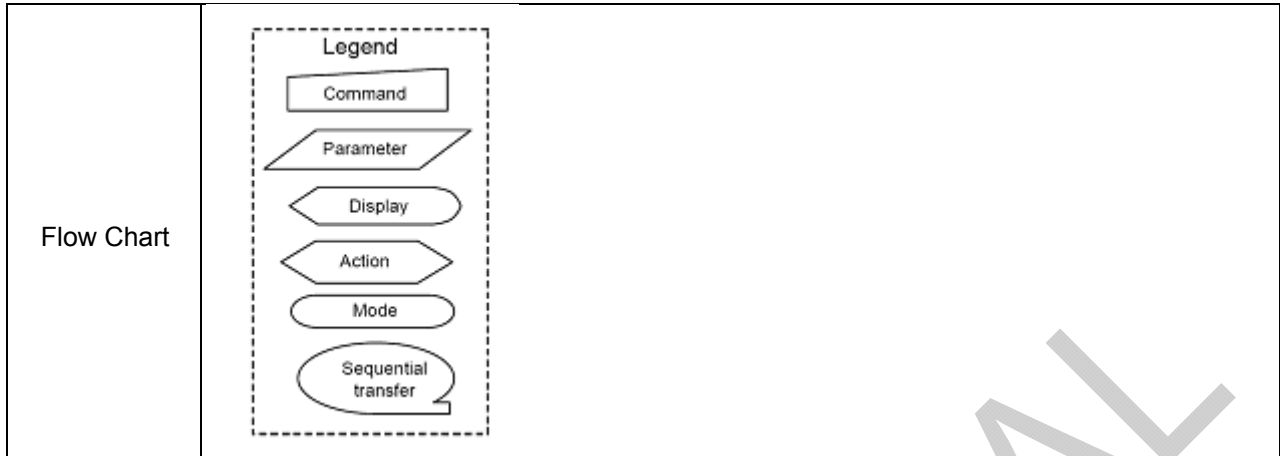
| DB00H                                     |                                                                                                                                                                                                                                                                                                                                                                                            |            |       | RDID2 |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|-------|------|------|------|------|------|------|------|------|-----|--------|---------------|------------------------------------------|-----------|-----------------------------------------|-------------------|-------------------------------------------|-----|------------------------------------------|-----------|----------|----------|-----------|-----|
| Instruction                               | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address    |       | D15-8 | D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   | HEX |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                                           |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI       | Other |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| RDID2                                     | R                                                                                                                                                                                                                                                                                                                                                                                          | DBh        | DB00h | x     | ID27 | ID26 | ID25 | ID24 | ID23 | ID22 | ID21 | ID20 | 80  |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Description                               | This read byte is used to track the Module/driver version. It is changed each time a version is made to the display, material or construction specifications.                                                                                                                                                                                                                              |            |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Restriction                               | Parameter Range: ID2 = 0x80h to 0xFFh                                                                                                                                                                                                                                                                                                                                                      |            |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Register Availability                     | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |            |       |       |      |      |      |      |      |      |      |      |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes       | Normal Mode On, Idle Mode On, Sleep Out | Yes               | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes       | Sleep In | Yes      |           |     |
| Status                                    | Availability                                                                                                                                                                                                                                                                                                                                                                               |            |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Normal Mode On, Idle Mode Off, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |            |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Normal Mode On, Idle Mode On, Sleep Out   | Yes                                                                                                                                                                                                                                                                                                                                                                                        |            |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Partial Mode On, Idle Mode Off, Sleep Out | Yes                                                                                                                                                                                                                                                                                                                                                                                        |            |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Partial Mode On, Idle Mode On, Sleep Out  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |            |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Sleep In                                  | Yes                                                                                                                                                                                                                                                                                                                                                                                        |            |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Default                                   | <table><tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr><tr><th>After MTP</th><th>Before MTP</th></tr><tr><td>Power On Sequence</td><td>MTP Value</td><td>80h</td></tr><tr><td>SW Reset</td><td>MTP Value</td><td>80h</td></tr><tr><td>HW Reset</td><td>MTP Value</td><td>80h</td></tr></table>                                                                       |            |       |       |      |      |      |      |      |      |      |      |     | Status | Default Value |                                          | After MTP | Before MTP                              | Power On Sequence | MTP Value                                 | 80h | SW Reset                                 | MTP Value | 80h      | HW Reset | MTP Value | 80h |
| Status                                    | Default Value                                                                                                                                                                                                                                                                                                                                                                              |            |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                                           | After MTP                                                                                                                                                                                                                                                                                                                                                                                  | Before MTP |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Power On Sequence                         | MTP Value                                                                                                                                                                                                                                                                                                                                                                                  | 80h        |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| SW Reset                                  | MTP Value                                                                                                                                                                                                                                                                                                                                                                                  | 80h        |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| HW Reset                                  | MTP Value                                                                                                                                                                                                                                                                                                                                                                                  | 80h        |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Flow Chart                                | <div><div>RDID2(DBh)</div><div>Send Parameter ID2[7:0]</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>                                                                                                                                                           |            |       |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |

**RDID3(DC00h) : Read ID3**

| DC00H                 |                                                                                                                                                                                                                                                                                                                                                                                            |               |            | RDID3 |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------|-------|------|------|------|------|------|------|------|------|-----|--------|---------------|------------------------------------------|-----------|-----------------------------------------|-------------------|-------------------------------------------|-----|------------------------------------------|-----------|----------|----------|-----------|-----|
| Instruction           | R/W                                                                                                                                                                                                                                                                                                                                                                                        | Address       |            | D15-8 | D7   | D6   | D5   | D4   | D3   | D2   | D1   | D0   | HEX |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                            | MIPI          | Other      |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| RDID3                 | R                                                                                                                                                                                                                                                                                                                                                                                          | DCh           | DC00h      | x     | ID37 | ID36 | ID35 | ID34 | ID33 | ID32 | ID31 | ID30 | 00  |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Description           | This parameter read byte identifies Module/driver.                                                                                                                                                                                                                                                                                                                                         |               |            |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Restriction           | -                                                                                                                                                                                                                                                                                                                                                                                          |               |            |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Register Availability | <table><tr><th>Status</th><th>Availability</th></tr><tr><td>Normal Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Normal Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode Off, Sleep Out</td><td>Yes</td></tr><tr><td>Partial Mode On, Idle Mode On, Sleep Out</td><td>Yes</td></tr><tr><td>Sleep In</td><td>Yes</td></tr></table> |               |            |       |      |      |      |      |      |      |      |      |     | Status | Availability  | Normal Mode On, Idle Mode Off, Sleep Out | Yes       | Normal Mode On, Idle Mode On, Sleep Out | Yes               | Partial Mode On, Idle Mode Off, Sleep Out | Yes | Partial Mode On, Idle Mode On, Sleep Out | Yes       | Sleep In | Yes      |           |     |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                     | Availability  |            |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                   | Yes           |            |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                    | Yes           |            |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                  | Yes           |            |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                   | Yes           |            |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                       | Sleep In                                                                                                                                                                                                                                                                                                                                                                                   | Yes           |            |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Default               | <table><tr><th rowspan="2">Status</th><th colspan="2">Default Value</th></tr><tr><th>After MTP</th><th>Before MTP</th></tr><tr><td>Power On Sequence</td><td>MTP Value</td><td>00h</td></tr><tr><td>SW Reset</td><td>MTP Value</td><td>00h</td></tr><tr><td>HW Reset</td><td>MTP Value</td><td>00h</td></tr></table>                                                                       |               |            |       |      |      |      |      |      |      |      |      |     | Status | Default Value |                                          | After MTP | Before MTP                              | Power On Sequence | MTP Value                                 | 00h | SW Reset                                 | MTP Value | 00h      | HW Reset | MTP Value | 00h |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                     | Default Value |            |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                       |                                                                                                                                                                                                                                                                                                                                                                                            | After MTP     | Before MTP |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                       | Power On Sequence                                                                                                                                                                                                                                                                                                                                                                          | MTP Value     | 00h        |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                       | SW Reset                                                                                                                                                                                                                                                                                                                                                                                   | MTP Value     | 00h        |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
|                       | HW Reset                                                                                                                                                                                                                                                                                                                                                                                   | MTP Value     | 00h        |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |
| Flow Chart            | <div><div>RDID3(DCh)</div><div>↓</div><div>Send Parameter ID3[7:0]</div></div> <div><div>Legend</div><div><div>Command</div><div>Parameter</div><div>Display</div><div>Action</div><div>Mode</div><div>Sequential transfer</div></div></div>                                                                                                                                               |               |            |       |      |      |      |      |      |      |      |      |     |        |               |                                          |           |                                         |                   |                                           |     |                                          |           |          |          |           |     |

**CMD\_Page (FE00h): CMD Mode Switch**

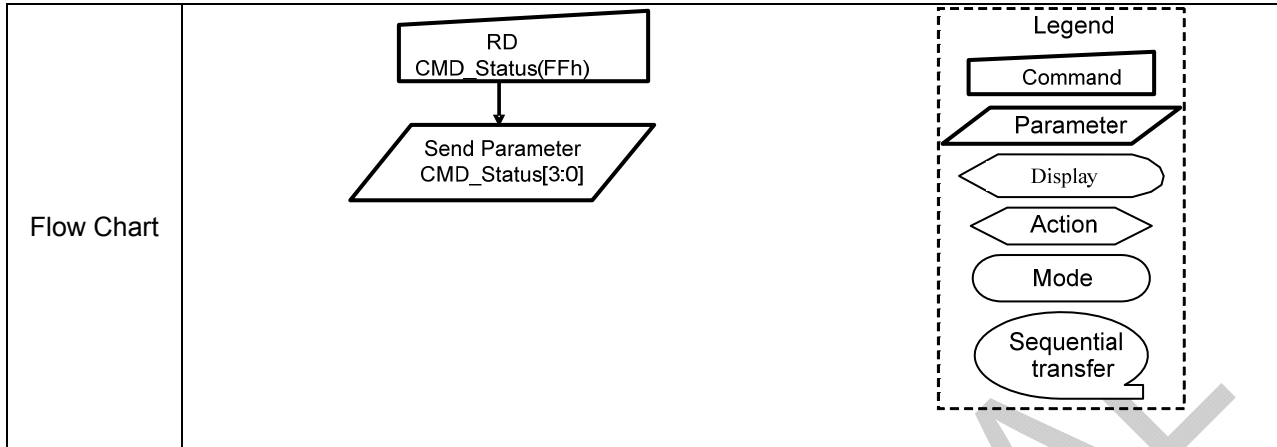
| FE00H                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | MAUCCTR (Manufacture Command Set Control) |               |               |                                          |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|---------------|---------------|------------------------------------------|----|----|----|---------------|----|----|----|-----|--------|--|--|--|--------------|--|--|--|------------------------------------------|--|--|--|-----|--|--|--|-----------------------------------------|--|--|--|-----|--|--|--|-------------------------------------------|--|--|--|-----|--|--|--|------------------------------------------|--|--|--|-----|--|--|--|----------|--|--|--|-----|--|--|--|
| Instruction           | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Address                                   |               | Parameter     |                                          |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | MIPI                                      | Others        | D15-D8        | D7                                       | D6 | D5 | D4 | D3            | D2 | D1 | D0 | HEX |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
| CMD_Page              | W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | FEh                                       | FE00h         | 00h           | 0                                        | 0  | 0  | 0  | CMD_Page[3:0] |    |    |    | 00  |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
| Description           | This command is used to switch the Manufacture Command Pages and User Commands sets.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                           |               |               |                                          |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
|                       | CMD_Page[3:0]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                           | Hex Value     |               | Description                              |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
|                       | 0000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                           | 00h (default) |               | User Command Set (UCS = CMD1)            |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
|                       | 0001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                           | 01h           |               | Manufacture Command Set Page0 ( CMD2 P0) |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
|                       | 0010                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                           | 02h           |               | Manufacture Command Set Page1 ( CMD2 P1) |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
|                       | 0011                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                           | 03h           |               | Manufacture Command Set Page2 ( CMD2 P2) |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
|                       | 0100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                           | 04h           |               | Manufacture Command Set Page3 ( CMD2 P3) |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
| Restriction           | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                           |               |               |                                          |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
| Register Availability | <table><tr><th colspan="4">Status</th><th colspan="4">Availability</th></tr><tr><td colspan="4">Normal Mode On, Idle Mode Off, Sleep Out</td><td colspan="4">Yes</td></tr><tr><td colspan="4">Normal Mode On, Idle Mode On, Sleep Out</td><td colspan="4">Yes</td></tr><tr><td colspan="4">Partial Mode On, Idle Mode Off, Sleep Out</td><td colspan="4">Yes</td></tr><tr><td colspan="4">Partial Mode On, Idle Mode On, Sleep Out</td><td colspan="4">Yes</td></tr><tr><td colspan="4">Sleep In</td><td colspan="4">Yes</td></tr></table> |                                           |               |               |                                          |    |    |    |               |    |    |    |     | Status |  |  |  | Availability |  |  |  | Normal Mode On, Idle Mode Off, Sleep Out |  |  |  | Yes |  |  |  | Normal Mode On, Idle Mode On, Sleep Out |  |  |  | Yes |  |  |  | Partial Mode On, Idle Mode Off, Sleep Out |  |  |  | Yes |  |  |  | Partial Mode On, Idle Mode On, Sleep Out |  |  |  | Yes |  |  |  | Sleep In |  |  |  | Yes |  |  |  |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                           |               |               | Availability                             |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                           |               |               | Yes                                      |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                           |               |               | Yes                                      |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                           |               |               | Yes                                      |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                           |               |               | Yes                                      |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
| Sleep In              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                           |               | Yes           |                                          |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
| Default               | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                           |               | Default Value |                                          |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                           |               | FEh / FE00h   |                                          |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
|                       | Power On Sequence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                           |               | 00h           |                                          |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
|                       | S/W Reset                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                           |               | 00h           |                                          |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |
|                       | H/W Reset                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                           |               | 00h           |                                          |    |    |    |               |    |    |    |     |        |  |  |  |              |  |  |  |                                          |  |  |  |     |  |  |  |                                         |  |  |  |     |  |  |  |                                           |  |  |  |     |  |  |  |                                          |  |  |  |     |  |  |  |          |  |  |  |     |  |  |  |



CONFIDENTIAL

**CMD\_Status(FF00h): Read CMD Page Status**

| FF00H                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CMD_Status (Read Command Page Status) |               |           |                                          |    |    |    |                 |    |    |    |     |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|---------------|-----------|------------------------------------------|----|----|----|-----------------|----|----|----|-----|--------|--|--|--|---------------|--|--|--|--|--|--|--|--|------------------------------------------|--|--|--|-----|--|--|--|--|--|--|--|--|-----------------------------------------|--|--|--|-----|--|--|--|--|--|--|--|--|-------------------------------------------|--|--|--|-----|--|--|--|--|--|--|--|--|------------------------------------------|--|--|--|-----|--|--|--|--|--|--|--|--|----------|--|--|--|-----|--|--|--|--|--|--|--|--|
| Instruction           | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Address                               |               | Parameter |                                          |    |    |    |                 |    |    |    |     |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | MIPI                                  | Others        | D15-D8    | D7                                       | D6 | D5 | D4 | D3              | D2 | D1 | D0 | HEX |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
| CMD_Status            | R                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | FFh                                   | FF00h         | 00h       | 0                                        | 0  | 0  | 0  | CMD_Status[3:0] |    |    |    | 00  |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
| Description           | This command is used to switch the Manufacture Command Pages and User Commands sets.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                       |               |           |                                          |    |    |    |                 |    |    |    |     |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
|                       | CMD_Status[3:0]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                       | Hex Value     |           | Description                              |    |    |    |                 |    |    |    |     |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
|                       | 0000                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                       | 00h (default) |           | User Command Set (UCS = CMD1)            |    |    |    |                 |    |    |    |     |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
|                       | 0001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                       | 01h           |           | Manufacture Command Set Page0 ( CMD2 P0) |    |    |    |                 |    |    |    |     |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
|                       | 0010                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                       | 02h           |           | Manufacture Command Set Page1 ( CMD2 P1) |    |    |    |                 |    |    |    |     |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
|                       | 0011                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                       | 03h           |           | Manufacture Command Set Page2 ( CMD2 P2) |    |    |    |                 |    |    |    |     |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
|                       | 0100                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                       | 04h           |           | Manufacture Command Set Page3 ( CMD2 P3) |    |    |    |                 |    |    |    |     |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
| Restriction           | -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                       |               |           |                                          |    |    |    |                 |    |    |    |     |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
| Register Availability | <table><tr><th colspan="4">Status</th><th colspan="9">Availability</th></tr><tr><td colspan="4">Normal Mode On, Idle Mode Off, Sleep Out</td><td colspan="9">Yes</td></tr><tr><td colspan="4">Normal Mode On, Idle Mode On, Sleep Out</td><td colspan="9">Yes</td></tr><tr><td colspan="4">Partial Mode On, Idle Mode Off, Sleep Out</td><td colspan="9">Yes</td></tr><tr><td colspan="4">Partial Mode On, Idle Mode On, Sleep Out</td><td colspan="9">Yes</td></tr><tr><td colspan="4">Sleep In</td><td colspan="9">Yes</td></tr></table> |                                       |               |           |                                          |    |    |    |                 |    |    |    |     | Status |  |  |  | Availability  |  |  |  |  |  |  |  |  | Normal Mode On, Idle Mode Off, Sleep Out |  |  |  | Yes |  |  |  |  |  |  |  |  | Normal Mode On, Idle Mode On, Sleep Out |  |  |  | Yes |  |  |  |  |  |  |  |  | Partial Mode On, Idle Mode Off, Sleep Out |  |  |  | Yes |  |  |  |  |  |  |  |  | Partial Mode On, Idle Mode On, Sleep Out |  |  |  | Yes |  |  |  |  |  |  |  |  | Sleep In |  |  |  | Yes |  |  |  |  |  |  |  |  |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                       |               |           | Availability                             |    |    |    |                 |    |    |    |     |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
|                       | Normal Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                       |               |           | Yes                                      |    |    |    |                 |    |    |    |     |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
|                       | Normal Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                       |               |           | Yes                                      |    |    |    |                 |    |    |    |     |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
|                       | Partial Mode On, Idle Mode Off, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                       |               |           | Yes                                      |    |    |    |                 |    |    |    |     |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
|                       | Partial Mode On, Idle Mode On, Sleep Out                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                       |               |           | Yes                                      |    |    |    |                 |    |    |    |     |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
|                       | Sleep In                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                       |               |           | Yes                                      |    |    |    |                 |    |    |    |     |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
| Default               | <table><tr><th colspan="4">Status</th><th colspan="9">Default Value</th></tr><tr><td colspan="4">Power On Sequence</td><td colspan="9">00h</td></tr><tr><td colspan="4">S/W Reset</td><td colspan="9">00h</td></tr><tr><td colspan="4">H/W Reset</td><td colspan="9">00h</td></tr></table>                                                                                                                                                                                                                                                 |                                       |               |           |                                          |    |    |    |                 |    |    |    |     | Status |  |  |  | Default Value |  |  |  |  |  |  |  |  | Power On Sequence                        |  |  |  | 00h |  |  |  |  |  |  |  |  | S/W Reset                               |  |  |  | 00h |  |  |  |  |  |  |  |  | H/W Reset                                 |  |  |  | 00h |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
|                       | Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                       |               |           | Default Value                            |    |    |    |                 |    |    |    |     |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
|                       | Power On Sequence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                       |               |           | 00h                                      |    |    |    |                 |    |    |    |     |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
|                       | S/W Reset                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                       |               |           | 00h                                      |    |    |    |                 |    |    |    |     |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |
| H/W Reset             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                       |               | 00h       |                                          |    |    |    |                 |    |    |    |     |        |  |  |  |               |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |                                         |  |  |  |     |  |  |  |  |  |  |  |  |                                           |  |  |  |     |  |  |  |  |  |  |  |  |                                          |  |  |  |     |  |  |  |  |  |  |  |  |          |  |  |  |     |  |  |  |  |  |  |  |  |



## 8. MCS Command

### 8.1 MCS Command List

| Command |      |       | R/W | Function                                                                 | D7                | D6              | D5         | D4           | D3             | D2              | D1             | D0    | Default (hex) | MTP |   |
|---------|------|-------|-----|--------------------------------------------------------------------------|-------------------|-----------------|------------|--------------|----------------|-----------------|----------------|-------|---------------|-----|---|
| Page    | Add. | Para. |     |                                                                          |                   |                 |            |              |                |                 |                |       |               |     |   |
| CMD2 P0 | 0Eh  | -     | R/W | Display and panel application information                                | -                 | CGM[2:0]        |            |              | CRGB           | CTB             | CRL            | CREV  | 21h           | 1   |   |
| CMD2 P0 | 0Fh  | -     | R/W | Set vertical scan line                                                   | NL[7:0]           |                 |            |              |                |                 |                |       | C8h           | 1   |   |
| CMD2 P0 | 10h  | -     | R/W | Set TE1's transient edge                                                 | STB_POSE_SEL[7:0] |                 |            |              |                |                 |                |       | 00h           | 1   |   |
| CMD2 P0 | 11h  | -     | R/W | Set TE1's period                                                         | STB_period[7:0]   |                 |            |              |                |                 |                |       | 00h           | 1   |   |
| CMD2 P0 | 12h  | -     | R/W | Set TE1's period polarity, and enable control.<br>Set Zig-zag panel type | STB_POSE_SEL[9:8] | STB_period[9:8] |            | FTE1_pol     | FTE1_EN        | ZZ_RL           | ZZ_E0          |       | 00h           | 1   |   |
| CMD2 P0 | 13h  | -     | R/W | Set temperature sensor                                                   | TMP_HYS_L[1:0]    | TMP_HYS_H[1:0]  |            | -            | TGIP_EN        | -               | TGEQ_L         |       | 50h           | 1   |   |
| CMD2 P0 | 14h  | -     | R/W | Set temperature sensor                                                   | -                 | TMP_TH_L[2:0]   |            |              | -              | TMP_TH_H[2:0]   |                |       | 33h           | 1   |   |
| CMD2 P0 | 17h  | -     | R/W | Source EQ setting                                                        | EQMOD             | EQS1[6:0]       |            |              |                |                 |                |       |               | 09h | 1 |
| CMD2 P0 | 18h  | -     | R/W | Source EQ setting                                                        | -                 | EQS2[6:0]       |            |              |                |                 |                |       |               | 09h | 1 |
| CMD2 P0 | 19h  | -     | R/W | Source EQ setting                                                        | -                 | EQS3[6:0]       |            |              |                |                 |                |       |               | 09h | 1 |
| CMD2 P0 | 1Bh  | -     | R/W | Set inversion type                                                       | -                 | NLA[2:0]        |            |              | -              | -               | -              | -     | 00h           | 1   |   |
| CMD2 P0 | 24h  | -     | R/W | Set power IC type                                                        | PWRIC_VER         | D017_SEL        | 0          | 0            | 0              | 0               | 0              | 0     | 40h           | 1   |   |
| CMD2 P0 | 25h  | -     | R/W | Set AVDD output from power IC                                            | 0                 | PIC_BT_DH[1:0]  |            | 1            | PIC_VDH[3:0]   |                 |                |       | 55h           | 1   |   |
| CMD2 P0 | 27h  | -     | R/W | Set AVDD output from PFM                                                 | -                 | -               | -          | VDP_FBA[4:0] |                |                 |                |       |               | 13h | 1 |
| CMD2 P0 | 26h  | -     | R/W | Set AVEE output from PFM                                                 | -                 | -               | -          | VDN_FBA[4:0] |                |                 |                |       |               | 13h | 1 |
| CMD2 P0 | 2Bh  | -     | R/W | Set DDVDL-pump control                                                   | DC2A[2:0]         |                 |            |              | VDLA[4:0]      |                 |                |       |               | 85h | 1 |
| CMD2 P0 | 2Fh  | -     | R/W | Set VGH-pump control                                                     | VGHSA[1:0]        | BT_GHA[1:0]     |            |              | VBHA[3:0]      |                 |                |       | 50h           | 1   |   |
| CMD2 P0 | 31h  | -     | R/W | Set VGH-pump clock                                                       | DC4A[2:0]         |                 |            | -            | -              | -               | -              | -     | 80h           | 1   |   |
| CMD2 P0 | 32h  | -     | R/W | Temperature compensation for VGH-ump output                              | VGH_OFFSET_L[2:0] |                 |            | -            | TMP_BTH_L[1:0] |                 | -              | -     | A4h           | 1   |   |
| CMD2 P0 | 33h  | -     | R/W | Temperature compensation for VGH-ump output                              | VGH_OFFSET_H[2:0] |                 |            | -            | TMP_BTH_H[1:0] |                 | -              | -     | A4h           | 1   |   |
| CMD2 P0 | 34h  | -     | R/W | Set VGL-pump control                                                     | VGLSA[1:0]        | BT_GLA[1:0]     |            |              | VBHA[3:0]      |                 |                |       | 55h           | 1   |   |
| CMD2 P0 | 36h  | -     | R/W | Set VGL-pump clock                                                       | DC5A[2:0]         |                 |            | -            | -              | -               | -              | -     | 80h           | 1   |   |
| CMD2 P0 | 37h  | -     | R/W | Temperature compensation for VGH-ump output                              | VGL_OFFSET_L[2:0] |                 |            | -            | TMP_BTL_L[1:0] |                 | -              | -     | 00h           | 1   |   |
| CMD2 P0 | 38h  | -     | R/W | Temperature compensation for VGH-ump output                              | VGL_OFFSET_H[2:0] |                 |            | -            | TMP_BTL_H[1:0] |                 | -              | -     | A4h           | 1   |   |
| CMD2 P0 | 39h  | -     | R/W | Set VCL-pump control                                                     | BTCL              | -               | DVCLA[1:0] |              | -              | DC3A[2:0]       |                |       | 04h           | 1   |   |
| CMD2 P0 | 3Dh  | -     | R/W | VCOM control                                                             |                   |                 |            |              |                |                 |                | VMSEL | 00h           | 1   |   |
| CMD2 P0 | 45h  | -     | R/W | VCOM output level control                                                | VCM_REG[8]        | VCM_REG_IDM[8]  | -          | -            | VCM2_REG[8]    | VCM2_REG_IDM[8] | -              | -     | 00h           | 3   |   |
| CMD2 P0 | 46h  | -     | R/W | VCOM output level control                                                | VCM_REG[7:0]      |                 |            |              |                |                 |                |       | 28h           | 3   |   |
| CMD2 P0 | 47h  | -     | R/W | VCOM output level control                                                | VCM2_REG[7:0]     |                 |            |              |                |                 |                |       | 00h           | 3   |   |
| CMD2 P0 | 48h  | -     | R/W | VCOM output level control                                                | VCM_REG_IDM[7:0]  |                 |            |              |                |                 |                |       | 00h           | 3   |   |
| CMD2 P0 | 49h  | -     | R/W | VCOM output level control                                                | VCM2_REG_IDM[7:0] |                 |            |              |                |                 |                |       | 00h           | 3   |   |
| CMD2 P0 | 52h  | -     | R/W | VGMN control                                                             | VGMNTR[7:0]       |                 |            |              |                |                 |                |       | 60h           | 1   |   |
| CMD2 P0 | 53h  | -     | R/W | VGSN control                                                             | VGSNTR[7:0]       |                 |            |              |                |                 |                |       | 00h           | 1   |   |
| CMD2 P0 | 54h  | -     | R/W | VGMP control                                                             | VGMPTR[7:0]       |                 |            |              |                |                 |                |       | 60h           | 1   |   |
| CMD2 P0 | 55h  | -     | R/W | VGSP control                                                             | VGSPTR[7:0]       |                 |            |              |                |                 |                |       | 00h           | 1   |   |
| CMD2 P0 | 56h  | -     | R/W | MSB of VGMN/VGSN/VGMP/VGSP                                               | VGMNTR[8]         | VGSNTR[8]       | VGMPTR[8]  | VGSPTR[8]    | -              | -               | -              | -     | 00h           | 1   |   |
| CMD2 P0 | 5Fh  | -     | R/W | Interface control                                                        | 0                 | PFM_TYPE[2:0]   |            |              | 0              | 0               | LANSEL_sw[1:0] |       |               | 13h | 1 |
| CMD2 P0 | E1h  | -     | R/W | Read/Write IC Supplier ID code information                               | SID[7:0]          |                 |            |              |                |                 |                |       | 00h           | 3   |   |
| CMD2 P0 | E2h  | -     | R/W | Read/Write IC ID code information                                        | SID[15:8]         |                 |            |              |                |                 |                |       | 00h           | 3   |   |
| CMD2 P0 | E3h  | -     | R/W | Read/Write IC Manufacture ID code information                            | MID[7:0]          |                 |            |              |                |                 |                |       | 00h           | 3   |   |

|         |     |   |     |                                               |               |               |                |              |              |              |              |                  |     |   |
|---------|-----|---|-----|-----------------------------------------------|---------------|---------------|----------------|--------------|--------------|--------------|--------------|------------------|-----|---|
| CMD2 P0 | E4h | - | R/W | Read/Write IC Manufacture ID code information | MID[15:8]     |               |                |              |              |              |              |                  | 00h | 3 |
| CMD2 P0 | E5h | - | R/W | Read/Write IC ID code information             | ID1[7:0]      |               |                |              |              |              |              |                  | 00h | 3 |
| CMD2 P0 | E6h | - | R/W | Read/Write IC ID code information             | ID2[7:0]      |               |                |              |              |              |              |                  | 80h | 3 |
| CMD2 P0 | E7h | - | R/W | Read/Write IC ID code information             | ID3[7:0]      |               |                |              |              |              |              |                  | 00h | 3 |
| CMD2 P0 | EEh | - | R   | Page status for MTP programming               | CMD2_3rd_Done | CMD2_2nd_Done | CMD2_P0_Done   | CMD2_P1_Done | CMD2_P2_Done | CMD2_P3_Done | CMD2_P4_Done | CMD2_GAM_Done    | 00h | 1 |
| CMD2 P0 | F0h | - | R/W | MTP program page enable control               | CMD_3rdEN     | CMD2_2ndEN    | CMD2_P0_PGM_EN | CMD2_P1_PGEN | CMD2_P2_PGEN | CMD2_P3_PGEN | CMD2_P4_PGEN | CMD2_gam_PGEN    | 00h | 1 |
| CMD2 P0 | F2h | - | R/W | MTP program page enable control               | MTP_LOAD_EN   | -             | -              | -            | -            | -            | EXT_MTP_PWR  | MTP_W            | 00h | 1 |
| CMD2 P0 | F3h | - | W   | Key check for MTP program                     | 1             | 0             | 1              | 0            | 0            | 1            | 0            | 1                | A5h | 1 |
| CMD2 P0 | F4h | - | W   | Key check for MTP program                     | 0             | 1             | 0              | 1            | 0            | 0            | 1            | 0                | 5Ah | 1 |
| CMD2 P0 | F5h | - | W   | Key check for MTP program                     | 0             | 0             | 1              | 1            | 1            | 1            | 0            | 0                | 3Ch | 1 |
| CMD2 P0 | F6h | - | R/W | MTP read mode selection                       | 0             | 0             | 0              | 0            | 0            | 0            | PTM[1:0]     |                  | 00h | 1 |
| CMD2 P3 | 60h | - | R/W | Gamma VP1                                     | -             | -             | P_V0[5:0]      |              |              |              |              |                  | 00h | 1 |
| CMD2 P3 | 61h | - | R/W | Gamma VP4                                     | -             | -             | P_V4[5:0]      |              |              |              |              |                  | 12h | 1 |
| CMD2 P3 | 62h | - | R/W | Gamma VP8                                     | -             | -             | P_V8[5:0]      |              |              |              |              |                  | 19h | 1 |
| CMD2 P3 | 63h | - | R/W | Gamma VP16                                    | -             | -             | -              | P_V16[4:0]   |              |              |              | 0Eh              | 1   |   |
| CMD2 P3 | 64h | - | R/W | Gamma VP24                                    | -             | -             | -              | P_V24[4:0]   |              |              |              | 06h              | 1   |   |
| CMD2 P3 | 65h | - | R/W | Gamma VP52                                    | -             | -             | -              | P_V52[4:0]   |              |              |              | 14h              | 1   |   |
| CMD2 P3 | 66h | - | R/W | Gamma VP80                                    | -             | -             | -              | P_V80[4:0]   |              |              |              | 10h              | 1   |   |
| CMD2 P3 | 67h | - | R/W | Gamma VP108                                   | -             | -             | -              | P_V108[4:0]  |              |              |              | 0Ch              | 1   |   |
| CMD2 P3 | 68h | - | R/W | Gamma VP147                                   | -             | -             | -              | P_V147[4:0]  |              |              |              | 16h              | 1   |   |
| CMD2 P3 | 69h | - | R/W | Gamma VP175                                   | -             | -             | -              | P_V175[4:0]  |              |              |              | 0Ch              | 1   |   |
| CMD2 P3 | 6Ah | - | R/W | Gamma VP203                                   | -             | -             | -              | P_V203[4:0]  |              |              |              | 0Eh              | 1   |   |
| CMD2 P3 | 6Bh | - | R/W | Gamma VP231                                   | -             | -             | -              | P_V231[4:0]  |              |              |              | 08h              | 1   |   |
| CMD2 P3 | 6Ch | - | R/W | Gamma VP239                                   | -             | -             | -              | P_V239[4:0]  |              |              |              | 0Fh              | 1   |   |
| CMD2 P3 | 6Dh | - | R/W | Gamma VP247                                   | -             | -             | P_V247[5:0]    |              |              |              |              |                  | 11h | 1 |
| CMD2 P3 | 6Eh | - | R/W | Gamma VP251                                   | -             | -             | P_V251[5:0]    |              |              |              |              |                  | 0Ch | 1 |
| CMD2 P3 | 6Fh | - | R/W | Gamma VP255                                   | -             | -             | P_V255[5:0]    |              |              |              |              |                  | 00h | 1 |
| CMD2 P3 | 70h | - | R/W | Gamma VN1                                     | -             | -             | N_V0[5:0]      |              |              |              |              |                  | 00h | 1 |
| CMD2 P3 | 71h | - | R/W | Gamma VN4                                     | -             | -             | N_V4[5:0]      |              |              |              |              |                  | 12h | 1 |
| CMD2 P3 | 72h | - | R/W | Gamma VN8                                     | -             | -             | N_V8[5:0]      |              |              |              |              |                  | 19h | 1 |
| CMD2 P3 | 73h | - | R/W | Gamma VN16                                    | -             | -             | -              | N_V16[4:0]   |              |              |              | 0Eh              | 1   |   |
| CMD2 P3 | 74h | - | R/W | Gamma VN24                                    | -             | -             | -              | N_V24[4:0]   |              |              |              | 06h              | 1   |   |
| CMD2 P3 | 75h | - | R/W | Gamma VN52                                    | -             | -             | -              | N_V52[4:0]   |              |              |              | 14h              | 1   |   |
| CMD2 P3 | 76h | - | R/W | Gamma VN80                                    | -             | -             | -              | N_V80[4:0]   |              |              |              | 10h              | 1   |   |
| CMD2 P3 | 77h | - | R/W | Gamma VN108                                   | -             | -             | -              | N_V108[4:0]  |              |              |              | 0Ch              | 1   |   |
| CMD2 P3 | 78h | - | R/W | Gamma VN147                                   | -             | -             | -              | N_V147[4:0]  |              |              |              | 16h              | 1   |   |
| CMD2 P3 | 79h | - | R/W | Gamma VN175                                   | -             | -             | -              | N_V175[4:0]  |              |              |              | 0Ch              | 1   |   |
| CMD2 P3 | 7Ah | - | R/W | Gamma VN203                                   | -             | -             | -              | N_V203[4:0]  |              |              |              | 0Eh              | 1   |   |
| CMD2 P3 | 7Bh | - | R/W | Gamma VN231                                   | -             | -             | -              | N_V231[4:0]  |              |              |              | 08h              | 1   |   |
| CMD2 P3 | 7Ch | - | R/W | Gamma VN239                                   | -             | -             | -              | N_V239[4:0]  |              |              |              | 0Fh              | 1   |   |
| CMD2 P3 | 7Dh | - | R/W | Gamma VP247                                   | -             | -             | N_V247[5:0]    |              |              |              |              |                  | 11h | 1 |
| CMD2 P3 | 7Eh | - | R/W | Gamma VP251                                   | -             | -             | N_V251[5:0]    |              |              |              |              |                  | 0Ch | 1 |
| CMD2 P3 | 7Fh | - | R/W | Gamma VP255                                   | -             | -             | N_V255[5:0]    |              |              |              |              |                  | 00h | 1 |
| CMD2 P3 | 80h | - | R/W | Force Gamma update                            | -             | -             |                |              |              |              |              | gam_for<br>ce_up | 00h | 1 |



## 8.2 MCS Command Description

## DPRSLCTR (0E40h)

| 0E40H       |                                                                                                                          |                                                                     |       | DOPCTR |      |          |                        |            |     |     |      |    |     |
|-------------|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-------|--------|------|----------|------------------------|------------|-----|-----|------|----|-----|
| Instruction | R/W                                                                                                                      | Address                                                             |       | R/W    |      |          |                        |            |     |     |      |    |     |
|             |                                                                                                                          | MIPI                                                                | Other | D15-D8 | D7   | D6       | D5                     | D4         | D3  | D2  | D1   | D0 | HEX |
| DPRSLCTR    | R/W                                                                                                                      | 0Eh                                                                 | 0E40h | 00h    | -    | CGM[2:0] |                        | CRGB       | CTB | CRL | CREV | 21 |     |
| Description | This command is used to setup Display and Panel bonding application.                                                     |                                                                     |       |        |      |          |                        |            |     |     |      |    |     |
|             | Bit                                                                                                                      | Description                                                         |       |        |      | Data     |                        |            |     |     |      |    |     |
|             | CGM[2:0]                                                                                                                 | Display resolution selection, as for NL[7:0] please refer to 0F40h. |       |        |      | CGM[2:0] |                        | Resolution |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        |      | 000      | 800RGBx(480+4*NL[7:0]) |            |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        |      | 001      | 768RGBx(480+4*NL[7:0]) |            |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        |      | 010      | 720RGBx(480+4*NL[7:0]) |            |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        |      | 011      | 640RGBx(480+4*NL[7:0]) |            |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        |      | 100      | 600RGBx(480+4*NL[7:0]) |            |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        |      | 101      | 540RGBx(480+4*NL[7:0]) |            |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        |      | 110      | Reserved               |            |     |     |      |    |     |
| 111         |                                                                                                                          |                                                                     |       |        |      | Reserved |                        |            |     |     |      |    |     |
| CRGB        | RGB-BGR Order selection. This bit is XOR operation with bit RGB of DCS command register 3600h                            |                                                                     |       |        | CRGB | RGB      | Order                  | Gamma      |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        | 0    | 0        | RGB                    | Normal     |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        | 0    | 1        | BGR                    | RB SWAP    |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        | 1    | 0        | BGR                    | RB SWAP    |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        | 1    | 1        | RGB                    | Normal     |     |     |      |    |     |
| CTB         | Vertical scanning direction selection for gate control signals. This bit is XOR with RSMY of DCS command register 3600h. |                                                                     |       |        | CTB  | RSMY     | Direction              |            |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        | 0    | 0        | T -> B                 |            |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        | 0    | 1        | B -> T                 |            |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        | 1    | 0        | B -> T                 |            |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        | 1    | 1        | T -> B                 |            |     |     |      |    |     |
| CRL         | Source driver data shift direction selection. This bit is XOR with RSMX of DCS command register 3600h command            |                                                                     |       |        | CRL  | RSMX     | Direction              |            |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        | 0    | 0        | S1 -> Sn               |            |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        | 0    | 1        | Sn -> S1               |            |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        | 1    | 0        | Sn -> S1               |            |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        | 1    | 1        | S1 -> Sn               |            |     |     |      |    |     |
| CREV        | Display Data Reversion selection for NB/NW panel type. This Bit is XOR with INVON/OFF commands.                          |                                                                     |       |        | CREV | INVON    | Display Data           |            |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        | 0    | 0        | No reversion (NW)      |            |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        | 0    | 1        | Reversion (NB)         |            |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        | 1    | 0        | Reversion (NB)         |            |     |     |      |    |     |
|             |                                                                                                                          |                                                                     |       |        | 1    | 1        | No reversion (NW)      |            |     |     |      |    |     |
| Restriction |                                                                                                                          |                                                                     |       |        |      |          |                        |            |     |     |      |    |     |

**DPRSLCTR (0F40h): DPRSLCTR**

| 0F40H       | DPRSLCTR                                                                     |         |           |           |                          |    |    |    |    |    |    |    |     |
|-------------|------------------------------------------------------------------------------|---------|-----------|-----------|--------------------------|----|----|----|----|----|----|----|-----|
| Instruction | R/W                                                                          | Address |           | Parameter |                          |    |    |    |    |    |    |    |     |
|             |                                                                              | MIPI    | Others    | D15-D8    | D7                       | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |
| DPRSLCTR    | R/W                                                                          | 0Fh     | 0F40h     | 00h       | NL[7:0]                  |    |    |    |    |    |    |    | C8  |
| Description | This command is used to define total vertical scan lines for normal display. |         |           |           |                          |    |    |    |    |    |    |    |     |
|             | Bit                                                                          |         | Value     |           | Display Scan Line Number |    |    |    |    |    |    |    |     |
|             | NL[7:0]                                                                      |         | 00h       | 0         | 480 lines                |    |    |    |    |    |    |    |     |
|             |                                                                              |         | 01h       | 1         | 484 lines                |    |    |    |    |    |    |    |     |
|             |                                                                              |         | 02h       | 2         | 488 lines                |    |    |    |    |    |    |    |     |
|             |                                                                              |         | :         | :         | = 480+4*NL[7:0]          |    |    |    |    |    |    |    |     |
|             |                                                                              |         | C7h       | 199       | 1276 lines               |    |    |    |    |    |    |    |     |
|             |                                                                              |         | C8h       | 200       | 1280 lines               |    |    |    |    |    |    |    |     |
|             |                                                                              |         | C9h - FFh | Reserved  | Reserved                 |    |    |    |    |    |    |    |     |
| Restriction |                                                                              |         |           |           |                          |    |    |    |    |    |    |    |     |

**STBCTR(1040h~1140h): TE1 Output Setting**

| 1040H                           |                                                                                                                                                                                                                                                                                                                   | STBCTR  |        |           |                   |          |    |                       |    |    |    |    |     |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------|-----------|-------------------|----------|----|-----------------------|----|----|----|----|-----|
| Instruction                     | R/W                                                                                                                                                                                                                                                                                                               | Address |        | Parameter |                   |          |    |                       |    |    |    |    |     |
|                                 |                                                                                                                                                                                                                                                                                                                   | MIPI    | Others | D15-D8    | D7                | D6       | D5 | D4                    | D3 | D2 | D1 | D0 | HEX |
| STBCTR                          | R/W                                                                                                                                                                                                                                                                                                               | 10h     | 1040h  | 00h       | STB_POSE_SEL[7:0] |          |    |                       |    |    |    |    | 00  |
| STBCTR                          | R/W                                                                                                                                                                                                                                                                                                               | 11h     | 1140h  | 00h       | STB_period[7:0]   |          |    |                       |    |    |    |    | 00  |
| Description                     | <p>These commands are to select transient edge as well as pulse period of signal TE1. The start point as timing reference for STB_POSE_SEL and STB_period are internal Hsync and STB's transient edge, respectively. Please also refer to register 0x1240h for detailed timing setting of Zig-Zag panel type.</p> |         |        |           |                   |          |    |                       |    |    |    |    |     |
|                                 | STB_POSE_SEL[9:0]                                                                                                                                                                                                                                                                                                 |         |        | Bit       |                   | Value    |    | Description           |    |    |    |    |     |
|                                 |                                                                                                                                                                                                                                                                                                                   |         |        | 0         |                   | 000h     |    | 0 Clocks              |    |    |    |    |     |
|                                 |                                                                                                                                                                                                                                                                                                                   |         |        | 1         |                   | 001h     |    | 1 Clock = 40ns        |    |    |    |    |     |
|                                 |                                                                                                                                                                                                                                                                                                                   |         |        | :         |                   | :        |    | 40 ns / Clock         |    |    |    |    |     |
|                                 |                                                                                                                                                                                                                                                                                                                   |         |        | 512       |                   | 200h     |    | 512 Clocks = 20480 ns |    |    |    |    |     |
|                                 | STB_period[9:0]                                                                                                                                                                                                                                                                                                   |         |        | Reserved  |                   | Reserved |    |                       |    |    |    |    |     |
|                                 |                                                                                                                                                                                                                                                                                                                   |         |        | 0         |                   | 000h     |    | 0 Clocks              |    |    |    |    |     |
|                                 |                                                                                                                                                                                                                                                                                                                   |         |        | 1         |                   | 001h     |    | 1 Clock = 40ns        |    |    |    |    |     |
|                                 |                                                                                                                                                                                                                                                                                                                   |         |        | :         |                   | :        |    | 40 ns / Clock         |    |    |    |    |     |
|                                 |                                                                                                                                                                                                                                                                                                                   |         |        | 512       |                   | 200h     |    | 512 Clocks = 20480 ns |    |    |    |    |     |
|                                 |                                                                                                                                                                                                                                                                                                                   |         |        | Reserved  |                   | Reserved |    |                       |    |    |    |    |     |
|                                 |                                                                                                                                                                                                                                                                                                                   |         |        |           |                   |          |    |                       |    |    |    |    |     |
| Restriction                     |                                                                                                                                                                                                                                                                                                                   |         |        |           |                   |          |    |                       |    |    |    |    |     |
| STB_POSE_SEL + STB_period < T1A |                                                                                                                                                                                                                                                                                                                   |         |        |           |                   |          |    |                       |    |    |    |    |     |

**STBCTR(1240h): TE1 Output Setting and Zig-Zag Connection**

| 0x1240h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                | STBCTR              |                                               |           |                       |    |                     |    |              |             |       |       |       |        |             |       |          |                          |                               |         |                    |                                               |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------|-----------|-----------------------|----|---------------------|----|--------------|-------------|-------|-------|-------|--------|-------------|-------|----------|--------------------------|-------------------------------|---------|--------------------|-----------------------------------------------|---|---|---|---|---|---|---|---|---|---|---|---|-----------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Instruction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | R/W                                                                                                                                                                                                                                                                            | Address             |                                               | Parameter |                       |    |                     |    |              |             |       |       |       |        |             |       |          |                          |                               |         |                    |                                               |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                | MIPI                | Others                                        | D15-D8    | D7                    | D6 | D5                  | D4 | D3           | D2          | D1    | D0    | HEX   |        |             |       |          |                          |                               |         |                    |                                               |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| STBCTR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | R/W                                                                                                                                                                                                                                                                            | 0x12h               | 0x1240h                                       | 00h       | STB_POSE_S<br>EL[9:8] |    | STB_period[9:<br>8] |    | FTE1_<br>pol | FTE1_<br>EN | ZZ_RL | ZZ_E0 | 0x00h |        |             |       |          |                          |                               |         |                    |                                               |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | This command is used to set TE1's output enable as well as polarity selection.                                                                                                                                                                                                 |                     |                                               |           |                       |    |                     |    |              |             |       |       |       |        |             |       |          |                          |                               |         |                    |                                               |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <table><tr><th>Bit</th><th>Description</th><th>Value</th></tr><tr><td>FTE1_pol</td><td>TE1 Output Polarity</td><td>0: Low Active<br/>1: Hi Active</td></tr><tr><td>FTE1_EN</td><td>TE1 Output Control</td><td>0: Disable TE1 Output<br/>1: Enable TE1 Output</td></tr></table> |                     |                                               |           |                       |    |                     |    |              |             |       |       |       | Bit    | Description | Value | FTE1_pol | TE1 Output Polarity      | 0: Low Active<br>1: Hi Active | FTE1_EN | TE1 Output Control | 0: Disable TE1 Output<br>1: Enable TE1 Output |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Bit                                                                                                                                                                                                                                                                            | Description         | Value                                         |           |                       |    |                     |    |              |             |       |       |       |        |             |       |          |                          |                               |         |                    |                                               |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | FTE1_pol                                                                                                                                                                                                                                                                       | TE1 Output Polarity | 0: Low Active<br>1: Hi Active                 |           |                       |    |                     |    |              |             |       |       |       |        |             |       |          |                          |                               |         |                    |                                               |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | FTE1_EN                                                                                                                                                                                                                                                                        | TE1 Output Control  | 0: Disable TE1 Output<br>1: Enable TE1 Output |           |                       |    |                     |    |              |             |       |       |       |        |             |       |          |                          |                               |         |                    |                                               |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| This command is used to select the Zig-Zag inversion driving type. It is active when Zig-Zag inversion is selected in INVCTR, register 0x1B40h.                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                |                     |                                               |           |                       |    |                     |    |              |             |       |       |       |        |             |       |          |                          |                               |         |                    |                                               |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| ZZ_RL, ZZ_EO: zigzag inversion driving type.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                |                     |                                               |           |                       |    |                     |    |              |             |       |       |       |        |             |       |          |                          |                               |         |                    |                                               |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| <table><tr><th>CRL</th><th>Zigzag</th><th>CRGB</th><th>ZZ_RL</th><th>ZZ_EO</th></tr><tr><td rowspan="4">0<br/>(IC on bottom side)</td><td>1</td><td>0</td><td>1</td><td>0</td></tr><tr><td>2</td><td>0</td><td>1</td><td>1</td></tr><tr><td>3</td><td>0</td><td>0</td><td>0</td></tr><tr><td>4</td><td>0</td><td>0</td><td>1</td></tr><tr><td rowspan="4">1<br/>(IC on top side)</td><td>1</td><td>1</td><td>0</td><td>0</td></tr><tr><td>2</td><td>1</td><td>0</td><td>1</td></tr><tr><td>3</td><td>1</td><td>1</td><td>0</td></tr><tr><td>4</td><td>1</td><td>1</td><td>1</td></tr></table> |                                                                                                                                                                                                                                                                                |                     |                                               |           |                       |    |                     |    |              |             |       |       | CRL   | Zigzag | CRGB        | ZZ_RL | ZZ_EO    | 0<br>(IC on bottom side) | 1                             | 0       | 1                  | 0                                             | 2 | 0 | 1 | 1 | 3 | 0 | 0 | 0 | 4 | 0 | 0 | 1 | 1<br>(IC on top side) | 1 | 1 | 0 | 0 | 2 | 1 | 0 | 1 | 3 | 1 | 1 | 0 | 4 | 1 | 1 | 1 |
| CRL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Zigzag                                                                                                                                                                                                                                                                         | CRGB                | ZZ_RL                                         | ZZ_EO     |                       |    |                     |    |              |             |       |       |       |        |             |       |          |                          |                               |         |                    |                                               |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 0<br>(IC on bottom side)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1                                                                                                                                                                                                                                                                              | 0                   | 1                                             | 0         |                       |    |                     |    |              |             |       |       |       |        |             |       |          |                          |                               |         |                    |                                               |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2                                                                                                                                                                                                                                                                              | 0                   | 1                                             | 1         |                       |    |                     |    |              |             |       |       |       |        |             |       |          |                          |                               |         |                    |                                               |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3                                                                                                                                                                                                                                                                              | 0                   | 0                                             | 0         |                       |    |                     |    |              |             |       |       |       |        |             |       |          |                          |                               |         |                    |                                               |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 4                                                                                                                                                                                                                                                                              | 0                   | 0                                             | 1         |                       |    |                     |    |              |             |       |       |       |        |             |       |          |                          |                               |         |                    |                                               |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| 1<br>(IC on top side)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1                                                                                                                                                                                                                                                                              | 1                   | 0                                             | 0         |                       |    |                     |    |              |             |       |       |       |        |             |       |          |                          |                               |         |                    |                                               |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 2                                                                                                                                                                                                                                                                              | 1                   | 0                                             | 1         |                       |    |                     |    |              |             |       |       |       |        |             |       |          |                          |                               |         |                    |                                               |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3                                                                                                                                                                                                                                                                              | 1                   | 1                                             | 0         |                       |    |                     |    |              |             |       |       |       |        |             |       |          |                          |                               |         |                    |                                               |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 4                                                                                                                                                                                                                                                                              | 1                   | 1                                             | 1         |                       |    |                     |    |              |             |       |       |       |        |             |       |          |                          |                               |         |                    |                                               |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
| Restriction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                |                     |                                               |           |                       |    |                     |    |              |             |       |       |       |        |             |       |          |                          |                               |         |                    |                                               |   |   |   |   |   |   |   |   |   |   |   |   |                       |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |

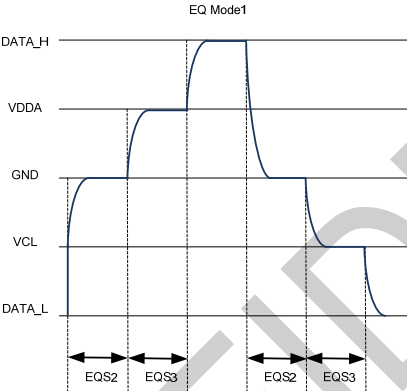
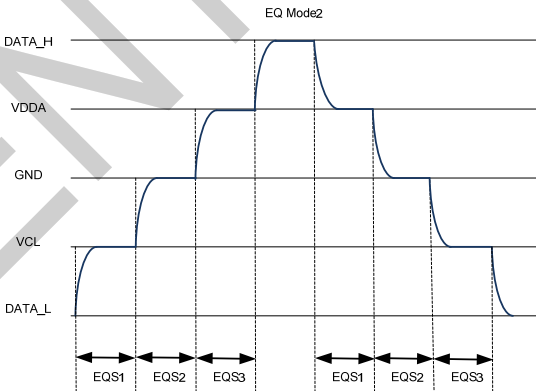
**TMPSCTR(1340h): Temperature Sensing Settings**

| 1340H                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                       | TMPSCTR                                     |                                                                 |           |                |                |    |         |    |        |    |    |                |            |             |             |         |                                             |                                                                 |             |                                           |                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-----------------------------------------------------------------|-----------|----------------|----------------|----|---------|----|--------|----|----|----------------|------------|-------------|-------------|---------|---------------------------------------------|-----------------------------------------------------------------|-------------|-------------------------------------------|-----------------------------------------|
| Instruction                                                                                                                                                                                                              | R/W                                                                                                                                                                                                                                                                                                                                                                                   | Address                                     |                                                                 | Parameter |                |                |    |         |    |        |    |    |                |            |             |             |         |                                             |                                                                 |             |                                           |                                         |
|                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                       | MIPI                                        | Others                                                          | D15-D8    | D7             | D6             | D5 | D4      | D3 | D2     | D1 | D0 | HEX            |            |             |             |         |                                             |                                                                 |             |                                           |                                         |
| TMP_SCTR                                                                                                                                                                                                                 | R/W                                                                                                                                                                                                                                                                                                                                                                                   | 13h                                         | 1340h                                                           | 00h       | TMP_HYS_L[1:0] | TMP_HYS_H[1:0] | -  | TGIP_EN | -  | TGEQ_L | -  | 50 |                |            |             |             |         |                                             |                                                                 |             |                                           |                                         |
| Description                                                                                                                                                                                                              | <table><thead><tr><th>Bit</th><th>Description</th><th>Value</th></tr></thead><tbody><tr><td>TGIP_EN</td><td>Temperature sensing function enable control</td><td>0: Disable temperature sensing<br/>1: Enable temperature sensing</td></tr><tr><td>TGEQ_L</td><td>Gate EQ enable control at low temperature</td><td>0: Enable Gate EQ<br/>1: Disable Gate EQ</td></tr></tbody></table> |                                             |                                                                 |           |                |                |    |         |    |        |    |    |                | Bit        | Description | Value       | TGIP_EN | Temperature sensing function enable control | 0: Disable temperature sensing<br>1: Enable temperature sensing | TGEQ_L      | Gate EQ enable control at low temperature | 0: Enable Gate EQ<br>1: Disable Gate EQ |
|                                                                                                                                                                                                                          | Bit                                                                                                                                                                                                                                                                                                                                                                                   | Description                                 | Value                                                           |           |                |                |    |         |    |        |    |    |                |            |             |             |         |                                             |                                                                 |             |                                           |                                         |
|                                                                                                                                                                                                                          | TGIP_EN                                                                                                                                                                                                                                                                                                                                                                               | Temperature sensing function enable control | 0: Disable temperature sensing<br>1: Enable temperature sensing |           |                |                |    |         |    |        |    |    |                |            |             |             |         |                                             |                                                                 |             |                                           |                                         |
|                                                                                                                                                                                                                          | TGEQ_L                                                                                                                                                                                                                                                                                                                                                                                | Gate EQ enable control at low temperature   | 0: Enable Gate EQ<br>1: Disable Gate EQ                         |           |                |                |    |         |    |        |    |    |                |            |             |             |         |                                             |                                                                 |             |                                           |                                         |
|                                                                                                                                                                                                                          | This command is to set hysteresis range of high/low temperature sensing.                                                                                                                                                                                                                                                                                                              |                                             |                                                                 |           |                |                |    |         |    |        |    |    |                |            |             |             |         |                                             |                                                                 |             |                                           |                                         |
| <table><thead><tr><th>TMP_HYS_L[1:0]</th><th>Hysteresis</th></tr></thead><tbody><tr><td>00</td><td>±2 degree C</td></tr><tr><td>01</td><td>±5 degree C</td></tr><tr><td>1X</td><td>±7 degree C</td></tr></tbody></table> |                                                                                                                                                                                                                                                                                                                                                                                       |                                             |                                                                 |           |                |                |    |         |    |        |    |    | TMP_HYS_L[1:0] | Hysteresis | 00          | ±2 degree C | 01      | ±5 degree C                                 | 1X                                                              | ±7 degree C |                                           |                                         |
| TMP_HYS_L[1:0]                                                                                                                                                                                                           | Hysteresis                                                                                                                                                                                                                                                                                                                                                                            |                                             |                                                                 |           |                |                |    |         |    |        |    |    |                |            |             |             |         |                                             |                                                                 |             |                                           |                                         |
| 00                                                                                                                                                                                                                       | ±2 degree C                                                                                                                                                                                                                                                                                                                                                                           |                                             |                                                                 |           |                |                |    |         |    |        |    |    |                |            |             |             |         |                                             |                                                                 |             |                                           |                                         |
| 01                                                                                                                                                                                                                       | ±5 degree C                                                                                                                                                                                                                                                                                                                                                                           |                                             |                                                                 |           |                |                |    |         |    |        |    |    |                |            |             |             |         |                                             |                                                                 |             |                                           |                                         |
| 1X                                                                                                                                                                                                                       | ±7 degree C                                                                                                                                                                                                                                                                                                                                                                           |                                             |                                                                 |           |                |                |    |         |    |        |    |    |                |            |             |             |         |                                             |                                                                 |             |                                           |                                         |
| <table><thead><tr><th>TMP_HYS_H[1:0]</th><th>Hysteresis</th></tr></thead><tbody><tr><td>00</td><td>±2 degree C</td></tr><tr><td>01</td><td>±5 degree C</td></tr><tr><td>1X</td><td>±7 degree C</td></tr></tbody></table> |                                                                                                                                                                                                                                                                                                                                                                                       |                                             |                                                                 |           |                |                |    |         |    |        |    |    | TMP_HYS_H[1:0] | Hysteresis | 00          | ±2 degree C | 01      | ±5 degree C                                 | 1X                                                              | ±7 degree C |                                           |                                         |
| TMP_HYS_H[1:0]                                                                                                                                                                                                           | Hysteresis                                                                                                                                                                                                                                                                                                                                                                            |                                             |                                                                 |           |                |                |    |         |    |        |    |    |                |            |             |             |         |                                             |                                                                 |             |                                           |                                         |
| 00                                                                                                                                                                                                                       | ±2 degree C                                                                                                                                                                                                                                                                                                                                                                           |                                             |                                                                 |           |                |                |    |         |    |        |    |    |                |            |             |             |         |                                             |                                                                 |             |                                           |                                         |
| 01                                                                                                                                                                                                                       | ±5 degree C                                                                                                                                                                                                                                                                                                                                                                           |                                             |                                                                 |           |                |                |    |         |    |        |    |    |                |            |             |             |         |                                             |                                                                 |             |                                           |                                         |
| 1X                                                                                                                                                                                                                       | ±7 degree C                                                                                                                                                                                                                                                                                                                                                                           |                                             |                                                                 |           |                |                |    |         |    |        |    |    |                |            |             |             |         |                                             |                                                                 |             |                                           |                                         |
| Restriction                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                       |                                             |                                                                 |           |                |                |    |         |    |        |    |    |                |            |             |             |         |                                             |                                                                 |             |                                           |                                         |

**TMPSCTR (1440h) Temperature Sensing Settings**

| 1440H       |                                                                                           | TMPSCTR |                                                   |           |    |               |    |    |               |    |    |    |     |
|-------------|-------------------------------------------------------------------------------------------|---------|---------------------------------------------------|-----------|----|---------------|----|----|---------------|----|----|----|-----|
| Instruction | R/W                                                                                       | Address |                                                   | Parameter |    |               |    |    |               |    |    |    |     |
|             |                                                                                           | MIPI    | Others                                            | D15-D8    | D7 | D6            | D5 | D4 | D3            | D2 | D1 | D0 | HEX |
| TMPSCTR     | R/W                                                                                       | 14h     | 1440h                                             | 00h       | -  | TMP_TH_L[2:0] |    | -  | TMP_TH_H[2:0] |    | 33 |    |     |
| Description | This command is used to set the threshold temperature for Low / High temperature sensing. |         |                                                   |           |    |               |    |    |               |    |    |    |     |
|             | TMP_TH_L[2:0]                                                                             |         | Threshold Temperature of Low Temperature Sensing  |           |    |               |    |    |               |    |    |    |     |
|             | 000                                                                                       |         | 10 degree C                                       |           |    |               |    |    |               |    |    |    |     |
|             | 001                                                                                       |         | 5 degree C                                        |           |    |               |    |    |               |    |    |    |     |
|             | 010                                                                                       |         | 0 degree C                                        |           |    |               |    |    |               |    |    |    |     |
|             | 011                                                                                       |         | -5 degree C                                       |           |    |               |    |    |               |    |    |    |     |
|             | 100                                                                                       |         | -10 degree C                                      |           |    |               |    |    |               |    |    |    |     |
|             | 101                                                                                       |         | -15 degree C                                      |           |    |               |    |    |               |    |    |    |     |
|             | 110~111                                                                                   |         | reserved                                          |           |    |               |    |    |               |    |    |    |     |
|             | TMP_TH_H[2:0]                                                                             |         | Threshold Temperature of High Temperature Sensing |           |    |               |    |    |               |    |    |    |     |
|             | 000                                                                                       |         | 65 degree C                                       |           |    |               |    |    |               |    |    |    |     |
|             | 001                                                                                       |         | 60 degree C                                       |           |    |               |    |    |               |    |    |    |     |
|             | 010                                                                                       |         | 55 degree C                                       |           |    |               |    |    |               |    |    |    |     |
|             | 011                                                                                       |         | 50 degree C                                       |           |    |               |    |    |               |    |    |    |     |
|             | 100                                                                                       |         | 45 degree C                                       |           |    |               |    |    |               |    |    |    |     |
|             | 101                                                                                       |         | 40 degree C                                       |           |    |               |    |    |               |    |    |    |     |
|             | 110~111                                                                                   |         | reserved                                          |           |    |               |    |    |               |    |    |    |     |
| Restriction |                                                                                           |         |                                                   |           |    |               |    |    |               |    |    |    |     |

**SDEQCTR(1740h~1940h): Source EQ Settings**

| 1740H ~ 1940H                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                   |                          | SDEQCTR               |           |       |           |    |    |    |    |    |    |       |       |                          |             |                                     |   |           |          |   |     |                |   |   |               |     |     |                       |  |          |  |          |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|-----------------------|-----------|-------|-----------|----|----|----|----|----|----|-------|-------|--------------------------|-------------|-------------------------------------|---|-----------|----------|---|-----|----------------|---|---|---------------|-----|-----|-----------------------|--|----------|--|----------|
| Instruction                                                                                                                                                                                                                                                                                                                                                                                                                          | R/W                                                                                                                                               | Address                  |                       | Parameter |       |           |    |    |    |    |    |    |       |       |                          |             |                                     |   |           |          |   |     |                |   |   |               |     |     |                       |  |          |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                   | MIPI                     | Others                | D15-D8    | D7    | D6        | D5 | D4 | D3 | D2 | D1 | D0 | HEX   |       |                          |             |                                     |   |           |          |   |     |                |   |   |               |     |     |                       |  |          |  |          |
| SDEQCTR                                                                                                                                                                                                                                                                                                                                                                                                                              | R/W                                                                                                                                               | 0x17h                    | 0x1740h               | 00h       | EQMOD | EQS1[6:0] |    |    |    |    |    |    | 0x09h |       |                          |             |                                     |   |           |          |   |     |                |   |   |               |     |     |                       |  |          |  |          |
| SDEQCTR                                                                                                                                                                                                                                                                                                                                                                                                                              | R/W                                                                                                                                               | 0x18h                    | 0x1840h               | 00h       | -     | EQS2[6:0] |    |    |    |    |    |    | 0x09h |       |                          |             |                                     |   |           |          |   |     |                |   |   |               |     |     |                       |  |          |  |          |
| SDEQCTR                                                                                                                                                                                                                                                                                                                                                                                                                              | R/W                                                                                                                                               | 0x19h                    | 0x1940h               | 00h       | -     | EQS3[6:0] |    |    |    |    |    |    | 0x09h |       |                          |             |                                     |   |           |          |   |     |                |   |   |               |     |     |                       |  |          |  |          |
| Description                                                                                                                                                                                                                                                                                                                                                                                                                          | This command is for Source EQ settings..                                                                                                          |                          |                       |           |       |           |    |    |    |    |    |    |       |       |                          |             |                                     |   |           |          |   |     |                |   |   |               |     |     |                       |  |          |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                      | <table><tr><th>EQMOD</th><th>Source EQ Mode Selection</th></tr><tr><td>0</td><td>EQ MODE 1</td></tr><tr><td>1</td><td>EQ MODE 2</td></tr></table> |                          |                       |           |       |           |    |    |    |    |    |    |       | EQMOD | Source EQ Mode Selection | 0           | EQ MODE 1                           | 1 | EQ MODE 2 |          |   |     |                |   |   |               |     |     |                       |  |          |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                      | EQMOD                                                                                                                                             | Source EQ Mode Selection |                       |           |       |           |    |    |    |    |    |    |       |       |                          |             |                                     |   |           |          |   |     |                |   |   |               |     |     |                       |  |          |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0                                                                                                                                                 | EQ MODE 1                |                       |           |       |           |    |    |    |    |    |    |       |       |                          |             |                                     |   |           |          |   |     |                |   |   |               |     |     |                       |  |          |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1                                                                                                                                                 | EQ MODE 2                |                       |           |       |           |    |    |    |    |    |    |       |       |                          |             |                                     |   |           |          |   |     |                |   |   |               |     |     |                       |  |          |  |          |
| <div><div><p>EQ Mode1</p></div><div><p>EQ Mode2</p></div></div>                                                                                                                                                                                                 |                                                                                                                                                   |                          |                       |           |       |           |    |    |    |    |    |    |       |       |                          |             |                                     |   |           |          |   |     |                |   |   |               |     |     |                       |  |          |  |          |
| <table><tr><th>Bit</th><th colspan="2">Value</th><th>Description</th></tr><tr><td rowspan="4">EQS1[6:0]<br/>EQS2[6:0]<br/>EQS3[6:0]</td><td>0</td><td>00h</td><td>0 Clocks</td></tr><tr><td>1</td><td>01h</td><td>1 Clock = 80ns</td></tr><tr><td>:</td><td>:</td><td>80 ns / Clock</td></tr><tr><td>128</td><td>7Fh</td><td>128 Clocks = 10240 ns</td></tr><tr><td></td><td colspan="2">Reserved</td><td>Reserved</td></tr></table> |                                                                                                                                                   |                          |                       |           |       |           |    |    |    |    |    |    | Bit   | Value |                          | Description | EQS1[6:0]<br>EQS2[6:0]<br>EQS3[6:0] | 0 | 00h       | 0 Clocks | 1 | 01h | 1 Clock = 80ns | : | : | 80 ns / Clock | 128 | 7Fh | 128 Clocks = 10240 ns |  | Reserved |  | Reserved |
| Bit                                                                                                                                                                                                                                                                                                                                                                                                                                  | Value                                                                                                                                             |                          | Description           |           |       |           |    |    |    |    |    |    |       |       |                          |             |                                     |   |           |          |   |     |                |   |   |               |     |     |                       |  |          |  |          |
| EQS1[6:0]<br>EQS2[6:0]<br>EQS3[6:0]                                                                                                                                                                                                                                                                                                                                                                                                  | 0                                                                                                                                                 | 00h                      | 0 Clocks              |           |       |           |    |    |    |    |    |    |       |       |                          |             |                                     |   |           |          |   |     |                |   |   |               |     |     |                       |  |          |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1                                                                                                                                                 | 01h                      | 1 Clock = 80ns        |           |       |           |    |    |    |    |    |    |       |       |                          |             |                                     |   |           |          |   |     |                |   |   |               |     |     |                       |  |          |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                      | :                                                                                                                                                 | :                        | 80 ns / Clock         |           |       |           |    |    |    |    |    |    |       |       |                          |             |                                     |   |           |          |   |     |                |   |   |               |     |     |                       |  |          |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                      | 128                                                                                                                                               | 7Fh                      | 128 Clocks = 10240 ns |           |       |           |    |    |    |    |    |    |       |       |                          |             |                                     |   |           |          |   |     |                |   |   |               |     |     |                       |  |          |  |          |
|                                                                                                                                                                                                                                                                                                                                                                                                                                      | Reserved                                                                                                                                          |                          | Reserved              |           |       |           |    |    |    |    |    |    |       |       |                          |             |                                     |   |           |          |   |     |                |   |   |               |     |     |                       |  |          |  |          |
| Restriction                                                                                                                                                                                                                                                                                                                                                                                                                          | Total EQ time must be less than 1H line period.<br>EQ Mode1: 2*(EQS1 + EQS2) < T1<br>EQ Mode2: 2*(EQS1 + EQS2 + EQS3) < T1                        |                          |                       |           |       |           |    |    |    |    |    |    |       |       |                          |             |                                     |   |           |          |   |     |                |   |   |               |     |     |                       |  |          |  |          |

**INVCTR(1B40h): Inversion Type**

| 1B40H       |                                                                                     | INVCTR  |        |           |    |                   |    |    |    |    |    |    |     |
|-------------|-------------------------------------------------------------------------------------|---------|--------|-----------|----|-------------------|----|----|----|----|----|----|-----|
| Instruction | R/W                                                                                 | Address |        | Parameter |    |                   |    |    |    |    |    |    |     |
|             |                                                                                     | MIPI    | Others | D15-D8    | D7 | D6                | D5 | D4 | D3 | D2 | D1 | D0 | HEX |
| INVCTR      | R/W                                                                                 | 1Bh     | 1B40h  | 00h       | -  | NLA[2:0]          |    | -  | -  | -  | -  | 00 |     |
| Description | This command is used to control the inversion mode of source driver in normal mode. |         |        |           |    |                   |    |    |    |    |    |    |     |
|             | NLA[2:0]                                                                            |         |        | Hex Value |    | Description       |    |    |    |    |    |    |     |
|             | 000                                                                                 |         |        | 00h       |    | Column Inversion  |    |    |    |    |    |    |     |
|             | 001                                                                                 |         |        | 01h       |    | 1-Dot Inversion   |    |    |    |    |    |    |     |
|             | 010                                                                                 |         |        | 02h       |    | 2-Dot Inversion   |    |    |    |    |    |    |     |
|             | 011                                                                                 |         |        | 03h       |    | 3-Dot Inversion   |    |    |    |    |    |    |     |
|             | 100                                                                                 |         |        | 04h       |    | 4-Dot Inversion   |    |    |    |    |    |    |     |
|             | 101                                                                                 |         |        | 05h       |    | Zig-zag Inversion |    |    |    |    |    |    |     |
|             | 110                                                                                 |         |        | 06h       |    | Reserved          |    |    |    |    |    |    |     |
|             | 111                                                                                 |         |        | 07h       |    | Reserved          |    |    |    |    |    |    |     |
| Restriction |                                                                                     |         |        |           |    |                   |    |    |    |    |    |    |     |



**EXT PWR IC Control (2440h~2540h) External PWR IC Control**

| 2440H<br>2540H     | External PWR IC Control                                                  |                                                                    |                                    |           |           |                |                                 |              |          |    |    |    |     |  |
|--------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------|------------------------------------|-----------|-----------|----------------|---------------------------------|--------------|----------|----|----|----|-----|--|
| Instruction        | R/W                                                                      | Address                                                            |                                    | Parameter |           |                |                                 |              |          |    |    |    |     |  |
|                    |                                                                          | MIPI                                                               | Others                             | D15-D8    | D7        | D6             | D5                              | D4           | D3       | D2 | D1 | D0 | HEX |  |
| EXT PWR IC Control | R/W                                                                      | 24h                                                                | 2440h                              | 00h       | PWRIC_VER | D017_SEL       | 0                               | 0            | 0        |    |    | 40 |     |  |
| EXT PWR IC Control | R/W                                                                      | 25h                                                                | 2540h                              | 00h       | 0         | PIC_BT_DH[1:0] | 1                               | PIC_VDH[3:0] |          |    | 55 |    |     |  |
| Description        | This command is used to select external power IC timing mode             |                                                                    |                                    |           |           |                |                                 |              |          |    |    |    |     |  |
|                    | Bit                                                                      |                                                                    | Description                        |           |           |                | Data                            |              |          |    |    |    |     |  |
|                    | PWRIC_VER                                                                |                                                                    | Power IC selection                 |           |           |                | 0: D016 mode<br>1: D017 mode    |              |          |    |    |    |     |  |
|                    | D017_TIM_SEL                                                             |                                                                    | Power IC timing selection          |           |           |                | 0: high active<br>1: low active |              |          |    |    |    |     |  |
|                    | This command is used to set the AVDD output option for external power IC |                                                                    |                                    |           |           |                |                                 |              |          |    |    |    |     |  |
|                    | Bit                                                                      |                                                                    | Description                        |           |           |                | Data                            |              |          |    |    |    |     |  |
|                    | PIC_BT_DH[1:0]                                                           |                                                                    | AVDD Pump Ratio Factor             |           |           |                | 00                              |              | Reserved |    |    |    |     |  |
|                    |                                                                          |                                                                    |                                    |           |           |                | 01                              |              | 1.5X     |    |    |    |     |  |
|                    |                                                                          |                                                                    |                                    |           |           |                | 10                              |              | 2X       |    |    |    |     |  |
|                    |                                                                          |                                                                    |                                    |           |           |                | 11                              |              | 3X       |    |    |    |     |  |
|                    | PIC_VDH[3:0]                                                             |                                                                    | AVDD Boost Voltage Level Selection |           |           |                | 0000                            |              | 6.0V     |    |    |    |     |  |
|                    |                                                                          |                                                                    |                                    |           |           |                | 0001                            |              | 5.9V     |    |    |    |     |  |
|                    |                                                                          |                                                                    |                                    |           |           |                | ... 0.1V/ code ...              |              |          |    |    |    |     |  |
|                    |                                                                          |                                                                    |                                    |           |           |                | 1101                            |              | 4.7V     |    |    |    |     |  |
|                    |                                                                          |                                                                    |                                    |           |           |                | 1110                            |              | 4.6V     |    |    |    |     |  |
|                    |                                                                          |                                                                    |                                    |           |           |                | 1111                            |              | 4.5V     |    |    |    |     |  |
|                    | Restriction                                                              | These settings will be updated only RM68200 is under sleep-in mode |                                    |           |           |                |                                 |              |          |    |    |    |     |  |

**SETAVDD(2740h): PFM Control for AVDD Output**

| 2740H       |                                                                                             |         |           | SETAVDD_PFM |                     |    |    |              |    |    |    |    |     |
|-------------|---------------------------------------------------------------------------------------------|---------|-----------|-------------|---------------------|----|----|--------------|----|----|----|----|-----|
| Instruction | R/W                                                                                         | Address |           | Parameter   |                     |    |    |              |    |    |    |    |     |
|             |                                                                                             | MIPI    | Others    | D15-D8      | D7                  | D6 | D5 | D4           | D3 | D2 | D1 | D0 | HEX |
| SETAVDD     | R/W                                                                                         | 27h     | 2740h     | 00h         | -                   | -  | -  | VDP_FBA[4:0] |    |    |    |    | 13h |
| Description | This command is used set AVDD voltage, generated by external boost circuit, in normal mode. |         |           |             |                     |    |    |              |    |    |    |    |     |
|             | Bit                                                                                         |         | Hex Value |             | AVDD Output Voltage |    |    |              |    |    |    |    |     |
|             | VDP_FBA[4:0]                                                                                |         | 00h       |             | 6.5V                |    |    |              |    |    |    |    |     |
|             |                                                                                             |         | 01h       |             | 6.4V                |    |    |              |    |    |    |    |     |
|             |                                                                                             |         | :         |             | 0.1V / Step         |    |    |              |    |    |    |    |     |
|             |                                                                                             |         | 14h       |             | 4.5V                |    |    |              |    |    |    |    |     |
|             |                                                                                             |         | 15h       |             | Reserved            |    |    |              |    |    |    |    |     |
|             |                                                                                             |         | 1Fh       |             | Reserved            |    |    |              |    |    |    |    |     |
| Restriction |                                                                                             |         |           |             |                     |    |    |              |    |    |    |    |     |

**SETAVEE(2940h): PFM Control for AVEE Output**

| 2940H       |                                                                                                |         |           | SETAVEE_PFM |                     |    |    |              |    |    |    |    |     |
|-------------|------------------------------------------------------------------------------------------------|---------|-----------|-------------|---------------------|----|----|--------------|----|----|----|----|-----|
| Instruction | R/W                                                                                            | Address |           | Parameter   |                     |    |    |              |    |    |    |    |     |
|             |                                                                                                | MIPI    | Others    | D15-D8      | D7                  | D6 | D5 | D4           | D3 | D2 | D1 | D0 | HEX |
| SETAVEE     | R/W                                                                                            | 29h     | 2940h     | 00h         | -                   | -  | -  | VDN_FBA[4:0] |    |    |    |    | 13h |
| Description | This command is used to set AVEE voltage, generated by external boost circuit, in normal mode. |         |           |             |                     |    |    |              |    |    |    |    |     |
|             | Bit                                                                                            |         | Hex Value |             | AVEE Output Voltage |    |    |              |    |    |    |    |     |
|             | VDN_FBA[4:0]                                                                                   |         | 00h       |             | -6.5V               |    |    |              |    |    |    |    |     |
|             |                                                                                                |         | 01h       |             | -6.4V               |    |    |              |    |    |    |    |     |
|             |                                                                                                |         | :         |             | -0.1V / Step        |    |    |              |    |    |    |    |     |
|             |                                                                                                |         | 14h       |             | -4.5V               |    |    |              |    |    |    |    |     |
|             |                                                                                                |         | 15h       |             | Reserved            |    |    |              |    |    |    |    |     |
|             |                                                                                                |         | 1Fh       |             | Reserved            |    |    |              |    |    |    |    |     |
| Restriction |                                                                                                |         |           |             |                     |    |    |              |    |    |    |    |     |

**BT2CTR(2B40h): DDVDL Charge Pump Control**

| 2B40H                                     |                                                                                                                                         | BT2CTR               |                              |           |           |    |    |                    |    |    |    |    |     |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------------|------------------------------|-----------|-----------|----|----|--------------------|----|----|----|----|-----|
| Instruction                               | R/W                                                                                                                                     | Address              |                              | Parameter |           |    |    |                    |    |    |    |    |     |
|                                           |                                                                                                                                         | MIPI                 | Others                       | D15-D8    | D7        | D6 | D5 | D4                 | D3 | D2 | D1 | D0 | HEX |
| BT2CTR                                    | R/W                                                                                                                                     | 2Bh                  | 2B40h                        | 00h       | DC2A[2:0] |    |    | VDLA[4:0]          |    |    |    |    | 85  |
| Description                               | This Command is used to set internal DDVDL's charge-pump clock, external power IC pumping clock, and DDVDL output level in normal mode. |                      |                              |           |           |    |    |                    |    |    |    |    |     |
|                                           | DC2A[2:0]                                                                                                                               |                      | DDVDL C-pump clock frequency |           |           |    |    | BCLK1 frequency    |    |    |    |    |     |
|                                           | 3'h0                                                                                                                                    |                      | Hsync / 32                   |           |           |    |    | 2.43KHz            |    |    |    |    |     |
|                                           | 3'h1                                                                                                                                    |                      | Hsync / 16                   |           |           |    |    | 4.86KHz            |    |    |    |    |     |
|                                           | 3'h2                                                                                                                                    |                      | Hsync / 8                    |           |           |    |    | 9.72KHz            |    |    |    |    |     |
|                                           | 3'h3                                                                                                                                    |                      | Hsync / 4                    |           |           |    |    | 19.44KHz           |    |    |    |    |     |
|                                           | 3'h4                                                                                                                                    |                      | Hsync / 2 (default)          |           |           |    |    | 38.88KHz (default) |    |    |    |    |     |
|                                           | 3'h5                                                                                                                                    |                      | Hsync                        |           |           |    |    | 77.76KHz           |    |    |    |    |     |
|                                           | 3'h6                                                                                                                                    |                      | 2 * Hsync                    |           |           |    |    | 155.52KHz          |    |    |    |    |     |
|                                           | 3'h7                                                                                                                                    |                      | 4 * Hsync                    |           |           |    |    | 311.04KHz          |    |    |    |    |     |
| ** Hsync = 60Hz x (1280+16) LINE=77.76KHz |                                                                                                                                         |                      |                              |           |           |    |    |                    |    |    |    |    |     |
| VDLA[4:0]                                 |                                                                                                                                         | DDVDL output voltage |                              |           |           |    |    |                    |    |    |    |    |     |
| 00h                                       |                                                                                                                                         | -6.5V                |                              |           |           |    |    |                    |    |    |    |    |     |
| 01h                                       |                                                                                                                                         | -6.4V                |                              |           |           |    |    |                    |    |    |    |    |     |
| 02h                                       |                                                                                                                                         | -6.3V                |                              |           |           |    |    |                    |    |    |    |    |     |
| :                                         |                                                                                                                                         | 0.1V step            |                              |           |           |    |    |                    |    |    |    |    |     |
| 05h(default)                              |                                                                                                                                         | -6.0V                |                              |           |           |    |    |                    |    |    |    |    |     |
| :                                         |                                                                                                                                         | 0.1V step            |                              |           |           |    |    |                    |    |    |    |    |     |
| 13h                                       |                                                                                                                                         | -4.6V                |                              |           |           |    |    |                    |    |    |    |    |     |
| 14h                                       |                                                                                                                                         | -4.5V                |                              |           |           |    |    |                    |    |    |    |    |     |
| 15h~1Fh                                   |                                                                                                                                         | Reserved             |                              |           |           |    |    |                    |    |    |    |    |     |
| Restriction                               |                                                                                                                                         |                      |                              |           |           |    |    |                    |    |    |    |    |     |

**BT3CTR (2F40h): VGH Charge Pump Control**

| 0x2F40h     |                                                                             |         |           | BT3CTR    |                                                         |             |           |    |                |    |    |    |     |  |
|-------------|-----------------------------------------------------------------------------|---------|-----------|-----------|---------------------------------------------------------|-------------|-----------|----|----------------|----|----|----|-----|--|
| Instruction | R/W                                                                         | Address |           | Parameter |                                                         |             |           |    |                |    |    |    |     |  |
|             |                                                                             | MIPI    | Others    | D15-D8    | D7                                                      | D6          | D5        | D4 | D3             | D2 | D1 | D0 | HEX |  |
| BT3CTR      | R/W                                                                         | 2Fh     | 2F40h     | 00h       | VGHSA[1:0]                                              | BT_GHA[1:0] | VBHA[3:0] |    |                | 50 |    |    |     |  |
| Description | This command is to control VGH charge pump function in Normal/Partial Mode. |         |           |           |                                                         |             |           |    |                |    |    |    |     |  |
|             | Bit                                                                         |         | Hex Value |           | VGH Output                                              |             |           |    |                |    |    |    |     |  |
|             | VGHSA[1:0]                                                                  |         | 0h        |           | VGH-pump output is un-regulated                         |             |           |    |                |    |    |    |     |  |
|             |                                                                             |         | 1h        |           | VGH's pump output is regulated and depends on VBHA[3:0] |             |           |    |                |    |    |    |     |  |
|             |                                                                             |         | 2h        |           | AVDD                                                    |             |           |    |                |    |    |    |     |  |
|             |                                                                             |         | 3h        |           | AVDD                                                    |             |           |    |                |    |    |    |     |  |
|             | Bit                                                                         |         | Hex Value |           | VGH Output                                              |             |           |    | Multiple-Ratio |    |    |    |     |  |
|             | BT_GHA[1:0]                                                                 |         | 0h        |           | 2*AVDD-AVEE                                             |             |           |    | VCI x 6        |    |    |    |     |  |
|             |                                                                             |         | 1h        |           | 2*AVDD-AVEE+VCI                                         |             |           |    | VCI x 7        |    |    |    |     |  |
|             |                                                                             |         | 2h        |           | 3*AVDD-AVEE                                             |             |           |    | VCI x 8        |    |    |    |     |  |
|             |                                                                             |         | 3h        |           | 3*AVDD-2*AVEE                                           |             |           |    | VCI x 10       |    |    |    |     |  |
|             | Bit                                                                         |         | Hex Value |           | VGH Output Voltage                                      |             |           |    |                |    |    |    |     |  |
|             | VBHA[3:0]                                                                   |         | 0h        |           | 13V                                                     |             |           |    |                |    |    |    |     |  |
|             |                                                                             |         | 1h        |           | 13.5V                                                   |             |           |    |                |    |    |    |     |  |
| 2h          |                                                                             |         | 14V       |           |                                                         |             |           |    |                |    |    |    |     |  |
| :           |                                                                             |         | 0.5V/step |           |                                                         |             |           |    |                |    |    |    |     |  |
| 8h          |                                                                             |         | 17V       |           |                                                         |             |           |    |                |    |    |    |     |  |
| :           |                                                                             |         | 0.5V/step |           |                                                         |             |           |    |                |    |    |    |     |  |
| Eh          |                                                                             |         | 20V       |           |                                                         |             |           |    |                |    |    |    |     |  |
| Fh          |                                                                             |         | 20.5V     |           |                                                         |             |           |    |                |    |    |    |     |  |
| Restriction |                                                                             |         |           |           |                                                         |             |           |    |                |    |    |    |     |  |

**BT3CTR(3140h): VGH Charge Pump Clock**

| 3140H                                     |                                                                 | BT3CTR  |                            |           |           |    |                    |    |    |    |    |    |       |
|-------------------------------------------|-----------------------------------------------------------------|---------|----------------------------|-----------|-----------|----|--------------------|----|----|----|----|----|-------|
| Instruction                               | R/W                                                             | Address |                            | Parameter |           |    |                    |    |    |    |    |    |       |
|                                           |                                                                 | MIPI    | Others                     | D15-D8    | D7        | D6 | D5                 | D4 | D3 | D2 | D1 | D0 | HEX   |
| BT3CTR                                    | R/W                                                             | 0x31h   | 0x3140h                    | 00h       | DC4A[2:0] |    |                    | -  | -  | -  | -  | -  | 0x80h |
| Description                               | This command is to set VGH-pump clock frequency in normal mode. |         |                            |           |           |    |                    |    |    |    |    |    |       |
|                                           | DC4A[2:0]                                                       |         | VGH C-pump clock frequency |           |           |    | BCLK3 frequency    |    |    |    |    |    |       |
|                                           | 3'h0                                                            |         | Hsync / 32                 |           |           |    | 2.43KHz            |    |    |    |    |    |       |
|                                           | 3'h1                                                            |         | Hsync / 16                 |           |           |    | 4.86KHz            |    |    |    |    |    |       |
|                                           | 3'h2                                                            |         | Hsync / 8                  |           |           |    | 9.72KHz            |    |    |    |    |    |       |
|                                           | 3'h3                                                            |         | Hsync / 4                  |           |           |    | 19.44KHz           |    |    |    |    |    |       |
|                                           | 3'h4                                                            |         | Hsync / 2 (default)        |           |           |    | 38.88KHz (default) |    |    |    |    |    |       |
|                                           | 3'h5                                                            |         | Hsync                      |           |           |    | 77.76KHz           |    |    |    |    |    |       |
|                                           | 3'h6                                                            |         | 2 * Hsync                  |           |           |    | 155.52KHz          |    |    |    |    |    |       |
|                                           | 3'h7                                                            |         | 4 * Hsync                  |           |           |    | 311.04KHz          |    |    |    |    |    |       |
| ** Hsync = 60Hz x (1280+16) LINE=77.76KHz |                                                                 |         |                            |           |           |    |                    |    |    |    |    |    |       |
| Restriction                               |                                                                 |         |                            |           |           |    |                    |    |    |    |    |    |       |

## TMPSVGHCTR(3240h~3340h):VGH Temperature Compensation

| 3240H<br>3349H                                                                            |                                                                                                                                                                                              | TMPSVGHCTR |        |                                           |                                           |    |    |                |    |    |    |    |     |
|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|--------|-------------------------------------------|-------------------------------------------|----|----|----------------|----|----|----|----|-----|
| Instruction                                                                               | R/W                                                                                                                                                                                          | Address    |        | Parameter                                 |                                           |    |    |                |    |    |    |    |     |
|                                                                                           |                                                                                                                                                                                              | MIPI       | Others | D15-D8                                    | D7                                        | D6 | D5 | D4             | D3 | D2 | D1 | D0 | HEX |
| TMPSVGHCTR                                                                                | R/W                                                                                                                                                                                          | 32h        | 3240h  | 00h                                       | VGH_OFFSET_L<br>[2:0]                     |    | -  | TMP_BTH_L[1:0] |    | -  | -  | -  | A4  |
| TMPSVGHCTR                                                                                | R/W                                                                                                                                                                                          | 33h        | 3340h  | 00h                                       | VGH_OFFSET_H<br>[2:0]                     |    | -  | TMP_BTH_H[1:0] |    | -  | -  | -  | A4  |
| Description                                                                               | This command is for temperature compensation control of VGH-pump output. TMP_BTH_L[1:0] /TMP_BTH_H[1:0] represents boosting times control of VGH-pump when low/high temperature is detected. |            |        |                                           |                                           |    |    |                |    |    |    |    |     |
|                                                                                           | TMP_BTH_L[1:0]<br>TMP_BTH_H[1:0]                                                                                                                                                             |            |        |                                           | Boosting times for VGH C-pump             |    |    |                |    |    |    |    |     |
|                                                                                           | 0h                                                                                                                                                                                           |            |        |                                           | 2*AVDD-AVEE                               |    |    |                |    |    |    |    |     |
|                                                                                           | 1h                                                                                                                                                                                           |            |        |                                           | 2*AVDD-AVEE+VCI                           |    |    |                |    |    |    |    |     |
|                                                                                           | 2h                                                                                                                                                                                           |            |        |                                           | 3*AVDD-1*AVEE                             |    |    |                |    |    |    |    |     |
|                                                                                           | 3h                                                                                                                                                                                           |            |        |                                           | 3*AVDD-2*AVEE                             |    |    |                |    |    |    |    |     |
|                                                                                           | VGH_OFFSET_L[2:0]: offset value for VGH voltage setting when low temperature is detected.                                                                                                    |            |        |                                           |                                           |    |    |                |    |    |    |    |     |
|                                                                                           | VGH_OFFSET_L[2:0]                                                                                                                                                                            |            |        |                                           | Code Offset Value for VGH Voltage Setting |    |    |                |    |    |    |    |     |
|                                                                                           | 0h                                                                                                                                                                                           |            |        |                                           | +0 (offset = 0V)                          |    |    |                |    |    |    |    |     |
|                                                                                           | 1h                                                                                                                                                                                           |            |        |                                           | +1 (offset = 0.5V)                        |    |    |                |    |    |    |    |     |
| 2h                                                                                        |                                                                                                                                                                                              |            |        | +2 (offset = 1.0V)                        |                                           |    |    |                |    |    |    |    |     |
| :                                                                                         |                                                                                                                                                                                              |            |        | :                                         |                                           |    |    |                |    |    |    |    |     |
| 7h                                                                                        |                                                                                                                                                                                              |            |        | +7 (offset = 3.5V)                        |                                           |    |    |                |    |    |    |    |     |
| Note: 0.5V/code                                                                           |                                                                                                                                                                                              |            |        |                                           |                                           |    |    |                |    |    |    |    |     |
| VGH_OFFSET_H[2:0]: offset value for VGH voltage setting when high temperature is detected |                                                                                                                                                                                              |            |        |                                           |                                           |    |    |                |    |    |    |    |     |
| VGH_OFFSET_H[2:0]                                                                         |                                                                                                                                                                                              |            |        | Code Offset Value for VGH Voltage Setting |                                           |    |    |                |    |    |    |    |     |
| 0h                                                                                        |                                                                                                                                                                                              |            |        | -0 (offset = 0V)                          |                                           |    |    |                |    |    |    |    |     |
| 1h                                                                                        |                                                                                                                                                                                              |            |        | -1 (offset = -0.5V)                       |                                           |    |    |                |    |    |    |    |     |
| 2h                                                                                        |                                                                                                                                                                                              |            |        | -2 (offset = -1.0V)                       |                                           |    |    |                |    |    |    |    |     |
| :                                                                                         |                                                                                                                                                                                              |            |        | :                                         |                                           |    |    |                |    |    |    |    |     |
| 7h                                                                                        |                                                                                                                                                                                              |            |        | -7 (offset = -3.5V)                       |                                           |    |    |                |    |    |    |    |     |
| Note: 0.5V/code                                                                           |                                                                                                                                                                                              |            |        |                                           |                                           |    |    |                |    |    |    |    |     |
| Restriction                                                                               | Under low-temperature case: 0h ≤ VBHA[3:0]+VGH_OFFSET_L[2:0] ≤ Fh<br>Under high-temperature case: 0h ≤ VBHA[3:0]+VGH_OFFSET_H[2:0] ≤ Fh                                                      |            |        |                                           |                                           |    |    |                |    |    |    |    |     |

**BT4CTR(3440h): VGL Charge Pump Control**

| 3440H       |                                                                             | BT4CTR  |                                                                                                                                |           |                                                         |    |             |    |                |    |    |       |     |
|-------------|-----------------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------|-----------|---------------------------------------------------------|----|-------------|----|----------------|----|----|-------|-----|
| Instruction | R/W                                                                         | Address |                                                                                                                                | Parameter |                                                         |    |             |    |                |    |    |       |     |
|             |                                                                             | MIPI    | Others                                                                                                                         | D15-D8    | D7                                                      | D6 | D5          | D4 | D3             | D2 | D1 | D0    | HEX |
| BT4CTR      | R/W                                                                         | 0x34h   | 0x3440h                                                                                                                        | 00h       | VGLSA[1:0]                                              |    | BT_GLA[1:0] |    | VBLA[3:0]      |    |    | 0x55h |     |
| Description | This command is to control VGL charge pump function in Normal/Partial Mode. |         |                                                                                                                                |           |                                                         |    |             |    |                |    |    |       |     |
|             | Bit                                                                         |         | Hex Value                                                                                                                      |           | VGL Output                                              |    |             |    |                |    |    |       |     |
|             | VGLSA[1:0]                                                                  |         | 0h                                                                                                                             |           | VGL-pump output is un-regulated                         |    |             |    |                |    |    |       |     |
|             |                                                                             |         | 1h                                                                                                                             |           | VGL's pump output is regulated and depends on VBLA[3:0] |    |             |    |                |    |    |       |     |
|             |                                                                             |         | 2h                                                                                                                             |           | AVEE                                                    |    |             |    |                |    |    |       |     |
|             |                                                                             |         | 3h                                                                                                                             |           | AVEE                                                    |    |             |    |                |    |    |       |     |
|             | Bit                                                                         |         | Hex Value                                                                                                                      |           | VGL Output                                              |    |             |    | Multiple-Ratio |    |    |       |     |
|             | BT_GLA[1:0]                                                                 |         | 0h                                                                                                                             |           | AVEE- AVDD                                              |    |             |    | VCI x (-4)     |    |    |       |     |
|             |                                                                             |         | 1h                                                                                                                             |           | 2*AVEE-VCI                                              |    |             |    | VCI x (-5)     |    |    |       |     |
|             |                                                                             |         | 2h                                                                                                                             |           | 2*AVEE-AVDD                                             |    |             |    | VCI x (-6)     |    |    |       |     |
|             |                                                                             |         | 3h                                                                                                                             |           | 2*AVEE-AVDD - VCI                                       |    |             |    | VCI x (-7)     |    |    |       |     |
|             | Bit                                                                         |         | Hex Value                                                                                                                      |           | VGL Output Voltage                                      |    |             |    |                |    |    |       |     |
|             | VBLA[3:0]                                                                   |         | 0h                                                                                                                             |           | -7V                                                     |    |             |    |                |    |    |       |     |
|             |                                                                             |         | 1h                                                                                                                             |           | -8V                                                     |    |             |    |                |    |    |       |     |
|             |                                                                             |         | 2h                                                                                                                             |           | -8.5V                                                   |    |             |    |                |    |    |       |     |
| 3h          |                                                                             |         | -9V                                                                                                                            |           |                                                         |    |             |    |                |    |    |       |     |
| 4h          |                                                                             |         | -9.5V                                                                                                                          |           |                                                         |    |             |    |                |    |    |       |     |
| 5h          |                                                                             |         | -10V                                                                                                                           |           |                                                         |    |             |    |                |    |    |       |     |
| 6h          |                                                                             |         | -10.5V                                                                                                                         |           |                                                         |    |             |    |                |    |    |       |     |
| :           |                                                                             |         | 0.5V/step                                                                                                                      |           |                                                         |    |             |    |                |    |    |       |     |
| Dh          |                                                                             |         | -14V                                                                                                                           |           |                                                         |    |             |    |                |    |    |       |     |
| Eh          |                                                                             |         | -14.5V                                                                                                                         |           |                                                         |    |             |    |                |    |    |       |     |
| Fh          |                                                                             |         | -15V                                                                                                                           |           |                                                         |    |             |    |                |    |    |       |     |
| Restriction |                                                                             |         | If C24 is not used for application, BT_GLA=3'h(2*AVEE-AVDD-VCI) can't be used. And, the minimum output voltage is 2*AVEE-AVDD. |           |                                                         |    |             |    |                |    |    |       |     |

**BT4CTR(3640h): VGL Charge Pump Clock**

| 3640H                                     |                                                                | BT4CTR  |                            |           |           |    |                    |    |    |    |    |    |     |
|-------------------------------------------|----------------------------------------------------------------|---------|----------------------------|-----------|-----------|----|--------------------|----|----|----|----|----|-----|
| Instruction                               | R/W                                                            | Address |                            | Parameter |           |    |                    |    |    |    |    |    |     |
|                                           |                                                                | MIPI    | Others                     | D15-D8    | D7        | D6 | D5                 | D4 | D3 | D2 | D1 | D0 | HEX |
| BT4CTR                                    | R/W                                                            | 36h     | 3640h                      | X         | DC5A[2:0] |    | -                  | -  | -  | -  | -  | -  | 80  |
| Description                               | This command is to set VGL-pump clock frequency in normal mode |         |                            |           |           |    |                    |    |    |    |    |    |     |
|                                           | DC5A[2:0]                                                      |         | VGL C-pump clock frequency |           |           |    | BCLK4 frequency    |    |    |    |    |    |     |
|                                           | 3'h0                                                           |         | Hsync / 32                 |           |           |    | 2.43KHz            |    |    |    |    |    |     |
|                                           | 3'h1                                                           |         | Hsync / 16                 |           |           |    | 4.86KHz            |    |    |    |    |    |     |
|                                           | 3'h2                                                           |         | Hsync / 8                  |           |           |    | 9.72KHz            |    |    |    |    |    |     |
|                                           | 3'h3                                                           |         | Hsync / 4                  |           |           |    | 19.44KHz           |    |    |    |    |    |     |
|                                           | 3'h4                                                           |         | Hsync / 2 (default)        |           |           |    | 38.88KHz (default) |    |    |    |    |    |     |
|                                           | 3'h5                                                           |         | Hsync                      |           |           |    | 77.76KHz           |    |    |    |    |    |     |
|                                           | 3'h6                                                           |         | 2 * Hsync                  |           |           |    | 155.52KHz          |    |    |    |    |    |     |
|                                           | 3'h7                                                           |         | 4 * Hsync                  |           |           |    | 311.04KHz          |    |    |    |    |    |     |
| ** Hsync = 60Hz x (1280+16) LINE=77.76KHz |                                                                |         |                            |           |           |    |                    |    |    |    |    |    |     |
| Restriction                               |                                                                |         |                            |           |           |    |                    |    |    |    |    |    |     |



**TMPSVGLCTR(3740h~3840h): VGL Temperature Compensation**

| 3740H<br>3840H    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | TMPSVGLCTR                                |        |           |                   |    |    |                |    |    |    |    |     |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------|-----------|-------------------|----|----|----------------|----|----|----|----|-----|----------------------------------|------------------------|----|-----------|----|------------|----|-------------|----|-----------------|-------------------|-------------------------------------------|----|------------------|----|--------------------|----|--------------------|---|---|----|--------------------|-------------------|-------------------------------------------|----|-------------------|----|---------------------|----|---------------------|---|---|----|---------------------|
| Instruction       | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Address                                   |        | Parameter |                   |    |    |                |    |    |    |    |     |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
|                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | MIPI                                      | Others | D15-D8    | D7                | D6 | D5 | D4             | D3 | D2 | D1 | D0 | HEX |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
| TMPSVGLCTR        | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 37h                                       | 3740h  | X         | VGL_OFFSET_L[2:0] |    | -  | TMP_BTL_L[1:0] |    | -  | -  | -  | 00  |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
| TMPSVGLCTR        | R/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 38h                                       | 3840h  | X         | VGL_OFFSET_H[2:0] |    | -  | TMP_BTL_H[1:0] |    | -  | -  | -  | A4  |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
| Description       | <p>This command is for temperature compensation control of VGL-pump output. TMP_BTL_L[1:0]/TMP_BTL_H[1:0] represents boosting times control of VGL-pump when low/high temperature is detected.</p> <table><tr><th>TMP_BTL_L[1:0]<br/>TMP_BTL_H[1:0]</th><th>Boosting times for VGL</th></tr><tr><td>0h</td><td>AVEE-AVDD</td></tr><tr><td>1h</td><td>2*AVEE-VCI</td></tr><tr><td>2h</td><td>2*AVEE-AVDD</td></tr><tr><td>3h</td><td>2*AVEE-AVDD-VCI</td></tr></table> <p>VGL_OFFSET_L[2:0]: offset value for VGL voltage setting when low temperature is detected.</p> <table><tr><th>VGL_OFFSET_L[2:0]</th><th>Code Offset Value for VGL Voltage Setting</th></tr><tr><td>0h</td><td>+0 (offset = 0V)</td></tr><tr><td>1h</td><td>+1 (offset = 0.5V)</td></tr><tr><td>2h</td><td>+2 (offset = 1.0V)</td></tr><tr><td>:</td><td>:</td></tr><tr><td>7h</td><td>+7 (offset = 3.5V)</td></tr></table> <p>VGL_OFFSET_H[2:0]: offset value for VGL voltage setting when high temperature is detected.</p> <table><tr><th>VGL_OFFSET_H[2:0]</th><th>Code Offset Value for VGL Voltage Setting</th></tr><tr><td>0h</td><td>- 0 (offset = 0V)</td></tr><tr><td>1h</td><td>-1 (offset = -0.5V)</td></tr><tr><td>2h</td><td>-2 (offset = -1.0V)</td></tr><tr><td>:</td><td>:</td></tr><tr><td>7h</td><td>-7 (offset = -3.5V)</td></tr></table> |                                           |        |           |                   |    |    |                |    |    |    |    |     | TMP_BTL_L[1:0]<br>TMP_BTL_H[1:0] | Boosting times for VGL | 0h | AVEE-AVDD | 1h | 2*AVEE-VCI | 2h | 2*AVEE-AVDD | 3h | 2*AVEE-AVDD-VCI | VGL_OFFSET_L[2:0] | Code Offset Value for VGL Voltage Setting | 0h | +0 (offset = 0V) | 1h | +1 (offset = 0.5V) | 2h | +2 (offset = 1.0V) | : | : | 7h | +7 (offset = 3.5V) | VGL_OFFSET_H[2:0] | Code Offset Value for VGL Voltage Setting | 0h | - 0 (offset = 0V) | 1h | -1 (offset = -0.5V) | 2h | -2 (offset = -1.0V) | : | : | 7h | -7 (offset = -3.5V) |
|                   | TMP_BTL_L[1:0]<br>TMP_BTL_H[1:0]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Boosting times for VGL                    |        |           |                   |    |    |                |    |    |    |    |     |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
|                   | 0h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | AVEE-AVDD                                 |        |           |                   |    |    |                |    |    |    |    |     |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
|                   | 1h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2*AVEE-VCI                                |        |           |                   |    |    |                |    |    |    |    |     |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
|                   | 2h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2*AVEE-AVDD                               |        |           |                   |    |    |                |    |    |    |    |     |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
|                   | 3h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2*AVEE-AVDD-VCI                           |        |           |                   |    |    |                |    |    |    |    |     |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
|                   | VGL_OFFSET_L[2:0]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Code Offset Value for VGL Voltage Setting |        |           |                   |    |    |                |    |    |    |    |     |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
|                   | 0h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | +0 (offset = 0V)                          |        |           |                   |    |    |                |    |    |    |    |     |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
|                   | 1h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | +1 (offset = 0.5V)                        |        |           |                   |    |    |                |    |    |    |    |     |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
|                   | 2h                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | +2 (offset = 1.0V)                        |        |           |                   |    |    |                |    |    |    |    |     |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
| :                 | :                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                           |        |           |                   |    |    |                |    |    |    |    |     |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
| 7h                | +7 (offset = 3.5V)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                           |        |           |                   |    |    |                |    |    |    |    |     |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
| VGL_OFFSET_H[2:0] | Code Offset Value for VGL Voltage Setting                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                           |        |           |                   |    |    |                |    |    |    |    |     |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
| 0h                | - 0 (offset = 0V)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                           |        |           |                   |    |    |                |    |    |    |    |     |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
| 1h                | -1 (offset = -0.5V)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                           |        |           |                   |    |    |                |    |    |    |    |     |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
| 2h                | -2 (offset = -1.0V)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                           |        |           |                   |    |    |                |    |    |    |    |     |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
| :                 | :                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                           |        |           |                   |    |    |                |    |    |    |    |     |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
| 7h                | -7 (offset = -3.5V)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                           |        |           |                   |    |    |                |    |    |    |    |     |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |
| Restriction       | Under low-temperature case: 0h ≤ VBLA[3:0]+VGL_OFFSET_L[2:0] ≤ Fh<br>Under high-temperature case: 0h ≤ VBLA[3:0]+VGL_OFFSET_H[2:0] ≤ Fh                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                           |        |           |                   |    |    |                |    |    |    |    |     |                                  |                        |    |           |    |            |    |             |    |                 |                   |                                           |    |                  |    |                    |    |                    |   |   |    |                    |                   |                                           |    |                   |    |                     |    |                     |   |   |    |                     |

**BT2CTR(3940h): VCL Charge Pump Control**

| 3940H       |                                                                                                       |         |                             | BT2CTR    |      |    |            |                    |           |    |    |    |     |
|-------------|-------------------------------------------------------------------------------------------------------|---------|-----------------------------|-----------|------|----|------------|--------------------|-----------|----|----|----|-----|
| Instruction | R/W                                                                                                   | Address |                             | Parameter |      |    |            |                    |           |    |    |    |     |
|             |                                                                                                       | MIPI    | Others                      | D15-D8    | D7   | D6 | D5         | D4                 | D3        | D2 | D1 | D0 | HEX |
| BT2CTR      | R/W                                                                                                   | 39h     | 3940h                       | X         | BTCL | -  | DVCLA[1:0] | -                  | DC3A[2:0] |    |    | 04 |     |
| Description | This command is to control VCL charge pump function in Normal/Partial Mode.                           |         |                             |           |      |    |            |                    |           |    |    |    |     |
|             | BTCL                                                                                                  |         | VCL-pump boosting times     |           |      |    |            |                    |           |    |    |    |     |
|             | 0 (default)                                                                                           |         | VCI x -1                    |           |      |    |            |                    |           |    |    |    |     |
|             | 1                                                                                                     |         | VCI x -2                    |           |      |    |            |                    |           |    |    |    |     |
|             | DVCLA[1:0]                                                                                            |         | VCL-pump output             |           |      |    |            |                    |           |    |    |    |     |
|             | 00h                                                                                                   |         | -2.5V                       |           |      |    |            |                    |           |    |    |    |     |
|             | 01h                                                                                                   |         | -3.0V                       |           |      |    |            |                    |           |    |    |    |     |
|             | 02h                                                                                                   |         | -3.5V                       |           |      |    |            |                    |           |    |    |    |     |
|             | 03h                                                                                                   |         | -4.0V                       |           |      |    |            |                    |           |    |    |    |     |
|             | DC3A[2:0]                                                                                             |         | VCL boost circuit frequency |           |      |    |            | BCLK2 frequency    |           |    |    |    |     |
|             | 3'h0                                                                                                  |         | Hsync / 32                  |           |      |    |            | 2.43KHz            |           |    |    |    |     |
|             | 3'h1                                                                                                  |         | Hsync / 16                  |           |      |    |            | 4.86KHz            |           |    |    |    |     |
|             | 3'h2                                                                                                  |         | Hsync / 8                   |           |      |    |            | 9.72KHz            |           |    |    |    |     |
|             | 3'h3                                                                                                  |         | Hsync / 4                   |           |      |    |            | 19.44KHz           |           |    |    |    |     |
|             | 3'h4                                                                                                  |         | Hsync / 2 (default)         |           |      |    |            | 38.88KHz (default) |           |    |    |    |     |
|             | 3'h5                                                                                                  |         | Hsync                       |           |      |    |            | 77.76KHz           |           |    |    |    |     |
|             | 3'h6                                                                                                  |         | 2 * Hsync                   |           |      |    |            | 155.52KHz          |           |    |    |    |     |
|             | 3'h7                                                                                                  |         | 4 * Hsync                   |           |      |    |            | 311.04KHz          |           |    |    |    |     |
|             | ** Hsync = 60Hz x (1280+16) LINE=77.76KHz                                                             |         |                             |           |      |    |            |                    |           |    |    |    |     |
| Restriction | If C42 is not used for application, only VCI x -1 can be set. And, the minimum output depends on VCI. |         |                             |           |      |    |            |                    |           |    |    |    |     |

**VCMCTR(3D40h): VCOM Control**

| 4540H       |                                                                                                                                |         |                  | VCMCTR    |       |               |                     |    |    |    |    |       |     |
|-------------|--------------------------------------------------------------------------------------------------------------------------------|---------|------------------|-----------|-------|---------------|---------------------|----|----|----|----|-------|-----|
| Instruction | R/W                                                                                                                            | Address |                  | Parameter |       |               |                     |    |    |    |    |       |     |
|             |                                                                                                                                | MIPI    | Others           | D15-D8    | D7    | D6            | D5                  | D4 | D3 | D2 | D1 | D0    | HEX |
| VCMCTR      | R/W                                                                                                                            | 3Dh     | 3D40h            | 00h       | -     | -             | -                   | -  |    |    |    | VMSEL | 00  |
| Description | This command is for VCOM output control depending on ambient temperature and gate scan direction.                              |         |                  |           |       |               |                     |    |    |    |    |       |     |
|             | VMSEL                                                                                                                          | TGIP_EN | Temperature      |           | IDMON | Scan          | VCOM Output Voltage |    |    |    |    |       |     |
|             | 0                                                                                                                              | 0       | All Temperature  |           | 0     | x             | VCM_REG[8:0]        |    |    |    |    |       |     |
|             |                                                                                                                                |         |                  |           | 1     | x             | VCM_REG_IDM[8:0]    |    |    |    |    |       |     |
|             | 0                                                                                                                              | 1       | Room Temperature |           | 0     | x             | VCM_REG[8:0]        |    |    |    |    |       |     |
|             |                                                                                                                                |         |                  |           | 1     | x             | VCM_REG_IDM[8:0]    |    |    |    |    |       |     |
|             |                                                                                                                                |         | Low Temperature  |           | x     | x             | VCM2_REG[8:0]       |    |    |    |    |       |     |
|             |                                                                                                                                |         | High Temperature |           | x     | x             | VCM2_REG_IDM[8:0]   |    |    |    |    |       |     |
|             | 1                                                                                                                              | x       | All Temperature  |           | 0     | FRD           | VCM_REG[8:0]        |    |    |    |    |       |     |
|             | BRD                                                                                                                            |         |                  |           |       | VCM2_REG[8:0] |                     |    |    |    |    |       |     |
|             | 1                                                                                                                              |         |                  |           | 1     | FRD           | VCM_REG_IDM[8:0]    |    |    |    |    |       |     |
|             |                                                                                                                                |         |                  |           |       | BRD           | VCM2_REG_IDM[8:0]   |    |    |    |    |       |     |
|             | *** TGIP_EN is the enable control for temperature sensing. Please refer to address 0x1340h for more information about TGIP_EN. |         |                  |           |       |               |                     |    |    |    |    |       |     |
| Restriction |                                                                                                                                |         |                  |           |       |               |                     |    |    |    |    |       |     |

**VCMCTR (4540h~4940h): VCOM Output Level Control**

| 4640H ~ 4940H |                                                                                                                                                                    | VCMCTR  |        |                   |                   |                |    |    |             |                 |    |    |     |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------|-------------------|-------------------|----------------|----|----|-------------|-----------------|----|----|-----|
| Instruction   | R/W                                                                                                                                                                | Address |        | Parameter         |                   |                |    |    |             |                 |    |    |     |
|               |                                                                                                                                                                    | MIPI    | Others | D15-D8            | D7                | D6             | D5 | D4 | D3          | D2              | D1 | D0 | HEX |
| VCMCTR        | R/W                                                                                                                                                                | 45h     | 4540h  | 00h               | VCM_REG[8]        | VCM_REG_IDM[8] | -  | -  | VCM2_REG[8] | VCM2_REG_IDM[8] | -  | -  | 00  |
| VCMCTR        | R/W                                                                                                                                                                | 46h     | 4640h  | 00h               | VCM_REG[7:0]      |                |    |    |             |                 |    |    | 28  |
| VCMCTR        | R/W                                                                                                                                                                | 47h     | 4740h  | 00h               | VCM2_REG[7:0]     |                |    |    |             |                 |    |    | 00  |
| VCMCTR        | R/W                                                                                                                                                                | 48h     | 4840h  | 00h               | VCM_REG_IDM[7:0]  |                |    |    |             |                 |    |    | 00  |
| VCMCTR        | R/W                                                                                                                                                                | 49h     | 4940h  | 00h               | VCM2_REG_IDM[7:0] |                |    |    |             |                 |    |    | 00  |
| Description   | These commands are for VCOM output level control. These settings are used for different purposes. For more detailed information, please refer to register 0x3D40h. |         |        |                   |                   |                |    |    |             |                 |    |    |     |
|               | VCM_REG[8:0]<br>VCM2_REG[8:0]<br>VCM_REG_IDM[8:0]<br>VCM2_REG_IDM[8:0]                                                                                             |         |        | VCOM Output level |                   |                |    |    |             |                 |    |    |     |
|               | 000h                                                                                                                                                               |         |        | 0V                |                   |                |    |    |             |                 |    |    |     |
|               | 001h                                                                                                                                                               |         |        | -0.0125V          |                   |                |    |    |             |                 |    |    |     |
|               | 002h                                                                                                                                                               |         |        | -0.0250V          |                   |                |    |    |             |                 |    |    |     |
|               | :                                                                                                                                                                  |         |        | :                 |                   |                |    |    |             |                 |    |    |     |
|               | :                                                                                                                                                                  |         |        | -0.0125V / Step   |                   |                |    |    |             |                 |    |    |     |
|               | :                                                                                                                                                                  |         |        | :                 |                   |                |    |    |             |                 |    |    |     |
|               | 116h                                                                                                                                                               |         |        | -3.4875V          |                   |                |    |    |             |                 |    |    |     |
|               | 117h                                                                                                                                                               |         |        | -3.5000V          |                   |                |    |    |             |                 |    |    |     |
|               | 118h ~ 1FFh                                                                                                                                                        |         |        | Reserved          |                   |                |    |    |             |                 |    |    |     |
|               | Restriction                                                                                                                                                        |         |        |                   |                   |                |    |    |             |                 |    |    |     |

## SETVGMN (5240h): VGMN Control

| 5240H       |                                                  | SETVGMN |        |           |             |           |    |                   |    |    |    |    |     |
|-------------|--------------------------------------------------|---------|--------|-----------|-------------|-----------|----|-------------------|----|----|----|----|-----|
| Instruction | R/W                                              | Address |        | Parameter |             |           |    |                   |    |    |    |    |     |
|             |                                                  | MIPI    | Others | D15-D8    | D7          | D6        | D5 | D4                | D3 | D2 | D1 | D0 | HEX |
| SETVGMN     | R/W                                              | 52h     | 5240h  | 00h       | VGMNTR[7:0] |           |    |                   |    |    |    |    | 60  |
| Description | This command is for VGMN output level selection. |         |        |           |             |           |    |                   |    |    |    |    |     |
|             |                                                  |         |        | Bit       |             | Value     |    | VGMN Output level |    |    |    |    |     |
|             | VGMNTR[8:0]                                      |         |        |           |             | 000h      |    | -3.0000V          |    |    |    |    |     |
|             |                                                  |         |        |           |             | 001h      |    | -3.0125V          |    |    |    |    |     |
|             |                                                  |         |        |           |             | :         |    | -0.0125V / Step   |    |    |    |    |     |
|             |                                                  |         |        |           |             | 060h      |    | -4.2000V          |    |    |    |    |     |
|             |                                                  |         |        |           |             | :         |    |                   |    |    |    |    |     |
|             |                                                  |         |        |           |             | 108h      |    | -6.3000V          |    |    |    |    |     |
|             |                                                  |         |        |           |             | 109h~1FFh |    | Reserved          |    |    |    |    |     |
|             |                                                  |         |        |           |             |           |    |                   |    |    |    |    |     |
|             |                                                  |         |        |           |             |           |    |                   |    |    |    |    |     |
| Restriction |                                                  |         |        |           |             |           |    |                   |    |    |    |    |     |

## SETVGSN (5340h): VGSN Control

| 0x5340h     |                                                  | SETVGSN |         |           |             |                   |                 |    |    |    |    |    |     |  |
|-------------|--------------------------------------------------|---------|---------|-----------|-------------|-------------------|-----------------|----|----|----|----|----|-----|--|
| Instruction | R/W                                              | Address |         | Parameter |             |                   |                 |    |    |    |    |    |     |  |
|             |                                                  | MIPI    | Others  | D15-D8    | D7          | D6                | D5              | D4 | D3 | D2 | D1 | D0 | HEX |  |
| SETVGSN     | R/W                                              | 0x53h   | 0x5340h | 00h       | VGSNTR[7:0] |                   |                 |    |    |    |    |    | 00  |  |
| Description | This command is for VGSN output level selection. |         |         |           |             |                   |                 |    |    |    |    |    |     |  |
|             | VGSNTR[8:0]                                      |         |         | Bit       | Value       | VGSN Output level |                 |    |    |    |    |    |     |  |
|             |                                                  |         |         |           |             | 000h              | 0.0000V         |    |    |    |    |    |     |  |
|             |                                                  |         |         |           |             | 001h              | -0.3000V        |    |    |    |    |    |     |  |
|             |                                                  |         |         |           |             | :                 | -0.0125V / Step |    |    |    |    |    |     |  |
|             |                                                  |         |         |           |             | 0A0h              | -2.2875V        |    |    |    |    |    |     |  |
|             |                                                  |         |         |           |             | :                 |                 |    |    |    |    |    |     |  |
|             |                                                  |         |         |           |             | 111h              | -3.7000V        |    |    |    |    |    |     |  |
|             |                                                  |         |         |           |             | 112h~1FFh         | Reserved        |    |    |    |    |    |     |  |
| Restriction |                                                  |         |         |           |             |                   |                 |    |    |    |    |    |     |  |

**SETVGMP (5440h): VGMP Control**

| 5440H       |                                                  |         |          | SETVGMP   |             |                |                   |    |    |    |    |    |     |  |
|-------------|--------------------------------------------------|---------|----------|-----------|-------------|----------------|-------------------|----|----|----|----|----|-----|--|
| Instruction | R/W                                              | Address |          | Parameter |             |                |                   |    |    |    |    |    |     |  |
|             |                                                  | MIPI    | Others   | D15-D8    | D7          | D6             | D5                | D4 | D3 | D2 | D1 | D0 | HEX |  |
| SETVGMP     | R/W                                              | 54h     | 5440h    | 00h       | VGMPTR[7:0] |                |                   |    |    |    |    |    | 60  |  |
| Description | This command is for VGMP output level selection. |         |          |           |             |                |                   |    |    |    |    |    |     |  |
|             | VGMPTR[8:0]                                      |         |          | Bit       | Value       |                | VGMP Output level |    |    |    |    |    |     |  |
|             |                                                  |         |          | 000h      |             | 3.0000V        |                   |    |    |    |    |    |     |  |
|             |                                                  |         |          | 001h      |             | 3.0125V        |                   |    |    |    |    |    |     |  |
|             |                                                  |         |          | :         |             | 0.0125V / Step |                   |    |    |    |    |    |     |  |
|             |                                                  |         |          | 060h      |             | 4.2000V        |                   |    |    |    |    |    |     |  |
|             |                                                  |         |          | :         |             |                |                   |    |    |    |    |    |     |  |
|             |                                                  |         |          | 108h      |             | 6.3000V        |                   |    |    |    |    |    |     |  |
|             | 109h~1FFh                                        |         | Reserved |           |             |                |                   |    |    |    |    |    |     |  |
| Restriction |                                                  |         |          |           |             |                |                   |    |    |    |    |    |     |  |

**SETVGSP (5540h): VGSP Control**

| 5540H       |                                                  | SETVGSP |           |           |                |                   |    |    |    |    |    |    |     |
|-------------|--------------------------------------------------|---------|-----------|-----------|----------------|-------------------|----|----|----|----|----|----|-----|
| Instruction | R/W                                              | Address |           | Parameter |                |                   |    |    |    |    |    |    |     |
|             |                                                  | MIPI    | Others    | D15-D8    | D7             | D6                | D5 | D4 | D3 | D2 | D1 | D0 | HEX |
| SETVGSP     | R/W                                              | 55h     | 5540h     | 00h       | VGSPTR[7:0]    |                   |    |    |    |    |    |    | 00  |
| Description | This command is for VGSP output level selection. |         |           |           |                |                   |    |    |    |    |    |    |     |
|             | VGSPTR[8:0]                                      |         | Bit       | Value     |                | VGSP Output level |    |    |    |    |    |    |     |
|             |                                                  |         | 000h      |           | 0.0000V        |                   |    |    |    |    |    |    |     |
|             |                                                  |         | 001h      |           | 0.3000V        |                   |    |    |    |    |    |    |     |
|             |                                                  |         | :         |           | 0.0125V / Step |                   |    |    |    |    |    |    |     |
|             |                                                  |         | 0A0h      |           | 2.2875V        |                   |    |    |    |    |    |    |     |
|             |                                                  |         | :         |           |                |                   |    |    |    |    |    |    |     |
|             |                                                  |         | 111h      |           | 3.7000V        |                   |    |    |    |    |    |    |     |
|             |                                                  |         | 112h~1FFh |           | Reserved       |                   |    |    |    |    |    |    |     |
| Restriction |                                                  |         |           |           |                |                   |    |    |    |    |    |    |     |

**SETVG (5640h): MSB of VGMP / VGSN / VGMN / VGSP**

| 5640H              | SETVG                                                                                                         |         |        |           |               |               |               |               |    |    |    |    |     |
|--------------------|---------------------------------------------------------------------------------------------------------------|---------|--------|-----------|---------------|---------------|---------------|---------------|----|----|----|----|-----|
| Instruction        | R/W                                                                                                           | Address |        | Parameter |               |               |               |               |    |    |    |    |     |
|                    |                                                                                                               | MIPI    | Others | D15-D8    | D7            | D6            | D5            | D4            | D3 | D2 | D1 | D0 | HEX |
| <b>SETVG</b>       | R/W                                                                                                           | 56h     | 5640h  | 00h       | VGMN<br>TR[8] | VGSN<br>TR[8] | VGMP<br>TR[8] | VGSP<br>TR[8] | -  | -  | -  | -  | 00  |
| <b>Description</b> | This command is used for VGMN/VGMP/VGSP/VGSN output level control. Please refer to registers 0x5240h~0x5540h. |         |        |           |               |               |               |               |    |    |    |    |     |
| <b>Restriction</b> |                                                                                                               |         |        |           |               |               |               |               |    |    |    |    |     |

**SW\_CTRL (5F40h): Interface Control for PFM and MIPI**

| 5F40H        |                                                                                                                                                                                                |         |        | PFMCTR                         |                                                                                                      |               |    |    |    |                |    |    |     |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------|--------------------------------|------------------------------------------------------------------------------------------------------|---------------|----|----|----|----------------|----|----|-----|
| Instruction  | R/W                                                                                                                                                                                            | Address |        | Parameter                      |                                                                                                      |               |    |    |    |                |    |    |     |
|              |                                                                                                                                                                                                | MIPI    | Others | D15-D8                         | D7                                                                                                   | D6            | D5 | D4 | D3 | D2             | D1 | D0 | HEX |
| SW_CTRL      | R/W                                                                                                                                                                                            | 5Fh     | 5F40h  | 00h                            | 0                                                                                                    | PFM_TYPE[2:0] |    | 0  | 0  | LANSEL_sw[1:0] |    | 13 |     |
| Description  | This command is used to setup PFM Mode as well as MIPI lane number selection for customer's application. The related power generation schemes under 2-power mode are below as following table. |         |        |                                |                                                                                                      |               |    |    |    |                |    |    |     |
|              | PFM_TYPE[2:0]                                                                                                                                                                                  |         |        |                                | Description                                                                                          |               |    |    |    |                |    |    |     |
|              | 000                                                                                                                                                                                            |         |        |                                | 2-power mode (VDDI / VCI)<br>AVDD and AVEE share the same inductor<br>- 2x power MOS + 1x inductor   |               |    |    |    |                |    |    |     |
|              | 001 (Default)                                                                                                                                                                                  |         |        |                                | 2-power mode (VDDI / VCI)<br>AVDD and AVEE use different inductors<br>- 2x power MOS + 2x inductors  |               |    |    |    |                |    |    |     |
|              | 011                                                                                                                                                                                            |         |        |                                | 2-power mode (VDDI / VCI)<br>AVDD: Generated by PFM<br>AVEE: Generated by internal C-pump            |               |    |    |    |                |    |    |     |
|              | 100                                                                                                                                                                                            |         |        |                                | 2-power mode (VDDI / AVDD)<br>AVDD: From external power supply<br>AVEE: Generated by internal C-pump |               |    |    |    |                |    |    |     |
|              | 101                                                                                                                                                                                            |         |        |                                | 2-power mode (VDDI / AVDD)<br>AVDD: From external power supply<br>AVEE: Generated by PFM             |               |    |    |    |                |    |    |     |
|              | 110                                                                                                                                                                                            |         |        |                                | 2-power mode (VDDI / AVDD)<br>AVDD: Generated by PFM<br>AVEE: Generated by external C-pump           |               |    |    |    |                |    |    |     |
|              | LANSEL_SW[1:0]                                                                                                                                                                                 |         |        |                                | Description                                                                                          |               |    |    |    |                |    |    |     |
|              | 00                                                                                                                                                                                             |         |        |                                | Reserved                                                                                             |               |    |    |    |                |    |    |     |
| 01           |                                                                                                                                                                                                |         |        | MIPI HS is 2-Lane input/output |                                                                                                      |               |    |    |    |                |    |    |     |
| 10           |                                                                                                                                                                                                |         |        | MIPI HS is 3-Lane input/output |                                                                                                      |               |    |    |    |                |    |    |     |
| 11 (Default) |                                                                                                                                                                                                |         |        | MIPI HS is 4-Lane input/output |                                                                                                      |               |    |    |    |                |    |    |     |
| Restriction  | (1) PFM_TYPE[2:0] is valid only when BSTM[2:0]=1x1                                                                                                                                             |         |        |                                |                                                                                                      |               |    |    |    |                |    |    |     |

**WRDDB (E140h~E440h): Read/Write Supplier ID Code Information**

| E140H       |                                                                                           | WRDDB   |        |                                   |           |    |    |    |    |    |    |    |     |
|-------------|-------------------------------------------------------------------------------------------|---------|--------|-----------------------------------|-----------|----|----|----|----|----|----|----|-----|
| Instruction | R/W                                                                                       | Address |        | Parameter                         |           |    |    |    |    |    |    |    |     |
|             |                                                                                           | MIPI    | Others | D15-D8                            | D7        | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |
| WRDDB       | R/W                                                                                       | E1h     | E140h  | 00h                               | SID[7:0]  |    |    |    |    |    |    |    | 00  |
| WRDDB       | R/W                                                                                       | E2h     | E240h  | 00h                               | SID[15:8] |    |    |    |    |    |    |    | 00  |
| WRDDB       | R/W                                                                                       | E3h     | E340h  | 00h                               | MID[7:0]  |    |    |    |    |    |    |    | 00  |
| WRDDB       | R/W                                                                                       | E4h     | E440h  | 00h                               | MID[15:8] |    |    |    |    |    |    |    | 00  |
| Description | These commands are for Supplier ID Code and Supplier elective data such as module number. |         |        |                                   |           |    |    |    |    |    |    |    |     |
|             | Bit                                                                                       |         |        | Description                       |           |    |    |    |    |    |    |    |     |
|             | SID[7:0]                                                                                  |         |        | LS Byte of Supplier ID            |           |    |    |    |    |    |    |    |     |
|             | SID[15:8]                                                                                 |         |        | MS Byte of Supplier ID            |           |    |    |    |    |    |    |    |     |
|             | MID[7:0]                                                                                  |         |        | LS Byte of Supplier Elective Data |           |    |    |    |    |    |    |    |     |
|             | MID[15:8]                                                                                 |         |        | MS Byte of Supplier Elective Data |           |    |    |    |    |    |    |    |     |
| Restriction |                                                                                           |         |        |                                   |           |    |    |    |    |    |    |    |     |

**WRDID (E540h~E740h): Read/Write Device ID Code**

| 0xE540h     |                                                                                                         | WRDID   |        |                              |          |    |    |    |    |    |    |    |     |
|-------------|---------------------------------------------------------------------------------------------------------|---------|--------|------------------------------|----------|----|----|----|----|----|----|----|-----|
| Instruction | R/W                                                                                                     | Address |        | Parameter                    |          |    |    |    |    |    |    |    |     |
|             |                                                                                                         | MIPI    | Others | D15-D8                       | D7       | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |
| WRDID       | R/W                                                                                                     | E5h     | E540h  | 00h                          | ID1[7:0] |    |    |    |    |    |    |    | 00  |
| WRDID       | R/W                                                                                                     | E6h     | E640h  | 00h                          | ID2[7:0] |    |    |    |    |    |    |    | 80  |
| WRDID       | R/W                                                                                                     | E7h     | E740h  | 00h                          | ID3[7:0] |    |    |    |    |    |    |    | 00  |
| Description | These commands are for Module Manufacture Number, Module/Driver Version Number, and Module / Driver ID. |         |        |                              |          |    |    |    |    |    |    |    |     |
|             | Bit                                                                                                     |         |        | Description                  |          |    |    |    |    |    |    |    |     |
|             | ID1[7:0]                                                                                                |         |        | Module Manufacture Number    |          |    |    |    |    |    |    |    |     |
|             | ID2[7:0]                                                                                                |         |        | Module/Driver Version Number |          |    |    |    |    |    |    |    |     |
|             | ID3[7:0]                                                                                                |         |        | Module /Driver ID            |          |    |    |    |    |    |    |    |     |
| Restriction |                                                                                                         |         |        |                              |          |    |    |    |    |    |    |    |     |



**RDMPFLG (EE40h): Page Status for MTP Programming**

| EE40H       |                                                           |         |                                    | RDMPFLG   |               |               |              |              |              |              |                                    |               |     |
|-------------|-----------------------------------------------------------|---------|------------------------------------|-----------|---------------|---------------|--------------|--------------|--------------|--------------|------------------------------------|---------------|-----|
| Instruction | R/W                                                       | Address |                                    | Parameter |               |               |              |              |              |              |                                    |               |     |
|             |                                                           | MIPI    | Others                             | D15-D8    | D7            | D6            | D5           | D4           | D3           | D2           | D1                                 | D0            | HEX |
| RDMPFLG     | R                                                         | EEh     | EE40h                              | 00h       | CMD2_3rd_Done | CMD2_2nd_Done | CMD2_P0_Done | CMD2_P1_Done | CMD2_P2_Done | CMD2_P3_Done | CMD2_P4_Done                       | CMD2_GAM_Done | 00h |
| Description | This command is to check MTP status of mapping CMD Pages. |         |                                    |           |               |               |              |              |              |              |                                    |               |     |
|             | Bit                                                       |         | Description                        |           |               |               |              |              |              |              | Data                               |               |     |
|             | CMD2_3rd_Done                                             |         | CMD2 P0 3 <sup>rd</sup> MTP Status |           |               |               |              |              |              |              | 0: Not Programmed<br>1: Programmed |               |     |
|             | CMD2_2nd_Done                                             |         | CMD2 P0 2nd MTP Status             |           |               |               |              |              |              |              | 0: Not Programmed<br>1: Programmed |               |     |
|             | CMD2P0_Done                                               |         | CMD2 P0 1 <sup>st</sup> MTP Status |           |               |               |              |              |              |              | 0: Not Programmed<br>1: Programmed |               |     |
|             | CMD2P1_Done                                               |         | CMD2 P1 MTP Status                 |           |               |               |              |              |              |              | 0: Not Programmed<br>1: Programmed |               |     |
|             | CMD2P2_Done                                               |         | CMD2 P2 MTP Status                 |           |               |               |              |              |              |              | 0: Not Programmed<br>1: Programmed |               |     |
|             | CMD2P3_Done                                               |         | CMD2 P3 MTP Status                 |           |               |               |              |              |              |              | 0: Not Programmed<br>1: Programmed |               |     |
|             | CMD2_gam_Done                                             |         | CMD2 Gamma voltages MTP Status     |           |               |               |              |              |              |              | 0: Not Programmed<br>1: Programmed |               |     |
|             |                                                           |         |                                    |           |               |               |              |              |              |              |                                    |               |     |
| Restriction |                                                           |         |                                    |           |               |               |              |              |              |              |                                    |               |     |

**MTPCTR (F040h): MTP Program Page Enable Control**

| F040H       |                                                                            |         |        | MTPCTR    |                                                    |            |              |              |                         |              |              |               |     |
|-------------|----------------------------------------------------------------------------|---------|--------|-----------|----------------------------------------------------|------------|--------------|--------------|-------------------------|--------------|--------------|---------------|-----|
| Instruction | R/W                                                                        | Address |        | Parameter |                                                    |            |              |              |                         |              |              |               |     |
|             |                                                                            | MIPI    | Others | D15-D8    | D7                                                 | D6         | D5           | D4           | D3                      | D2           | D1           | D0            | HEX |
| MTPCTR      | R/W                                                                        | F0h     | F040h  | 00h       | CMD2_3rdEN                                         | CMD2_2ndEN | CMD2_P0_PGEN | CMD2_P1_PGEN | CMD2_P2_PGEN            | CMD2_P3_PGEN | CMD2_P4_PGEN | CMD2_gam_PGEN | 00  |
| Description | This command is to enable MTP program function of corresponding CMD pages. |         |        |           |                                                    |            |              |              |                         |              |              |               |     |
|             | Bit                                                                        |         |        |           | Description                                        |            |              |              | Data                    |              |              |               |     |
|             | CMD2_3rdEN                                                                 |         |        |           | Enable CMD2 Page0 3 <sup>rd</sup> program function |            |              |              | 0: Disable<br>1: Enable |              |              |               |     |
|             | CMD2_2ndEN                                                                 |         |        |           | Enable CMD2 Page0 2 <sup>nd</sup> program function |            |              |              | 0: Disable<br>1: Enable |              |              |               |     |
|             | CMD2_P0_PGEN                                                               |         |        |           | Enable CMD2 Page0 1 <sup>st</sup> program function |            |              |              | 0: Disable<br>1: Enable |              |              |               |     |
|             | CMD2_P1_PGEN                                                               |         |        |           | Enable CMD2 Page1 program function                 |            |              |              | 0: Disable<br>1: Enable |              |              |               |     |
|             | CMD2_P2_PGEN                                                               |         |        |           | Enable CMD2 Page2 program function                 |            |              |              | 0: Disable<br>1: Enable |              |              |               |     |
|             | CMD2_P3_PGEN                                                               |         |        |           | Enable CMD2 Page3 program function                 |            |              |              | 0: Disable<br>1: Enable |              |              |               |     |
|             | CMD2_gam_PGEN                                                              |         |        |           | Enable CMD2 Gamma voltage program function         |            |              |              | 0: Disable<br>1: Enable |              |              |               |     |
| Restriction |                                                                            |         |        |           |                                                    |            |              |              |                         |              |              |               |     |

**WMTP (F240h): MTP Programming Start and Power Selection**

| F240H       |                                                  | WMTP    |                                 |           |                                                       |    |    |    |    |    |            |       |       |
|-------------|--------------------------------------------------|---------|---------------------------------|-----------|-------------------------------------------------------|----|----|----|----|----|------------|-------|-------|
| Instruction | R/W                                              | Address |                                 | Parameter |                                                       |    |    |    |    |    |            |       |       |
|             |                                                  | MIPI    | Others                          | D15-D8    | D7                                                    | D6 | D5 | D4 | D3 | D2 | D1         | D0    | HEX   |
| WMTP        | R/W                                              | 0xF2h   | 0xF240h                         | 00h       | MTP_LOAD_EN                                           | -  | -  | -  | -  | -  | EXT_MTPPWR | MTP_W | 0x02h |
| Description | This command is to control MTP Program function. |         |                                 |           |                                                       |    |    |    |    |    |            |       |       |
|             | Bit                                              |         | Description                     |           | Data                                                  |    |    |    |    |    |            |       |       |
|             | MTP_W                                            |         | MTP Program function control    |           | 0 : No Programming<br>1 : Execute Programming         |    |    |    |    |    |            |       |       |
|             | EXT_MTPPWR                                       |         | MTP Programming power selection |           | 0 : Internal trim power<br>1 : External trim power    |    |    |    |    |    |            |       |       |
| Restriction | MTP_LOAD_EN                                      |         | MTP data reload option          |           | 0 : Normal reload scheme<br>1 : Force MTP data reload |    |    |    |    |    |            |       |       |

**MTPW (F340h~F540h): Key Check for MTP Programming**

| F340H ~ F540H |                                 | MTPW    |        |           |    |    |    |    |    |    |    |    |     |
|---------------|---------------------------------|---------|--------|-----------|----|----|----|----|----|----|----|----|-----|
| Instruction   | R/W                             | Address |        | Parameter |    |    |    |    |    |    |    |    |     |
|               |                                 | MIPI    | Others | D15-D8    | D7 | D6 | D5 | D4 | D3 | D2 | D1 | D0 | HEX |
| MTPW          | W                               | F3h     | F340h  | 00h       | 1  | 0  | 1  | 0  | 0  | 1  | 0  | 1  | A5  |
| MTPW          | W                               | F4h     | F440h  | 00h       | 0  | 1  | 0  | 1  | 1  | 0  | 1  | 0  | 5A  |
| MTPW          | W                               | F5h     | F540h  | 00h       | 0  | 0  | 1  | 1  | 1  | 1  | 0  | 0  | 3C  |
| Description   | Key check for MTP write command |         |        |           |    |    |    |    |    |    |    |    |     |
| Restriction   |                                 |         |        |           |    |    |    |    |    |    |    |    |     |

**MGRDMTP (F640h): MTP Read Mode Selection**

| F640H       |                                               |         |                            | MGRDMTP   |    |    |    |      |                  |    |          |    |     |
|-------------|-----------------------------------------------|---------|----------------------------|-----------|----|----|----|------|------------------|----|----------|----|-----|
| Instruction | R/W                                           | Address |                            | Parameter |    |    |    |      |                  |    |          |    |     |
|             |                                               | MIPI    | Others                     | D15-D8    | D7 | D6 | D5 | D4   | D3               | D2 | D1       | D0 | HEX |
| MGRDMTP     | R/W                                           | F6h     | F640h                      | 00h       | 0  | 0  | 0  | 0    | 0                | 0  | PTM[1:0] |    | 00  |
| Description | This command is to select MTP read condition. |         |                            |           |    |    |    |      |                  |    |          |    |     |
|             | Bit                                           |         | Description                |           |    |    |    | Data |                  |    |          |    |     |
|             | PTM[1:0]                                      |         | Select MTP read condition. |           |    |    |    | 00   | Normal Read      |    |          |    |     |
|             |                                               |         |                            |           |    |    |    | 01   | Reserved         |    |          |    |     |
|             |                                               |         |                            |           |    |    |    | 10   | Reserved         |    |          |    |     |
|             |                                               |         |                            |           |    |    |    | 11   | Margin Read Mode |    |          |    |     |
|             |                                               |         |                            |           |    |    |    |      |                  |    |          |    |     |
| Restriction |                                               |         |                            |           |    |    |    |      |                  |    |          |    |     |

## Gamma VP1~VP16(6080h~6F80h): Positive Gamma Setting

| 6080H ~ 6F80H |                                                                                                                                                                                                               | Positive Gamma |        |           |    |    |             |             |    |    |    |    |     |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------|-----------|----|----|-------------|-------------|----|----|----|----|-----|
| Instruction   | R/W                                                                                                                                                                                                           | Address        |        | Parameter |    |    |             |             |    |    |    |    |     |
|               |                                                                                                                                                                                                               | MIPI           | Others | D15-D8    | D7 | D6 | D5          | D4          | D3 | D2 | D1 | D0 | HEX |
| Gamma VP1     | R/W                                                                                                                                                                                                           | 60h            | 6080h  | 00h       | -  | -  | P_V0[5:0]   |             |    |    |    |    | 00  |
| Gamma VP2     | R/W                                                                                                                                                                                                           | 61h            | 6180h  | 00h       | -  | -  | P_V4[5:0]   |             |    |    |    |    | 12  |
| Gamma VP3     | R/W                                                                                                                                                                                                           | 62h            | 6280h  | 00h       | -  | -  | P_V8[5:0]   |             |    |    |    |    | 19  |
| Gamma VP4     | R/W                                                                                                                                                                                                           | 63h            | 6380h  | 00h       | -  | -  | -           | P_V16[4:0]  |    |    |    |    | 0E  |
| Gamma VP5     | R/W                                                                                                                                                                                                           | 64h            | 6480h  | 00h       | -  | -  | -           | P_V24[4:0]  |    |    |    |    | 06  |
| Gamma VP6     | R/W                                                                                                                                                                                                           | 65h            | 6580h  | 00h       | -  | -  | -           | P_V52[4:0]  |    |    |    |    | 14  |
| Gamma VP7     | R/W                                                                                                                                                                                                           | 66h            | 6680h  | 00h       | -  | -  | -           | P_V80[4:0]  |    |    |    |    | 10  |
| Gamma VP8     | R/W                                                                                                                                                                                                           | 67h            | 6780h  | 00h       | -  | -  | -           | P_V108[4:0] |    |    |    |    | 0C  |
| Gamma VP9     | R/W                                                                                                                                                                                                           | 68h            | 6880h  | 00h       | -  | -  | -           | P_V147[4:0] |    |    |    |    | 16  |
| Gamma VP10    | R/W                                                                                                                                                                                                           | 69h            | 6980h  | 00h       | -  | -  | -           | P_V175[4:0] |    |    |    |    | 0C  |
| Gamma VP11    | R/W                                                                                                                                                                                                           | 6Ah            | 6A80h  | 00h       | -  | -  | -           | P_V203[4:0] |    |    |    |    | 0E  |
| Gamma VP12    | R/W                                                                                                                                                                                                           | 6Bh            | 6B80h  | 00h       | -  | -  | -           | P_V231[4:0] |    |    |    |    | 08  |
| Gamma VP13    | R/W                                                                                                                                                                                                           | 6Ch            | 6C80h  | 00h       | -  | -  | -           | P_V239[4:0] |    |    |    |    | 0F  |
| Gamma VP14    | R/W                                                                                                                                                                                                           | 6Dh            | 6D80h  | 00h       | -  | -  | P_V247[5:0] |             |    |    |    |    | 11  |
| Gamma VP15    | R/W                                                                                                                                                                                                           | 6Eh            | 6E80h  | 00h       | -  | -  | P_V251[5:0] |             |    |    |    |    | 0C  |
| Gamma VP16    | R/W                                                                                                                                                                                                           | 6Fh            | 6F80h  | 00h       | -  | -  | P_V255[5:0] |             |    |    |    |    | 00  |
| Description   | These commands are used to adjust nodal voltages for Positive Gamma Node0, Node4, Node8, Node16, Node24, Node52, Node80, Node108, Node147, Node175, Node203, Node231, Node239, Node247, Node251, and Node255. |                |        |           |    |    |             |             |    |    |    |    |     |
| Restriction   | User needs to update these parameters under sleep-in mode.                                                                                                                                                    |                |        |           |    |    |             |             |    |    |    |    |     |

## Gamma VN1~VN16(7080h~7F80h): Negative Gamma Setting

| 7080H        |                                                                                                                                                                                                               |         |        | Negative Gamma |    |    |             |             |    |    |    |              |     |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------|----------------|----|----|-------------|-------------|----|----|----|--------------|-----|
| Instruction  | R/W                                                                                                                                                                                                           | Address |        | Parameter      |    |    |             |             |    |    |    |              |     |
|              |                                                                                                                                                                                                               | MIPI    | Others | D15-D8         | D7 | D6 | D5          | D4          | D3 | D2 | D1 | D0           | HEX |
| Gamma VN1    | R/W                                                                                                                                                                                                           | 70h     | 7080h  | 00h            | -  | -  | N_V0[5:0]   |             |    |    |    |              | 00  |
| Gamma VN2    | R/W                                                                                                                                                                                                           | 71h     | 7180h  | 00h            | -  | -  | N_V4[5:0]   |             |    |    |    |              | 12  |
| Gamma VN3    | R/W                                                                                                                                                                                                           | 72h     | 7280h  | 00h            | -  | -  | N_V8[5:0]   |             |    |    |    |              | 19  |
| Gamma VN4    | R/W                                                                                                                                                                                                           | 73h     | 7380h  | 00h            | -  | -  | -           | N_V16[4:0]  |    |    |    |              | 0E  |
| Gamma VN5    | R/W                                                                                                                                                                                                           | 74h     | 7480h  | 00h            | -  | -  | -           | N_V24[4:0]  |    |    |    |              | 06  |
| Gamma VN6    | R/W                                                                                                                                                                                                           | 75h     | 7580h  | 00h            | -  | -  | -           | N_V52[4:0]  |    |    |    |              | 14  |
| Gamma VN7    | R/W                                                                                                                                                                                                           | 76h     | 7680h  | 00h            | -  | -  | -           | N_V80[4:0]  |    |    |    |              | 10  |
| Gamma VN8    | R/W                                                                                                                                                                                                           | 77h     | 7780h  | 00h            | -  | -  | -           | N_V108[5:0] |    |    |    |              | 0C  |
| Gamma VN9    | R/W                                                                                                                                                                                                           | 78h     | 7880h  | 00h            | -  | -  | -           | N_V147[4:0] |    |    |    |              | 16  |
| Gamma VN10   | R/W                                                                                                                                                                                                           | 79h     | 7980h  | 00h            | -  | -  | -           | N_V175[4:0] |    |    |    |              | 0C  |
| Gamma VN11   | R/W                                                                                                                                                                                                           | 7Ah     | 7A80h  | 00h            | -  | -  | -           | N_V203[4:0] |    |    |    |              | 0E  |
| Gamma VN12   | R/W                                                                                                                                                                                                           | 7Bh     | 7B80h  | 00h            | -  | -  | -           | N_V231[4:0] |    |    |    |              | 08  |
| Gamma VN13   | R/W                                                                                                                                                                                                           | 7Ch     | 7C80h  | 00h            | -  | -  | -           | N_V239[4:0] |    |    |    |              | 0F  |
| Gamma VN14   | R/W                                                                                                                                                                                                           | 7Dh     | 7D80h  | 00h            | -  | -  | N_V247[5:0] |             |    |    |    |              | 11  |
| Gamma VN15   | R/W                                                                                                                                                                                                           | 7Eh     | 7E80h  | 00h            | -  | -  | N_V251[5:0] |             |    |    |    |              | 0C  |
| Gamma VN16   | R/W                                                                                                                                                                                                           | 7Fh     | 7F80h  | 00h            | -  | -  | N_V255[5:0] |             |    |    |    |              | 00  |
| Gamma VN16   | R/W                                                                                                                                                                                                           | 7Fh     | 7F80h  | 00h            | -  | -  | N_V255[5:0] |             |    |    |    |              | 00  |
| Gamma Update | R/W                                                                                                                                                                                                           | 80h     | 8080h  | 00h            | -  | -  |             |             |    |    |    | gam_force_up | 00  |
| Description  | These commands are used to adjust nodal voltages for Negative Gamma Node0, Node4, Node8, Node16, Node24, Node52, Node80, Node108, Node147, Node175, Node203, Node231, Node239, Node247, Node251, and Node255. |         |        |                |    |    |             |             |    |    |    |              |     |
| Restriction  | User needs to update these parameters under sleep-in mode.                                                                                                                                                    |         |        |                |    |    |             |             |    |    |    |              |     |

## 9. Electrical Characteristics

### 9.1 Absolute Maximum Ratings

The absolute maximum rating is listed on following table. When device is operated beyond the absolute maximum ratings specified in this table, it may be permanently damaged. It is strongly recommended to use this device within the specified limits for normal operation. If these electrical characteristic conditions are exceeded during normal operation, this driver IC will malfunction and cause poor reliability.

| Item                       | Symbol            | Value   | Unit |
|----------------------------|-------------------|---------|------|
| Power supply voltage       | VDDI, VDDAM       | 3.6     | V    |
| Power supply voltage       | VDDA, VDDDB, VDDR | 6.6     | V    |
| Supply voltage (MV)        | AVDD-AVSS         | 6.6     | V    |
|                            | AVEE-AVSS         | 6.6     | V    |
| Supply voltage (HV)        | VGH – VGL         | 32      | V    |
| Differential input voltage | HSSI_CLK_P/N      | 1.8     | V    |
|                            | HSSI_DATA0_P/N    |         |      |
|                            | HSSI_DATA1_P/N    |         |      |
| Operating temperature      | Topr              | -40~85  | °C   |
| Storage temperature        | Tstg              | -40~125 | °C   |

### 9.2 ESD Protection Level

| Model           | Test Condition            | Level |
|-----------------|---------------------------|-------|
| Human Body Mode | R = 1.5 kohm / C = 100 pF | ≥ 2KV |
| Machine Mode    | R = 0 ohm / C = 200 pF    | ≥200V |

### 9.3 Latch-Up Protection Level

The device will not latch up at trigger current levels less than  $\pm 200\text{mA}$ .

## 9.4 DC Characteristics

### 9.4.1 Basic Characteristics

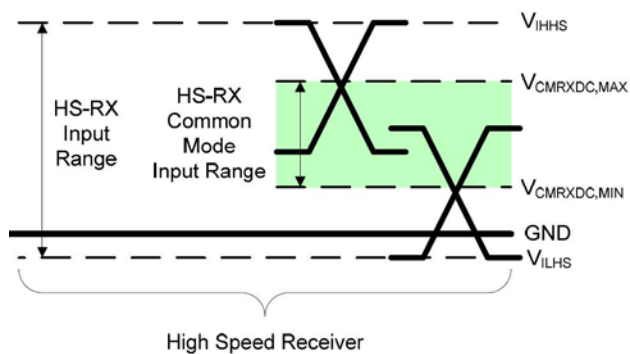
| Parameter                                    | Symbol               | Condition                              | Min.           | Typ. | Max.               | Unit |
|----------------------------------------------|----------------------|----------------------------------------|----------------|------|--------------------|------|
| Analog Power Supply Voltage                  | VCI                  | Operation Voltage                      | 2.5            | 3.3  | 4.8                | V    |
| I/O pin Power Supply Voltage                 | VDDI                 | I/O supply voltage                     | 1.65           | 1.8  | 3.3                | V    |
|                                              | MVDDL                | I/O supply voltage                     | 1.1            | 1.2  | 1.3                | V    |
| Logic High level input voltage               | VIH                  | VDDI = 1.65V ~ 3.3V                    | 0.7* VDDI      | -    | VDDI               | V    |
| Logic Low level input voltage                | VIL                  | VDDI = 1.65V ~ 3.3V                    | 0.0            | -    | 0.3* VDDI          | V    |
| Logic High level Output voltage              | VOH                  | Iout = -1 mA                           | 0.8* VDDI      | -    | VDDI               | V    |
| Logic Low level Output voltage               | VOL                  | Iout = +1 mA                           | 0.0            | -    | 0.2* VDDI          | V    |
| Logic High level input current (Except MIPI) | IIHD                 | Vin=0~VDDI                             |                |      | 1                  | uA   |
| Logic Low level input current (Except MIPI)  | IILD                 | Vin=0~VDDI                             | -1             |      |                    | uA   |
| Logic High level input current (MIPI)        | IIHD                 | Vin=0~MVDDL                            |                |      | 1                  | uA   |
| Logic Low level input current (MIPI)         | IILD                 | Vin=0~MVDDL                            | -1             |      |                    | uA   |
| AVDD booster voltage                         | AVDD                 |                                        | 4.5            |      | 6.5                | V    |
| AVEE booster voltage                         | AVEE                 |                                        | -6.5           |      | -4.5               | V    |
| VCL booster voltage                          | VCL                  |                                        | -2.5           |      | -4                 | V    |
| VGH booster voltage                          | VGH                  |                                        | 2AVDD<br>-AVEE |      | 3AVDD<br>-2AVEE    | V    |
| VGL booster voltage                          | VGL                  |                                        | AVEE<br>-AVDD  |      | 2AVEE<br>-AVDD-VCI | V    |
| Maximum voltage between VGH and VGL          | VGHL                 | VGH-VGL                                |                |      | 30                 | V    |
| Gamma reference voltage                      | VGMP                 |                                        | 3.0            |      | AVDD-0.1           | V    |
|                                              | VGSP                 |                                        | 0.0            |      | 3.7                | V    |
|                                              | VGMN                 |                                        | AVEE+0.1       |      | -3.0               | V    |
|                                              | VGSN                 |                                        | -3.7           |      | 0.0                | V    |
| Source output offset voltage                 | V <sub>OFFSET</sub>  |                                        |                |      | 45                 | mV   |
| Output deviation voltage (Positive output)   | V <sub>DEV,POS</sub> | Sout ≥ AVDD-1.0V, and 0V < Sout ≤ 1.0V |                |      | 20                 | mV   |
|                                              |                      | 1.0V < Sout < AVDD-1.0V                |                |      | 10                 | mV   |
| Output deviation voltage (Negative output)   | V <sub>DEV,NEG</sub> | Sout ≤ AVEE+1.0V, and -1V ≤ Sout ≤ 0V  |                |      | 20                 | mV   |
|                                              |                      | -1.0V > Sout > AVEE+1.0V               |                |      | 10                 | mV   |

Notes:

1. VCI means VDDA, VDDR, and Vddb.
2. Ta(ambient temperature) ranges from -30°C to 85 °C.
3. VSS means VSSA, VSSR, VSSB, and AVSS.

## 9.5 MIPI Characteristics

### 9.5.1 DC Specifications High-Speed Receiver Specification



| Parameter | Description                         | Min | Nom | Max | Units    | Note |
|-----------|-------------------------------------|-----|-----|-----|----------|------|
| VCMRX(DC) | Common-mode voltage for HS receiver | 70  |     | 330 | mV       | 1,2  |
| VIDTH     | Differential input high threshold   |     |     | 70  | mV       |      |
| VIDTL     | Differential input low threshold    | -70 |     |     | mV       |      |
| VIHHS     | Single-ended input high voltage     |     |     | 460 | mV       | 1    |
| VILHS     | Single-ended input low voltage      | -40 |     |     | mV       | 1    |
| ZID       | Differential input impedance        | 80  | 100 | 125 | $\Omega$ |      |

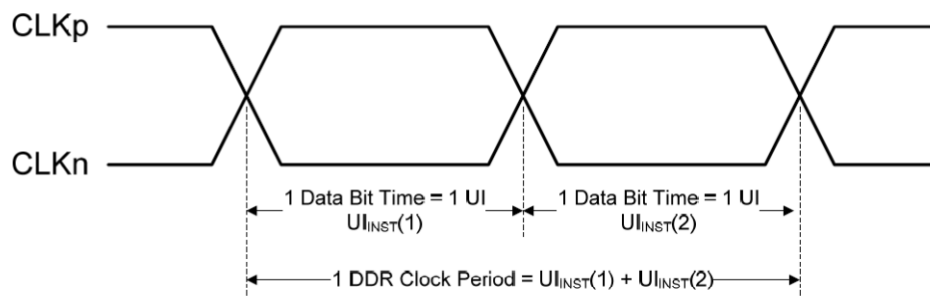
Notes:

1. Excluding possible additional RF interference of 100mV peak sine wave beyond 450MHz.
2. Values in this table include a ground difference of 50mV between the transmitter and the receiver, the static common-mode level tolerance and variations below 450MHz



## 9.5.2 Forward high speed transmissions

### DDR Clock Definition



| Clock Parameter  | Symbol      | Min | Typ | Max  | Units | Notes |
|------------------|-------------|-----|-----|------|-------|-------|
| UI instantaneous | $UI_{INST}$ |     |     | 12.5 | ns    | 1,2   |

#### Notes:

1. This value corresponds to a minimum 80 Mbps data rate.
2. The minimum UI shall not be violated for any single bit period, i.e., any DDR half cycle within a data burst.

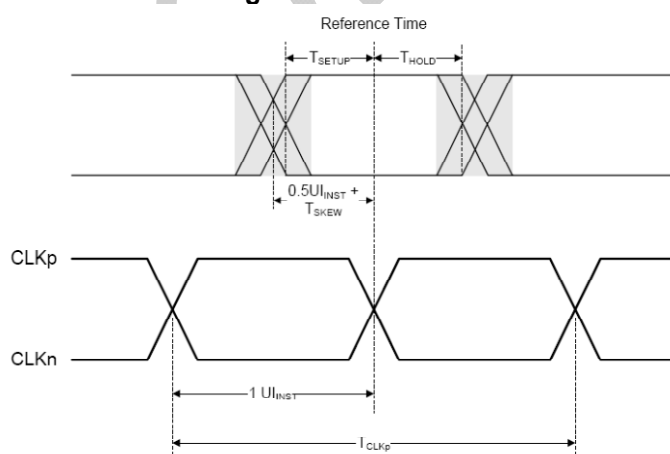
### Data-Clock Timing Specifications

| Parameter                                    | Symbol          | Min   | Typ | Max  | Units       | Notes |
|----------------------------------------------|-----------------|-------|-----|------|-------------|-------|
| Data to Clock Skew [measured at Transmitter] | $T_{SKEW[TX]}$  | -0.15 |     | 0.15 | $UI_{INST}$ | 1     |
| Data to Clock Setup Time [receiver]          | $T_{SETUP[RX]}$ | 0.15  |     |      | $UI_{INST}$ | 2     |
| Clock to Data Hold Time [receiver]           | $T_{HOLD[RX]}$  | 0.15  |     |      | $UI_{INST}$ | 2     |

#### Notes:

1. Total silicon and package delay budget of  $0.3 \cdot UI_{INST}$
2. Total setup and hold window for receiver of  $0.3 \cdot UI_{INST}$

### Data to Clock Timing Definitions

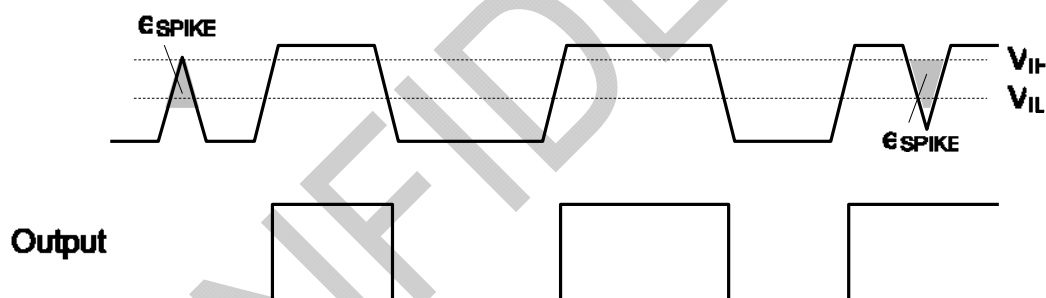


## 9.5.3 Low power transceiver specifications

| Parameters                     | Symbol   | Condition                              | Min | Typ | Max  | Unit |
|--------------------------------|----------|----------------------------------------|-----|-----|------|------|
| Logic high level input voltage | VIHCD    | Contention Detection (Lane_D0)         | 450 |     | 1350 | mV   |
| Logic low level input voltage  | VILCD    | Contention Detection (Lane_D0)         | 0   |     | 200  | mV   |
| Logic high level input voltage | VIH-LPRX | LP-Rx (Lane_CK, Lane_D0, Lane_D1)      | 880 | -   | 1350 | mV   |
| Logic low level input voltage  | VIL-LPRX | LP-Rx (Lane_CK, Lane_D0, Lane_D1)      | 0   |     | 550  | mV   |
| Logic low level input voltage  | VIL-ULPS | LP-Rx ULPS (Lane_CK, Lane_D0, Lane_D1) | 0   |     | 300  | mV   |
| Logic high level input voltage | VOH-LPTX | Contention Detection (Lane_D0)         | 1.1 | 1.2 | 1.3  | V    |
| Logic low level input voltage  | VOL-LPTX | Contention Detection (Lane_D0)         | -50 | 0   | 50   | mV   |
| eSPIKE <sup>(1,2,3)</sup>      | Fig. 2   | Input pulse rejection                  |     |     | 300  | V.ps |

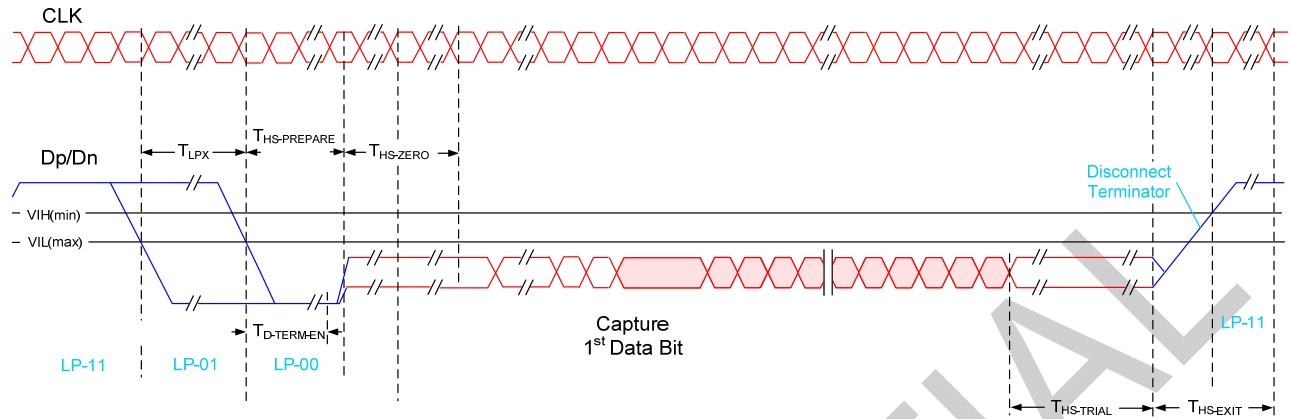
Notes:

- (1) Time-voltage integration of a spike above VIL when being in LP-0 state or below VIH when being in LP-1 state an impulse less than this will not change the receiver state.
- (2) In addition to the required glitch rejection, implementers shall ensure rejection of known RF-interferers. Input Glitch Rejection of Low Power Receivers as follow.

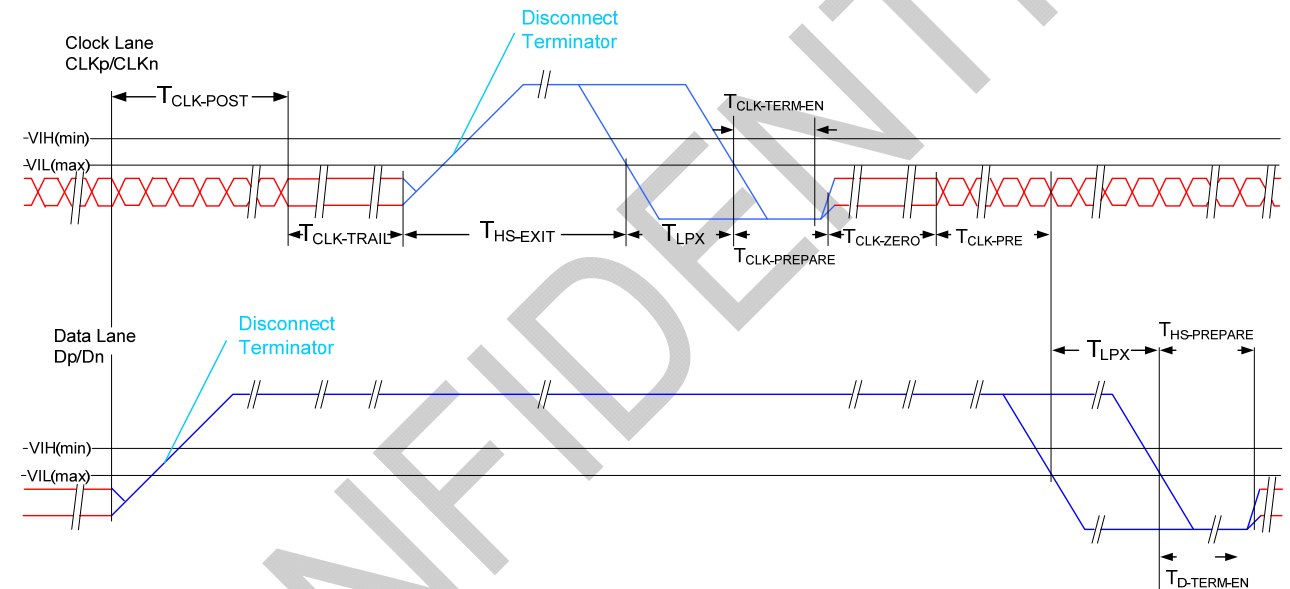


## 9.5.4 DSI Timing Characteristics

### HS Data Transmission Burst



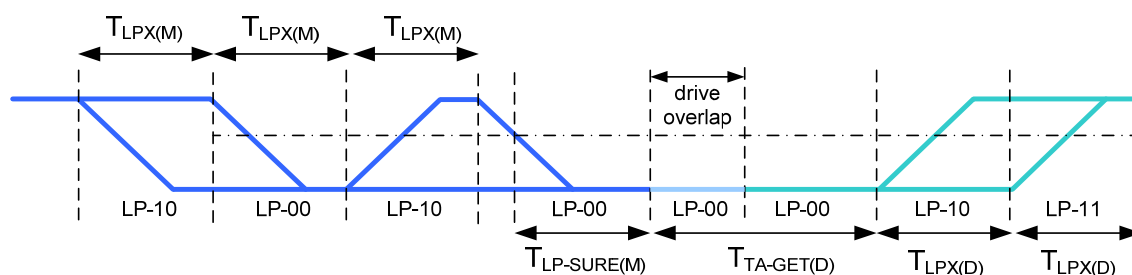
### HS clock transmission



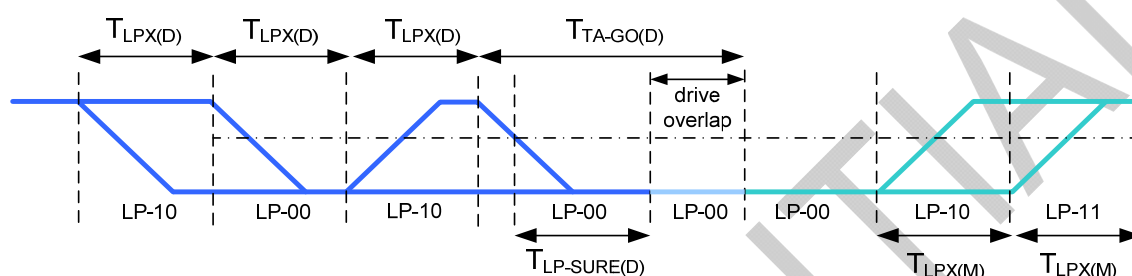
## Timing Parameters:

| Parameter                        | Description                                                                                                                                                                                                                  | Min                                | Typ | Max            | Unit |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|-----|----------------|------|
| $T_{CLK-POST}$                   | Time that the transmitter continues to send HS clock after the last associated Data Lane has transitioned to LP Mode. Interval is defined as the period from the end of $T_{HS-TRAIL}$ to the beginning of $T_{CLK-TRAIL}$ . | $60ns + 52*UI$                     |     |                | ns   |
| $T_{CLK-TRAIL}$                  | Time that the transmitter drives the HS-0 state after the last payload clock bit of a HS transmission burst.                                                                                                                 | 60                                 |     |                | ns   |
| $T_{HS-EXIT}$                    | Time that the transmitter drives LP-11 following a HS burst.                                                                                                                                                                 | 300                                |     |                | ns   |
| $T_{CLK-TERM-EN}$                | Time for the Clock Lane receiver to enable the HS line termination, starting from the time point when Dn crosses $V_{IL,MAX}$ .                                                                                              | Time for Dn to reach $V_{TERM-EN}$ |     | 38             | ns   |
| $T_{CLK-PREPARE}$                | Time that the transmitter drives the Clock Lane LP-00 Line state immediately before the HS-0 Line state starting the HS transmission.                                                                                        | 38                                 |     | 95             | ns   |
| $T_{CLK-PRE}$                    | Time that the HS clock shall be driven by the transmitter prior to any associated Data Lane beginning the transition from LP to HS mode.                                                                                     | 8                                  |     |                | UI   |
| $T_{CLK-PREPARE} + T_{CLK-ZERO}$ | $T_{CLK-PREPARE}$ + time that the transmitter drives the HS-0 state prior to starting the Clock.                                                                                                                             | 300                                |     |                | ns   |
| $T_{D-TERM-EN}$                  | Time for the Data Lane receiver to enable the HS line termination, starting from the time point when Dn crosses $V_{IL,MAX}$ .                                                                                               | Time for Dn to reach $V_{TERM-EN}$ |     | $35 ns + 4*UI$ |      |
| $T_{HS-PREPARE}$                 | Time that the transmitter drives the Data Lane LP-00 Line state immediately before the HS-0 Line state starting the HS transmission                                                                                          | $40ns + 4*UI$                      |     | $85 ns + 6*UI$ | ns   |
| $T_{HS-PREPARE} + T_{HS-ZERO}$   | $T_{HS-PREPARE}$ + time that the transmitter drives the HS-0 state prior to transmitting the Sync sequence.                                                                                                                  | $145ns + 10*UI$                    |     |                | ns   |
| $T_{HS-TRAIL}$                   | Time that the transmitter drives the flipped differential state after last payload data bit of a HS transmission burst                                                                                                       | $60ns + 4*UI$                      |     |                | ns   |

## Turnaround Procedure



## Bus turnaround (BAT) from MPU to display module timing



## Bus turnaround (BAT) from display module to MPU timing

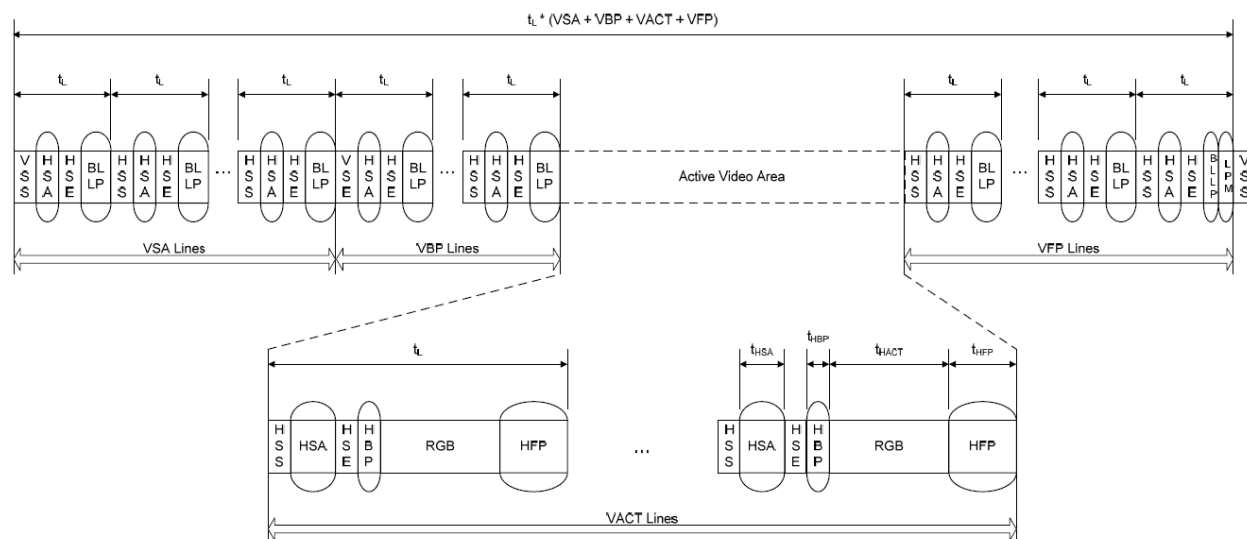
## Low Power Mode:

| Parameter        | Description                                                                                                                     | Min          | Typ                  | Max                  | Unit | Notes |
|------------------|---------------------------------------------------------------------------------------------------------------------------------|--------------|----------------------|----------------------|------|-------|
| $T_{LPX(M)}$     | Transmitted length of any Low-Power state period of MCU to display module                                                       | 50           |                      | 150                  | ns   | 1,2   |
| $T_{TA-SURE(M)}$ | Time that the display module waits after the LP-10 state before transmitting the Bridge state (LP-00) during a Link Turnaround. | $T_{LPX(M)}$ |                      | $2 \cdot T_{LPX(M)}$ | ns   | 2     |
| $T_{LPX(D)}$     | Transmitted length of any Low-Power state period of display module to MCU                                                       | 50           |                      | 150                  | ns   | 1,2   |
| $T_{TA-GET(D)}$  | Time that the display module drives the Bridge state (LP-00) after accepting control during a Link Turnaround.                  |              | $5 \cdot T_{LPX(D)}$ |                      | ns   | 2     |
| $T_{TA-GO(D)}$   | Time that the display module drives the Bridge state (LP-00) before releasing control during a Link Turnaround.                 |              | $4 \cdot T_{LPX(D)}$ |                      | ns   | 2     |
| $T_{TA-SURE(D)}$ | Time that the MPU waits after the LP-10 state before transmitting the Bridge state (LP-00) during a Link Turnaround.            | $T_{LPX(D)}$ |                      | $2 \cdot T_{LPX(D)}$ | ns   | 2     |

## NOTE:

1.  $T_{LPX}$  is an internal state machine timing reference. Externally measured values may differ slightly from the specified values due to asymmetrical rise and fall times.
2. Transmitter-specific parameter

## 9.5.5 DSI Video Mode Interface Timing



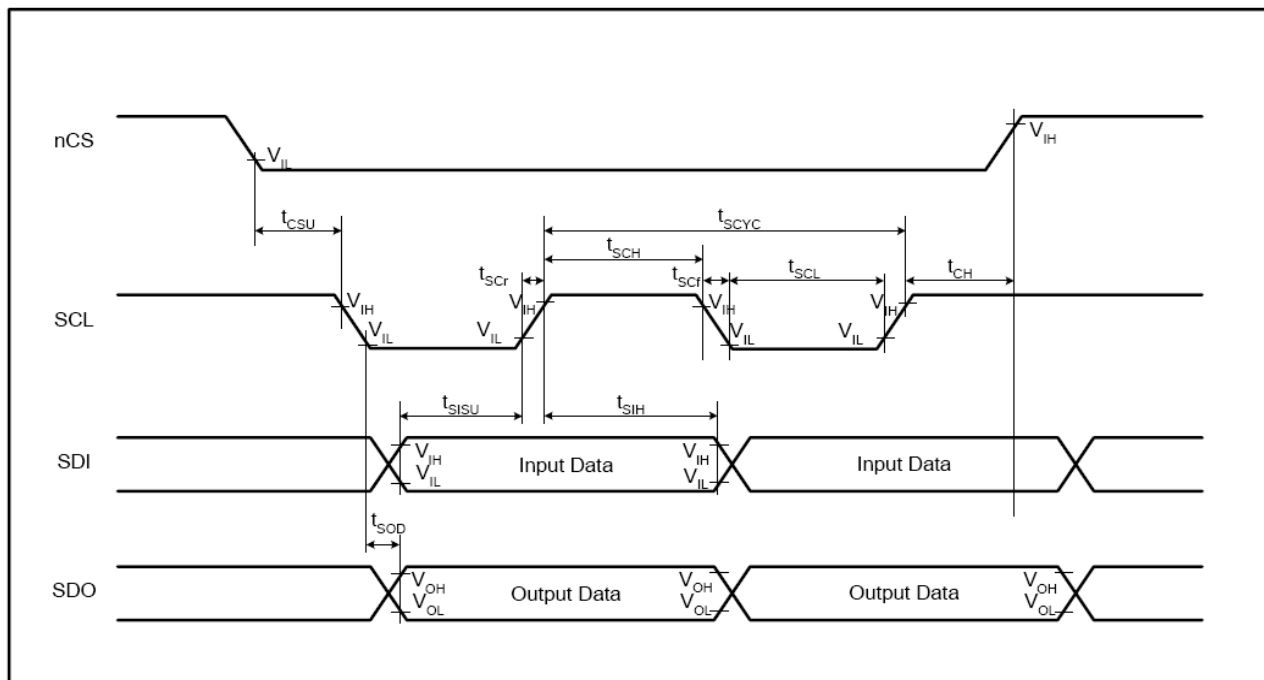
| Symbol     | Parameter                   | Condition          | Min        | Typ         | Max  | Units  |
|------------|-----------------------------|--------------------|------------|-------------|------|--------|
| BRPHY      | Bit rate total on all Lanes | WXGA               | 250        | -           | 850  | Mbps   |
| TBR34      | Total bit rate              | WXGA, 3 or 4 lanes | -          | -           | 2    | Gbps   |
| TBR2       | Total bit rate              | WXGA, 2 lanes      | -          | -           | 1.7  | Gbps   |
| $t_L$      | Line time                   | WXGA               | -          | 12.8, note1 | -    | us     |
| $t_{HBP}$  | Horizontal back porch       | WXGA               | 0.5        | -           | -    | us     |
| $t_{HACT}$ | Time for image data         | 4 lanes            | 9.6, note3 | -           | 19.2 | us     |
| $t_{HACT}$ | Active pixel per line       | WXGA               | -          | 800         | -    | pixels |
| $t_{HFP}$  | Horizontal front porch      | -                  | 0.5        | -           | -    | us     |
| VSA        | Vertical sync active        | -                  | 1          | -           | -    | H      |
| VBP        | Vertical back porch         | -                  | Note2      | 8           | -    | H      |
| VACT       | Active lines per frame      | WXGA               | -          | 1280        | -    | H      |
| VFP        | Vertical front porch        | -                  | 6          | 8           | -    | H      |

Note1: Frame rate (Typ)=60Hz, VBP=8, and VFP=8.

Note2: VBP(min) value depends on GOA timing.

Note3: 800pixels,  $t_{HACT}=24*800/2GHz=9.6us$

## 9.6 SPI Timing Characteristics

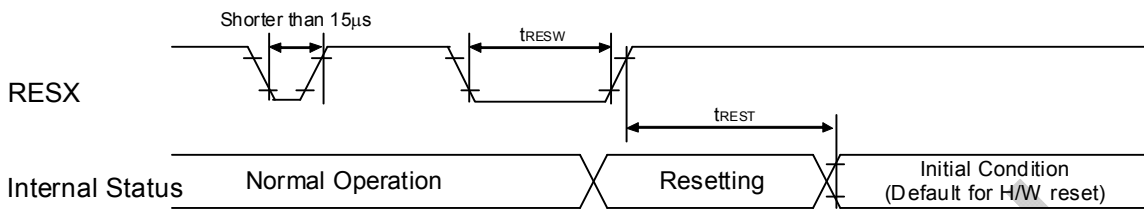


| Signal | Symbol     | Parameter                     | MIN | MAX | Unit | Description |
|--------|------------|-------------------------------|-----|-----|------|-------------|
| SCL    | $T_{SCYC}$ | Clock cycle (Write)           | 100 |     | ns   | -           |
|        | $T_{SCYC}$ | Clock cycle (Read)            | 300 |     | ns   |             |
|        | $T_{SCH}$  | Clock "H" pulse width (Write) | 40  |     | ns   |             |
|        | $T_{SCH}$  | Clock "H" pulse width (Read)  | 140 |     | ns   |             |
|        | $T_{SCL}$  | Clock "L" pulse width (Write) | 40  |     | ns   |             |
|        | $T_{SCL}$  | Clock "L" pulse width (Read)  | 140 |     | ns   |             |
|        | $T_{SCHr}$ | Clock rise time               |     | 5   | ns   |             |
|        | $T_{SCHf}$ | Clock fall time               |     | 5   | ns   |             |
| nCS    | $T_{CSU}$  | Chip select setup time        | 20  |     | ns   | -           |
|        | $T_{CH}$   | Chip select hold time         | 50  |     | ns   |             |
| SDI    | $T_{SISU}$ | Data input setup time         | 20  |     | ns   | -           |
|        | $T_{SIH}$  | Data input hold time          | 20  |     | ns   |             |
| SDO    | $T_{SOD}$  | Data output setup time        |     | 120 | ns   | -           |
|        | $T_{SOH}$  | Data output hold time         | 5   |     | ns   |             |

Note: Logic high and low levels are specified as 20% and 80% of IOVCC for Input signals.

Note:  $T_a = -30$  to  $70\text{ }^{\circ}\text{C}$ , IOVCC=1.65V to 3.3V, VCI=2.5V to 4.8V, GND=0V

## 9.7 Reset Timing Characteristics



Reset timing:

IOVCC=1.65V to 3.6V, AGND=DGND=0V, Ta=-40 to 85°C

| Symbol            | Parameter                 | Related Pins | MIN | TYP | MAX | Note                                     | Unit |
|-------------------|---------------------------|--------------|-----|-----|-----|------------------------------------------|------|
| t <sub>RESW</sub> | *1) Reset low pulse width | RESX         | 15  | -   | -   |                                          | μs   |
| t <sub>REST</sub> | *2) Reset complete time   | -            | -   | -   | 5   | When reset applied during sleep-in mode  | ms   |
|                   |                           | -            | -   | -   | 120 | When reset applied during sleep-out mode | ms   |

| RESX Pulse           | Action                                                             |
|----------------------|--------------------------------------------------------------------|
| Shorter than 5μs     | Reset Rejected                                                     |
| Longer than 15μs     | IC Reset                                                           |
| Between 5μs and 15μs | Reset starts<br>(It depends on voltage and temperature condition.) |

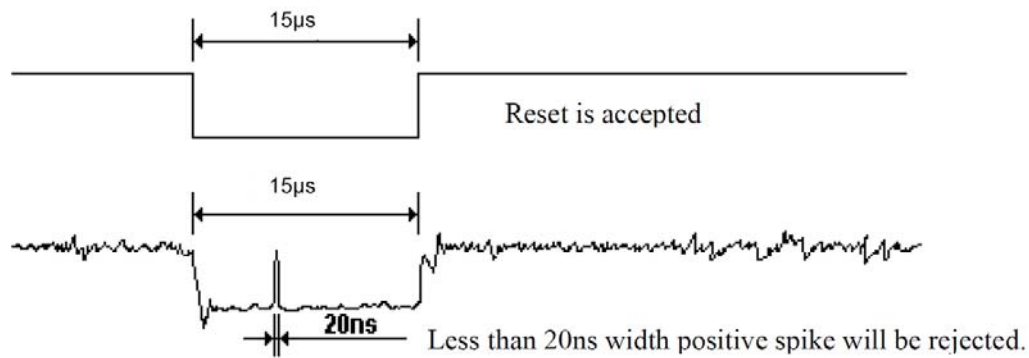
Note 1) Spike due to an electrostatic discharge on RESX line does not cause irregular system reset according to the table below.

Note 2. During the resetting period, the display will be blanked (The display is entering blanking sequence, which maximum time is 120 ms, when Reset Starts in Sleep Out –mode. The display remains the blank state in Sleep In –mode) and then return to Default condition for H/W reset.

Note 3. During Reset Complete Time, data in MTP will be latched to internal register during this period. This loading is done every time when there is H/W reset complete time (t<sub>REST</sub>) within 5ms after a rising edge of RESX.

Note 4. Spike Rejection also applies during a valid reset pulse as shown below:





*Note 5. It is necessary to wait 5msec after releasing RESX before sending commands. Also Sleep Out command cannot be sent for 120msec.*